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UNIVERSITY OF CALIFORNIA SAN DIEGO

(A)symmetries in the morphosyntax of embedded clauses in Khalkha Mongolian

A dissertation submitted in partial satisfaction of the
requirements for the degree Doctor of Philosophy

in

Linguistics

by

Jun Jie Lim

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2025

The dissertation of Jun Jie Lim is approved, and it is acceptable in quality and form for publication on microfilm and electronically.

University of California San Diego

2025

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LIST OF ABBREVIATIONS

1/2/3	first/second/third person	NEG	negation
ABL	ablative	NEG.COP	negative copula
ACC	accusative	NFN	nonfinite
ADN	adnominal	NFNOM	nonfactive nominalisation
ANAPH	anaphor	NMLZ	nominaliser
ASP	aspect	NOM	nominative
C	complementiser	NPST	nonpast
C.NML	nominalised complementiser	PASS	passive
CAUS	causative	PFV	perfective
CAUS.CVB	causative converb	PL	plural
CCMT.CVB	concomitant converb	POSS	possessive
CCSV.CVB	concessive converb	PQ	polar question marker
COM	comitative	PRES	present
COMPL	completive	PRF	perfect
CON	conclusive	PROG	progressive
COND.CVB	conditional converb	PROX	proximal
COP	copula	PST	past
CVB	converb	PTCP	participle
DAT	dative	Q	question marker
DIR	directive	RP	reflexive possessive
DIST	distal	SG	singular
EXCL	exclusive	SOC	sociative
EXCLM	exclamative	SUCC.CVB	successive converb
EVID.COP	evidential copula	SBJV	subjunctive
F/M	feminine/masculine	TEMP.CVB	temporal converb
FNOM	factive nominalisation	TERM	terminative (case)
FRML	formal	TERM.CVB	terminative converb
FUT	future	TOP	topic marker
GEN	genitive	VRN	verbal noun
GER	gerund	WHQ	<i>wh</i> -question marker
HAB	habitative		
INCL	inclusive		
IND	indicative		
INF	infinitive		
IFML	informal		
IMM.CVB	immediative converb		
IMPF	imperfect		
INSTR	instrumental		
IPFV	imperfective		

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ABSTRACT OF THE DISSERTATION

(A)symmetries in the morphosyntax of embedded clauses in Khalkha Mongolian

by

Jun Jie Lim

Doctor of Philosophy in Linguistics

University of California San Diego, 2025

Professor Emily Clem, Co-Chair

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This dissertation draws on original fieldwork data from two Mongolic languages – Khalkha Mongolian (henceforth Khalkha) and Kalmyk Oirat (henceforth Kalmyk) – to investigate issues pertaining to clausal nominalisation, syntactic locality, differential argument marking (DAM), and case assignment. The empirical focus is on four embedded clause types in Khalkha, supplemented by comparative data from Kalmyk. These clause types include three types of complement clauses – finite bare CPs, finite nominalised clauses (NCs), and

nonfinite NCs – and one type of adjunct clause – converbial adjuncts. Chapter 2 provides a brief introduction to the Mongolic language family, fieldwork context, and the basic morphosyntactic profiles of Khalkha and Kalmyk.

Chapter 3 shows that each of the four embedded clause types under consideration in Khalkha can be uniquely characterised along three dimensions: finiteness, nominalisation, and the presence or absence of a DP layer within the nominal superstructure. I argue that converbial adjuncts are *small nominalised clauses* in that they show evidence of both being nominalised and lacking a DP layer within the nominal superstructure above the clause. I further argue that, assuming some form of parallelism between clausal and nominal domains, NCs projecting a “smaller” nominal structure above a “larger” clausal structure are unattested both in Khalkha and cross-linguistically. I propose a novel explanation that links their absence to more general constraints on structure building formalised via the Williams Cycle.

Chapter 4 investigates the distribution of embedded subjects across the four clause types, using diagnostics from long-distance extraction and cross-clausal binding. I argue that accusative subjects occupy the highest phase edge of embedded (nominalised) clauses, genitive subjects occupy Spec,nP within a full DP superstructure, and case-unmarked (“nom-inative”) subjects occupy Spec,TP of the embedded clause. A key finding is that binding and extraction differ in their locality domains: while extraction appears to obey phase-locality, binding appears to be possible out of nominal domains irrespective of the presence of nominal phases. This suggests that locality domains for extraction and binding may not uniformly align with locality domains imposed by a standard conception of phases.

Chapter 5 shows that accusative-based DAM in Khalkha adheres to two similar but ultimately distinct referentiality scales. I propose a movement-based account of DAM that results in accusative objects being located in Spec,vP and accusative subjects being located in the highest phase edge of an embedded (nominalised) clause. Building on previous work, I argue that accusative is best analysed as a dependent case in Khalkha and show how it

interacts with the movement-based account of DAM for dependent accusative case assignment on objects and subjects. Next, I discuss novel data from subjects of subject clauses in Khalkha that reveal that containment relations between nominals, in addition to local c-command ones, are relevant for dependent accusative case assignment. That containment relations are relevant for dependent accusative case assignment offers novel support for an Agree-based approach to dependent case over a configurational one. Under the Agree-based approach, accusative arises as the spell out of the features of a previous goal of Agree on another DP. I extend this analysis to account for genitive subjects in Khalkha, and show how it may offer an underlying explanation for the absence of case marking on certain objects and subjects in the language.

Taken together, this dissertation demonstrates that a detailed investigation of clausal nominalisation not only contributes to our understanding of Mongolic morphosyntax, but also yields important insights into issues of broader theoretical significance in syntax, such as the interactions between clausal and nominal structure, syntactic locality, and dependent case theory.

Chapter 1

Introduction

This dissertation investigates the morphosyntactic properties of embedded clauses and embedded subjects, with a particular focus on four distinct types of embedded (nominalised) clauses in Khalkha Mongolian (henceforth Khalkha), the standardised variety of Mongolian spoken in Mongolia. It also draws comparisons with Kalmyk Oirat (henceforth Kalmyk), a Mongolic language mainly spoken in the Republic of Kalmykia, Russia, where it holds co-official status with Russian. Through a detailed investigation of Khalkha and Kalmyk embedded (nominalised) clauses, this dissertation contributes insights into the morphosyntax of nominalisation, the locality domains for extraction and binding, differential argument marking, and the nature of dependencies involved in dependent case assignment.

This dissertation is organised as follows. In Chapter 2, I provide some background on two Mongolic languages: Khalkha, the empirical focus of this dissertation, and Kalmyk, a closely-related Mongolic language. In this chapter, I also introduce the four embedded clause types that I focus on in Khalkha, as well as their counterparts in Kalmyk. Chapter 3 examines the structure of the four embedded clause types in Khalkha – finite bare CPs, finite nominalised clauses (NCs), nonfinite NCs, and converbial adjuncts – with some comparative discussion on three of their counterparts in Kalmyk. Chapter 4 investigates the positions of differentially case-marked subjects across the four embedded clause types by considering

how they fare with respect to long-distance extraction into the matrix clause and binding by the matrix subject. Chapter 5 is divided into two parts. The first part studies the referential properties associated with differential accusative case marking on matrix objects and embedded subjects, and proposes a movement-based analysis of DAM in Khalkha. The second part ties together the main findings in the previous chapters and, in light of novel data from subjects of subject clauses, provides an account of accusative and genitive case assignment and the treatment of case-unmarked nominals in Khalkha. Chapter 6 concludes. In the rest of this chapter, I provide a brief overview of Chapters 2-5, as well as the main findings of each chapter.

1.1 Chapter 2: Background on Khalkha and Kalmyk

Chapter 2 provides some background on the Mongolic language family and introduces the two Mongolic varieties – Khalkha and Kalmyk – that form the empirical foundation of this dissertation. I establish key structural properties of Khalkha that will inform the main arguments of later chapters. These properties include subject movement to Spec,TP; the distinction between overt tense markers, aspect markers, and converbs; binding properties; case marking; and possessive enclitics. I also introduce the four embedded clause types, along with the subjects contained within them, that form the empirical focus of this dissertation. An example of each clause type is shown below; they comprise three types of argument clauses (1)-(3) and one type of adjunct clause called a converbial adjunct (4). While subjects in matrix clauses may only be case-unmarked (traditionally labelled “nominative”), the subjects of the four embedded clause types may be case-unmarked, accusative, and in finite (2) and nonfinite (3) nominalised clauses, genitive.

(1) *Finite bare CP*

Bat changaar [**nohoi-g/nohoi** gaihal-tai gej] hel-sen.

B. loudly dog-ACC/dog wonder-COM C say-PST

‘Bat said loudly that dogs are wonderful.’

(Fong 2019, p.3, ex.5)

- (2) *Finite complement nominalised clause*
 Naraa [**Bilgüün-iig/Bilgüün-ii/Bilgüün** ger-iin daalgavr-Ø-aa duusga-san
 N. B.-ACC/B.-GEN/B. home-GEN task-ACC-RP finish-PST
 gedg]-iig sana-san.
 C.NML -ACC remember-PST
 ‘Naraa remembered that Bilguun finished his homework.’
- (3) *Nonfinite complement nominalised clause*
 Bi [**Bat-iig/Bat-iin/Bat** sugalaa-nd hoj-son]-iig med-ne.
 1SG B.-ACC/B.-GEN/B. lottery-DAT win-PFV -ACC know-NPST
 ‘I know that Bat won the lottery.’ (Aravind 2021, p.379, ex.6)
- (4) *Converbial adjunct*
 Bi [**tüüniig/ter** hicheel-Ø=ee hii-j bai-had] unt-san.
 1SG 3SG.ACC/3SG lesson-ACC=RP do-PROG COP-TEMP.CVB sleep-PST
 ‘When she/he was studying, I fell asleep.’

This chapter also shows that Kalmyk largely parallels Khalkha in its clausal, verbal, and nominal structure, with one key difference: Kalmyk exhibits overt subject ϕ -agreement on predicates, whereas Khalkha does not. Of the four embedded clause types that I focus on in Khalkha, I also discuss three of their counterparts in Kalmyk, in anticipation of their discussion in Chapter 3.

1.2 Chapter 3: Finiteness and nominalisation in Khalkha embedded clauses

In Chapter 3, I show that each of the four embedded clause types in Khalkha can be uniquely understood in terms of their finiteness – which correlates with their clausal structure – and nominal structure. I begin by establishing that case-unmarked subjects in the four clause types move to Spec,TP, as evidenced by scope and adverb placement diagnostics. Next, I demonstrate that the clause types differ systematically in three aspects: (i) whether they are finite or nonfinite, which correlates with the presence/absence of defective/non-defective T^0 and a full CP structure; (ii) whether they are nominalised, which implies the

presence of nP and PossP; and (iii) if nominalised, whether a DP layer is present in the nominal superstructure. I also show that overt subject agreement in Kalmyk is useful for understanding the distinction between finite and nonfinite clauses in the language.

I argue that while converbial adjuncts in Khalkha exhibit evidence of being nominalised, they also exhibit evidence that the nominal superstructure projected above the clause does not contain a DP layer. Drawing a connection to small nominals (Pereltsvaig 2006, *et seq.*), I analyse converbial adjuncts as small nominalised clauses whose nominal superstructure consists of nP and PossP but not DP. This yields a distribution of Khalkha NCs as shown in Table 1.1, where the unattested cell (a CP nominalised by a PossP superstructure without a DP layer) represents a gap in the empirical landscape:

Table 1.1: Preview of types of NCs in Khalkha

Clause size	Nominal superstructure	
	PossP	DP
TP	Converbial adjunct	Nonfinite argument
CP	(Unattested)	Finite argument

I argue that this gap is not accidental but follows from general principles of structure building, specifically under the Williams Cycle (Williams 2003, 2011, 2013). Assuming that some form of parallelism holds between the nominal and clausal domains, I propose a typology of NCs in which a smaller nominal superstructure cannot embed a larger clausal structure. This finding provides new insights into how the interaction of nominal and clausal structure acts as a constraint on clausal nominalisation, and offers a new perspective on the parallelism of nominal and clausal domains.

1.3 Chapter 4: The position of embedded subjects

Building on the findings in Chapter 3, which will have established key structural distinctions among the four embedded clause types under consideration in Khalkha, Chapter

4 seeks to establish the syntactic positions of accusative objects in matrix clauses, and accusative, genitive (where available), and case-unmarked subjects within the four embedded clauses. First, I motivate the idea that CP, vP, DP, and nP constitute phases in Khalkha. Next, I propose that accusative objects move to Spec,vP, accusative subjects occupy the highest phase edge of the embedded clause (which may include Spec,DP in nominalised clauses) (e.g., Fong 2019; Aravind 2021; Gong 2022; Peters 2024), genitive subjects occupy Spec,nP within the nominal superstructure of NCs (Aravind 2021), and case-unmarked subjects occupy Spec,TP (e.g., Fong 2019; Aravind 2021; Gong 2022; Peters 2024). Using diagnostics from binding and extraction, I provide evidence to support these subject positions in Khalkha. Specifically, I show that i) accusative subjects are accessible for both long-distance extraction and binding by the matrix subject, which provides support that they are located at the highest phase edge of the (nominalised) clause; ii) case-unmarked subjects are inaccessible for both operations, which suggests that they are lower than accusative subjects; and iii) genitive subjects are accessible for binding by the matrix subject but not long-distance extraction, which supports the idea that they are located below accusative subjects but above case-unmarked subjects. These findings are summarised in Table 1.2.

Table 1.2: Preview of extraction and binding patterns across embedded clauses in Khalkha

	Embedded subject		
	ACC	GEN	UNM
	(in finite & nonfinite NCs)		
May be extracted into matrix clause?	✓	✗	✗
Visible to matrix subject for binding?	✓	✓	✗

Based on the findings in Table 1.2, I show that long-distance extraction in Khalkha is phase-local: extraction targets only embedded subjects that have raised to the highest phase edge of the embedded (nominalised) clause. In contrast, binding may target not only accusative subjects at the highest phase edge, but also genitive subjects located in Spec,nP within a DP superstructure. This suggests that, in Khalkha, binding is sensitive to clausal boundaries rather than to phase boundaries per se. More broadly, the asymmetric behaviour

of genitive subjects with respect to extraction and binding indicates that syntactic locality cannot be fully reduced to standard phase-based spellout domains. I briefly discuss an alternative approach, building on Hicks (2009), under which phase heads may trigger spellout to PF, LF, or both interfaces, potentially accounting for the non-isomorphism between extraction and binding domains observed in Khalkha.

1.4 Chapter 5: Differential argument marking and case assignment in Khalkha

The final content chapter of this dissertation comprises two interrelated parts. The first part examines the similarities and differences between two accusative-based differential argument marking (DAM) systems in Khalkha: differential object marking (DOM), and differential subject marking (DSM) in embedded clauses. I show that DOM and DSM follow similar but ultimately distinct referentiality scales, as shown in Table 1.3. UNM refers to case-unmarked subjects and objects. The % symbol in the bottom right of Table 1.3 represents interspeaker variation.

Table 1.3: Preview of DOM and DSM in Khalkha

	DOM	DSM			
		Finite bare CP	Finite NC	Nonfinite NC	Converbial adjunct
1/2P pronoun	ACC	ACC	ACC	ACC	ACC
3P pronoun	ACC	ACC/UNM	ACC/UNM	ACC/UNM	ACC/UNM
Proper name	ACC	ACC/UNM	ACC/UNM	ACC/UNM	ACC/UNM
Anaphoric	ACC	ACC/UNM	ACC/UNM	ACC/UNM	ACC/UNM
Bridging	ACC	ACC/UNM	ACC/UNM	ACC/UNM	ACC/UNM
Unique	ACC/UNM	ACC/UNM	ACC/UNM	ACC/UNM	ACC/UNM
Wide scope indefinite	ACC/UNM	ACC/UNM	ACC/UNM	ACC/UNM	ACC/UNM
Narrow scope indefinite	UNM	UNM	UNM	UNM	%ACC/%UNM

The findings from Table 1.3 contribute to a finer-grained picture of DOM and DSM in Khalkha than was previously reported by Guntsetseg (2016). This is made possible by

considering a wider range of nominal types and definiteness subtypes as factors in triggering DOM and DSM in Khalkha.

The second part of Chapter 5 ties together the previous findings on embedded clause structure, subject position, and DAM, to provide an account of case assignment in Khalkha. I discuss evidence that supports a dependent analysis of accusative case as opposed to a case assigned by transitive Voice⁰ in Khalkha. Given that accusative objects have moved to Spec,vP and accusative subjects have moved to the highest phase edge of the embedded (nominalised) clause they are contained in, I demonstrate how a movement-based account of DAM interacts with dependent accusative case assignment. Specifically, I argue that objects located in Spec,vP and subjects located in the highest phase edge of the embedded (nominalised) clause are assigned accusative case under proximity to the matrix subject.

Finally, while there are two prominent approaches to modelling dependent case – a configurational approach (e.g. Yip et al. 1987; Marantz 1991; McFadden 2004; Baker and Vinokurova 2010; Baker 2015; Levin and Preminger 2015) vs. an Agree-based approach (Clem and Deal 2024; Poole 2024) – I present novel evidence from subjects of subject clauses in Khalkha that slightly favours an Agree-based approach. Specifically, I show that my consultants accept, to varying degrees, accusative subjects in subject clauses. I argue that this offers preliminary support for an Agree-based implementation of dependent case following Clem and Deal 2024, in which the accusative case morpheme is the spellout of another nominal’s ϕ -features as the result of a second cycle of Agree. Under the view that case is not a primitive syntactic feature but the contextual spellout of another nominal’s features, I show how genitive case can be modelled under the same set of assumptions I proposed for accusative in Khalkha, at the same time deriving the core intuition that genitive subjects depend on the presence of a DP layer within the nominal superstructure of NCs. Finally, I propose that surface morphological similarities between case-unmarked objects and case-unmarked subjects in Khalkha are due to the lack of a specific vocabulary item for case that

corresponds to the features found on those nominals. In this sense, I argue that the absence of case marking on nominals is a morphological default in Khalkha.

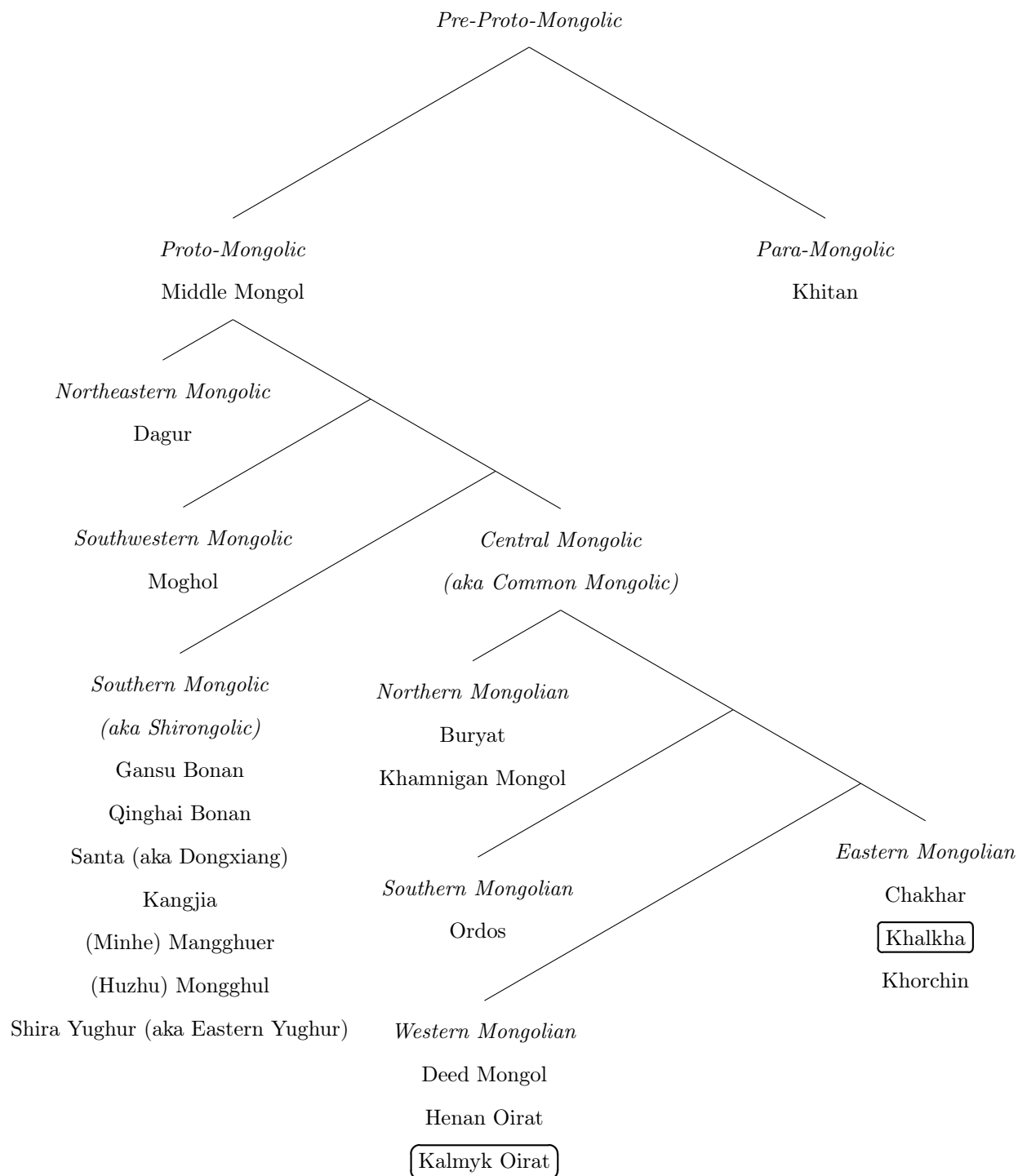
More broadly, the availability of accusative subjects in subject clauses in Khalkha attests to the existence of a previously-unnoticed dependency that may license dependent case – that in which a nominal is *contained* within another nominal. Khalkha thus shows that local c-command as well as containment relations are relevant for understanding accusative case assignment in the language. Further, as containment relations are predicted to be irrelevant for dependent case assignment under configurational approaches but relevant under Agree-based approaches, they have the potential to adjudicate between configurational vs. Agree-based theories of dependent case assignment. By showing that containment configurations are relevant for dependent case assignment, Khalkha thus offers novel support that dependent case is mediated by Agree – an core, independently-needed syntactic operation. This further suggests that configurational rules which have been independently proposed for dependent case assignment may not be necessary.

Chapter 2

Background on Khalkha and Kalmyk

This chapter provides a brief introduction to the two Mongolic languages investigated in this dissertation, Khalkha Mongolian (henceforth Khalkha) and Kalmyk Oirat (henceforth Kalmyk). Khalkha and Kalmyk belong to the Central Mongolic branch of the Mongolic language family, as shown in (5) on the next page. While Proto-Mongolic refers to the reconstructed ancestor of the modern Mongolic languages, derived through comparative analysis, Middle Mongol refers to the earliest attested stage of the language, documented in texts from the 13th to 16th centuries that reflect the spoken varieties of that period. Despite this distinction, Proto-Mongolic and Middle Mongol are considered to be nearly identical (Janhunen 2020, 2024). Recent advances in the study of the extinct Khitan language – the official language of the Liao dynasty in Manchuria (907–1125), preserved in a corpus of partially-deciphered inscriptions – have led scholars to propose that Khitan constitutes a sister lineage to Proto-Mongolic (Janhunen 2024, pp.85-86). As such, scholars have classified Khitan as Para-Mongolic, a designation that implies the existence of an earlier common ancestor, Pre-Proto-Mongolic, from which both Proto-Mongolic/Middle Mongol and Para-Mongolic/Khitan are thought to have descended.

(5) *The Mongolic language family* (based on Janhunen 2020, 2024)



The diversification of Mongolic, which gave rise to approximately 18 distinct modern Mongolic languages, occurred over a relatively brief period of 700–800 years following the expansion of the Mongol Empire in the 13th and 14th centuries across East and Central

Asia, the Middle East, and Eastern Europe (Janhunen 2020, 2024). As shown in (5), this diversification unfolded through a series of successive splits: Northeastern Mongolic (i.e. Dagur) was the first to branch off, followed by Southwestern Mongolic (i.e. Moghol), then Southern Mongolic (also known as Shirongolic), and finally Central Mongolic (also known as Common Mongolic). According to Janhunen, both geographic separation and language contact played crucial roles in driving the gradual differentiation of the Mongolic languages.

While *Khalkha* is sometimes used as a cover term for various Mongolic dialects spoken across Mongolia and China (Janhunen 2012, p.9), in this work I use *Khalkha* to refer specifically to the standardised variety and official language of Mongolia. According to Svantesson (2020), Khalkha has approximately 2.6 million speakers in Mongolia.

Kalmyk, by contrast, refers to the variety of Oirat spoken in the Republic of Kalmykia, Russia, where it holds co-official status with Russian. It is also spoken to a lesser extent in diaspora communities in the United States and Europe. According to Baranova et al. (2021), citing a 2010 Russian census, fewer than half of 180,000 ethnic Kalmyks in Kalmykia reported speaking Kalmyk at that time – a sharp decline from 2002, when nearly twice as many speakers were recorded. According to Bläsing (2003, p.229), there are two main dialects of Kalmyk: Dörbet, spoken mainly in western Kalmykia, and Torghut, spoken mainly in eastern Kalmykia; a third dialect, Buzava, is considered a mixture of Dörbet and Torghut. Bläsing (2003, p.229) also writes that “[t]he modern literary [Kalmyk] language is mainly based on the Torghut dialect, though it incorporates a large number of concessions to Dörbet”.

All uncited data in this dissertation come from original fieldwork conducted with three Khalkha consultants from Fall 2020 to Spring 2025, and with two Kalmyk consultants from Fall 2022 to Spring 2025. My three Khalkha consultants are: Garid Erdenechuluun (male, 25; currently based in Singapore), Becky Gantumur (female, 38; currently based in Australia) and Maralaa Munkhbat (female, 24; currently based in Mongolia). My two Kalmyk consultants are: Darina Gedeeva (Torghut; female, 72) and Valentina Bevikova (Dörbet; female, 68), both based in the Republic of Kalmykia, Russia. Data collection

primarily involved translation tasks, elicited production tasks, and acceptability judgment tasks (see Bochnak and Matthewson 2020 for discussion).

With my Khalkha consultants, the tasks were administered in individual sessions lasting 1–2 hours each. The sessions were conducted both in person and over Zoom, using English as the metalanguage. All three Khalkha consultants possess advanced to near-native proficiency in English. On occasion, I also elicited translations and judgments from my Khalkha consultants via text messaging.

Meetings with my Kalmyk consultants were conducted exclusively over Zoom. As my two main Kalmyk consultants are Kalmyk–Russian bilinguals, I worked with Ghilyana Dordzhieva (Torghut; female, 58; based in the US), who is proficient in English, Kalmyk, and Russian, and who served as an interpreter in our meetings. Each session thus involved three participants – myself, Ghilyana, and a Kalmyk consultant – and lasted 1–2 hours. English and Russian were used as the primary metalanguages.

In addition, some Kalmyk data were obtained through the development of learning resources created in collaboration with Darina Gedeeva and Ghilyana Dordzhieva, as part of ongoing efforts to preserve and revitalise Kalmyk within the Kalmyk-American diaspora community. These materials include online lessons, a Living Dictionary (Gedeeva et al. 2025b) and a children’s storybook (Gedeeva et al. 2025a).

Both Khalkha and Kalmyk remain relatively understudied in terms of formal syntactic analysis. For Khalkha, an early transformational grammar can be found in Binnick (1979). In recent years, there has been a growing body of work investigating various syntactic phenomena in Khalkha from a generative perspective. These include studies on movement (e.g. Fong 2019; Aravind 2021; Gong 2022, 2023a, 2025; Öztürk 2013), case assignment (e.g. Peters 2020, 2024; Lim 2022b, 2024a), nominal structure (e.g. Driemel 2020, 2023; Barrie and Kang 2022; Kang 2023; Barrie 2024; Gong and Despić 2024; Jang and Lim 2024; Lim 2025), and sluicing (e.g. Sakamoto 2014).

An early linguistic grammar of Kalmyk is provided in Street 1959. In contrast to Khalkha, there is virtually no generative syntactic literature on Kalmyk, with the notable exceptions of Knyazev 2015, 2016, which examine forms of Kalmyk complementisers, and Lim 2024b, which explores the interaction between agreement and nominative case assignment.

The empirical focus of this dissertation is on four types of embedded clauses in Khalkha, as well as the morphosyntactic properties of subjects found inside these clauses. The primary goal of this chapter is to provide background on the morphosyntax of Khalkha (§2.1) and Kalmyk (§2.2) in order to establish key similarities and differences between the two languages and to highlight interesting areas of intra-Mongolic variation. I will represent the Khalkha data, both from my fieldwork and secondhand sources, using a romanised version of the standardised Cyrillic orthography (see Appendix A). For Kalmyk data collected through my own fieldwork, I will represent it using IPA transcription, while Kalmyk data cited from secondhand sources will be presented in the source orthography.

2.1 Background on Khalkha morphosyntax

In this section, I provide some background on the syntax of Khalkha. Following Guntsetseg (2016, p.24), the basic word order of an intransitive clause is S-V (6), that of a transitive clause is S-DO-V (7), and that of a ditransitive clause is S-IO-DO-V (8).

- (6) *Intransitive clause: S-V*

Tuyaa yav-san.

T. go-PST

‘Tuyaa went.’

(Guntsetseg 2016, p.24, ex.41)

- (7) *Transitive clause: S-DO-V*

Tuyaa Dorj-iig üns-sen.

T. D.-ACC kiss-PST

‘Tuyaa kissed Dorj.’

(Guntsetseg 2016, p.24, ex.42)

- (8) *Ditransitive clause: S-IO-DO-V*
 Tuyaa Dorj-id neg nom ög-sön.
 T. D.-DAT one book give-PST
 ‘Tuyaa gave Dorj a book.’

(Guntsetseg 2016, p.107, ex.290a)

Khalkha displays head-final word order (Guntsetseg 2016, p.20). As seen above, verb inflections follow the verb stem, while the verb follows the subject and direct object (6)-(8). Relative clauses precede the head noun as shown in (9) and adpositions follow their complements as shown in (10) – both typical properties of head-final languages.¹

- (9) *Relative clause precedes head noun*
 Tuyaa [[Dorj-iin zövlö-sön] **nom**]-iig unsh-san.
 T. D.-GEN recommend-PFV book -ACC read-PST
 ‘Tuyaa read the book which Dorj recommended.’

(Guntsetseg 2016, p.20, ex.30)

- (10) *Adposition follows complement*
 Naraa [[minii shiree-n] **deer**] ayga tav'-san.
 N. 1SG.GEN table-N on cup put-PST
 ‘Naraa placed a cup on my table.’

I discuss further aspects of Khalkha clause structure in §2.1.1, the verbal domain in §2.1.2 and the nominal domain in §2.1.3.

2.1.1 Khalkha clause structure

As Guntsetseg (2008, p.20) and Gong (2022, pp.3-4) note, Khalkha is a language that displays relatively free word order due to the availability of scrambling. Word order is thus not always a reliable diagnostic for determining the structural relations between arguments in Khalkha. Nevertheless, in this subsection, I argue that in Khalkha the subject moves to Spec,TP, based on evidence from scope effects and adverb placement. I also argue

¹The stem-final nasal (glossed N) in (10) has been variously called “unstable-*n*”, “fleeting-*n*”, or “hidden-*n*” in the Mongolic literature, due to its appearance on a heterogeneous class of nouns only in the context of certain case markers and postpositions. In this sense, it is best understood as both a lexically-determined and a morphosyntactically-conditioned stem alternation. For consistency, I will refer to the stem-final nasal in this dissertation as hidden-*n*. For a synchronic and diachronic account of hidden-*n* in Khalkha, see Thompson 2008. See also Rákos 2015 and Janhunen 2012, pp.110-112 for a description of some grammatical functions of hidden-*n* in Khalkha.

that subjects asymmetrically c-command indirect objects and direct objects, and that indirect objects asymmetrically c-command direct objects. As support, I discuss evidence from binding.

2.1.1.1 S moves to Spec,TP

Since Khalkha is head-final, other diagnostics besides word order are needed to determine whether the external argument has raised to Spec,TP. One diagnostic, discussed by Miyagawa (2001) for Japanese, uses scope effects to ascertain the position of subjects. I begin with a discussion of the Japanese facts, before showing how Miyagawa’s diagnostic can be extended to Khalkha.

In the Japanese sentence (11), the verb complex possesses the morpheme order $V^0 < \text{Neg}^0 < T^0$, which indicates that T^0 -*ta* PST is higher than Neg^0 -*nakat*, which in turn is higher than the verb root *home* ‘to praise’ (Mirror Principle, Baker 1985). In (11), the universal quantifier direct object (DO) can only be interpreted within the scope of negation, i.e. it is not the case that Taroo praised everyone; he may have praised some.²

- (11) Taroo-ga **zen’in-o** home-**nakat**-ta (yo / to omou). $\checkmark \neg > \forall, (*)\forall > \neg^3$
 Taro-NOM all-ACC praise-NEG-PST EXCLM / C think
 ‘(I think that) Taro didn’t praise all (!).’ (Miyagawa 2001, p.298, ex.7)

Miyagawa (2001) argues that if external arguments A-move to Spec,TP in Japanese, then a universal quantifier subject should not be interpreted within the scope of negation. This prediction is borne out in (12): the sentence means that no one – not even one person – took that test.

- (12) **Zen’in-ga** sono tesuto-o uke-**nakat**-ta (yo / to omou). $*\neg > \forall, \checkmark \forall > \neg$
 all-NOM that test-ACC take-NEG-PAST EXCLM / C think
 ‘All did not take that test.’ (Miyagawa 2001, p.298, ex.8)

²As Miyagawa (2001, p.329) explains, there can be multiple ways to end the sentence (11): *yo* produces an exclamative, while *to omou* ‘I think that’ turns it into a subordinate clause.

³Miyagawa (2001) uses (*) to indicate that some speakers accept the ALL>NOT reading, which he attributes to a collective reading of ‘all’. The fact that such an interpretation is unavailable in Khalkha (14) suggests that the collective reading of ‘all’ is unavailable.

In short, in a head-final language like Japanese, where $V^0 < \text{Neg}^0 < T^0$, whether subjects move to Spec,TP or not can be diagnosed by the interpretation of a quantified subject vis-à-vis a quantified direct object under negation.

Turning our attention to Khalkha, in matrix root clauses, Khalkha exhibits the same order of morphemes within the verb complex as in Japanese, as seen in (13). By the Mirror Principle (Baker 1985), this implies that $V^0 < \text{Neg}^0 < T^0$ in Khalkha. See §2.1.2 for further discussion on the Khalkha verb complex, including the use of the copula base *bai* in (13).

- (13) *Khalkha verb complex: $V^0 < \text{Asp}^0 < \text{Neg}^0 < T^0$*
 Odgerel huushuur hii-gee-güi bai-na.
 O. khuushuur do-IPFV-NEG COP-NPST
 ‘Odgerel hasn’t made khuushuur (fried meat pastry).’

(14) shows that a universally quantified DO must be interpreted within the scope of negation. This is expected as the DO *büh huushuur* ‘all khuushuur’ occupies the complement position of V^0 , where it is c-commanded by Neg^0 . Conversely, (15) shows that a universally-quantified subject must take scope outside of negation (note that the verb *tents-eh* ‘to pass’ assigns dative case to its internal argument). This suggests that the universally-quantified subject *büh oyutnuud* ‘all students’ must be located in a position above NegP, where it is not c-commanded by Neg^0 . A natural candidate for this landing site above NegP is Spec,TP.^{4, 5}

⁴Strictly speaking, it is not logically possible to construct an $\forall > \neg$ scenario that is incompatible with a $\neg > \forall$ reading, as every scenario in which all students did not pass the exam (verifying $\forall > \neg$) is also a scenario in which not all students passed the exam (verifying $\neg > \forall$). It would appear that the weaker reading – the reading that is logically entailed by the stronger reading which matches the scope relation between the quantifier and negation – is dispreferred by my consultants, perhaps due to pragmatic competition with the stronger reading.

⁵The past tense translations in (14) and (15) represent the target sentences I used to elicit these sentences, and also the back-translations provided by my consultants. The translations are somewhat unexpected given the presence of the nonpast tense suffix *-na* in these sentences. However, previous work has noted that the imperfective suffix *-AA* is only used in conjunction with a present tense interpretation, referring to an action or event “initiated in the past but continuing into the present” (Janhunen 2012, pp.161-162; see also Binnick 2012, p.29). Thus, a more accurate translation for (14) may be ‘Bat has not eaten all the khuushuur’, and for (15), ‘All the students have not passed the exam.’. This in turn suggests that the past tense interpretations in (14)-(15) could be pragmatically derived.

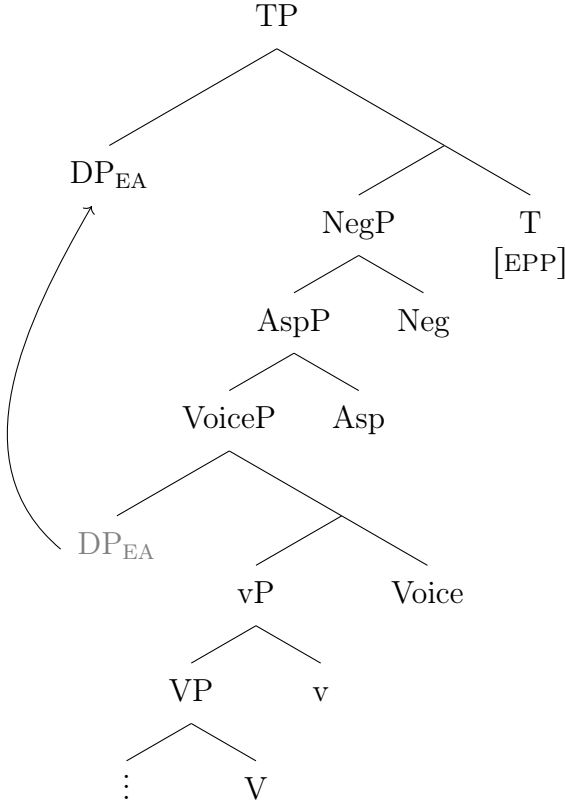
- (14) Bat **büh huushuur-iig** id-ee-**güi** bai-na
 B. all khuushuur-ACC eat-IPFV-NEG COP-NPST
 ‘Bat did not eat all the khuushuur.’
 #Context: There are 10 khuushuurs and Bat didn’t eat any. $*\forall > \neg$
 ✓Context: There are 10 khuushuur and Bat ate one. $\checkmark\neg > \forall$
- (15) **Büh oyuutn-uud** ter shalgaltan-d tents-ee-**güi** bai-na.
 all student-PL that exam-DAT pass-IPFV-NEG COP-NPST
 ‘All the students didn’t pass that exam.’
 ✓Context: There are 10 students in the class and no one passed that exam. $\checkmark\forall > \neg$
 #Context: There are 10 students in the class and only two passed that exam. $*\neg > \forall$

Following Peters 2024, I adopt the idea that external arguments are introduced by Voice⁰ – a distinct head than v⁰, which verbalises a base (e.g. Pyllkkänen 2008; Harley 2013; Legate 2014). In line with the Uniformity of Theta Assignment Hypothesis (UTAH) (Baker 1988), I assume that the grammatical subject in Khalkha may originate as the external argument of a transitive or unergative verb in Spec,VoiceP (16a), or as the internal argument of an unaccusative verb from the complement position of V (16b).⁶ Given that unaccusatives lack an external argument, I also assume that VoiceP is not projected in such structures. I propose that subjects A-move to Spec,TP to satisfy an EPP feature on T⁰. As such, subjects are interpreted in a position higher than negation, which is below TP. This is schematised in (16).

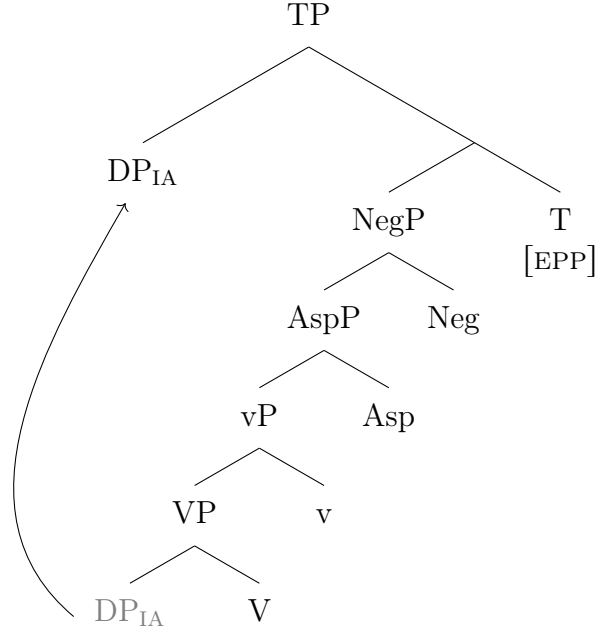
⁶To the best of my knowledge, there are no reliable diagnostics in Khalkha to distinguish unaccusative subjects from unergative ones.

(16) *Subject moves to Spec,TP above NegP*

a. *Transitive/Unergative subjects*



b. *Unaccusative subjects*



An interesting difference arises between Japanese and Khalkha concerning the nature of EPP on T^0 . In Japanese, scrambling the direct object over the subject results in a sentence that is ambiguous between two scopal readings (17). Miyagawa (2001) argues that the ambiguity is due to two different derivations of (17) involving two types of scrambling. The $\forall > \neg$ reading in (17a) is expected: the external argument A-moves to Spec,TP and is interpreted outside the scope of negation; A'-scrambling of the DO to a position above Spec,TP – call it Spec,FP – reconstructs in its base position for scope at LF. The $\neg > \forall$ reading in (17b) comes about because the EPP feature on T^0 can also be satisfied by A-scrambling of the direct object to Spec,TP. In this scenario, the universal quantifier subject remains in its base position (i.e. Spec,vP) and is interpreted within the scope of negation.

- (17) **Sono tesuto-o₁** zen'in-ga ₁ uke-nakat-ta (yo / to omou).
 That test-ACC₁ all-NOM take-NEG-PAST EXCLM / COMP think
 'That test, all didn't take.' (Miyagawa 2001, p.299, ex.11b)
- a. A'-scrambling of object
 $[_{FP} \text{ Sono tesuto-o}_1 [_{TP} \text{ zen'in-ga}_2 [_{NegP} [_{vP} \text{ }_{-2} [_{VP} \text{ }_{-1} \dots]]]]] \quad \forall > \neg$
- b. A-scrambling of object
 $[_{TP} \text{ Sono tesuto-o}_1 [_{NegP} [_{vP} \text{ zen'in-ga} [_{VP} \text{ }_{-1} \dots]]]] \quad \neg > \forall$

While it is possible to scramble the DO to clause-initial position in Khalkha (18), the sentence is unambiguously interpreted as $\forall > \neg$, suggesting that the DO reconstructs for scope. I take this to indicate that Khalkha lacks A-scrambling of DOs to Spec,TP (though short scrambling to a position below Spec,TP is possible; see discussion surrounding (40) in §2.1.1.3). Thus, I take any derivation that results in the DO preceding the subject, such as that in (18), to involve A'-scrambling of the DO to a position above Spec,TP.

- (18) **Ter shalgaltan-d₁** büh oyut-nuud ₁ tents-ee-güi bai-na.
 that exam-DAT all student-PL pass-IPFV-NEG COP-NPST
 'All the students didn't pass that exam.'
 ✓Context: There are 10 students in the class and no one passed that exam. $\forall > \neg$
 #Context: There are 10 students in the class and only two passed that exam.
 $*\neg > \forall$

Additional evidence that subjects move to Spec,TP comes from the placement of AspP and vP adverbs. First, Guntsetseg (2016) shows that the adverb *dandaa* 'always' – an AspP adverb – may immediately precede the verb or the dative-marked internal argument of *tusal-ah* 'to help' (which assigns dative case to its internal argument). However, it cannot appear to the left of the external argument subject, as shown (19). Given that external arguments are introduced in Spec,VoiceP, which is below AspP, the inability of *dandaa* to precede an external argument subject can be explained by the fact that subjects move to Spec,TP. On this view, since the subject is located in Spec,TP, it cannot appear to the right of *dandaa*. In (19), I take the order *dandaa* > DO to reflect the base order, while DO > *dandaa* may be derived via object scrambling to a position below Spec,TP.

- (19) { *Dandaa } Tuyaa { **dandaa** } Dorj-id { **dandaa** } tusal-dag.
 { *Always } T. { always } D.-DAT { always } help-HAB
 ‘Tuyaa always helps Dorj.’ (Guntsetseg 2016, p.25, ex.46)

While DOs that are not assigned dative case may be case-unmarked or accusative-marked (see, e.g., Guntsetseg 2008, 2016), Peters (2024) shows that a DO that precedes a manner adverb such as *uurtaigaar* ‘angrily’ must be marked with accusative case, as shown in (20). Given that manner adverbs adjoin to vP, the surface position of the accusative-marked DO *hool* ‘food’ in (20), which is first merged in the complement position of V, suggests that it has shifted out of VP, e.g. to Spec,vP. See Chapter 5 for a movement-based account of differentially accusative-marked objects in Khalkha.

- (20) Bi hool*(-iig) **uurtaigaar** id-sen.
 1SG food*(-ACC) angrily eat-PST
 ‘I ate the food angrily.’ (Peters 2024, p.15, ex.30)

Given my account which takes external arguments to be introduced in Spec,VoiceP, which is directly above vP, it is unsurprising that a manner adverb such as *hurdan* ‘quickly’ – which adjoins to vP – may not precede an external argument subject as shown in (21). The placement of manner adverbs with respect to external argument subjects is thus uninformative as to whether subjects move to Spec,TP.⁷

- (21) { *Hurdan } Bilgüün { **hurdan** } ene baishin-g { **hurdan** } bari-san.
 { *Quickly } B. { quickly } this building-ACC { quickly } build-PST
 ‘Bilguun built this building quickly.’

⁷Interestingly, Gong (2023a) reports that examples such as (ib), where a manner adverb precedes an external argument subject, are acceptable:

- (i) a. Bat **hurdan** güi-j bai-na.
 B. quickly run-PROG COP-NPST
 ‘Bat is running quickly.’
 b. **Hurdan**₁ Bat ₁ güi-j bai-na.
 Quickly B. run-PROG COP-NPST
 ‘Bat is running quickly.’ (Gong 2023a, p.417, fn.2)

Gong suggests that *hurdan* ‘quickly’ is a VP-level adverb and that it may undergo clause-internal scrambling, deriving (ib). Such data contradict the judgements of my consultants, e.g. (21), as well as those previously reported in Guntsetseg 2016; Fong 2019; Peters 2024. I assume that Khalkha speakers may vary as to whether they accept clause-internal scrambling of low adverbs.

However, if Spec,TP is a derived position, we might expect that non-agentive arguments may also raise to satisfy the EPP feature on T^0 . In passive constructions, the external argument is absent and the internal argument of the verb functions as the theme subject. If subjects raise to Spec,TP to check T^0 's EPP feature, we predict that a manner adverb such as *hurdan*, which adjoined to vP, should not be able to precede a passivised, theme subject in Spec,TP. This prediction is borne out, as seen in (22).

- (22) { *Hurdan } ene baishin { **hurdan** } bari-gd-san.
 { *Quickly } this building { quickly } build-PASS-PST
 'This building was built quickly.'

Compare the passive sentence in (22) with its active counterpart in (21). In (21), the manner adverb *hurdan* may immediately precede the verb or the accusative-marked theme object but not the agentive subject. However, in (22), while *hurdan* may precede the verb, it may not precede the theme subject. This suggests that in the absence of an external argument, the theme argument raises to Spec,TP to satisfy the EPP feature on T^0 . The high position of the theme argument in passives thus reinforces the idea that subjects move to Spec,TP and that Spec,TP is a derived position.

While aspectual and manner adverbs are useful diagnostics for subject position, temporal adverbs such as *margaash* 'tomorrow' are less reliable in this regard as they display more flexibility in their distribution. For instance, *margaash* may appear before or after the clausal subject, as shown in (23).

- (23) { **Margaash** } Naraa { **margaash** } ene baishin-g nuraa-na.
 { Tomorrow } N. { tomorrow } this building-ACC tear.down-NPST
 'Naraa will tear down this building tomorrow.'

I assume that temporal adverbs adjoin to TP, and that the order in which *margaash* precedes a case-unmarked subject reflects the base order, while the order in which a case-unmarked

subject precedes *margaash* is derived via short scrambling of the subject to a position above TP.⁸

Taken together, the adverb distribution facts provide support for the claim that subjects raise to Spec,TP in Khalkha, as they cannot be preceded by low adverbs such as aspectual or manner adverbs, which adjoin below TP.

In sum, the evidence from the interpretation of universally quantified subjects under negation, along with the facts from adverb placement, indicates that subjects occupy Spec,TP – above NegP, AspP, VoiceP, vP, and VP. I have argued that this is a derived A-position, supported by (i) the absence of reconstruction effects and (ii) the ability of non-agentive arguments to raise to this position. In the next section, I turn to binding in Khalkha, which supports the idea that subjects asymmetrically c-command both indirect and direct objects, and that indirect objects asymmetrically c-command direct objects.

2.1.1.2 S c-commands IO and DO

I show using binding effects that subjects c-command indirect objects (IOs) and direct objects (DOs). A partial inventory of pronominal forms in Khalkha is shown in Table 2.1. The case-unmarked (UNM) column represents subject forms that are traditionally described as bearing “nominative” case, which is morphologically unmarked in Khalkha. For reasons that will become clear in Chapter 5, I will refer to “nominative” subjects as case-unmarked subjects and gloss them without any case in this dissertation.

Khalkha contains three types of anaphors – reflexives, reciprocals, and reflexive possessives – all of which are morphologically complex and share a superficial similarity in containing the reflexive possessive (RP) enclitic. Reflexive anaphors comprise a ‘self’-pronoun

⁸As I show in the following chapters, temporal adverbs may also precede accusative and genitive subjects, which I analyse as having moved to higher positions within the clausal periphery. This suggests that temporal adverbs may also scramble to the edge of the embedded (nominalised) clause.

⁹The first plural inclusive/exclusive distinction is only expressed in the genitive.

¹⁰A singular possessor reading is possible with *manai*, e.g. *manai aav* ‘our/my father’, but not *bidnii*, e.g. *bidnii aav* ‘our/*my father’.

¹¹One consultant reports the use of an alternative colloquial form for 3SG.GEN: *ternii*.

Table 2.1: Selected pronouns in Khalkha

	UNM	ACC	GEN	DAT	INSTR
1SG	bi	namaig	minii	nadad	nadaar
1PL(.INCL)	bid	bidniig	bidnii ⁹	bidend	bidneer
1PL.EXCL	-	-	manai ¹⁰	-	-
2SG.IFML	chi	chamaig	chinii	chamd	chamaar
2SG.FRML	ta	taniig	tanii	tand	tanaar
2PL.IFML	ta nar	ta nariig	ta nariin	ta nart	ta naraar
2PL.FRML	ta büh	ta bühiig	ta bühnii	ta bühend	ta büheer
3SG	ter	tüüniig	tüünii ¹¹	tüünd	tüüneer
3PL	ted	tedniig	tednii	tedend	tedneer

öör and an obligatory RP enclitic. To express coreference between the subject and a direct object, the pronominal forms cannot be used due to Binding Condition B, as shown in (24a) and (25a). Instead, the accusative-marked reflexive anaphor must be used, as shown in (24b) and (25b).

(24) Context: I saw a photo of myself.

- a. *Bi **namaig** har-san.
1SG 1SG.ACC see-PST
Intended: ‘I saw me.’
- b. Bi **öör-iig=öö** har-san.
1SG self-ACC=RP see-PST
‘I saw myself.’

(25) Context: Naraa saw a photo of herself.

- a. *Naraa_j **tüüniig_j** har-san.
N. 3SG.ACC see-PST
Intended: ‘Naraa_j saw her_j.’
- b. Naraa **öör-iig=öö** har-san.
N. self-ACC=RP see-PST
‘Naraa saw herself.’

A reflexive anaphor may not serve as a main clause subject as shown in (26a), suggesting that its distribution is regulated by Binding Condition A. However, a case-unmarked reflexive anaphor may be used as an emphatic reflexive, as shown in (26b), which is surface identical to (26a), and (27), which contains an unaccusative verb. These examples are interpreted as

containing a *pro* subject. Since the emphatic reflexive modifies the subject, I will assume that they adjoin to the subject DP.

- (26) a. ***Öör=öö** namaig har-san.
 Self=RP 1SG.ACC see-PST
 Int. ‘Myself saw me.’
 b. *pro* **öö**r=öö namaig har-san.
pro self=RP 1SG.ACC see-PST
 ‘Someone saw me herself/himself. (e.g. of their own volition, desire, etc.)’
- (27) *pro* **öö**r=öö ir-sen.
pro self=RP come-PST
 ‘Someone came herself/himself. (e.g. of their own volition, desire, etc.)’

Turning to reciprocal pronouns, Khalkha contains two forms, one containing the stem *biye.biye* ‘each other (lit. body body)’ as shown in (28a), and one containing the stem *neg.neg* ‘each other/one another (lit. one one)’ as shown in (28b). Both forms also contain an obligatory RP enclitic. To my knowledge, the two forms mean roughly the same and are interchangeable. As shown in (29), reciprocal subjects in main clauses are not possible.

- (28) a. Bid **biye.biye-n-iig=ee** har-san.
 1PL each.other-N-ACC=RP see-PST
 ‘We saw each other.’
 b. Bid **neg.neg-n-iig=ee** har-san.
 1PL each.other-N-ACC=RP see-PST
 ‘We saw each other.’
- (29) ***Biye.biy=ee/Neg.neg=ee** bidniig har-san.
 Each.other=RP/each.other=RP 1PL.ACC see-PST
 Int. ‘Each other saw us.’

Finally, to express coreference between the subject and the object’s possessor in Khalkha, a referential pronoun cannot be used as shown in (30b). Instead, the genitive-marked reflexive anaphor (30c) or the RP enclitic (30d) is used. (31) shows that RP-marking on subjects, which indicates the presence of a reflexive possessor within the subject, is not possible in main clauses.

- (30) a. Naraa **minii** nom-iig unsh-san
 N. 1SG.GEN book-ACC read-PST
 ‘Naraa read my book.’
 b. Naraa_j **tüünii**_{k/*j} nom-iig unsh-san.
 N. 3SG.GEN book-ACC read-PST
 ‘Naraa_j read her_{k/*j} book.’
 c. Naraa **öör-iin-h=öö** nom-iig unsh-san.
 N. self-GEN-H=RP book-ACC read-PST
 ‘Naraa_j read her_{j/*k} (own) book.’¹²
 d. Naraa nom-Ø=**oo**¹³ unsh-san.
 N. book-ACC=RP read-PST
 ‘Naraa_j read her_{j/*k} book.’

- (31) * **Egch=ee** namaig har-san.
 Older.sister=RP 1SG.ACC see-PST
 Int. ‘My older sister saw me.’

All three types of anaphoric DOs – RP-marked reflexives, RP-marked reciprocals, and RP-marked possessa – can only be bound by the local subject and not the IO (assuming for now that it asymmetrically c-commands the DO), as shown in (32) and (33) (see also Gong and Despić 2024; Guntsetseg 2016, pp.56-58, 159-160). This is due to the subject-oriented property of the reflexive possessive anaphor in Khalkha.

- (32) *RP-marked reflexive can only be bound by local subject*
 Emch(=n’) Dorj-id (tolin-d) **öör-iig=öö** haruul-san.
 Doctor(=3.POSS) D.-DAT mirror-DAT self-ACC=RP show-PST
 ‘The doctor_j showed to Dorj_k himself_{j/*k} (in the mirror).’
 (Gong and Despić 2024, p.2, ex.2)

- (33) *RP-marked reciprocal can only be bound by local subject*
 Bagsh-nar oyutn-uud-ad **biye.biye-n-iig=ee** taniltsuul-san.
 teacher-PL student-PL-DAT each.other-N-ACC=RP introduce-PST
 ‘The teachers_j introduced to the students_k each other_{j/*k}.’

¹²There is a slight difference in meaning between (30c) and (30d): (30c) can only refer to a book owned by Naraa while (30d) can be used to refer to a book that is associated with Naraa, e.g. a book she owns or a book she borrowed from the library (and thus does not technically own).

¹³As noted by Guntsetseg (2016, pp.58-59, 82-83), ACC may be morphologically zero on non-kinship nouns in the context of the RP enclitic. For further discussion, see §4.4.1.

- (34) *RP-marked possessum can only be bound by local subject*

Bi Naraa-d **zurg-Ø=aa** ilgee-ne.
 1SG N.-DAT picture-ACC=RP send-NPST
 ‘I will send Naraa_j my/*her_j picture.’

(35) also shows that an RP-marked IO can only be bound by the subject and not the DO, resulting in the pragmatically strange reading that ‘I’ belong to an ‘owner’. The binding facts in (35) are thus not instructive of the relationship between IO and DO, a point to which I will return shortly.

- (35) Bi bömbög-iig **ezen-d=ee** butsaa-na.
 1SG ball-ACC owner-DAT=RP return-NPST
 ‘I will return the ball_j to my/*its_j owner.’

Gong 2023b; Gong and Despić 2024 further show that it is the RP enclitic that encodes the local-subject-bound orientation. When the ‘self’-pronoun *öör* is attached with the 3.POSS enclitic, as shown in (36) below, the complex reflexive pronoun can only be bound by a nonsubject argument but not the local subject. This contrasts with (32), in which the RP-marked *öör* can only be bound by the local subject.

- (36) *3.POSS-marked reflexive can only be bound by nonsubject argument*
 Emch(=n’) Dorj-id (tolin-d) **öör-iig=n’** haruul-san
 Doctor(=3.POSS) D.-DAT mirror-DAT self-ACC=3.POSS show-PST
 ‘The doctor_j showed Dorj_k himself_{k/*j} (in the mirror).’

(Gong and Despić 2024, p.2, ex.2)

Gong 2023b; Gong and Despić 2024 provide an analysis of subject-oriented binding in Khalkha, in which RP-marked arguments are bound by a subject located in the specifier of a verbal functional head, v^0 or Voice⁰; under the present analysis in this dissertation, I take the relevant head to be Voice⁰. Their analysis thus provides a unified account of reflexive pronouns and reflexive possessives in Khalkha. Given the local-subject-bound orientation of RP-marked reciprocal pronouns, I suggest that they may also be accounted for under Gong’s (2023b) and Gong and Despić’s (2024) analysis.

Finally, a proper name DO (37) and IO (38) cannot be coreferential with the subject due to Binding Condition C, which states that R-expressions cannot be bound. This provides further evidence that subjects c-command DOs and IOs.

- (37) Ter_{j/*k} **Odgerel-iig**_k har-san.
 3SG O.-ACC see-PST
 ‘She_{j/*k} saw Odgerel_k.’
- (38) Ter_{j/*k} **Dulmaa-d**_k beleg hudaldan av-san.
 3SG D.-DAT gift purchase take-PST
 ‘She_{j/*k} bought Dulmaa_k a gift.’

2.1.1.3 IO c-commands DO

Two pieces of evidence support the idea that IOs c-command DOs. First, as seen in (39a), the proper name possessor of a DO cannot be coreferential with a pronominal IO due to Binding Condition C. The reverse is possible, however, as shown in (39b): a pronominal possessor of a DO can be coreferential with a proper name IO.

- (39) a. Bi tüünd_{j/*k} **Odgerel-iin**_k zurg-iig ög-sön.
 1SG 3SG.DAT O.-GEN picture-ACC give-PST
 ‘I gave her_{j/*k} Odgerel_k’s photo.’
- b. Bi Odgerel-d_j **tüünii**_{j/k} zurg-iig ög-sön.
 1SG O.-DAT 3SG.GEN picture-ACC give-PST
 ‘I gave Odgerel_j her_{j/k} photo.’

Second, Guntsetseg (2016, p.24) reports that DO-IO order in ditransitive clauses is marked. This observation is corroborated by Gong (2022, pp.65-66), who shows that S-IO-DO-V (40a) is unambiguously interpreted with surface scope, whereas S-DO-IO-V (40b) is scopally ambiguous:

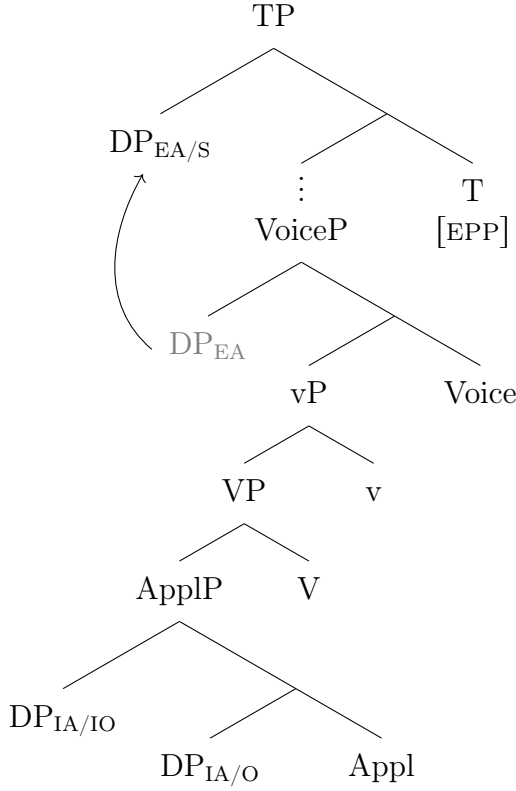
- (40) a. Bat [_{IO} neg hün-d] [_{DO} boodol bolgon-iig] ög-sön
 B. one person-DAT package every-ACC give-PST
 ‘Bat gave a person every package.’ $\exists > \forall, * \forall > \exists$
- b. Bat [_{DO} boodol bolgon-iig] [_{IO} neg hün-d] ög-sön
 B. package every-ACC one person-DAT give-PST
 ‘Bat gave a person every package.’ $\exists > \forall, \forall > \exists$
- (Gong 2022, p.66, exx.95-96)

Gong argues that the ambiguity in (40b) is due to short scrambling of the DO over the IO, such that the DO can be interpreted in its base position (below the IO) or in its surface position (above the IO). Gong (2022, pp.73-88) shows that short scrambling of the DO above the IO has properties of A-movement, e.g. it feeds anaphor binding, variable binding and remedies weak crossover effects. Thus, I adopt Gong’s proposal that DO-IO order is derived from the base IO-DO order via short A-scrambling of the DO above IO, and assume a low applicative structure merged below VP (cf. Pykkänen 2008).

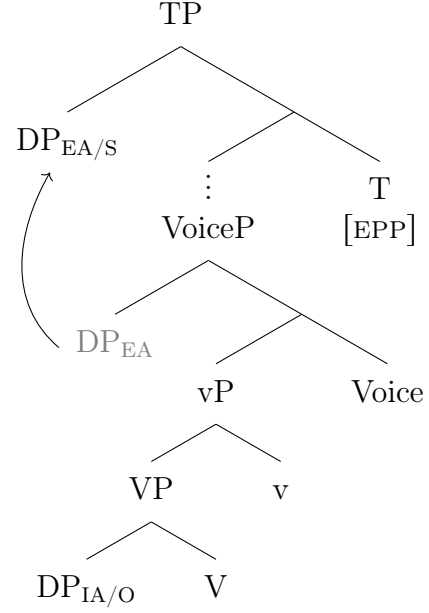
2.1.1.4 Summary

To summarise, I have shown that the behaviour of arguments with respect to binding falls out from the structural positions of these arguments. Specifically, I have shown that the subject (S) c-commands IO and DO, while IO c-commands DO. I propose that in transitive clauses, external arguments are base-merged in Spec,VoiceP before raising to subject position in Spec,TP to check the EPP feature on T^0 . I also propose that IOs are introduced in Spec,ApplP below VP, and that DOs are introduced in the complement position of Appl⁰. In transitives, DOs are simply introduced in the complement position of V^0 . The structures of a ditransitive (41a) and a transitive (41b) are schematised below.

(41) a. *The structure of a distransitive*



b. *The structure of a transitive*



2.1.2 The Khalkha verbal domain

In this subsection, I discuss the elements that compose the Khalkha verb complex, and the distinction between ‘finite’ and ‘nonfinite’ verb endings used in the traditional Mongolic literature. An understanding of verbal structure will be important in establishing the size of embedded clauses in Chapter 3. To preface the discussion, I will show that the verb complex follows the template in (42).

- (42) *Khalkha verb complex*
 VERB STEM (-VOICE) (-ASPECT) (-NEG) (COP) (-CONVERB/-TENSE)

As shown in (43), voice suffixes – such as the passive suffix – appear closest to the verb stem, and precede aspect and negation suffixes. (13), repeated below, shows that the copula base and tense suffixes follow negation. For concreteness, I also assume the presence

of a null past tense suffix in (43), given that it is interpreted in the past tense (see, however, footnote 5 in this chapter).

- (43) Tüünii nom, devter ol-**d-son-güi-Ø**.
 3SG.GEN book notebook find-PASS-PFV-NEG-NPST
 ‘His books and notebooks were not found.’ (Tserenpil and Kullmann 2008, p.149)
- (13) Odgerel huushuur hii-**gee-güi bai-na**.
 O. khuushuur do-IPFV-NEG COP-NPST
 ‘Odgerel hasn’t made khuushuur (fried meat pastry).’

In the rest of this section, I set aside voice suffixes and negation, and focus on the properties of aspect, tense, and converb markers in Khalkha. I first discuss tense and aspect markers, before turning to converb markers. The functions and glossing of these markers listed in the tables that follow are adapted from the terminology used in Janhunen (2012).

2.1.2.1 Tense and aspect markers

A partial list of tense markers, also called finite indicative tense-aspect endings in the Mongolic literature, is given in Table 2.2. I assume that there are three null T^0 ’s, one which is used in certain nonfinite contexts, one which is an allomorph of the nonpast suffix $-nA$,¹⁴ and one which is an allomorph of the general past tense suffix $-sAn$.¹⁵ Nonfinite T^0 will be further discussed in Chapter 3. Uppercase letters indicate vowels that undergo harmony: ‘A’ represents backness harmony, while ‘U’ represents rounding harmony.

¹⁴The null NPST allomorph surfaces in examples like (44) below, where the sentence is interpreted in the present tense, but there is no overt tense morphology. Further research is needed to understand the distribution of the forms of the NPST suffix.

¹⁵The null past tense suffix is commonly found after negation, as seen in (43).

¹⁶I follow Tserenpil and Kullmann (2008, pp.190-191) in analysing the $-nA$ suffix as having both a present and future interpretation, and thus it seems more accurate to describe it as a nonpast (NPST) marker. The authors also write that the present tense interpretation is “only used for actions that remain the same or for lasting truths” (p.190). It appears to me that the present interpretation possibly arises with stative and/or atelic verbs, although further research is required to confirm this hypothesis.

¹⁷Binnick (2012) argues that there is evidence that the perfective participle $-sAn$ has developed into a past tense marker. Since the two markers are homophonous, it is not usually obvious which one is used when only one tense or aspect marker is present. Thus, I only gloss $-sAn$ as PFV when it is clearly a perfective participle, e.g. when it is used with another tense marker or in non-finite contexts.

Table 2.2: Selected tense markers in Khalkha

Suffix	Function	Gloss
Ø	nonfinite	NFN
-nA~-Ø	(durative) present imperfect ¹⁶	NPST
-lAA	(confirmative) present perfect	PRES.PRF
-v	(terminative) past imperfect	PST.IMPF
-jee	(resultative) past perfect	PST.PRF
-sAn ¹⁷ ~-Ø	general past	PST

Aspect markers are also called participles (Janhunen 2012), verbal nouns (Binnick 2012) or nonfinite verbal markers (Svantesson 2003) in the Mongolic literature. A partial list of aspect markers is shown in Table 2.3.

Table 2.3: Selected aspect markers in Khalkha

Suffix	Function	Gloss
-h	futuritive/infinitive	INF ¹⁸
-dAg	habitive	HAB
-sAn	perfective	PFV
-AA	imperfective	IPFV
-j~-ch ¹⁹	progressive	PROG

It can be seen that aspect markers are structurally lower than Tense endings by their relative positioning with respect to negation, i.e. $\text{Asp}^0 < \text{Neg}^0 < \text{T}^0$. First, observe that the habitive Aspect suffix precedes negation (44).

- (44) Odgerel huushuur hii-**deg**-güi-Ø.
 O. khuushuur do-HAB-NEG-NPST
 ‘Odgerel does not make khuushuur.’

¹⁸The Aspect suffix *-h*, glossed as INF(itive), is also glossed as FUT(uritive) in works such as Janhunen 2012. Binnick (2012, p.83) writes that *-h* is most accurately described as a nonpast marker – “it is interpreted as a present with stative expressions [...] but as a future with active and eventive predicates”. It appears to be described as “infinitival” as it cannot appear as the main verb ending of a declarative sentence. Importantly, it does not serve to mark that a clause is nonfinite, and I simply adopt the gloss INF by convention. See also Gong (2022, p.23) and Janhunen (2012, pp.60-61) for discussion.

¹⁹As Binnick (2012, p.xxi) notes, there is no consensus in the Mongolic literature on what to call *-j/-ch*, although it seems to be historically derived from a converb. There is, however, general agreement that it serves to mark the continuative/progressive aspect, which also reflects how my consultants use this form. Hence, I have chosen to gloss it as PROG and to include it as one of the aspect endings.

However, (45) shows that the overt nonpast tense marker follows negation. In the formal register, when both an aspect and a tense marker are overt, the aspect marker attaches to the verb stem, while the tense marker is attached to a copula base, e.g. *bai*. This seems to be related to a general restriction on having both an overt aspect suffix and an overt tense suffix co-occur on the same verb stem, at least in the formal, standardised variety of Khalkha. The negative suffix *-güi* may only follow aspect markers, as seen in (45a). In the colloquial spoken register, it is possible for aspect, negation and tense to co-occur on the same lexical verb stem, as shown in (45b). In this case, the ordering of aspect and tense relative to the negative suffix *-güi* remains the same.

- (45) a. Odgerel huushuur hii-h-güi bai-**na**.
 O. khuushuur do-INF-NEG COP-NPST
 ‘Odgerel will not make khuushuur.’
- b. Odgerel huushuur hii-h-güi-**n**-ee.
 O. khuushuur do-INF-NEG-NPST-EMPH
 ‘Odgerel will not make khuushuur.’

In the absence of negation, the relative ordering of aspect and tense remains the same – aspect markers precede tense markers – as shown in (46).

- (46) Bi sur-**ch** bai-**san**.
 1SG study-PROG COP-PST
 ‘I was studying.’

In the rest of this dissertation, I will represent only overt tense suffixes in all Khalkha examples. I will represent null tense suffixes only when I wish to draw attention to their presence.

2.1.2.2 Converb markers

Another class of tense-aspect markers are the so-called converbs, which Janhunen (2012, p.163) describes as a class of nonfinite verb endings that appear in dependent clauses, turning them into adverbial clauses that modify the main clause. A converbial clause typi-

cally functions as a temporal modifier to the main clause, but may express other modifying functions as well, e.g. conditional. A partial list of converb markers is given in Table 2.4.

Table 2.4: Selected converb markers in Khalkha

Suffix	Function	Gloss
-ngUUt	‘as soon as’ (concomitant)	CCMT.CVB
-vAl	‘if’ (conditional)	COND.CVB
-mAgts	‘immediately after’ (immediative)	IMM.CVB
-hAd	‘when, while’ (temporal)	TEMP.CVB
-tAl	‘until’ (terminative)	TERM.CVB

As seen in (47), converb markers appear after aspect suffixes and negation.

- (47) Naraa [nadad chamaig itg-**ee-güi** bai-**had**] itge-sen.
 N. 1SG.DAT 2SG.ACC believe-IPFV-NEG COP-TEMP.CVB believe-PST
 ‘When you didn’t trust me, Naraa trusted me.’

Additionally, converb markers do not co-occur with overt tense markers within the same clause. They also trigger the presence of the copula base *bai* when there is an overt aspect marker on the verb – a parallel with tense markers. As I argue in Chapter 3, I treat converb markers as a type of T^0 . If true, then the fact that they do not co-occur with tense markers within the same clause is unsurprising.

2.1.3 The Khalkha nominal domain

In this subsection, I discuss the order of elements and some properties of nominal phrases in Khalkha. Possessor DPs, demonstratives, numerals and adjectives typically precede the head noun in Khalkha, while plural marking on the head noun is optional.²⁰ The canonical order of these elements, when all are present, is exemplified in (48). I assume that this ordering reflects the hierarchy of projections in the nominal spine, with the modifiers

²⁰Plural marking is ungrammatical if the noun is modified by a numeral:

- (i) Tuyaa nom-iin san-d dörvön nom(*-nuud) unsh-san.
 T. book-GEN centre-DAT four book(*-PL) read-PST
 ‘Tuyaa read four books at the library.’

occupying specifiers of dedicated functional projections (e.g. NumP, DemP, PossP), thus deriving their leftward placement despite the otherwise uniformly right-headed nature of the nominal spine.

- (48) *Possessor > Dem > Num > Adj > Noun*
 minii en gurban höörhön nohoi
 1SG.GEN this three cute dog
 ‘these three cute dogs of mine’

Certain quantifiers appear before the head noun, while others appear after. The universal quantifier *büh* ‘all’ precedes the head noun as shown in (49), while the distributive quantifier *bolgon* ‘each/every’ follows the head noun as shown in (50). Assuming that quantifiers are located in QP, I propose that *büh* is a quantificational adjective in Spec,QP, while quantifiers like *bolgon* occupy Q⁰.

- (49) *Büh ‘all’ precedes head noun*
 Büh oyutn-uud
 all student-PL
 ‘all students’

- (50) *Bolgon ‘each’ follows head noun*
 hüühed bolgon
 child each
 ‘each/every child’

Other elements that follow the head noun include plural marking, case suffixes and possessive (POSS) enclitics, all shown in (51). I propose that plural markers occupy Num⁰, possessive enclitics occupy Poss⁰, and case markers occupy D⁰. I return to the relative surface positioning of possessive enclitics and case markers shortly.

- (51) *Plural markers, case markers, and possessive enclitics follow N*
 Dorj noh-**d-iig=**min’ hoollo-son.
 D. dog-PL-ACC=1SG.POSS feed-PST
 ‘Dorj fed my dogs.’

Table 2.5 illustrates the inventory of case suffixes in Khalkha; where a case suffix contains more than one allomorph, the alternation is phonologically-conditioned.

Table 2.5: Case markers in Khalkha

Case	Gloss	Suffix
Accusative	ACC	-iig, -g
Genitive	GEN	-(g)iin, -ii, -n
Dative	DAT	-d, -t
Ablative	ABL	-(g)AAs
Instrumental	INSTR	-(g)AAr
Comitative	COM	-tAi
Directive	DIR	rUU, lUU ²¹

Finally, Table 2.6 displays the inventory of POSS enclitics in Khalkha. The 1st/2nd person POSS enclitics appear to be reduced forms of the full genitive pronouns.

Table 2.6: GEN pronouns and POSS enclitics in Khalkha

PERSON	PRONOUN	ENCLITIC
1SG	minii	min'
1PL.INCL	bidnii	maan'
1PL.EXCL	manai	maan' ²²
2SG.IFML	chinii	chin'
2SG.FRML	tanii	tan'
2PL.IFML	tanai	tan'
2PL.FRML	ta bühnii	tan' ²³
3SG	tüünii	n'
3PL	tednii	n' ²⁴

²¹In Mongolian Cyrillic, the directive case marker is written as a separate word. As such, scholars debate whether it is a true case suffix. I will assume it is a case marker.

²²The distinction between 1PL.INCL and 1PL.EXCL is neutralised in the clitic form.

²³The distinctions between 2SG.FRML, 2PL.IFML and 2PL.FRML are neutralised in the clitic form.

²⁴The distinction between 3SG and 3PL is neutralised in the clitic form.

A pronominal possessor may be expressed using a genitive pronoun (52a), its corresponding POSS enclitic (52b), or with both a genitive pronoun and a POSS enclitic (52c).

(52) *Two ways to express pronominal possession in Khalkha*

- a. Chi **minii** nom-iig unsh-san.
2SG 1SG.GEN book-ACC read-PST
'You read my book.'
- b. Chi nom-iig=**min'** unsh-san.
2SG book-ACC=1SG.POSS read-PST
'You read my book.'
- c. Chi **minii** nom-iig=**min'** unsh-san.
2SG 1SG.GEN book-ACC=1SG.POSS read-PST
'You read my book.'

The acceptability of clitic doubling of the pronominal possessor (52c) is debated in the literature. Gong (2021, 2022) and Guntsetseg (2016, p.51) report that the genitive pronoun and possessive enclitic do not usually co-occur, showing that the clitic-doubled sequence *minii ... min'* is ungrammatical. While my consultants do find (52c) strange to say, they note that it is sometimes used under emphatic circumstances. A search in the Mongolian National Corpus (MNC) also turns up many examples of the clitic-doubled sequence *minii ... min*, e.g. (53).

- (53) ... Chi **minii** gants ohin-iig=**min'** al-san ...
2SG 1SG.GEN only daughter-ACC=1SG.POSS kill-PST
'... You killed my only daughter ...' (example from MNC, retrieved 2022/06/09)

My consultants also find clitic doubling of 2SG (e.g. *chinii... chin'*) and 3SG (e.g. *tüünii... n'*) much more acceptable (see Hammar 1983, p.86 and Brosig et al. 2018 for similar observations). I assume that clitic doubling of possessors is possible, albeit licensed only in certain pragmatic (e.g. emphatic) contexts. I defer an investigation of the exact conditions that license possessor clitic doubling to future research.

As discussed in §2.1.1.2, reflexive possession can be expressed using a GEN-marked anaphoric pronoun (54a) or the RP enclitic (54b) (see footnote 12 in this chapter for some

discussion of the differences in meaning between using the reflexive possessive pronoun vs. the RP enclitic).

(54) Two ways to express reflexive possession in Khalkha

- a. Bi **öör-iin-h=öö** nom-iig unsh-san.
1SG self-GEN-H=RP book-ACC read-PST
'I read my (own) book.'
- b. Bi nom-Ø=**oo** unsh-san.
1SG book-ACC=RP read-PST
'I read my book.'

I propose that the surface position of POSS enclitics is derived, rather than reflective of the Mirror Principle (Baker 1985). When case marking is present, POSS follows case, e.g. (52b), which is unexpected given that case is typically marked on the whole nominal. Further, assuming that case markers occupy D^0 , the surface position of POSS enclitics would imply that $D^0 < \text{Poss}^0$, which is at odds with the idea that D^0 is the highest functional category in nominal structures. However, if we assume $\text{Poss}^0 < D^0$, the surface order in (52b) would violate the Mirror Principle (Baker 1985).

The same Mirror Principle-violating order is found in case-marked quantified nominals, as seen in (55) and (56).

(55) Bilgüün [_{PP} shiree bolgon-ii=**min**' urd] ayga tav'-san.
B. table each-GEN=1SG.POSS before cup put-PST
'Bilguun put a cup in front of each of my tables.'

(56) Bi [_{IO} hüühed bolgon-d=**oo**] neg neg tsunh av-san.
1SG child each-DAT=RP one one bag buy-PST
'I bought a bag for each of my children.' (Tserenpil and Kullmann 2008, p.286)

In the above examples, the reflexive possessor is interpreted within the scope of the postnominal quantifier *bolgon* 'each', which implies $\text{Poss}^0 < Q^0$. Yet in the surface order, the POSS enclitic follows both the postnominal quantifier and case marking, thus violating the scope-obeying Mirror Principle order.

Interestingly, in PPs, the position of POSS enclitics depends on the presence of case-marking. I demonstrate this using the minimal pair in (57) containing the postposition *deer*

‘on’, which may select for a genitive-marked or a case-unmarked nominal complement. If the postposition takes a case-marked nominal complement, POSS surfaces after case marking and before the postposition, as shown in (57a). If the postposition takes a case-unmarked nominal complement, POSS surfaces after the postposition, as shown in (57b) (note that *shiree* ‘table’ is marked with hidden-*n*).

(57) ‘Bilguun put the cup on my table.’

a. *PP with case-marked NP*

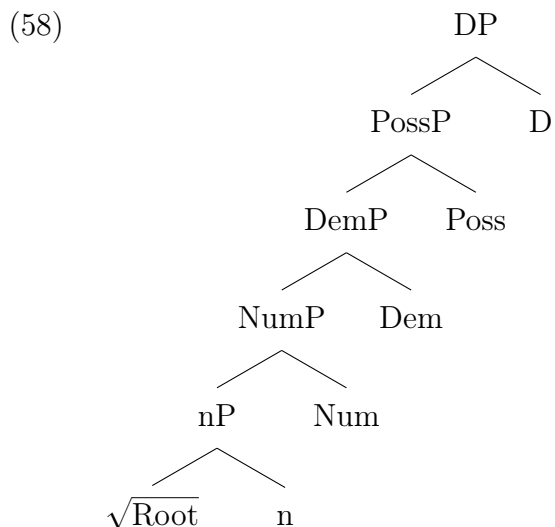
Bilgüün ayag-iig [PP shiree-nii(=min’) deer(*=min’)] tav’-san.
 B. cup-ACC table-GEN=1SG.POSS on*=1SG.POSS put-PST

b. *PP with case-unmarked NP*

Bilgüün ayag-iig [PP shiree-n(*=min’) deer(=min’)] tav’-san.
 B. cup-ACC table-N*=1SG.POSS on=1SG.POSS put-PST

The Mirror Principle-violating and variable position of POSS suggests that its position may be derived via postsyntactic morphological reordering. Thus, I propose that the surface position of POSS enclitics is derived, rather than reflective of the Mirror Principle. For a Lowering analysis based on similar facts in another Mongolic language, Dagur, see Gong 2021. In this dissertation, I will assume that D^0 (which hosts case morphology) and P^0 (which hosts postpositions) may Lower onto Poss^0 , leaving a fuller analysis to future work.

Based on the evidence discussed above, I propose the following hierarchy of projections for Khalkha nominals in (58).



2.1.4 Khalkha embedded clauses

This subsection introduces embedded clauses in Khalkha, with a focus on the four types that constitute the empirical focus of this dissertation. These four embedded clause types comprise three types of argument clauses – finite bare CPs (59), finite nominalised clauses (NCs) (60), and nonfinite NCs (61) – and one type of adjunct clause – converbial adjuncts (62).

(59) *Finite bare CP*

Bat changaar [**nohoi-g/nohoi** gaihal-tai **gej**] hel-sen.
 B. loudly dog-ACC/dog wonder-COM C say-PST
 ‘Bat said loudly that dogs are wonderful.’ (Fong 2019, p.3, ex.5)

(60) *Finite complement NC*

Naraa [**Bilgüün-iig/Bilgüün-ii/Bilgüün** ger-iin daalgavr-Ø-aa duusga-san
 N. B.-ACC/B.-GEN/B. home-GEN task-ACC-RP finish-PST
gedg]-iig sana-san.
 C.NML -ACC remember-PST
 ‘Naraa remembered that Bilguun finished his homework.’

(61) *Nonfinite complement NC*

Bi [**Bat-iig/Bat-iin/Bat** sugalaa-nd hoj-son]-iig med-ne.
 1SG B.-ACC/B.-GEN/B. lottery-DAT win-PFV -ACC know-NPST
 ‘I know that Bat won the lottery.’ (Aravind 2021, p.379, ex.6)

(62) *Converbial adjunct*

Bi [**tüüniig/ter** hicheel-Ø=ee hii-j bai-had] unt-san.
 1SG 3SG.ACC/3SG lesson-ACC=RP do-PROG COP-TEMP.CVB sleep-PST
 ‘When she/he was studying, I fell asleep.’

Finite bare CPs (59) contain an overt complementiser *gej*, lack case marking, and may contain accusative and case-unmarked subjects. Finite NCs (60) contain an overt

complementiser *ged(e)g*,²⁵ bear case marking, and may contain accusative, genitive, and case-unmarked subjects. Nonfinite NCs (61) lack an overt complementiser, bear case marking, and may contain accusative, genitive, and case-unmarked subjects. Finally, converbial adjuncts (62) lack an overt complementiser and case marking, and may contain accusative and case-unmarked subjects.

I propose that converbial adjuncts adjoin to vP – a position below VoiceP where the external argument is introduced.²⁶ This is supported by two observations. First, as shown in (63), a converbial adjunct cannot follow a manner adverb like *hurdan* ‘quickly’, which adjoins to vP. However, in (64), a converbial adjunct may precede or follow an aspectual adverb like *dandaa* ‘always’, which adjoins to AspP.

- (63) Bat {*hurdan} [eej-iig=ee har'-had]
 B. {*quickly} mother-ACC=RP return-TEMP.CVB
 {hurdan} ger-iin daagavr-Ø=aa hii-sen.
 {quickly} home-GEN assignment-ACC=RP do-PST
 ‘When his_j mother returned (home), Bat_j quickly did his homework.’

²⁵The variation in complementiser forms can be traced to the fact that *ge* is historically the root of the verb ‘to say’; in modern usage, it has mostly been replaced by *hel* ‘to say’. As evidence of their verbal provenance, it is possible to deconstruct *gej* as comprising the root *ge* + the converb *-j*, and *ged(e)g* as comprising *ge* + the habitive aspect suffix *-d(e)g*. Other complementiser forms involving different aspect suffixes are attested, e.g. *ge* + the futuritive/infinitive suffix *-h* and *ge* + the perfective suffix *-s(e)n*. *Ge* itself may still be used as a verb, e.g. *gene* ‘say(s); will say’, which comprises the root *ge* + the nonpast tense suffix *-ne*. In fact, *gene* cannot co-occur with another ‘say’-verb, implying that it is not used as a complementiser.

Different forms of the complementiser may be associated with slightly different meanings regarding the nature of the information reported in the embedded clause. For example, one consultant suggests that *ges(e)n* is used when the information reported in the embedded clause is secondhand information (see also Tserenpil and Kullmann 2008, p.309). Another consultant, however, does not share this intuition, and reports that *ged(e)g* and *ges(e)n* can be used interchangeably. An in-depth investigation of these meanings is beyond the scope of this dissertation. However, many authors note that *gej* and *ged(e)g* are most commonly used; see, e.g. Tserenpil and Kullmann 2008, p.309; Janhunen 2012, pp.283-284; Gong 2022, pp.24-27. This is consistent with how my consultants use them. Thus, I adopt the view that *gej* and *ged(e)g* have been grammaticalised into complementisers in modern Khalkha, and I analyse *ged(e)g* as a nominalised form of the complementiser; see Chapter 3 for further details.

²⁶In contrast, Peters (2024, p.19) proposes that converbial adjuncts adjoin to a position above VoiceP but below TP. Under this analysis, converbial adjunct subjects are never locally c-commanded by a matrix subject argument in Spec,VoiceP, and the fact that converbial adjunct subjects and possessors located within such subjects may be anaphorically bound by matrix subjects (see Chapter 4 for further data) would require an alternative explanation than that proposed by Gong 2023b; Gong and Despić 2024.

- (64) Misheel {dandaa} [eej-iig=ee har'-tal]
 M. {always} mother-ACC=RP return-TERM.CVB
 {dandaa} video togloom toglo-dog.
 {always} video game play-HAB
 ‘Misheel_j always plays video games until her_j mother returns (home).’

The contrast between (63) and (64) suggests that converbial adjuncts adjoin to a position at least as high as vP.²⁷ Further, (63) and (64) also show that the possessor of the embedded subject within the converbial adjunct can be bound by the matrix subject via reflexive-possessive marking. Assuming that subject-oriented anaphoric binding in Khalkha requires local c-command by a subject located in Spec,VoiceP (Gong 2023b; Gong and Despić 2024) (also see §2.1.1.2), this suggests that converbial adjuncts adjoin below VoiceP. Taken together, I assume that converbial adjuncts uniformly adjoin to vP, and that they may scramble higher within the main clause, e.g., in (64), where the converbial adjunct precedes the aspectual adverb *dandaa* ‘always’.

Additionally, Khalkha contains relative clauses (RCs) and postpositional adjuncts. While these embedded clause types fall outside of the scope of this dissertation, they are shown here for completeness. Relative clauses may be classified as subject RCs (65), object RCs (66) and adjunct RCs (67). In object position, as shown in the (b) examples below, an RC-modified noun obligatorily bears accusative case. The RC-internal subject may be genitive or case-unmarked but not accusative, as shown in (66) and (67).

(65) *Subject relative clause*

- a. [[Bilgüün-iig har-san] hün] namaig har-san.
 B.-ACC see-PFV person 1SG.ACC see-PST
 ‘The person who saw Bilgüün saw me.’
 b. Bi [[Bilgüün-iig har-san] hün-iig] har-san.
 1SG B.-ACC see-PFV person-ACC see-PST
 ‘I saw the person who saw Bilgüün.’

²⁷Given that both *hurdan* ‘quickly’ and converbial adjuncts adjoin to vP, in principle, we predict that the positions of these constituents would be interchangeable. However, *hurdan* may not precede the converbial adjunct, as shown in (63). I take this to imply that other factors may be involved in determining the order of manner adverbs and converbial adjuncts.

(66) *Object relative clause*

- a. [[Minii/Bi/*Namaig öchigdör har-san] hün] Dorj-iig üns-sen.
 1SG.GEN/1SG/*1SG.ACC yesterday see-PFV person D.-ACC kiss-PST
 ‘The person I saw yesterday kissed Dorj.’
- b. Dorj [[minii/bi/*namaig öchigdör har-san] hün-iig] üns-sen.
 D. 1SG.GEN/1SG/*1SG.ACC yesterday see-PFV person-ACC kiss-PST
 ‘Dorj kissed the person I saw yesterday.’

(67) *Adjunct relative clause*

- a. [[Naraa-giin/Naraa/*Naraa-g bich-sen] üzeg] ald-agd-san.
 N.-GEN/N./*N.-ACC write-PFV pen lose-PASS-PST
 ‘The pen that Naraa wrote with is lost.’
- b. Bi [[Naraa-giin/Naraa/*Naraa-g bich-sen] üzg-iig] ol-son.
 1SG N.-GEN/N./*N.-ACC write-PFV pen-ACC find-PST
 ‘I found the pen that Naraa wrote with.’

Postpositional adjuncts are headed by an overt postposition and function as modifiers to the matrix clause. Depending on the postposition, its clausal complement may be case-unmarked or case-marked. (68) shows an adjunct clause headed by *hürtel* ‘until’, which takes a case-unmarked clausal complement. (69) shows an adjunct clause headed by *daraa* ‘after’, which takes a genitive-marked clausal complement. (70) shows an adjunct clause headed by *ömnö* ‘before’ which takes an ablative-marked clausal complement. The overt subject of postpositional adjunct clauses may be accusative or case-unmarked, but not genitive.

(68) *Postpositional adjunct headed by хүртел ‘until’*

Bat [Dulmaa-g/Dulmaa/*Dulmaa-giin hool hii-h **hürtel**] video togloom
 B. D.-ACC/D./*D.-GEN food make-INF until video game
 toglo-son.
 play-PST
 ‘Bat played video games until Dulmaa cooked.’

(69) *Postpositional adjunct headed by daraa ‘after’*

Bat [Dulmaa-g/Dulmaa/*Dulmaa-giin hool hii-sn-ii **daraa**] video togloom
 B. D.-ACC/D./*D.-GEN food make-PFV-GEN after video game
 toglo-son.
 play-PST
 ‘Bat played video games after Dulmaa cooked.’

- (70) *Postpositional adjunct headed by ömnö ‘before’*
 Bat | Dulmaa-g/Dulmaa/*Dulmaa-giin hool hii-h-ees **ömnö** | video togloom
 B. D.-ACC/D./*D.-GEN food make-INF-ABL before video game
 toglo-son.
 play-PST
 ‘Bat played video games before Dulmaa cooked.’

2.1.5 Summary

In this section, I have outlined key aspects of clause structure, verb structure, and nominal structure in Khalkha. Based on evidence from scope interactions and adverb placement, I have argued that subjects move to Spec,TP, which constitutes a derived A-position. Drawing on binding data, I have further argued that subjects asymmetrically c-command both DOs and IOs, and that IOs asymmetrically c-command DOs.

Within the verbal domain, I have shown that Khalkha possesses three distinct classes of verbal suffixes: tense markers, aspect markers, and converb markers. Within the nominal domain, I have assumed a uniformly head-final structure, based on the order of various nominal modifiers which I take to be merged in the specifiers of dedicated functional projections. I also adopt the view that the surface position of POSS enclitics is postsyntactically derived (Gong 2021).

Finally, I have introduced the four embedded clause types that form the empirical focus of this dissertation. They are finite bare CPs, finite NCs, nonfinite NCs, and converbial adjuncts.

In the next section, I provide some background on the morphosyntax of Kalmyk while noting similarities and differences in comparison with the morphosyntax of Khalkha.

2.2 Background on Kalmyk morphosyntax

While the Kalmyk data comes from two speakers who each speak a different dialect – Dörbet and Torghut – the examples presented in this dissertation are accepted by both speakers and do not contain any significant differences across dialect. Like in Khalkha, the basic word order in Kalmyk is S-V in an intransitive clause (71), S-DO-V in a transitive clause (72), and S-IO-DO-V in a ditransitive clause (73).

- (71) *ky:kŋ gara-ɕa-na*
 girl jump-PROG-NPST
 ‘The girl is jumping.’²⁸

- (72) *badma mandɕ-igə tsok-əv*
 B. M.-ACC hit-PST
 ‘Badma hit Manj.’

- (73) *badma tandə zætə bilgəl-xə-Ø*
 B. 2SG.FRML.DAT nutmeg gift-INF-NPST
 ‘Badma will gift you (polite) nutmeg.’

(71)-(73) also show that verb inflections follow the verb stem and the verb follows its arguments, indicating that Kalmyk possesses a head-final word order. Typical of head-final languages, in Kalmyk relative clauses precede their head nouns (74), and postpositions are used (75).

- (74) *Relative clause precedes head noun*
 [*xœ xærul-dəg*] *noxə-s*
 sheep guard-HAB dog-PL
 ‘dogs who guard sheep’

²⁸Kalmyk also possesses hidden-*n* on certain nouns, as seen in *ky:kŋ* ‘girl’. However, the distribution of hidden-*n* in Kalmyk is quite different from hidden-*n* in Khalkha. In Khalkha, hidden-*n* surfaces only in the context of certain case markers and postpositions, whereas in Kalmyk, hidden-*n* is preserved in the absence of any overt case markers (e.g. case-unmarked subjects) and disappears in the context of certain case markers, e.g. *ky:k-igə* ‘girl-ACC’. Bläsing (2003, p.232) writes that in Kalmyk “the [hidden] nasal is preserved in the nominative case, but it is lost in the indefinite, accusative, instrumental, and possessive cases”.

- (75) *Adposition precedes nominal complement*
 bagsh [[shköl] tal] yow-j yow-na
 teacher school to.side.of go-PROG go-NPST
 ‘The teacher is going in the direction of the school.’ (Bläsing 2003, p.236)

2.2.1 Kalmyk clause structure

I will assume the same clause structure for Kalmyk as I argued in Khalkha, namely, that S c-commands DO (and IO). Supporting evidence for this view comes from Condition B effects displayed by pronominal DOs and pronominal possessors of direct objects.

Note that Kalmyk possesses the same order of elements in the verb complex as Khalkha – $Asp^0 < Neg^0 < T^0$ – as shown in (76).

- (76) *Kalmyk verb complex: $Asp^0 < Neg^0 < T^0$*
 bi cetskøldɾ en dektɾ umf-sɳ uga bi-læ-v
 1SG yesterday this book read-PFV NEG COP-PRF.PST-1SG
 ‘I did not read this book yesterday.’

First, observe that a pronominal DO cannot be coreferential with its clausemate subject, as shown in (77)-(79). Note that in these examples, the DO has scrambled over the IO, resulting in S-DO-IO-V order.

- (77) a. bi bijan nyr yzdæg-tə yz-ɕa-na^w-v
 1SG REFL.ACC mirror-DAT see-PROG-NPST-1SG
 ‘I see myself in the mirror.’
 b. *bi namagə nyr yzdæg-tə yz-ɕa-na^w-v
 1SG 1SG.ACC mirror-DAT see-PROG-NPST-1SG
 Int. ‘I see me in the mirror.’
- (78) a. ʈi bijan nyr yzdæg-tə yz-ɕa-na-ʈɪ
 2SG REFL.ACC mirror-DAT see-PROG-NPST-2SG
 ‘You see yourself in the mirror.’
 b. *ʈi ʈamagə nyr yzdæg-tə yz-ɕa-na-ʈɪ
 2SG 2SG.ACC mirror-DAT see-PROG-NPST-2SG
 Int. ‘You see you in the mirror.’

- (79) a. *damba bijan nyr yzdæg-tə yz-ɟa-na*
 D. REFL.ACC mirror-DAT see-PROG-NPST
 ‘Damba sees himself in the mirror.’
- b. *damba_j terygə_{k/*j} nyr yzdæg-tə yz-ɟa-na*
 D. 3SG.ACC mirror-DAT see-PROG-NPST
 ‘Damba_j sees him/her_{k/*j} in the mirror.’

The (b) examples above show that a pronominal DO cannot be bound by its clausemate subject, otherwise it violates Binding Condition B. To express coreference between the subject and DO, the accusative form of the reflexive anaphor *bijan* must be used, as seen in the (a) examples.

Further, like in Khalkha, pronominal possessors are also subject to Binding Condition B in Kalmyk. As shown in (80b) and (81b), the possessor of the DO and IO respectively cannot be bound by the clausemate subject, otherwise resulting in a Binding Condition B violation. Instead, the (reflexive possessive) RP enclitic is used to express coreference, as seen in the (a) examples below.

- (80) a. *bi noxa-g=an il-læ-v*
 1SG dog-ACC=RP stroke-PRF.PST-1SG
 ‘I stroked my dog.’
- b. **bi mini noxa-gə il-læ-v*
 1SG 1SG.GEN dog-ACC stroke-PRF.PST-1SG
 Int. ‘I stroked my dog.’
- (81) a. *bi ɕerulyar noxa-d=an xotə ɕəgə-læ-v*
 1SG in.morning dog-DAT=RP food give-PRF.PST-1SG
 ‘I gave food to my dog in the morning.’
- b. **bi ɕerulyar mini noxa-d xotə ɕəgə-læ-v*
 1SG in.morning 1SG.GEN dog-DAT food give-PRF.PST-1SG
 Int. ‘I gave food to my dog in the morning.’

Thus, I conclude that subjects c-command both DOs and IOs in Kalmyk. In the next subsection, I discuss the verb complex in Kalmyk, which differs from Khalkha in one key aspect – the presence of subject ϕ -agreement.

2.2.2 The Kalmyk verb complex

This subsection discusses the elements that compose the Kalmyk verb complex, which follows the following general template in (82). I discuss converb markers in §2.2.2.3.

- (82) *Kalmyk verb complex*
 VERB STEM (-VOICE) (-ASPECT) (-NEG) (-CONVERB/-TENSE) (-SUBJ.AGR)

(83) shows that Kalmyk contains a distinct causative morpheme *-ul*, which I assume realises a distinct v^0 above verbalising v^0 . The passive morpheme *-gd*, which realises Voice⁰, follows both the verb root and the causative morpheme. Finally, the passive morpheme is followed by the progressive aspect suffix *-ja*, the tense suffix *-na*, and 2SG subject agreement morphology *-ch*.

- (83) Chi, Ivanov, taldan ghazr-t ghar-h-in tölöd,
 2SG I. another place-DAT get.out-INF-GEN in.order.to,
 til tal yov-**ul-gd-ja-na-ch**.
 enemy.rear to.side.of go-CAUS-PASS-PROG-NPST-2SG
 ‘You, Ivanov, in order to go to another country, are being made to go to the rear side
 of the enemy.’ (Kukanova 2016, p.24)

(84) shows that negation follows the causative, passive, and aspect morphemes, but precedes tense and subject agreement morphemes.

- (84) tʃi nanar jov-**ul-gd-ɬa-x-ʃə-Ø-tʃ**
 2SG 1SG.INSTR go-CAUS-PASS-PROG-INF-NEG-NPST-2SG
 ‘You are not being sent by me.’

2.2.2.1 Subject-predicate agreement

One of the most striking differences between Kalmyk and Khalkha is the presence of subject-predicate ϕ -agreement in Kalmyk. In the Mongolic literature, such agreement morphology is referred to as “predicative personal endings”. In addition to Kalmyk (Street 1959, pp.55-56), subject agreement is also found in other Oirat varieties, Buryat, Khamnigan Mongol, Dagur and Moghol (Janhunen 2020, 2024); see, e.g., Poppe 1960, p.123 on Buryat. Janhunen (2020, p.16) writes that subject agreement morphology in these Mongolic

languages developed from “postposed pronominal nominatives” and that “[t]here is no doubt that these developments took place under the influence of the neighbouring Turkic and Tungusic languages, which have similar systems of person marking”. For syntactic accounts of the diachronic development of agreement morphology in Mongolic, see Julien 2002 and Fuß 2005.

As seen in (82), subject agreement morphology is always located in final position of the verb complex. Table 2.7 displays the subject agreement morphemes found in Kalmyk (see also Bläsing 2003, p.245 and Birtalan 2020, p.366).

Table 2.7: Subject agreement morphemes in Kalmyk

	SG	PL
1	-v~-β/-b ²⁹	-vdŋ~-βdŋ
2	-tʃ	-t
3		Ø

²⁹The alternation from /v~β/ to [b] is due to /v~β/ undergoing fortition when preceded by a nasal consonant /n/ or /m/. Additionally, if the preceding consonant is /n/, it then assimilates in place with [b], becoming [m]. These two processes are given in (i) and can be seen in (iib) on a predicate noun. Further, /ə/-epenthesis applies in two contexts. First, in general, coda clusters in Mongolic languages adhere to a sonority hierarchy. For example, in Khalkha, Svantesson (1995) and Svantesson et al. (2005) propose that stops/affricates < fricatives < sonorants, and /ə/-epenthesis applies to remedy violations of falling sonority in consonant clusters. I assume that a similar hierarchy also holds in Kalmyk. Syllabification proceeds from right to left. In (iic), /mb/ obeys falling sonority and /ə/-epenthesis does not apply. However, /gm/ violates falling sonority and ə-epenthesis applies between /gm/. Second, as shown in (iid), /ə/-epenthesis may also apply after word-final consonants (Indjieva 2009, p.26). For discussion of syllabification in Mongolic, see Svantesson 1995; Svantesson et al. 2005 on Khalkha, and Indjieva 2009, pp.25-26 on Kalmyk.

- (i) a. /v~β/ → [b] / [+nasal] _
b. /n/ → [m] / _ [+labial]
- (ii) a. bi mergn-v
1SG M.-1SG
‘I am Mergen.’
b. (i) *applies*
/bi mergnv/ → /bi mergmb/
c. /mb/ does not violate falling sonority but /gm/ does; /ə/ inserted between /gm/
/bi mergmb/ → /bi mergəmb/
d. /ə/ inserted after word-final C
/bi mergəmb/ → /bi mergəmbə/

The examples in (85) illustrate subject-verb agreement in root transitive clauses. Note that the pronunciation of the tense-aspect markers may vary (e.g. /la^w/~/la/~/læ/) depending on the agreement markers (see also Bläsing 2003).

(85) Kalmyk: *φ-agreement in finite root clauses*

- a. **bi** tʃamagə yz-**la^w-v**
1SG 2SG.ACC see-PRF.PST-1SG
'I saw you (sg).'
- b. **bidŋ/madŋ** tʃamagə yz-**læ-vidŋ**
1PL.INCL/1PL.EXCL 2SG.ACC see-PRF.PST-1PL
'We saw you (sg).'
- c. **tʃi** namagə yz-**la-tʃ**
2SG 1SG.ACC see-PRF.PST-2SG
'You (casual, sg) saw me.'
- d. **ta** namagə yz-**læ-t**
2SG.FRML 1SG.ACC see-PRF.PST-2PL
'You (polite, sg) saw me.'
- e. **tadŋ** namagə yz-**læ-t**
2PL 1SG.ACC see-PRF.PST-2PL
'You (pl) saw me.'
- f. **terə** namagə yz-**læ**
3SG 1SG.ACC see-PRF.PST
'She/he saw me.'
- g. **tedŋ** namagə yz-**læ**
3PL 1SG.ACC see-PRF.PST
'They saw me.'

First person subjects trigger agreement morphology that distinguishes between the singular (85a) and plural (85b). The polite second person singular /ta/ (85d) triggers the same subject agreement morphology as the second person plural /tadŋ/ (85e) and thus it may be seen a formally plural but referentially singular form (Wang 2023). Third person subjects, both singular (85f) and plural (85g), do not trigger overt agreement on the verb. While I do not gloss third person agreement in the Kalmyk sentences in this dissertation, I remain agnostic as to whether the lack of overt agreement in (85f) and (85g) represents a total absence of agreement or null agreement morphology on the verb.

While the case-unmarked subject may be dropped, subject agreement morphology on the verb is obligatory (86). It is not possible, for instance, to have default 3rd person verb forms with 1st person singular case-unmarked subjects, e.g. (87).

- (86) (bi) tʃamagə yz-la^w-v
 1SG 2SG.ACC see-PRF.PST-1SG
 ‘I saw you (sg).’

- (87) *bi tʃamagə yz-læ
 1SG 2SG.ACC see-PRF.PST
 Int. ‘I saw you (sg).’

I assume, like in Khalkha, that unergative and transitive subjects are first merged in Spec,VoiceP while unaccusative and passive subjects are first merged in the complement position of the verb. As shown by the minimal pair in (88)-(89), grammatical subjects control agreement on the verb. Given the verb-final position of subject agreement morphology, I propose that it arises due to Agree between T⁰ and the subject in Kalmyk. I further assume that Kalmyk T⁰ bears an [EPP] feature, and that subject-verb agreement in Kalmyk is accompanied by movement of the subject to Spec,TP.

- (88) bi tʃamagə jov-ul-sɳ bi-læ-v
 1SG 2SG.ACC go-CAUS-PFV COP-PRF.PST-1SG
 ‘I sent you.’

- (89) tʃi nanar jov-ul-gd-sɳ bi-læ-tʃ
 2SG 1SG.INSTR go-CAUS-PASS-PFV COP-PRF.PST-2SG
 ‘You were sent by me.’

2.2.2.2 Tense and aspect markers

As shown in (83), Kalymk contains distinct tense and aspect markers. Tense markers, also called “finite indicative” tense endings, are described as having the properties of appearing in finite clauses and bearing subject agreement morphology (Bläsing 2003, p.244; Birtalan 2003, p.366). Table 2.8 provides a list of some Tense markers in Kalmyk. As reported in Bläsing 2003, p.244 and Birtalan 2020, p.366, *-nA* may function as a present or

future tense marker. As such, I treat it as a nonpast marker with a null allomorph, as shown in (73).

Table 2.8: Selected tense markers in Kalmyk

Suffix	Function	Gloss
-nA~ $\emptyset$ ³⁰	present-future	NPST
-v(ə)~ $\emptyset$ ³¹	narrative past	PST
-lA	pluperfect	PRF.PST
-ɖʒ(ə)	evidential past	EVID.PST

Aspect markers are also called participles in the Kalmyk/Oirat literature (Bläsing 2003; Birtalan 2003, 2020). Table 2.9 lists some aspect markers in Kalmyk.

Table 2.9: Selected aspect markers in Kalmyk

Suffix	Function	Gloss
-s(n)	perfective	PFV
-x(ə)	futuritive/infinitive	INF
-d(ə)g	habitive	HAB
-AA	imperfective	IPFV
-ɖʒA~ $\emptyset$	progressive	PROG

I have only found one possible combination in Kalmyk where both an aspect and a tense marker occupy the same verb complex – the progressive aspect marker $/-ɖʒA~\emptyset/$ and the nonpast tense marker nA – as shown in (90).³² Other combinations of aspect and tense markers are used with a copula base such as *bi* (an allomorph of $/bæ:/$, cognate with

³⁰As with Khalkha, I use capital *A* to represent vowels that undergo front-back harmony in Kalmyk.

³¹The (narrative) past tense suffix $-v(ə)$ becomes $-u$ if it precedes another $/v/$ -initial morpheme, e.g. 1SG subject agreement $-v$ or the polar interrogative (PQ) particle $-v$:

- (i) xon-**v**-v → xon**uv**
sleep-PST-1SG
‘I slept.’
- (ii) sæn xon-**v**-v-ɖʒ → sæn xon**uv**ɖʒ
good sleep-PST-PQ-2SG
‘Did you sleep well?’

³²Bläsing (2003, p.244) lists $/-ɖʒAnA~\emptyset/$ as a standalone finite tense marker in Kalmyk, and analyses it as being derived from $/-ɖʒA~\emptyset/$ -PROG + $/bæ:næ/$ COP-NPST.

Khalkha *bai*); see, e.g., (88)-(90). Further research is required to ascertain whether other combinations of tense and aspect can be pronounced without the use of a copula base.

- (90) bi tsevɾ aɣar ki:l-ɕa-na^w-v
 1SG clean air breathe-PROG-NPST-1SG
 ‘I breathe in clean air.’

2.2.2.3 Converb markers

Kalmyk also possesses an inventory of converb markers, which attach to verbs of dependent clauses and turn them into adverbial clauses that modify the main clause. A non-exhaustive list of converb markers, some cognate with the Khalkha markers in Table 2.10, are given below (based on Bläsing 2003, p.243-244):

Table 2.10: Selected converb markers in Kalmyk

Suffix	Function	Gloss
-xlA	‘when, while’ (successive)	SUCC.CVB
-tl	‘until’ (terminative)	TERM.CVB
-sAr ³³	‘since, before’ (causal)	CAUS.CVB
-vsn	‘although’ (concessive)	CCSV.CVB

An example of each of the converbs in Table 2.10 used in a clause is given below:

- (91) [camag ir-**xlä**] bi kino-d od-na-w
 2SG.ACC come-SUCC.CVB 1SG cinema-DAT go-NPST-1SG
 ‘As soon as you come, I will go to the cinema.’ (Bläsing 2003, p.244)
- (92) [namag ir-**tl**] enünd suu-ja
 1SG.ACC come-TERM.CVB this.DAT stay-IMP
 ‘Stay here until I come!’ (Bläsing 2003, p.244)
- (93) [söö xarngghu bol-**sar**] bi töör-ü-w
 night dark become-CAUS.CVB 1SG be.lost-PRF.PST-1SG
 ‘Because the night was dark, I got lost.’ (Bläsing 2003, p.244)

³³Also called the abtemporal converb in the Kalmyk literature.

- (94) xalymg xön [muu noos-ta bol-**wcn**] ik max ik öök ög-nä
 Kalmyk sheep bad wool-SOC become-CCSV.CVB a.lot meat a.lot fat give-NPST
 ‘Although it has bad wool, the Kalmyk sheep gives a lot of meat and a lot of fat.’
 (Bläsing 2003, p.243)

2.2.2.4 Negation markers

Kalmyk possesses a complex inventory of negation markers, some of which are rarely used and only found in historical written texts. For diachronic accounts of these markers, see Baranova 2015, 2019; Baranova and Mishchenko 2022. I will briefly introduce the two most common negative suffixes used in verbal predicates, *-go* and *-fə*, both of which are derived from the negative copulas *uga* and *bifə* respectively.

According to Baranova and Mishchenko 2022, p.421, using the negative copula *uga* is the only way to negate past predicates. Three examples are shown below. (96) shows that overt subject agreement morphology surfaces on the negative copula itself in the absence of any tense markers. (97) shows that when the negative copula co-occurs with overt tense morphology, the tense marker is hosted on the copula base *bi*.

- (95) däkčə zarɁ-də od-sən **uga**
 again court-DAT go-PFV NEG.COP
 ‘He hasn’t gone to court again.’ (Baranova and Mishchenko 2022, p.421, ex.44)
- (96) a madən tüündə tas mu ke-Ɂäd **uga**-vidn
 but 1PL.INCL 3SG.DAT very bad do-CVB.PRF NEG.COP-1PL
 ‘But we did not do anything awfully bad to him.’
 (Baranova and Mishchenko 2022, p.421, ex.45)
- (97) nanla ädl surhul’ sur-sn **uga** bi-lä
 1SG.COM like sciences study-PFV NEG.COP COP-PRF.PST
 ‘He did not study as much as me.’ (Skribnik and Seesing 2014, p.163, ex.41)

To negate nonpast events, the negative suffixes *-go* and *-fə*, and the negative copula *bifə*, are used (Baranova and Mishchenko 2022, pp.422-423). From my observations, *bifə* is only used to negate nominal predicates, as shown in (98).

- (98) bi ʃin ir-sɯ kyn **bifə-v**
 1SG new come-PFV person NEG.COP-1SG
 ‘I am not a newcomer.’

Baranova (2019) and Baranova and Mishchenko (2022) note that *-fə* expresses present negation (99), while *-go* expresses future negation (100), even when both occur in the (surface identical) context of the futuritive/infinitive aspect marker *-x(ə)*.

(99) *Present negation with -fə*

- a. nanta xamdan unt-ča-x-šə
 1SG.SOC together sleep-PROG-INF-NEG.PRES
 ‘(She) is not (at this moment) sleeping together with me.’
 (Baranova and Mishchenko 2022, p.423, ex.48)
- b. Badma, či tasə songs-ča-x-šə-č!
 B. 2SG almost listen-PROG-INF-NEG.PRES-2SG
 ‘Badma, you are not listening to me at all!’ (Baranova 2015, p.6, ex.8)

(100) *Future negation with -go*

- a. meklä-tä us-ar xotə kežə bol-x-**go** (= bolšgo)³⁴
 frog-COM water-INSTR food do-PROG COP-INF-NEG.FUT
 ‘(She/he) cannot cook with water containing frogs.’
 (Baranova and Mishchenko 2022, p.423, ex.49)
- b. bi tadn-d kel-x-**go-v** (=kelšgov)
 1SG 2PL-DAT say-INF-NEG.FUT-1SG
 ‘I won’t tell you (pl.).’ (Seesing 2014, p.18, ex.20)

From my observations, this characterisation of *-go* and *-fə* appears to be correct, as shown by the minimal pair in (101).

- (101) a. bi oda en dektɹ umʃ-tʃa-x-**fə-v**
 1SG now this book read-PROG-INF-NEG.PRES-1SG
 ‘I’m not reading this book now.’
- b. bi oda en dektɹ umʃ-x-**go^w-v**
 1SG now this book read-INF-NEG.FUT-1SG
 ‘I’m not going to read this book now.’

³⁴The futuritive/infinitive aspect morpheme */x(ə)/* surfaces as the allomorph *[ʃ]* in the context of the negative suffix *-go* (see, e.g., Baranova 2019, p.26).

I leave a fuller analysis of the alternation between *-fə* and *-go* for future research. For instance, they may suggest that there are instead two distinct null tense suffixes – a present tense and a future tense suffix – which condition the surface form of negation. If we accept that there are two distinct null tense suffixes, *-fə* and *-go* may also be portmanteau forms that spell out negation in combination with a present or future tense feature.

2.2.3 The Kalmyk nominal domain

Kalmyk is reported to have 10 distinct nominal cases (see, e.g., Bläsing 2003, pp.235-237; Birtalan 2020, pp.359-360), listed in Table 2.11. Two of these are not found in Khalkha: the sociative (also called sociative-possessive or associative) and the terminative case.

Table 2.11: Case markers in Kalmyk

Case	Gloss	Suffix
Accusative	ACC	-ig(ə), -g(ə)
Genitive	GEN	-A, -(i)n, -gin, -yin
Dative	DAT	-d, -t
Ablative	ABL	-(y)As(ə)
Instrumental	INSTR	-(y)Ar
Comitative	COM	-lA
Sociative	SOC	-tA
Directive	DIR	-(y)Ur
Terminative	TERM	-tsA

According to my consultants, the comitative case (also called connective case) is used to express joint action, e.g. (102), while the sociative case can either be used to express joint action or a property or thing associated with the modified NP, e.g. (103).

- (102) bi ekf-**lä** jov-na^w-v
 1SG older.sister-COM go-NPST-1SG
 ‘I go with my older sister.’

- (103) a. ajuka-**ta** xamdan futbol naad-v-vidŋ (=naaduvidŋ)
 A.-SOC together football play-PST-1PL
 ‘Ayuka (and I) played football together.’
- b. œ:k-**tæ** maxŋ
 fat-SOC meat
 ‘meat with fat’

Table 2.12 lists the case-unmarked (subject), accusative, genitive, and dative forms of Kalmyk pronouns. An interesting difference between Kalmyk and Khalkha can be found in the 3rd person pronominal forms in Kalmyk. Diachronically, 3rd person pronouns in Khalkha and Kalmyk are derived from proximal and distal demonstratives. In modern Khalkha, however, the distinction has been neutralised in the pronominal paradigm, and *ter* (homophonous with the distal demonstrative) is used for 3SG reference. In Kalmyk, the proximal /enə/ is used to refer to someone or something physically close to the speaker, while the distal /terə/ is used to refer to someone or something physically far away from the speaker.

Table 2.12: Selected pronouns in Kalmyk

	UNM	ACC	GEN	DAT
1SG	bi	namagə	mini	nandə
1PL.INCL	bidŋ	(bidnigə)	(bidna)	(bidŋdə) ³⁵
1PL.EXCL	madŋ	madnigə	mana/madna	mandə/madŋdə
2SG.IFML	ʃi	ʃamagə	ʃini	ʃamdə
2SG.FRML	ta	tanigə	tana	tandə
2PL	tadŋ	tadnigə	tadna	tadŋdə
3SG.PROX	enə	enygə	enynə/y:nə	enyndə/y:ndə
3PL.PROX	edŋ	ednigə	ednə	edŋdə
3SG.DIST	terə	terygə	terynə/ty:nə	teryndə/ty:ndə
3PL.DIST	tedŋ	tednigə	tednə	tedŋdə

³⁵Bläsing (2003, pp.238-239) writes that in modern Kalmyk, the stem *bidŋ* seems only to be preserved in the nominative, and that the distinction between inclusive and exclusive first plural is “synchronically lost”.

Finally, Table 2.13 lists the forms of the possessive enclitics in Kalmyk. The 1st/2nd person possessive enclitics may be seen as derived from the corresponding genitive pronouns.

Table 2.13: GEN pronouns and POSS enclitics in Kalmyk

	PRONOUN	ENCLITIC
1SG	mini	m
1PL	mana/madna	mdn̩
2SG	tʃini/tana	tʃn̩
2PL	tadna	tn̩
3SG	terynæ/ty:næ	n̩ ^j
3PL	tednii	n̩ ^j

It is also possible to use either the genitive pronoun (104a), the possessive enclitic (104b), or both (104c), in Kalmyk to express possession. In contrast to the relatively marked status of having both a genitive pronoun and a POSS enclitic in a possessive construction in Khalkha (see discussion around (52) and (53) in §2.1.3), examples such as (104c) are perfectly natural in Kalmyk. Note that in Kalmyk, POSS enclitics also unexpectedly follow case markers, suggesting that their surface position is derived.

- (104) a. *terə ærulyar mini noxa-də xotə ægə-læ-v*
 3SG in.morning 1SG.GEN dog-DAT food give-PRF.PST-1SG
 ‘He/She gave food to my dog in the morning.’
 b. *terə ærulyar noxa-də=m xotə ægə-læ-v*
 3SG in.morning dog-DAT-1SG food give-PRF.PST-1SG
 ‘He/She gave food to my dog in the morning.’
 c. *terə ærulyar mini noxa-də=m xotə ægə-læ-v*
 3SG in.morning 1SG.GEN dog-DAT-1SG food give-PRF.PST-1SG
 ‘He/She gave food to my dog in the morning.’

2.2.4 Kalmyk embedded clauses

This subsection introduces embedded clauses in Kalmyk, with a focus on finite bare CPs, nonfinite NCs, and converbial adjuncts.³⁶ An example of each embedded clause type is provided below; see §2.2.2.3 for additional examples of converbial adjuncts. Note that in most of the examples below, the embedded clause appears before the matrix subject, which I take to be the result of scrambling.

(105) *Finite bare CP*

- a. [maŋdurtan **bi** badma-la xarya-na^w-v **giɖʒə**]₁ maŋɖə ₁ kel-əv
 tomorrow 1SG B.-COM meet-PST-1SG C M. say-PST
 ‘Manj said that I will meet with Badma tomorrow.’
- b. [maŋdurtan **namagə** badma-la xarya-na(*-v) **giɖʒə**]₁ maŋɖə ₁ kel-əv
 tomorrow 1SG.ACC B.-COM meet-PST(*-1SG) C M. say-PST
 ‘Manj said that I will meet with Badma tomorrow.’

(106) *Nonfinite NC*

- a. [**namagə/bi** badmigə tsok-s]-igə₁ maŋɖə ₁ yz-lə
 1SG.ACC/1SG B.-ACC hit-PFV -ACC M. see-PST
 ‘Manj saw that I hit Badma.’
- b. [**mini** tsok-s]-igə₁ maŋɖə ₁ yz-lə
 1SG.GEN hit-PFV -ACC M. see-PRF.PST
 ‘Manj saw that I hit someone.’

(107) *Converbial adjunct*

kema [**namagə/bi** dektər umf-ɖʒa-xla]
 K. 1SG.ACC/1SG book read-PROG-SUCC.CVB
 bitʃik bitʃ-ɖʒæ-lə
 letter write-PROG-PRF.PST
 ‘When I was reading a book, Kema was writing a letter.’

In finite bare CPs (105), the subject may be case-unmarked or accusative, and the embedded clause contains an overt complementiser /giɖʒə/ (cognate with Khalkha *gej*; IPA: /giɖʒ/) and no case marking. Interestingly, subject ϕ -agreement in a finite bare CP is only possible when the embedded subject is case-unmarked (105a), but not when it is accusative

³⁶I have not been able to find a structure that is parallel to finite NCs in Khalkha. Further investigation is required to determine if finite NCs exist in Kalmyk.

(105b).³⁷ In nonfinite NCs (106), the subject may be case-unmarked, accusative, or genitive, and the embedded clause lacks an overt complementiser but bears case marking. In converbial adjuncts (107), the subject may be case-unmarked or accusative, and the embedded clause lacks both an overt complementiser and case marking.

2.2.5 Summary

This section has provided some background on the morphosyntax of Kalmyk, focusing on key aspects of clause structure, verb structure, and nominal structure. Like in Khalkha, I have argued that subjects c-command both DOs and IOs in Kalmyk. I have also shown that Kalmyk possesses three distinct classes of verbal suffixes: tense markers, aspect markers, and converb markers. I have also highlighted some interesting differences between Kalmyk and Khalkha, such as the presence of subject-predicate agreement, the different forms of negation used with present and future tense, and the existence of additional case markers in Kalmyk. Finally, I have introduced three embedded clause types in Kalmyk that also exist in Khalkha: finite bare CPs, nonfinite NCs, and converbial adjuncts. In Chapter 3, I show how subject ϕ -agreement in these embedded clauses are useful for understanding the distinction between finite and nonfinite clauses in Kalmyk.

2.3 Summary

In this chapter, I have provided a brief introduction to the Mongolic language family and the two Mongolic languages – Khalkha and Kalmyk – that I investigate in this dissertation. I have argued that the basic order of arguments in Khalkha and Kalmyk is S-IO-DO-V, and that subjects are located in Spec,TP. I have discussed several elements that comprise the Khalkha and Kalmyk verb complexes, showing that the order $\text{Asp}^0 < \text{Neg}^0 < \text{T}^0$ holds true

³⁷This suggests that subject ϕ -agreement in Kalmyk may discriminate between the case of subjects according to a case accessibility hierarchy (Bobaljik 2008; Preminger 2014). For some discussion of this analysis for Kalmyk, see Lim 2024b.

in both languages. I have also discussed several elements that constitute the Khalkha and Kalmyk nominal domains, with a focus on case markers, genitive pronouns, and possessive suffixes/enclitics. Finally, I have introduced four embedded clause types in Khalkha, and three of their counterparts in Kalmyk. I have shown that these embedded clauses share some superficial similarities and differences with regards to the case-marking of subjects contained within them, whether they contain an overt complementiser, and whether they bear case marking. In the next chapter, I will show how these differences across these clause types may be understood in terms of differences in clausal and nominal structure, which further serve as a window into understanding the distinction between finiteness and nonfiniteness in Mongolic.

Chapter 3

Finiteness and nominalisation in Khalkha embedded clauses

The previous chapter has introduced some embedded clause types in Khalkha and Kalmyk, some of which appear to be nominalised, e.g. due to the presence of case marking on the embedded clause. Nominalised clauses (NCs) are often taken to reflect the presence of both clausal and nominal functional structure. This chapter discusses the idea of a *mixed extended projection* – an extended projection where nominal functional structure is projected immediately above clausal structure (Borsley and Kornfilt 2000; Kornfilt and Whitman 2011) – and applies it to Khalkha NCs to account for their mixed clausal and nominal properties.

The literature on mixed extended projections has typically focused on two empirical questions: (i) What is the structure of the clause below the nominal functional structure?; and (ii) Does the nominal structure consist of a single projection or multiple subprojections? With respect to (i), researchers largely agree that across and within languages, nominalisation may take place at various levels of clausal structure, e.g. VP, vP, VoiceP, AspP, TP, CP (see, e.g., Alexiadou 2001; Alexiadou et al. 2007; Kornfilt and Whitman 2011; Alexiadou and Borer 2020). With respect to (ii), however, researchers differ as to whether the clause is directly nominalised by a DP layer (the so-called “direct-DP analysis”) or whether interme-

diate nominal functional projections such as nP, PossP, NumP etc. are present within the nominal superstructure.

Mixed extended projections have been proposed in many analyses of NCs, such as in Modern Greek (Roussou 1991), Turkish (e.g. Borsley and Kornfilt 2000), Ndebele (Pietraszko 2019), Hill Mari (Voznesenskaia 2019), Khalkha Mongolian (Aravind 2021), Kazakh (Ótott-Kovács 2023), and Japanese (Nasu et al. 2024). The previous analyses are similar in that they propose the projection of a nominal layer above the clausal layer in NCs. With the exception of Nasu et al. 2024, previous analyses also uniformly analyse NCs as containing a DP projection. However, they differ in whether the clause is directly nominalised by DP (e.g. Roussou 1991; Borsley and Kornfilt 2000; Pietraszko 2019; Ótott-Kovács 2023), or whether the clause is first nominalised by n^0 (Aravind 2021) - a nominaliser in the sense of Marantz 2001 or a category-switching head in the sense of Rouveret 2023. Above nP, intermediate projections such as PossP may be projected before DP is projected (e.g. Voznesenskaia 2019). Recent work by Nasu et al. 2024, however, has argued that Japanese NCs are nominalised by n^0 , and that they may or may not contain a DP layer, i.e. Japanese contains NCs that only contain a nP layer. In sum, the previous literature has discussed three possible structures of NCs, schematised in (108).

(108) *Three analyses of mixed extended projections in NCs*

- a. *NC directly nominalised by DP*
DP > clausal structure
- b. *NC nominalised by nP and contains DP*
DP > (...) nP > clausal structure
- c. *NC nominalised by nP and lacks DP*
(no DP) (...) > nP > clausal structure

Against this background, I investigate four types of embedded clauses in Khalkha, introduced in Chapter 2, three of which exhibit both clausal and nominal properties, and thus provide evidence of containing mixed extended projections. I adopt a view of mixed extended projections in which the presence or absence of verbal or nominal properties is the direct result of the presence or absence of dedicated functional projections (see, e.g.,

Alexiadou 2001; Alexiadou et al. 2010). I show that each embedded clause type can be uniquely understood as to whether it is finite or nonfinite, nominalised or non-nominalised, and if nominalised, the type of nominal properties it displays. Specifically, I show that finite and nonfinite NCs fully parallel possessive nominals with respect to an additional set of nominal properties: the availability of genitive case assignment within the clause and the clause’s ability to bear structural case marking. In contrast, converbial adjuncts do not display these nominal properties.

To account for the variation in nominal properties across Khalkha NCs, I argue that Khalkha provides evidence for two types of nominal structure in mixed extended projections: one that projects a full DP containing PossP and nP, and one that projects a smaller PossP containing nP but which lacks a DP layer. These two structures are schematised in (109).

- (109) *Two types of mixed extended projections in Khalkha*
- a. *Argument NC*
DP > PossP > nP > clausal structure
 - b. *Converbial adjunct NC*
PossP > nP > clausal structure

Building on Nasu et al. 2024 which argues for (108c) (although see footnote 11 in this chapter and §3.6), I propose that NCs may not always contain a DP layer. Crucially, I propose that clauses in Khalkha are first nominalised by nP and that variation in nominal properties can be attributed to the presence or absence of dedicated nominal functional projections above nP. Without further stipulations, the direct-DP shell analysis in (108a) would not be able to account for why argument NCs in Khalkha display more nominal properties than converbial adjunct NCs, despite both showing evidence of being nominalised.

My analysis is also inspired by parallel observations in languages such as Tatar, where possessive nominals have been argued to vary as to whether they are full DPs or Small Nominals (SNs) that project PossP but lack DP (Pereltsvaig and Lyutikova 2014; Lyutikova and Pereltsvaig 2015). In this vein, I will refer to NCs that lack DP as small nominalised clauses (SNCs). As alluded to above, the evidence for a full DP-analysis of argument NCs

and a SNC-analysis of converbial adjuncts in Khalkha is straightforward: argument NCs display the same set of nominal properties as possessive nominals, while converbial adjuncts display only a subset of those properties.

Below the level of nominalisation, I also show that the four types of embedded clauses in Khalkha vary as to whether they are finite or nonfinite. Adopting the view that finiteness is an abstract property of clauses that may be inferred from a constellation of morphosyntactic properties (Kornfilt 2007; Nikolaeva 2007, 2010, 2012; McFadden and Sundaresan 2014; Rouveret 2023), I show that finite embedded clauses contain a nondefective (i.e. tensed) T^0 and a full CP structure, while nonfinite clauses contain a defective T^0 and lack CP structure. This is schematised in (110).

- (110) *Finite and nonfinite embedded clauses in Khalkha*
- a. *Finite embedded clause*
(nominal structure >) CP > TP_{nondefective}
 - b. *Nonfinite embedded clause*
nominal structure > TP_{defective}

By the end of my analysis of Khalkha NCs, I note an interesting gap that emerges with respect to the possible combinations of nominal and clausal structure within a mixed extended projection. While Khalkha contains finite (i.e. full CP) clauses nominalised by a full DP and nonfinite (i.e. reduced TP) clauses nominalised by a full DP or reduced PossP, there is no evidence within the language of a structure comprising a finite clause nominalised by a reduced PossP. I will refer to this unattested structure, represented by the bottom left cell in Table 3.1 below, as an example of an *improper mixed extended projection*.

Table 3.1: Types of NCs in Khalkha

	PossP	DP
TP	Converbial adjunct	Nonfinite argument
CP	(Unattested)	Finite argument

The nonexistence of the bottom left cell in Table 3.1 represents the idea that a reduced nominal layer cannot nominalise a full(er) clausal structure – an idea that has been discussed in previous work (e.g., Alexiadou et al. 2010; Alexiadou 2020; Iordăchioaia 2020; Svenonius 2021). In particular, Svenonius (2021) makes a similar but distinct prediction regarding the nonexistence of such structures. Building on his analysis, I show that the asymmetry in Table 3.1 can be understood in terms of the Williams Cycle (Williams 2003, et seq.) and related work.

This chapter is organised as follows. In §3.1, I provide some theoretical background on clausal finiteness and nonfiniteness, focusing on two recurring concepts of (non)finiteness in the literature: whether T^0 is defective (e.g. tensed) and whether the clause is truncated below CP. In §3.2, I establish that all four embedded clause types under consideration contain TP. In §3.3, I show that the four embedded clause types can be differentiated according to whether they are finite or nonfinite by further discussing two other morphosyntactic cues of (non)finiteness: whether T^0 is defective or nondefective and whether CP is present or absent in the embedded clause. In §3.4, I probe the amount of nominal structure present in embedded clauses by considering two other nominal properties – the presence of a genitive subject, and the presence of case marking on the clause. I show that variation in the nominal properties of the four embedded clause types can be understood in terms of variation in the amount of nominal structure they contain. In §3.5, I synthesise the findings from §3.3 and §3.4 and provide the structures of the four types of embedded clauses in Khalkha. In §3.6, I unify the findings from the previous sections to arrive at a micro-typology of embedded clause types in Khalkha. I propose to account for an unattested NC type by building on prior work by Svenonius (2021) and Williams (2003, 2011, 2013). §3.7 concludes.

3.1 Theoretical perspectives on (non)finiteness

As many previous works have noted, finiteness itself does not form a uniform grammatical category cross-linguistically (see, e.g., collection of articles in Nikolaeva 2007; Nikolaeva 2010, 2012; McFadden and Sundaresan 2014; Rouveret 2023). However, finiteness can be understood as an abstract property of clauses that may be detected via the expression of other morphosyntactic phenomena. Because languages differ greatly in their morphosyntax, different languages have different means of expressing finiteness distinctions. A useful way to compare the types of grammatical phenomena that may encode or be associated with (non)finiteness can be seen in Table 3.2:

Table 3.2: Canonical (non)finiteness (Nikolaeva 2012)

Criteria	More finite	Less finite
Morphological	Tense marking	No tense marking
	Subject agreement	No subject agreement
	Mood/illocutionary force marking	No mood/illocutionary force marking
	Politeness marking	No politeness marking
	Evidential marking	No evidential marking
Syntactic	No switch-reference marking	Switch-reference marking
	Nominative subject	Non-nominative subject
	Independent clause	Dependent clause
	Subject licensing	No subject
	Morphosyntactic expression of informational structure	No morphosyntactic expression of informational structure
Semantic	Assertion	No assertion
	Independent temporal anchoring	No independent temporal anchoring
	Information structuring	No information structuring

Surveying multiple languages, Nikolaeva (2012) proposes a set of morphological, syntactic and semantic criteria to diagnose (non)finiteness under the comparative framework of Canonical Typology. The properties under the column ‘More finite’ are ones that are commonly found in finite clauses, while the properties under ‘Less finite’ are commonly found

in nonfinite clauses. As Nikolaeva notes, it is important to recognise that these criteria are neither exhaustive nor strictly universal, and are only meant to serve as typological tools for describing (non)finiteness.

For my present purposes, I focus on the correlation between finiteness on the one hand, and the presence of tense marking, subject agreement and nominative subjects (bolded in Table 3.2) on the other. There are two prominent ideas in the literature as to how finiteness is expressed morphosyntactically. Early conceptions of finiteness have tried to tie together the expression of overt tense distinctions, subject agreement and nominative case assignment (e.g. Chomsky 1981). The basic idea is that a finite clause is one that contains a tensed T^0 that is able to agree with and assign nominative case to subject arguments. Conversely, nonfinite clauses are often taken to contain a “defective” version of T^0 that does not agree with and assign nominative case to subject arguments. In addition, nonfinite clauses are commonly assumed to be structurally smaller than finite clauses (e.g. Wurmbrand 2001; Adger 2007; McFadden and Sundaresan 2014). For instance, finite clauses often contain a full CP structure while nonfinite clauses lack CP structure. These distinctions can be seen in English (111): (111a) shows a finite embedded clause with nominative case on the subject, subject agreement on the auxiliary verb, and an overt complementiser; (111b) shows a nonfinite embedded clause (also called an infinitival complement) where there is non-nominative case on the subject, no subject agreement on ‘have’, and no overt complementiser.

- (111) a. I believe [that **he has** played the piano].
 b. I believe [**him**/*he **to** have played the piano].

This idea has been subsequently refined in a few different ways, which I discuss below. For instance, Raposo (1987) shows that infinitival clauses in European Portuguese, which lack tense but contain overt subject agreement on the verb, are able to license nominative subjects (112). This has led to a modification of the previous idea that while there is a tight connection between agreement and nominative case assignment (also noted earlier by George and Kornfilt 1981), whether a clause is said to be finite or nonfinite ultimately depends on

the presence of (overt) tense marking or distinctions.¹ This is related to Chomsky’s (2000) idea that an infinitival clause selected by matrix V^0 contains defective T^0 .

- (112) European Portuguese
 Será difícil [**eles** aprovar-**em** a proposta].
 It difficult 3PL.NOM to.approve-3PL the proposal
 ‘It will be difficult for them to approve the proposal.’
 (modified from Raposo 1987, p.86, ex.2a)

Thus, Raposo (1987) proposes that while tensed clauses are finite cross-linguistically, languages may differ as to whether clauses that lack tense – nonfinite clauses – display agreement. However, McFadden and Sundaresan (2014) have cast doubt on Raposo’s (1987) proposal that tense is what regulates finiteness distinctions, as the verb in Tamil gerundive clauses, e.g. (113), is inflected for tense but uninflected for agreement, unlike a finite verb in a matrix clause, which is inflected for both tense and agreement. By Raposo’s proposal, the embedded clause in (113) should be finite, but McFadden and Sundaresan argue that it is nonfinite as it may not stand independently.

- (113) Tamil
 Raman; [EC_i Seetha-væ naa|ekki paar-pp-adaagæ] so-nn-aan.
 R. EC S.-ACC tomorrow see-FUT-GER.ACC say-PST-3.M.SG
 ‘Raman spoke of seeing Seetha tomorrow.’
 (McFadden and Sundaresan 2014, p.8, ex.16)

Thus, McFadden and Sundaresan (2014) argue that neither tense nor agreement determines finiteness cross-linguistically. McFadden and Sundaresan welcome the localisation of finiteness distinctions on a dedicated Fin^0 head in the CP domain, which allows for the possibility of mismatches between finiteness and tense and agreement. This idea comes from Rizzi (1997), who proposes that the CP layer can be further split into several distinct functional projections, with Fin^0 occupying the lowest projection in the expanded CP domain (114). Given that tense is independently encoded on T^0 , agreement on a dedicated functional

¹Adger 2007 proposes that the lack of tense in nonfinite clauses can be instantiated either as a featural specification on T^0 , e.g. [TENSE:–], or truncation of the clause at VP, resulting in the total absence of TP (and CP). He argues that the latter is found in Scottish Gaelic verbal noun clauses.

head, e.g. Agr⁰, and finiteness on Fin⁰, it is predicted that these properties may not always co-occur.

- (114) *The Expanded CP domain (Rizzi 1997)*
 ForceP > TopP > FocP > TopP > FinP > TP

The move to locate finiteness distinctions on a dedicated functional head has several advantages in explaining the properties of nonfinite clauses. First, as McFadden and Sundaresan (2014) note, there seems to be a correlation between finiteness and the phasehood of a clause, as seen in Icelandic (115).

- (115) Icelandic
- a. *Nonfinite subjunctive CP*
 Jón_i heyr-ð-i [að ég **hef**-ð-i svikið sig_{i/*j}].
 Jon hear-IND.PST-3SG that I have.SBJV-PST-3SG betrayed.PTCP ANAPH
 ‘Jon_i heard that I had betrayed him_{i/*j}.’
 - b. *Finite indicative CP*
 * Jón_i heyr-ð-i [að ég **haf**-ð-i svikið sig_i].
 Jon hear-IND.PST-3SG that I have.IND-PST-3SG betrayed.PTCP ANAPH
 Int.: ‘Jon_i heard that I had betrayed him_i.’
 (modified from McFadden and Sundaresan 2014, p.11, exx.27-28)

While anaphor binding is typically clause-bound, we observe that in Icelandic, long-distance anaphoric binding is possible across a nonfinite subjunctive CP (115a), but not a finite indicative CP (115b). McFadden and Sundaresan suggest that while FinP in finite CPs normally constitutes a phase boundary, nonfinite CPs are either headed by a defective Fin⁰ or lack Fin⁰ entirely, such that an element within these clause types is accessible for further syntactic operations in the matrix clause.

The distinction between finite and nonfinite domains may also correlate with the amount of structure in the CP domain. Supporting evidence comes from Italian, where the position of topics differs in finite (116) and nonfinite (117) clauses. In finite clauses, a topicalisation site is available under complementiser *che* (116a) but not above it (116b).

Conversely, in nonfinite clauses, a topicalisation site is not available above the complementiser *di* (117a) but available under it (117b).

(116) Italian

- a. Credo **che**, **il tuo libro**, loro lo apprezzerebbero molto.
 I.think that_[+FIN] the your book them it will.appreciate much
 ‘I think that they will appreciate your book very much.’

- b. *Credo, il tuo libro, che loro lo apprezzerebbero molto.
 I.think the your book that_[+FIN] them it will.appreciate much
 ‘I think that they will appreciate your book very much.’

(Adger 2007, p.30, exx.14-15)

(117) Italian

- a. *Credo di, il tuo libro, apprezzar-lo molto.
 I.think that_[-FIN] the your book appreciate-it much
 ‘I think that they will appreciate your book very much.’

- b. Credo, **il tuo libro**, **di** apprezzar-lo molto.
 I.think the your book that_[-FIN] appreciate-it much
 ‘I think that they will appreciate your book very much.’

(Adger 2007, p.31, exx.16-17)

Adger (2007) proposes that Italian possesses a clausal periphery as shown in (118).

(118) Credo [_{ForceP} [_{TopP} [_{FinP} [_{TP} ...]]]] (modified from Adger 2007, p.31, ex.18)

He argues that the contrast between (116) and (117) is predicted if *che* instantiates Force⁰ in finite clauses and *di* instantiates Fin⁰ in nonfinite clauses. Thus, finite *che*-clauses are ForcePs containing TopP and finite FinP, while nonfinite *di*-clauses are smaller TopPs that contain nonfinite FinP but lack ForceP.

Adger (2007) also points out a parallel in English: while a topic position is available below *that* (119), it is not available below *for* (120) or in raising constructions (121). He proposes that *that*-CPs project ForcePs (119) and *for*-CPs project FinPs (120), and that *to*-clauses are TPs (121). The takeaway is that the embedding predicate *to propose* and the raising predicate *to seem* select for nonfinite argument clauses with reduced structures –

FinP and TP respectively – and that this is relevant for the observed lack of topicalisation sites in these embedded clause types.

(119) I decided that, your book, we should all read first. (Adger 2007, p.31, ex.19a)

(120) * I propose for, these books, John to read. (Adger 2007, p.31, ex.22b)

(121) * John seems, your book, to like best. (Adger 2007, p.32, ex.24)

In sum, while there are a number of ways crosslinguistically to analyse finiteness distinctions, two main ideas stand out: defectivity and size. On one hand, nonfinite clauses might be said to contain defective T^0 (e.g. lacking tense specifications or agreement) or defective Fin^0 (e.g. not constituting a phasal boundary). On the other hand, it has also been noted that nonfinite clauses tend to be truncated structures, projecting only up to the level of TopP, FinP, TP, or VP (Adger 2007), and possibly other intermediate truncation sites between TP and VP (e.g. AspP, VoiceP, and vP).

While I do not discount the relevance of Fin^0 in understanding finiteness distinctions in Khalkha, for the purposes of this dissertation, T^0 and C^0 suffice in drawing out such distinctions in my following analysis of clausal structure in Khalkha embedded clauses. Specifically, I will show that in Khalkha, i) finite embedded clauses that exhibit the full range of tense endings contain nondefective T^0 , while nonfinite embedded clauses that do not exhibit tense contain defective T^0 ; ii) finite embedded clauses containing the complementiser *gej/ged(e)g* possess a full CP structure, while nonfinite embedded clauses lacking *gej/ged(e)g* possess only TP and lack CP structure. Additionally, I show that the availability of overt subject ϕ -agreement is useful for understanding the distinction between finiteness and nonfiniteness in Kalmyk. Finally, I show that case-unmarked (i.e. “nominative”) subjects in Khalkha and Kalmyk are available in both finite and nonfinite clauses, suggesting that “nominative” case assignment is not relevant as a diagnostic for finiteness in Mongolic. In the next section, I will establish, as a baseline, that TP is present in all embedded clause types, regardless of whether overt finite tense endings are present.

As a brief aside before I end this discussion, it is important to note that I am using the terms “finite” and “nonfinite” slightly differently than how they are used in the Mongolic literature (see, e.g., Binnick 2012; Janhunen 2012). In that literature, “finite” and “nonfinite” are terms that describe types of verb inflections and the clauses that those inflectional markers may appear in. Inflectional markers that are “finite” are negatively defined as being unable to appear in some dependent clauses, such as certain types of NCs, converbial adjuncts and relative clauses. This accords with the definition of finiteness I adopt, in that “finite” markers are indeed only found in finite clauses. Conversely, inflectional markers that may appear in dependent clauses are described as “nonfinite”, even though “nonfinite” markers may themselves appear in “finite” clauses that permit the use of “finite” markers. This distributional definition of finiteness differs from that in the theoretical syntactic literature, where finiteness refers to an abstract and structural property of clauses tied to the properties of functional heads (e.g. of T^0) and clause size (e.g. whether a full CP is projected). It is this latter notion of finiteness that I will concern myself with in the rest of this dissertation.

3.2 The presence of TP in four types of embedded clauses

In this section, I will argue that the four embedded clause types under investigation in Khalkha contain at least a TP layer. I will show that subjects in all embedded clause types move to Spec,TP based on i) scope effects and ii) the placement of adverbs. An example of each embedded clause type is shown below:

(122) *Finite bare CP*

Bi [Naraa margaash ir-ne gej] hel-sen.
 1SG N. tomorrow come-NPST C say-PST
 ‘I said that Naraa will come tomorrow.’

(123) *Finite complement NC*

Bi [Naraa-giin margaash ir-ne gedg]-iig(=n’) mart-san.
 1SG N.-GEN tomorrow come-NPST C.NML ACC=3.POSS forget-PST
 ‘I forgot that Naraa will come tomorrow.’

- (124) *Nonfinite complement NC*
 Bi [Naraa-giin margaash ire-h]-iig(=n') mart-san.
 1SG N.-GEN tomorrow come-INF -ACC=3.POSS forget-PST
 'I forgot that Naraa is coming tomorrow.'
- (125) *Converbial adjunct*
 Bi [ter hicheel-Ø=ee hii-j bai-had](=n') unt-san.
 1SG 3SG lesson-ACC=RP do-PROG COP-TEMP.CVB =3.POSS sleep-PST
 'When she/he was studying, I fell asleep.'

As I discussed for main clause subjects in Chapter 2, since Khalkha is head-final, I will use scope effects and adverb placement to show that subjects in each embedded clause type also raise to Spec,TP. This implies that TP is projected in each clause type under consideration. In §3.2.1, I discuss evidence from the interpretation of embedded subjects with respect to clausal negation, and in §3.2.2, I discuss evidence from adverb placement in the four embedded clause types.

3.2.1 The interpretation of subjects with respect to clausal negation

Consider first the interpretation of subjects with respect to clausal negation. A universally quantified subject in a finite bare CP (126), finite NC (127), and nonfinite NC (128) can only be interpreted outside the scope of embedded negation.

- (126) *Universally quantified subject takes scope above negation in finite bare CP*
 Bilgüün [**büh oyuut-nuud** ter shalgaltan-d tents-ee-**güi** bai-na gej] hel-sen.
 B. all student-PL that exam-DAT pass-IPFV-NEG COP-NPST C say-PST
 'Bilguun said that all students didn't pass that exam.'
 ✓Context: There are 10 students in the class and no one passed that exam. $\checkmark \forall > \neg$
 #Context: There are 10 students in the class and only two passed that exam. $*\neg > \forall$
- (127) *Universally quantified subject takes scope above negation in finite NC*
 Bilgüün [**büh oyuut-nuud** ter shalgaltan-d tents-ee-**güi** bai-na gedg
 B. all student-PL that exam-DAT pass-IPFV-NEG COP-NPST C.NML
]-iig hel-sen.
 -ACC say-PST
 'Bilguun said that all students didn't pass that exam.'
 ✓Context: There are 10 students in the class and no one passed that exam. $\checkmark \forall > \neg$
 #Context: There are 10 students in the class and only two passed that exam. $*\neg > \forall$

- (128) *Universally quantified subject takes scope above negation in nonfinite NC*
 Bilgüün [**büh oyuut-nuud** ter shalgaltan-d tents-ee-**güi**]-g hel-sen.
 B. all student-PL that exam-DAT pass-IPFV-NEG -ACC say-PST
 ‘Bilguun said that all students didn’t pass that exam.’
 ✓Context: There are 10 students in the class and no one passed that exam. $\checkmark \forall > \neg$
 #Context: There are 10 students in the class and only two passed that exam. $*\neg > \forall$
- (129) and (130) show adjuncts headed by the conditional verb *-vAl* (glossed COND.CVB) and the temporal verb *-hAd* (glossed TEMP.CVB) respectively. These sentences, while judged to be grammatically acceptable, have pragmatically strange readings as the embedded subject *büh oyuutnuud* ‘all students’ take scope above embedded negation within each adjunct clause. In (129), the reading that the teacher will be happy if all their students fail the exam obtains, while in (130), the reading that the teacher explained when all (not at least some) students did not understand the question obtains.
- (129) *Universally quantified subject takes scope above negation in -vAl converbial adjunct*
 Bagsh [**büh oyuut-nuud** ter shalgaltan-d tents-ee-**güi** bai-val]
 Teacher all student-PL that exam-DAT pass-IPFV-NEG COP-COND.CVB
 bayarla-h bol-no.
 be.happy-INF become-NPST
 ‘The teacher will be happy if all students do not pass the exam.’
 ✓Context: The teacher is unkind and wants all students to fail the exam. $\checkmark \forall > \neg$
 #Context: The teacher has low expectations of their students, and will be happy if at least one student passes the exam. $*\neg > \forall$
- (130) *Universally quantified subject takes scope above negation in -hAd converbial adjunct*
 Bagsh [**büh oyuut-nuud** asuult-iig oilgo-h-**güi** bai-had]
 Teacher all student-PL question-ACC understand-INF-NEG COP-TEMP.CVB
 tailbarla-san.
 explain-PST
 ‘The teacher explained when all the students did not understand the question.’
 ✓Context: In a class of 10, the teacher explained when all 10 students did not understand the question. $\checkmark \forall > \neg$
 #Context: In a class of 10, the teacher explained when not all 10 students (i.e. some students) understood the question. $*\neg > \forall$

In sum, the examples above show that universally quantified subjects are interpreted outside the scope of negation, supporting the idea that subjects move to Spec,TP in all four embedded clause types discussed here.

3.2.2 Adverb placement

Recall from §2.1.1.1 that in Khalkha main clauses, manner adverbs (which adjoin to vP) may not precede theme subjects (first introduced in the complement position of V^0), suggesting that subjects move to Spec,TP. In this subsection, I discuss evidence showing that in the four embedded clause types under consideration i) theme subjects obligatorily precede manner adverbs, suggesting that they have moved to Spec,TP, and ii) each embedded clause type may contain an independent temporal adverb, which I take to adjoin to TP. Together, these facts provide evidence that the four embedded clause types contain a TP layer.

First, Fong (2019) shows that finite bare CPs may contain a temporal adverb such as *margaash* ‘tomorrow’, which is unambiguously interpreted as modifying the embedded clause (131). This suggests that finite bare CPs contain a TP layer.

- (131) *Finite bare CPs may contain temporal adverbs*
 Bat [**margaash** Dulmaa nom unshi-na gej] hel-sen.
 B. tomorrow D. book read-NPST C say-PST
 ‘Bat said that Dulmaa will read a book tomorrow.’ (Fong 2019, p.2, ex.3)

Fong (2019) also shows that manner adverbs such as *hurdan* ‘quickly’ cannot precede a case-unmarked subject (132). However, given that manner adverbs adjoin to vP and do not scramble higher (however, see footnote 7 in Chapter 2 for conflicting judgements reported by Gong 2023a), it is expected that they cannot precede an external argument subject, which I take to be introduced in Spec,VoiceP.

- (132) *Manner adverbs cannot precede subjects in finite bare CPs*
 Dorj [{*hurdan} Naraa {**hurdan**} baishin bari-san gej] hel-sen.
 D. {*quickly} N. {quickly} house build-PST C say-PST
 ‘Dorj said that Naraa quickly built a house.’ (Fong 2019, p.16, ex.48)

Nevertheless, it is also not possible for a manner adverb to precede a theme subject in a finite bare CP, as shown in (133). Since the theme subject is first introduced in the complement position of V^0 , the fact that it cannot follow *hurdan* suggests that it has raised to Spec,TP within the embedded clause.

- (133) *Manner adverb cannot precede theme subject in finite bare CP*
 Dorj [{*hurdan} ene baishin {**hurdan**} bari-gd-san gej] hel-sen.
 D. {*quickly} this building {quickly} build-PASS-PST C say-PST
 ‘Dorj said that this building was built quickly.’

Next, I show that the same adverb distribution facts also hold in finite NCs, nonfinite complement NCs and converbial adjuncts. Consider the pair of finite NCs in (134) and (135). (134) shows that a finite NC may contain a temporal adverb *margaash* ‘tomorrow’, suggesting the presence of TP in the embedded clause. (135) shows that a theme subject may not follow the manner adverb *hurdan* ‘quickly’, suggesting that the embedded subject has moved to Spec,TP.

- (134) *Finite NC may contain temporal adverbs*
 Dorj [**margaash** Naraa tögöldör huur toglo-no gedg]-iig hel-sen.
 D. tomorrow N. piano play-NPST C.NML -ACC say-PST
 ‘Dorj said that Naraa will play the piano tomorrow.’

- (135) *Manner adverb cannot precede theme subject in finite NC*
 Dorj [{*hurdan} ene baishin {**hurdan**} bari-gd-san gedg]-iig sons-sen.
 D. {*quickly} this building {quickly} build-PASS-PST C.NML -ACC hear-PST
 ‘Dorj heard that this building was built quickly.’

In (136), the nonfinite NC may contain the temporal adverb *margaash* ‘tomorrow’, suggesting the presence of TP within the embedded clause. In (137), the theme subject may not follow the manner adverb *hurdan* ‘quickly’, suggesting that the embedded subject has moved to Spec,TP.

- (136) *Nonfinite NCs may contain temporal adverbs*
 Dorj [**margaash** Naraa ene baishin-g nuraa-h]-iig med-sen.
 D. tomorrow N. this building-ACC tear.down-INF -ACC know-PST
 ‘Dorj knew that tomorrow Naraa will tear down this building.’

- (137) *Manner adverb cannot precede theme subject in nonfinite NC*
 Dorj [{*hurdan} ene baishin {**hurdan**} bari-gd-sn]-iig med-sen.
 D. {*quickly} this building {quickly} build-PASS-PFV -ACC know-PST
 ‘Dorj knew that this building was built quickly.’

Turning to converbial adjuncts, (138) shows that a converbial adjunct may contain the temporal adverb *margaash* ‘tomorrow’, given that the case-unmarked embedded subject may precede *margaash* (which I assume is due to the availability of scrambling). (139) shows that the theme subject may not follow the manner adverb *hurdan* ‘quickly’, suggesting that the embedded subject has moved to Spec,TP.

- (138) *Temporal adverbs may precede subjects in converbial adjuncts*
 Bi [{Naraa} **margaash** {Naraa} ir-tel] huushuur hii-h-güi.
 1SG {N.} tomorrow {N.} come-TERM.CVB khuushuur do-INF-NEG
 ‘I will not make khuushuur until Naraa comes tomorrow.’
- (139) *Manner adverb cannot precede theme subject in converbial adjunct*
 Bilgüün [{*hurdan} ene nom {**hurdan**} ol-d-vol]
 B. {*quickly} this book {quickly} find-PASS-COND.CVB
 uurla-h-güi.
 become.angry-INF-NEG
 ‘If this book is found quickly, Bilguun will not be angry.’

Finally, a temporal adverb in the four embedded clause types can mismatch with a temporal adverb in the matrix clause. In other words, each of the four embedded clauses can have a different tense interpretation than the matrix clause. This is seen in (140)-(143), and provides further evidence for TP structure in all four embedded clause types.

- (140) *Temporal mismatch in finite bare CP*
Öchigdör bi [**margaash** Naraa ir-ne gej] hel-sen.
 Yesterday 1SG tomorrow N. come-NPST C say-PST.
 ‘Yesterday I said that Naraa will come tomorrow.’
- (141) *Temporal mismatch in finite NC*
Öchigdör bi [**margaash** Naraa ir-ne gedg]-iig mart-san.
 Yesterday 1SG tomorrow N. come-NPST C.NML -ACC forget-PST.
 ‘Yesterday I forgot that Naraa will come tomorrow.’
- (142) *Temporal mismatch in nonfinite complement NC*
Öchigdör Bi [**margaash** Naraa ire-h]-iig mart-san.
 Yesterday 1SG Tomorrow N. come-INF -ACC forget-PST.
 ‘Yesterday I forgot that Naraa is coming tomorrow.’

(143) *Temporal mismatch in -vAl converbial adjunct*

Margaash bid [**öchigdör** Naraa tasalbar av-san bai-val] togloIt ruu
 Tomorrow 1PL yesterday N. ticket buy-PFV COP-COND.CVB concert DIR
 yav-na.

go-NPST

‘If Naraa bought the tickets yesterday, we will go to the concert tomorrow.’

To summarise the adverb placement facts in the four embedded clause types under consideration, each embedded clause type may contain an independent temporal adverb that differs from the tense interpretation in the matrix clause. Further, a theme subject in each embedded clause type may not follow a manner adverb, suggesting that the subject has moved to Spec,TP. Together, these two facts support the idea that the four embedded clause types contain a TP layer.

3.2.3 Summary

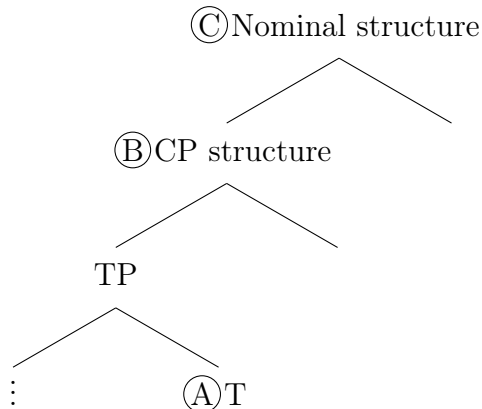
To briefly summarise the findings of this subsection, I have argued, using evidence from i) scope effects and ii) adverb placement, that the four embedded clause types under consideration in Khalkha contain a TP layer. This narrows the space of possibilities in which the four embedded clauses may differ in their clausal structure: the type of T^0 they contain and the amount of clausal structure present above TP. I investigate these issues in the next section.

3.3 The structure of (non)finiteness in Khalkha embedded clauses

In §3.2, I have established, as a baseline, that both matrix clauses and the four types of embedded clauses contain TP. Potential sources of variation across the four embedded clause types can thus be attributed to (A) the variety of T^0 (defective vs nondefective), (B)

the amount of CP structure above the clause, and ③ the amount of nominal structure above TP. This is illustrated in (144).

(144) *Potential sources of variation in Khalkha embedded clauses*



I will investigate ① and ② in the remainder of this section, returning to ③ in §3.4. I show that in Khalkha, finite clauses contain a full CP as well as a TP containing nondefective T^0 , while nonfinite clauses contain a TP containing defective T^0 but lack CP structure. This is schematically represented in (145).²

(145) Clausal structure of embedded clauses in Khalkha

- a. Finite clauses (finite bare CP, finite NC)
 (nominal structure) > **CP** > **TP (nondefective)**
- b. Nonfinite clauses (nonfinite NC, converbial adjunct)
 nominal structure > **TP (defective)**

I first connect the ability to contain overt finite tense markers to the presence of nondefective T^0 in finite embedded clauses, and the inability to do so to the presence of defective T^0 in nonfinite embedded clauses (§3.3.1). Further, the ability to express overt tense seems to be strongly connected to the presence of the overt complementiser *gej/ged(e)g*. Discussing Kalmyk in §3.3.2, I show a further connection between the presence of overt subject agreement and the presence of an overt complementiser in embedded clauses, which suggests that subject ϕ -agreement is also linked to the type of T^0 in Kalmyk. Next, I

²I leave open the possibility that there may be additional structure above TP within the CP domain, e.g. FinP, which may directly encode finiteness or nonfiniteness as discussed in §2.1. This is consistent with the view that clauses contain a more articulated left periphery (e.g. Rizzi 1997). It also does not affect the main argument in this section: that nonfinite clauses contain less clausal structure than finite clauses.

argue that the presence of the overt complementiser *gej/ged(e)g* in Khalkha finite clauses indicates the presence of a fuller CP structure above TP, while the lack thereof in nonfinite clauses indicates reduced or no structure above TP. Supporting evidence comes from i) the distribution of the wh-interrogative marker *be/ve* in embedded clauses (§3.3.3), and ii) the availability of thematic topic positions in embedded clauses (§3.3.4).

3.3.1 The distribution of overt tense endings & (non)defective T^0

In §2.1.2.1, I introduced selected tense and aspect markers in Khalkha. Previewing the findings in this section, I show that finite bare CPs and finite NCs may contain overt tense markers, while nonfinite NCs and converbial adjuncts may not. I propose that the ability to express overt tense is tied to the presence of a nondefective or defective T^0 . Since finite bare CPs and finite NCs may express overt tense, this suggests that they contain nondefective T^0 . On the other hand, since nonfinite NCs and converbial adjuncts may not express overt tense, this suggests that they contain defective T^0 .

Consider the distribution of the nonpast tense marker *-nA*, which is allowed in finite bare CPs (146) and finite NCs (147) but not nonfinite NCs (148) or converbial adjuncts (149).

- (146) *Finite bare CP may contain nonpast -nA*
 Bi [Naraa margaash ir-**ne** gej] mart-san.
 1SG N. tomorrow come-NPST C forget-PST
 ‘I forgot that Naraa will come tomorrow.’
- (147) *Finite complement NC may contain nonpast -nA*
 Bi [Naraa margaash ir-**ne** gedg]-iig mart-san.
 1SG N. tomorrow come-NPST C.NML ACC forget-PST
 ‘I forgot that Naraa will come tomorrow.’
- (148) *Nonfinite complement NC cannot contain nonpast -nA*
 Bi [Naraa margaash { ire-**h** / *-**ne** }]-iig mart-san.
 1SG N. tomorrow { come-INF / *-NPST } -ACC forget-PST
 ‘I forgot that Naraa is coming tomorrow.’

- (149) *-hAd converbial adjunct cannot contain nonpast -nA*
 Bi [Naraa margaash { ire-**hed** / *ir-**ne** bai-had }]
 1SG N. tomorrow { come-TEMP.CVB / *come-NPST COP-TEMP.CVB }
 huushuur hii-ne.
 khuushuur do-NPST
 ‘I will make khuushuur when Naraa comes tomorrow.’

To reconcile the fact that nonfinite NCs and converbial adjuncts cannot contain an overt tense suffix with the finding in §3.2 that they contain a TP layer, I propose that the TP layer in nonfinite argument clauses is headed by a phonologically null, defective T^0 . In Khalkha, the defectivity of T^0 is equivalent to the inability of T^0 to be occupied by any of the finite tense markers listed in Table 3.3.³ I further propose that in converbial adjuncts, defective T^0 is instantiated by the elements that are traditionally called ‘converbs’ in the Mongolic literature (e.g. Janhunen 2012). The defective nature of converbial T^0 is further supported by the observation that converbial adjuncts typically depend on the main clause for their tense interpretation. Under this view, ‘converbs’ are a type of phonologically overt defective T^0 found in nonfinite adjunct clauses. However, I will continue to use the term ‘converb’, reserving the term ‘overt tense suffix’ only for finite tense suffixes. A revised, partial inventory of tense suffixes in Khalkha is shown in Table 3.3.

Table 3.3: Selected finite and nonfinite tense markers in Khalkha

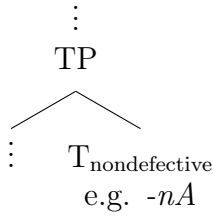
Finite		Nonfinite	
-nA-~-Ø	nonpast	-Ø	defective
-lAA	present imperfect	-ngUUt	concomitant verb
-v	past imperfect	-vAl	conditional verb
-jee	past perfect	-mAgts	immediative verb
-sAn~-Ø	general past	-hAd	temporal verb
		-tAl	terminative verb

We are now in a position to see how the four types of embedded clauses vary as to the type of T^0 they contain. Finite embedded clauses, e.g. (146) and (147), contain

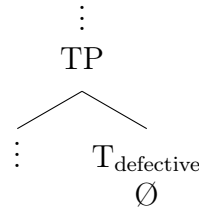
³That tense is limited in NCs that contain TP but lack CP is not unusual. For instance, Borsley and Kornfilt (2000) report that factive nominalisations in Turkish, which they argue contain TP and lack CP, only express a nonfuture vs. future tense distinction (see also Kornfilt and Whitman 2011).

nondefective T^0 as shown in (150), which may be occupied by finite tense markers and may be phonologically overt. Nonfinite argument clauses, e.g. (148), contain a defective T^0 that cannot be occupied by finite tense markers and is always phonologically null, as shown in (151). Converbial adjuncts, e.g. (149), also contain a defective T^0 that cannot be occupied by finite tense markers but is phonologically overt (aka *converbs* in the Mongolic literature), as shown in (152).

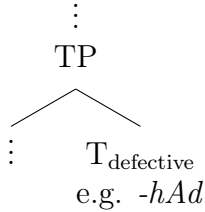
(150) T^0 in finite clauses



(151) T^0 in nonfinite TPs



(152) T^0 in converbial adjuncts



Recall from §2.1 that defective T^0 is often associated with non-nominative subjects and the absence of ϕ -agreement, e.g. as seen in the English examples (108). Given that case-unmarked (i.e. “nominative”) subjects are available in the four embedded clause types under consideration, I conclude that subject case is not an appropriate diagnostic either for the defectivity of T^0 or for understanding (non)finiteness in Khalkha. As Khalkha does not exhibit overt ϕ -agreement in the verbal domain, it is also not straightforwardly possible to tell whether T^0 in nonfinite NCs and converbial adjuncts Agrees with the subject, although I argue in Chapter 5 that both finite and nonfinite T^0 Agrees. Kalmyk, on the other hand, exhibits overt ϕ -agreement on verbs, which may shed light on whether the correlation between the presence of nondefective T^0 and subject ϕ -agreement holds in the language. I turn next to a discussion of the Kalmyk data.

3.3.2 Subject ϕ -agreement in Kalmyk as an indicator of finiteness

As I discussed in Chapter 2, Kalmyk displays overt subject ϕ -agreement on verbs and predicate nouns and adjectives. Kalmyk also contains three embedded clause types found in Khalkha: finite bare CPs headed by the overt complementiser /giɖə/, and nonfinite NCs and converbial adjuncts which lack an overt complementiser. Note that in some of the examples below, the embedded clause has been fronted – representing my consultants’ preference to do so – but this is immaterial to the present discussion.

Observe that in finite bare CPs ϕ -agreement is possible with the case-unmarked embedded subject (153).

- (153) Kalmyk: *ϕ -agreement possible in finite bare CP*
 [maɣdurtan bi badma-la xarya-**na^w-v** giɖə]₁ manɖə __₁ kel-əv
 tomorrow 1SG B.-COM meet-PST-1SG C M. say-PST
 ‘Manj said that I will meet with Badma tomorrow.’

However, ϕ -agreement with the case-unmarked subject is not possible in a nonfinite complement NC (154), or a converbial adjunct, e.g., one headed by the successive converb /-xla/ as shown in (155).

- (154) Kalmyk: *ϕ -agreement not possible in nonfinite NC*
 [bi badmigə tsok-**s/*-la^w-v**]-igə₁ manɖə __₁ yz-lə
 1SG B.-ACC hit-PFV/*-PRF.PST-1SG -ACC M. see-PST
 ‘Manj saw that I hit Badma.’

- (155) Kalmyk: *ϕ -agreement not possible in converbial adjunct*
 kema [bi dektər umf-ɖəa-**xla/*-xla-v**]
 K. 1SG book read-PROG-SUCC.CVB/*-SUCC.CVB-1SG
 bitʃik bitʃ-ɖæ-lə
 letter write-PROG-PRF.PST
 ‘When I was reading a book, Kema was writing a letter.’

In the previous subsection, I have argued that finite bare CPs contain a nondefective T⁰ while nonfinite NCs and converbial adjuncts contain a defective T⁰. In Kalmyk, the defectivity of T⁰ may also be understood as whether it permits overt subject ϕ -agreement with the embedded subject. Since subject ϕ -agreement is possible in finite bare CPs (153),

this suggests that these clause types contain nondefective T^0 . On the other hand, since subject ϕ -agreement is impossible in nonfinite NCs (154) and converbial adjuncts (155), this suggests that these clause types contain defective T^0 .

Note that while Kalmyk shows that there is a correlation between overt agreement and the defectivity of T^0 , I argue in Chapter 5 that both nondefective and defective T^0 in Khalkha Agree with arguments that T^0 locally c-commands, and that this Agree operation is responsible for dependent case assignment. Thus, a priori, I assume that the operation that yields overt subject ϕ -agreement can be independent from a separate abstract Agree operation by T^0 . Thus, the Kalmyk data in (153)-(155) should not be interpreted as evidence that nondefective T^0 does not Agree, but as supporting the cross-linguistic observation that clauses that are less finite also tend to lack overt subject agreement, as shown in Table 3.2.

Finally, while it is commonly noted that less finite clauses tend to contain non-nominative subjects (see Table 3.2), Kalmyk – like Khalkha in the previous subsection – shows that this correlation does not hold. Case-unmarked subjects (traditionally referred to as “nominative”) are available in both finite bare CPs that contain overt subject ϕ -agreement (153) and nonfinite NCs and converbial adjuncts that lack subject ϕ -agreement (154)-(155). This suggests that the source of “nominative” case in Khalkha and Kalmyk is not tied to the presence of finite T^0 , as is commonly assumed under a Chomskyan view of nominative case assignment (Chomsky 2000, 2001). In Chapter 5, I provide an analysis that ties the absence of case on “nominative” subjects and case-unmarked objects in Khalkha to the same underlying source.

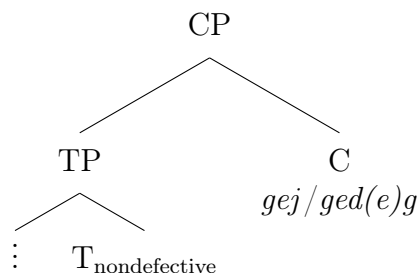
To briefly summarise what we have seen for Khalkha, finite clauses contain nondefective T^0 , which correlates with their ability to contain overt tense markers, while nonfinite clauses contain defective T^0 , which correlates with their inability to contain overt tense markers. In Kalmyk, nondefective T^0 also correlates with the presence of overt subject ϕ -agreement, while defective T^0 correlates with the lack of overt subject ϕ -agreement. I have proposed that in Khalkha, defective T^0 is non-overt in nonfinite NCs but overt in converbial

adjuncts; these overt defective tense markers are what are commonly known as ‘converbs’. Having established that Khalkha and Kalmyk embedded clauses contain different types of T^0 , in the next subsection, I turn my attention to the presence of CP structure above TP in each of the four embedded clause types.

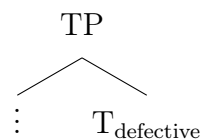
3.3.3 The distribution of *wh*-interrogative marker *be/ve*

While I have not explicitly pointed this out, in both Khalkha and Kalmyk, there is a strong connection between the presence of nondefective T^0 and the presence of an overt complementiser. Since finite bare CPs and finite NCs (in Khalkha) contain an overt complementiser while nonfinite NCs and converbial adjuncts do not, this suggests that finite clauses contain more structure above TP than nonfinite NCs and converbial adjuncts. Specifically, I argue that finite clauses contain a full CP structure above TP, while nonfinite clauses lack a full CP structure. This is schematised in (156) and (157).

(156) *CP layer in finite clauses*



(157) *No CP layer in nonfinite clauses*



One piece of evidence that supports this size distinction between finite and nonfinite clauses in Khalkha comes from the distribution of interrogative markers in embedded clauses. This diagnostic, which detects the presence of CP structure, is inspired by Saito (2012), who uses the distribution of the interrogative marker *ka* in Japanese embedded clauses to investigate the structure of their clausal periphery. Khalkha is a *wh*-in-situ language, and *wh*-questions contain the *wh*-question marker *be/ve*,⁴ as seen in (158).

⁴The form *be* is used after words ending with /m/, /n/ and /v/, while *ve* is used elsewhere.

- (158) Chi yuu id-sen **be**?
 2SG what eat-PST WHQ
 ‘What did you eat?’

As it is not possible to form an embedded question with a converbial adjunct, I will set them aside in this discussion. (159)-(161) show that complement clauses may serve as embedded questions. However, while finite clauses may optionally contain the *wh*-question marker (159)-(160), nonfinite complement clauses may not (162) cf. (161). For concreteness, I will assume that the overt complementiser *gej/ged(e)g* is in Force⁰, while the *wh*-question marker is in Foc⁰, given that *gej/ged(e)g* follows *be/ve* in the clause.⁵ See Rizzi 1997 for a discussion of different kinds of complementisers in the expanded CP periphery.

- (159) *Finite bare CP can contain wh-question marker be*
 Bi [hen tsuivan id-sen (**be**) gej] med-ne.
 1SG [who tsuivan eat-PST (WHQ) C] know-NPST
 ‘I know who ate tsuivan (fried noodles).’
- (160) *Finite NC can contain wh-question marker be*
 Bi [hen tsuivan id-sen (**be**) gedg]-iig med-ne.
 1SG [who tsuivan eat-PST (WHQ) C.NML]-ACC know-NPST
 ‘I know who ate tsuivan.’
- (161) *Nonfinite argument NC can be embedded question*
 Bi [hen tsuivan id-sn]-iig med-ne.
 1SG who tsuivan eat-PFV -ACC know-NPST
 ‘I know who ate tsuivan.’
- (162) *Nonfinite argument NC cannot contain wh-question marker be/ve*
 *Bi [hen tsuivan id-sen **be/ve**]-g med-ne.
 1SG who tsuivan eat-PFV WHQ -ACC know-NPST
 Int.: ‘I know who ate tsuivan.’

The inability of nonfinite complement NCs to contain the *wh*-question marker *be/ve* (and the absence of an overt complementiser *gej/ged(e)g* in them) suggests that nonfinite

⁵Gong 2023a analyses the *wh*-interrogative particle *be/ve* as instantiating Force⁰, while *gej* occupies another C⁰ above Force⁰. The exact categorial identity of the particle does not affect my main argument that finite embedded clauses contain a fuller CP structure, while nonfinite argument NCs lack CP structure. Note as well that the grammaticality of (161) suggests that it is not the presence of the *wh*-interrogative particle that licenses embedded questions, contrary to Gong 2023a. I leave this issue open for future research.

clauses are truncated below CP, i.e. they lack CP structure. Conversely, the ability of finite clauses to contain *be/ve* (and *gej/ged(e)g*) suggests that they contain a full CP structure. The size of the clausal periphery is thus a source of difference between finite and nonfinite clauses.

3.3.4 The availability of thematic topic positions

Another piece of evidence that points towards a size difference between the two types of embedded clauses is the availability of thematic (or aboutness) topicalisation, as suggested by Gong (2022, 2023a). Gong (2022, 2023a) shows that a thematic topic position is available in a finite bare CP (163), but not in a nonfinite complement NC (164).⁶

- (163) *Thematic topic reading available in finite bare CP*
 Bagsh [Bat bol offis-t=n' hoyor nom-iig unshi-j duusga-san gej]
 Teacher B. TOP office-DAT=3.POSS two book-ACC read-CVB finish-PST C
 hel-sen.
 say-PST
 ✓Contrastive reading: ‘The teacher_i said that Bat (but not other people) finished reading two books in his_i office.’
 ✓Thematic topic reading: ‘The teacher_i said that as for Bat, he he finished reading two books in his_i office.’
 (Gong 2022, p.199, ex.288)

- (164) *Thematic topic reading unavailable in nonfinite argument NC*
 Bagsh [Bat bol tavan nom unshi-h]-iig hel-sen.
 Teacher B. TOP five book read-INF -ACC say-PST
 ✓Contrastive reading: ‘The teacher said that Bat (but not other people) read 5 books.’
 ✗Thematic topic reading: ‘The teacher said that as for Bat, he read five books.’
 (Gong 2022, p.199-200, ex.289)

Assuming that a thematic topic position is located in the expanded CP periphery, e.g. TopicP (Rizzi 1997) or ThemeP (Gong 2023a), the asymmetry between (163) and (164)

⁶Gong (2023a) does not discuss what licenses a contrastive topic reading, although she notes that contrastive topics are licensed in a wider range of embedded contexts than thematic topics. Thus, while she argues that thematic topics need to be licensed by raising to a dedicated position within the CP periphery (ThemeP), she assumes that contrastive topics can be licensed in-situ.

can be interpreted to suggest that finite clauses containing the overt complementiser *gej* are full CPs, while nonfinite clauses lacking *gej* lack CP structure. Gong does not discuss thematic topicalisation in converbial adjuncts and finite NCs. Based on her analysis, one would predict that the behaviour of thematic topicalisation in finite NCs should parallel that of finite bare CPs, since both clause types contain an overt complementiser *gej/ged(e)g*, while thematic topicalisation in converbial adjuncts should parallel that in nonfinite NCs, since both clause types lack *gej/ged(e)g*.⁷

3.3.5 Summary

In this section, I have argued that all four embedded clause types in Khalkha contain TP, thereby narrowing the space of variation in clausal structure to the type of T⁰ and the presence or absence of CP structure contained in embedded clauses. I have provided additional arguments for the finite vs. nonfinite distinction in Khalkha embedded clauses: finite clauses contain a nondefective T⁰ and a full CP structure, while nonfinite clauses contain a defective T⁰ and lack any CP structure. Table 3.4 summarises the findings of this section:

Table 3.4: Clausal properties of Khalkha embedded clauses

Morphosyntactic properties	Clause type			
	Finite bare CP	Finite NC	Nonfinite NC	Converbial adjunct
<u>Correlates of finiteness</u>				
Contains overt C <i>gej/ged(e)g</i>	✓	✓	✗	✗
Able to contain overt tense suffixes	✓	✓	✗	✗
Able to contain WHQ <i>be/ve</i>	✓	✓	✗	– ⁸
Able to contain thematic topics	✓	–	✗	– ⁹

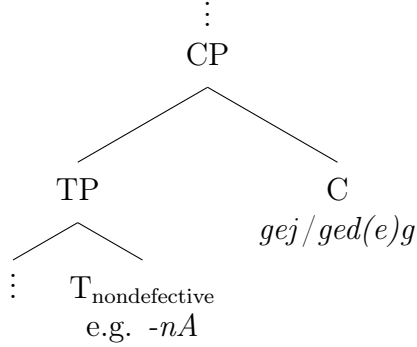
⁷Unfortunately, I have not been successful at eliciting the contrasts between a contrastive vs. thematic topic readings in embedded clauses with my consultants.

⁸The dash here represents the fact that embedded question formation is not possible with converbial adjuncts.

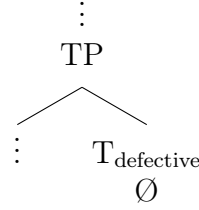
⁹The dashes in this row represent the two clause types for which Gong (2022, 2023a) does not provide data regarding thematic topicalisation.

The clausal structures of finite clauses (including finite bare CPs and finite NCs), nonfinite NCs, and converbial adjuncts are shown in (165), (166) and (167) respectively:

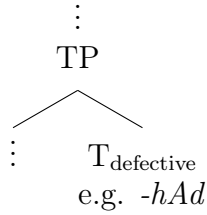
(165) *Clausal structure in finite clauses*



(166) *Clausal structure in nonfinite TPs*



(167) *Clausal structure in converbial adjuncts*

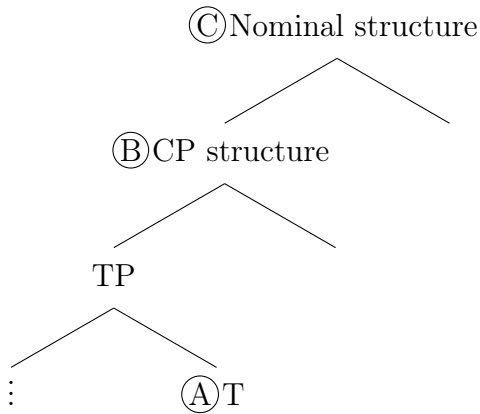


In the next section, I will probe further into the structure of the nominal layers in three of the four embedded clause types in Khalkha.

3.4 The structure of nominalisation in Khalkha embedded clauses

Up to this point, I have discussed how the four embedded clause types in Khalkha may differ as to (A) the type of T^0 they contain, and (B) the amount of CP structure they contain (see (144), repeated below). In this section, I further argue that argument NCs and converbial adjunct NCs actually differ in (C) the amount of nominal structure they contain.

(144) *Potential sources of variation in Khalkha embedded clauses*



As I will show below, while finite NCs, nonfinite NCs, and converbial adjuncts may host possessive enclitics, only finite and nonfinite NCs may additionally bear structural case marking and contain genitive subjects. Following works such as Alexiadou 2001, 2020, Alexiadou et al. 2010 and Baker 2011, I will take the presence or absence of a nominal property to reflect the presence or absence of a dedicated functional layer. It follows that, while both clause types are nominalised, argument NCs contain more nominal structure than converbial adjunct NCs.

My analysis is not without precedent; while the exact details differ, Kornfilt 2006 also argues that adjunct NCs in Turkish lack some nominal structure found in argument NCs. My analysis is also supported by parallel observations from the literature on so-called Small Nominals, where it is also proposed that variation in nominal properties reflects variation in the amount of nominal functional structure present (Pereltsvaig 2006, et seq.) I propose that, just as nominals may vary as to whether they are full DPs or small nominals that lack DP (Pereltsvaig and Lyutikova 2014; Lyutikova and Pereltsvaig 2015), nominalised clauses may also vary as to whether they are full DPs or small nominalised clauses (SNCs) which lack DP. Specifically, I propose that argument NCs are full DPs that contain an intermediate nominal layer PossP and nominalising nP, while converbial adjuncts are SNCs – smaller PossPs that contain nominalising nP but lack DP. This is schematised in (168).

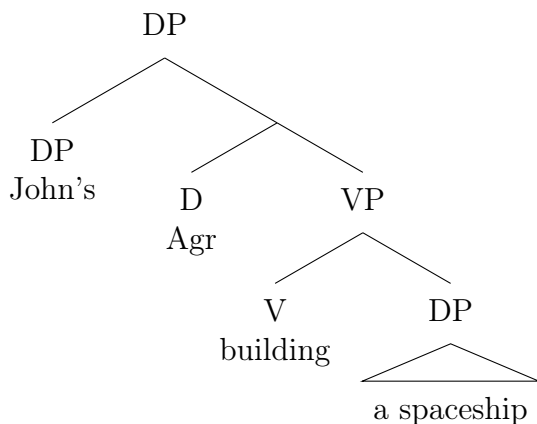
- (168) Nominal structure of embedded clauses in Khalkha
- a. Argument NCs (finite NC, nonfinite NC)
DP > **PossP** > **nP** > clausal structure
 - b. Converbial adjuncts
PossP > **nP** > clausal structure

I begin with a brief background on nominalisation in mixed extended projections in §3.4.1 and on small nominals in §3.4.2. In §3.4.3, I discuss how the insights from the analysis of small nominals are applicable towards an understanding of nominal structure in embedded clauses in Khalkha. I end the section in §3.4.4 with a short discussion of how the findings are relevant for understanding genitive case assignment in Khalkha.

3.4.1 Theoretical perspectives on nominal structure

In current minimalist syntax, it has been argued that there are multiple functional projections within the nominal domain (Abney 1987; Szabolcsi 1983, 1994; Travis and Lamontagne 1992; Ritter 1991, 1992; Marantz 2001, a.o.) that parallel the expanded CP domain (Rizzi 1997). Abney (1987), building on work by Lees (1960), argues that certain types of clauses, such as English gerunds, should be analysed as comprising a nominal exterior, e.g., (169).

- (169) *Structure of ‘John’s building a spaceship’ (Abney 1987, adapted in Punske 2014)*



Such an analysis of NCs – where a nominal functional structure is projected immediately above clausal structure – has come to be known as a *mixed extended projection*,

following work by Borsley and Kornfilt 2000. Building on this previous work, Kornfilt and Whitman (2011) propose the Functional Nominalization Thesis (FNT)¹⁰ to account for the mixed nominal and clausal properties of NCs:

- (170) *The Functional Nominalization Thesis (FNT) (Kornfilt and Whitman 2011)*
 Nominal properties of a nominalization are contributed by a nominal functional projection. The nominalization has verbal properties below the nominal functional projection, nominal properties above it.

The FNT makes two interrelated claims: i) first, that the nominal properties of nominalisations are directly related to the presence of nominal functional structure within them, and ii) that the verbal properties of the nominalisation depend on where in the clause nominalisation is introduced. Kornfilt and Whitman (2011) discuss cross-linguistic evidence that nominalisation may take place at various levels of clausal structure. For example, in Polish (171), the presence of the determiner *to* on the embedded clause suggests that it is nominalised, while the presence of the complementiser *ze* in the embedded clause suggests that the clause below the level of nominalisation is a CP (Kornfilt and Whitman 2011).

- (171) Polish
 Jan oznajmil [to ze Maria zmienia prace].
 Jan announced that_{DP} that_{CP} Maria is-changing job
 ‘Jan announced that Mary is changing her job.’
 (modified from Kornfilt and Whitman 2011, p.1299, ex.6)

TP nominalisations, where nominal functional structure is projected immediately above TP, are exemplified by Turkish factive (FNOM) and nonfactive (NFNOM) nominalisations. Both embedded clauses in (172) contain a genitive subject, possessor agreement morphology, and accusative case marking on the clause, suggesting the presence of nominal structure. The clauses differ in the following ways: (172a) contains a FNOM marker and expresses nonfuture tense and indicative mood, while the (172b) contains a NFNOM marker and expresses future tense and subjunctive mood (see also Kornfilt 2001). Kornfilt and Whitman

¹⁰Kornfilt and Whitman (2011)’s FNT stands in direct opposition to lexicalist theories of nominalisations. For example, Kiparsky 2017 rejects the FNT and proposes that the nominal-like distribution and properties of gerundive/participial clauses is due to the fact that their Infl⁰ bears an unvalued Case feature.

argue that in (172) T^0 is present but defective as the subject does not receive nominative case and finer tense distinctions beyond future/nonfuture cannot be expressed.

(172) Turkish

a. *Factive nominalisation*

Hasan [uşag-ın oda-yı temizle-diğ-i]-i söyle-di
H. servant-GEN room-ACC clean-FNOM-3SG.POSS -ACC say-PST.3SG
‘Hasan said that the servant cleaned the room.’

b. *Nonfactive nominalisation*

Hasan [uşag-ın oda-yı temizle-me-sin]-i söyle-di
H. servant-GEN room-ACC clean-NFNOM-3SG.POSS -ACC say-PST.3SG
‘Hasan said that the servant should clean the room.’

(Kornfilt and Whitman 2011, p.1300, exx.11-12)

Italian (173) provides an example where nominal structure (which can be detected by the presence of a determiner, possessor and adjective) encapsulates a vP-sized clause. Kornfilt and Whitman assume that there is no additional structure above vP (for instance, no defective T^0 responsible for the infinitival form of the verb), but that v^0 must be present to assign accusative case to the object.

(173) Italian

[_{DP} il suo continuo [_{vP} eseguire la canzone impeccabilmente]]
the his/her continual perform.INF the song impeccably
‘her/his continuous performing the song impeccably’

(Zucchi 1993, cited in Kornfilt and Whitman 2011, p.1299, ex.5)

Last but not least, VP and AspP nominalisations are also attested, as discussed by Alexiadou et al. (2010) for Romanian. Romanian contains two types of nominalisations, the supine (174a) and the infinitive (174b). Alexiadou et al. argue that the supine projects an AspP containing VP, while the infinitive projects VP and not AspP. One straightforward piece of evidence comes from the fact that the supine may contain aspectual adverbs such as *constantly* (174a) while the infinitive may not (174b).

(174) Romanian

a. *Supine nominalisation*

citi-t-ul {constant al} ziarelor {constant}
 read-ASP-the {constantly of} newspapers.GEN {constantly}
 ‘constantly reading newspapers’

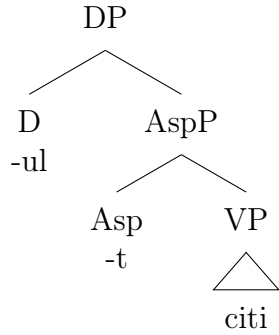
b. *Infinitive nominalisation*

omite-r-e-a {*constant a} unor informații {??constant}
 omit-NMLZ-F.SG-the {*constantly of} some information.GEN {??constantly}
 ‘omitting of some information’ (translation mine)

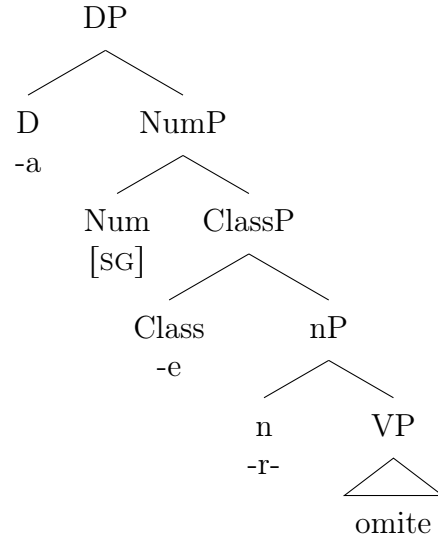
(Alexiadou et al. 2010, p.550, ex.24)

Notice that not only do the two types of Romanian nominalisations contain different amounts of clausal structure, they also appear to contain different amounts of nominal structure. Specifically, Alexiadou et al. (2010) argue that the supine (174a), whose nominal status is only indicated by a determiner on the verb (i.e. *-ul*) and genitive case on its complement, comprises only a DP projection immediately above VP. The infinitive (174b) on the other hand, appears to contain more nominal structure, as indicated by the presence of a nominaliser (i.e. *-r*), nominal gender (i.e. feminine singular *-e*), and a determiner (i.e. *-a*) on the verb. The nominal layers of the two types of nominalisations in Romanian are represented in (175a) and (175b) respectively (from Alexiadou et al. 2010, p.550). They differ in that in (175a) the AspP-sized clause is directly nominalised by DP, while in (175b) the VP-sized clause is nominalised by nP and involves intermediate nominal functional layers below DP. A key assumption in Alexiadou et al.’s analysis of Romanian is couched within the framework of Distributed Morphology (Halle and Marantz 1993, 1994; Marantz 1997; Harley and Noyer 1999), under which it is proposed that dedicated nominal functional projections are responsible for different types of nominal morphology, e.g. Num(ber)P for number morphology.

(175) a. *Structure of (174a)*



b. *Structure of (174b)*



Finally, recent work by Nasu et al. (2024) has proposed that Japanese contains two kinds of NCs, both of which contain nP. They assume that the demonstrative *ano* ‘that’ is located in Spec,DP, and argue that embedded questions selected by an inquiry-type predicate, e.g. (176a), are nPs that lack a DP layer as they cannot contain *ano*. On the other hand, they argue that embedded questions selected by other types of question-embedding predicates, e.g. (176b), are DPs that contain nP as they may contain *ano*.¹¹

(176) Japanese

a. John-wa [(***ano**) nani-o Mary-ga katta ka]-o {tazuneta/touta}.
 J.-TOP (*that) what-ACC M.-NOM bought Q -ACC {asked/questioned}
 ‘John {asked/questioned} what Mary had bought.’

b. Kyoo-no kaigi-de [**ano** dooyatte kakaku-o osaeru ka]-ga
 today-GEN meeting-in that how price-ACC control Q -NOM
 hanasiaw-are-ta.
 discuss-PASS-PST
 ‘That how we control the price was discussed in today’s meeting’

(Nasu et al. 2024, p.13, exx.28-29)

¹¹However, as I discuss in §3.6.3, there are several properties of the Japanese sentences in (176) that are unaccounted for and which suggest that the sentences may not be directly equivalent. For one, (176a) is a complement clause while (176b) is a subject clause. As such, (176a) may not truly represent a NC in which a nP that lacks a DP layer is projected above a full CP structure – a type of NC that I argue in §3.6 should not be possible following more general principles of structure building under the Williams Cycle (Williams 2003, 2011, 2013).

In sum, in addition to showing that different amounts of clause structure can be nominalised across languages, the previous literature has also proposed three distinct analyses of nominal structure in NCs. These are schematised in (108), repeated below:

(108) *Three analyses of mixed extended projections in NCs*

- a. *NC directly nominalised by DP*
DP > clausal structure
- b. *NC nominalised by nP and contains DP*
DP > (...) nP > clausal structure
- c. *NC nominalised by nP and lacks DP*
(no DP) (...) > nP > clausal structure

As I will ultimately argue that argument NCs and converbial adjuncts in Khalkha contain different amounts of nominal structure above the clause, I will set aside direct-DP shell analyses of NCs (108a) in my following discussion as such approaches do not straightforwardly allow for variation in nominal functional structure. Specifically, I argue that argument NCs instantiate (108b), and that converbial adjuncts instantiate (108c). I call the structure of converbial adjuncts in Khalkha a *small nominalised clause* (SNC), under similarity with small nominals (Pereltsvaig 2006, et seq.). Since there is no morphosyntactic evidence that Khalkha NCs contain other nominal projections (e.g. NumP) other than DP and PossP, such projections will not figure in the following discussion. I turn next to some background on small nominals.

3.4.2 Small Nominals

The idea that a single language may contain full DPs and smaller-sized nominals is familiar from the literature on noun incorporation and pseudo-noun incorporation (for recent overviews, see Massam 2009, 2017; Johns 2017). However, a different version of this idea comes from Pereltsvaig (2006), who argues that standalone (non-pseudo-incorporated) nominals may also vary as to whether they are full DPs or small nominals – nominals that lack a DP layer. Pereltsvaig argues that not all small nominals are incorporated or pseudo-incorporated nouns; for instance, small nominals need not be adjacent to the verb and can

be displaced by certain operations that target non-referential nominals, i.e. nominals which lack a DP layer. For example, an operation known as Approximative Inversion in Russian may only target small nominals, e.g. *čelovek pjat* ‘five people’ in (177). Pereltsvaig argues that the ill-formedness of (177) with the referential pronoun *ix* ‘them’ lends support to the idea that the topicalised argument is non-referential, and hence a small nominal.

- (177) Russian
Čelovek pjat’, my eščë ne videlo (*ix).
 people five we still not saw (*them)
 ‘As for a group of approximately five people, we haven’t seen such a group yet.’
 (Pereltsvaig 2006, p.474, fn.32)

The view that a single language may contain both full DPs and small nominals stands in opposition to two influential views of nominal structure: that argument nominals in all languages must be DPs, following Longobardi (1994); or that languages can be classified according to whether they are a DP-language or NP-language, following Chierchia (1998) and Bošković (2008).¹²

Another language that has been argued to contain both full DPs and small nominals is Tatar (Turkic), where the variation can be observed in the possessive domain (Pereltsvaig and Lyutikova 2014; Lyutikova and Pereltsvaig 2015). In the so-called *ezafe-3* possessive construction in (178a), the possessor is marked with genitive case and possessor agreement is reflected on the head noun. (178a) denotes an apple that belongs to children. In the *ezafe-2* possessive construction in (178b), morphology that is homophonous with the third singular possessive agreement suffix (-sı) is also present on the head noun, but the possessor does not receive any case marking. (178b) denotes an apple that is associated with children.

¹²Even within such an approach, it is sometimes murky how to classify individual languages, such as Turkish. For example, Bošković and Şener (2014) argue that Turkish should be classified as an NP-language, while their view has been criticised by Kornfilt 2018a,b.

(178) Tatar

a. *Ezafe-3 possessive nominal*

bala-lar-**niŋ** alma-**sı**
 child-PL-GEN apple-3.POSS
 ‘(the) children’s apple’

b. *Ezafe-2 possessive nominal*

bala-lar alma-**sı**
 child-PL apple-3.POSS
 ‘children’s apple (i.e. an apple for children)’

(Lyutikova and Pereltsvaig 2015, p.300, ex.20)

Further, Pereltsvaig and Lyutikova show that while *ezafe-3* objects are obligatorily accusative-marked (179a), *ezafe-2* objects may be case-unmarked (179b) or accusative-marked (179c).

(179) Tatar

a. *Ezafe-3 object with obligatory accusative marking*

Min [bala-lar-niŋ tabib-i*(-**n**)] ezl-i-m.
 I child-PL-GEN doctor-3.POSS*(-ACC) look.for-PRES-1SG
 ‘I am looking for a/the children’s doctor.’

b. *Ezafe-2 object without accusative marking*

Min [bala-lar tabib-i] ezl-i-m.
 I child-PL doctor-3.POSS look.for-PRES-1SG
 ‘I am looking for a pediatrician.’

c. *Ezafe-2 object with accusative marking*

Min [bala-lar tabib-i-**n**] ezl-i-m.
 I child-PL doctor-3.POSS-ACC look.for-PRES-1SG
 ‘I am looking for {a certain/the} pediatrician.’

(Lyutikova and Pereltsvaig 2015, p.316, ex.60)

Finally, Pereltsvaig and Lyutikova (2014, p.206) assume that case markers are found on K^0 , which selects only for DPs, and that DPs in Tatar are subject to the Case Filter.

For *ezafe-3* constructions, Pereltsvaig and Lyutikova propose that they are full DPs which contain full DP possessors. Since the *ezafe-3* construction is a full DP, it is subject to the Case Filter and is obligatorily accusative marked. Since the *ezafe-3* possessor is also a full DP, it is also subject to the Case Filter and is obligatorily genitive marked. The authors argue that the source of genitive case is D^0 of the *ezafe-3* construction, which Agrees with

the DP possessor, raises it to Spec,DP,¹³ and assigns it genitive case. They treat the *ezafe-3* marker on the head noun (e.g. *-si* in (178a)) as ϕ -agreement on D⁰.

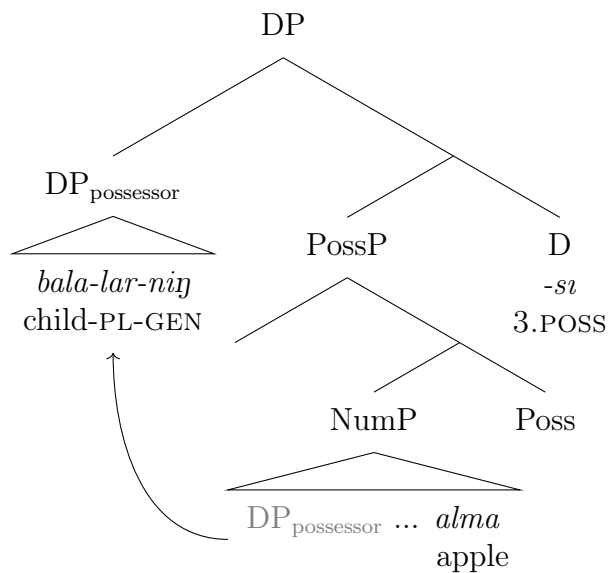
For *ezafe-2* constructions, Pereltsvaig and Lyutikova propose that these may either be PossPs lacking a DP layer (i.e. small nominals) or PossPs embedded under a null DP. They also propose that *ezafe-2* possessors are PossPs lacking a DP layer. They analyse case-unmarked *ezafe-2* objects as bare PossPs (e.g. (179b)), and accusative-marked *ezafe-2* objects as full DPs (e.g. (179c)). Although the authors are not entirely explicit on this point, they seem to suggest that even when PossPs are selected by a null D⁰, this head does not Agree with the possessor and does not assign genitive case. Since *ezafe-2* possessors cannot be assigned any Case – given the lack of (genitive-assigning) D⁰ – they cannot be full DPs due to the Case Filter. As support for this view, they note that *ezafe-2* possessors cannot be pronouns or proper names, which are full DPs and are subject to the Case Filter. Thus, *ezafe-2* possessors must themselves be small nominals, i.e. PossPs. The authors propose that in *ezafe-2*, the possessor occupies Spec,PossP, and that the suffix on the head noun (e.g. *-si* in (178b) and (179b)-(179c)) is an *ezafe-2* marker realised on Poss⁰.¹⁴ (180a)-(180c) illustrate the structures of the objects in (179).

¹³Pereltsvaig and Lyutikova (2014) suggest two possible positions where the *ezafe-3* possessor is first merged: in Spec,NP, which corresponds to an Agent/Creator/Owner interpretation of the possessor, or as a complement to N⁰, which corresponds to an internal argument interpretation of the possessor. For *ezafe-2* possessors, the authors argue that they are first merged in Spec,PossP.

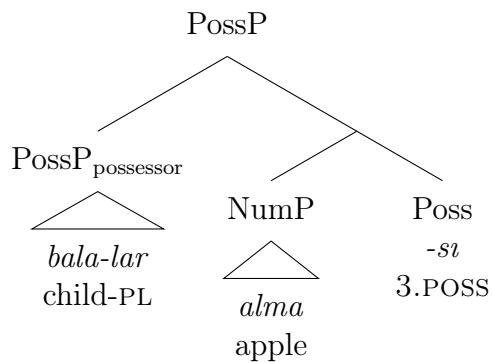
¹⁴The authors gloss the *ezafe-2* as 3rd person possessive agreement, but are agnostic as to whether the *ezafe-2* marker represents true ϕ -agreement with the third person possessor argument. As *ezafe-2* constructions cannot contain first and second person possessor arguments, it is not possible to test for covarying agreement.

(180) *Three possessive structures in Tatar (from Pereltsvaig and Lyutikova 2014)*

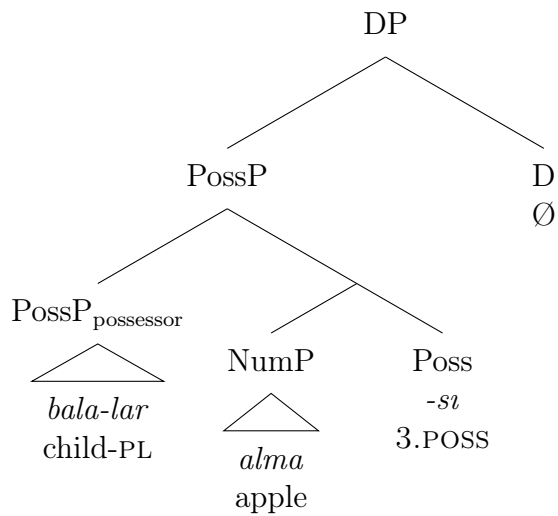
a. *Ezafe-3 DP*



b. *Ezafe-2 PossP*



c. *Null DP merged above ezafe-2*



That the possessors are located in different structural positions is supported by evidence such as adjective placement, as seen in (181). In (181a), the *ezafe-3* possessor, marked with genitive case, obligatorily precedes the adjective *kük* ‘blue’, while in (181b) the *ezafe-2* case-unmarked possessor obligatorily follows the same adjective. While the authors do not specify the position of the adjective, they assume that it occupies the same position in (181a) and (181b). The difference in the distribution of the adjective thus suggests that the genitive-marked possessor is structurally higher (e.g. in Spec,DP) than the case-unmarked one (e.g. located in Spec,PossP).

(181) Tatar

a. *Ezafe-3 possessor obligatorily precedes adjective*

(*kük) bala-lar-nıñ (kük) itek-lär-e
 blue child-PL-GEN blue boot-PL-3.POSS
 ‘(the) children’s blue boots’

b. *Ezafe-2 possessor obligatorily follows adjective*

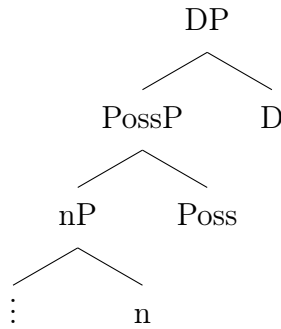
(kük) bala-lar (*kük) itek-lär-e
 blue child-PL blue boot-PL-3.POSS
 ‘blue children’s boots’ (Lyutikova and Pereltsvaig 2015, pp.300-301, ex.21)

In sum, possessive nominals in Tatar come in two sizes: full DPs (with Agreeing D^0 or null, non-Agreeing D^0) which contain PossPs, and reduced PossPs which lack DP. In the next subsection, I show how Pereltsvaig and Lyutikova’s analysis of small nominals in Tatar can be applied to understand a similar structural variation in Khalkha NCs.

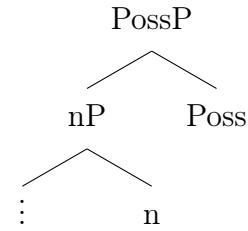
3.4.3 Nominal structure in Khalkha embedded clauses

In this subsection, I argue that embedded clauses in Khalkha differ as to whether their nominal structure comprises a full DP containing PossP and nP (182), or a PossP containing nP but lacking a DP layer (183).

(182) *Nominal structure in complement NC*



(183) *Nominal structure in converbial adjunct*



As alluded to earlier, possessive nominals in Khalkha resemble *ezafe-3* constructions in Tatar in that they may contain a genitive-marked pronoun (184a), a ϕ -agreeing possessive (POSS) enclitic (184b), or both (184c). This was also discussed in §2.1.3. Notice further that possessive DP objects obligatorily bear accusative case marking.

(184) *Three ways to express possession in Khalkha*

a. GEN *pronoun*

Chi **minii** nom-**iig** unsh-san.
 2SG 1SG.GEN book-ACC read-PST

b. POSS *enclitic*

Chi nom-**iig=min'** unsh-san.
 2SG book-ACC=1SG.POSS read-PST

c. GEN *pronoun* + POSS *enclitic*

Chi **minii** nom-**iig=min'** unsh-san.
 2SG 1SG.GEN book-ACC=1SG.POSS read-PST
 'You read my book.'

To the best of my knowledge, Khalkha does not appear to have small nominals resembling *ezafe-2* possessives in Tatar. Barrie and Kang 2022, Kang 2023, and Barrie 2024 discuss prosodic evidence for both full DPs and smaller nPs in Khalkha,¹⁵ but they analyse reduced nPs as pseudo-incorporated nominals rather than standalone small nominals as defined by Pereltsvaig 2006. Pending further investigation, this paper assumes the absence of non-pseudo-incorporated small nominals in Khalkha's purely nominal domain. However, I argue that the full DP-small nominal distinction emerges in the domain of nominalised

¹⁵Their analysis focuses on simplex nominals. They cite Guntsetseg (2016), who observes that pseudo-incorporated nouns in Khalkha lack demonstratives, possessors, postpositions, case markers, and plural marking. See also Driemel 2020, 2023 on pseudo-noun incorporation in Khalkha.

clauses. To do this, I will examine how the four embedded clause types in Khalkha pattern with respect to the three nominal properties found on possessive nominals: i) the availability of possessive enclitics, ii) the availability of genitive case assignment, and iii) the ability to bear structural case marking. Previewing my findings, I show that finite CPs do not exhibit any of the three abovementioned nominal properties, complement NCs display all three nominal properties, and converbial adjuncts display only one of the three nominal properties – namely, the ability to host possessive enclitics.

Following Borsley and Kornfilt 2000; Alexiadou 2001, 2020; Alexiadou et al. 2007, 2010; Kornfilt and Whitman 2011, a.o., I adopt the idea that the clausal properties of NCs stem from clausal functional structure (e.g. vP, VoiceP, AspP, TP, CP) below the level of nominalisation, while their nominal properties arise from nominal functional structure (e.g., nP, PossP, DP) above the clausal level. Nominalisation is mediated by a categorising (Marantz 2001) or “category-switching” (Rouveret 2023) head, n^0 , projected directly above the clausal level, which serves to nominalise the clause. I assume that n^0 is null but triggers allomorphy of C^0 in finite NCs, as represented in (185).

(185) *Vocabulary Items for C^0 in Khalkha*

- a. $C^0 \leftrightarrow gej$
- b. $C^0 \leftrightarrow ged(e)g / _ n^0$

I take the variation in nominal properties across different types of NCs to reflect differences in the amount of nominal functional structure above n^0 . This analysis argues against a direct DP-shell approach to Khalkha NCs, which posits DP as the sole nominal layer above the clause (e.g. Borsley and Kornfilt 2000; Alexiadou et al. 2010; Ótrott-Kovács 2023). Such an approach would be ill-equipped to account for the variation in nominal properties across NC types without permitting differences in the extent of nominal functional structure.

Let’s consider each embedded clause type in turn. Finite bare CPs do not exhibit any of the three nominal properties of possessive nominals (186), which I take to indicate

the lack of any nominal structure in them. Thus, I will exclude finite bare CPs from the following discussion to focus on the other three embedded clause types in Khalkha.

- (186) *Finite bare CP may not contain a GEN subject, bear case marking, or host a POSS enclitic*

Bi [Naraa-g/Naraa/***Naraa-giin** unta-j bai-san gej](*-iig)(*=**n'**)
 1SG N.-ACC/N./*N.-GEN sleep-PROG COP-PST C (*-ACC)(*=3.POSS)
 med-ne
 know-NPST
 'I know that Naraa was sleeping.'

Finite (187) and nonfinite (188) NCs fully parallel possessive nominals in exhibiting all three nominal properties: they may host possessive enclitics, contain a genitive subject and bear structural case marking. Thus, I take both finite and nonfinite NCs to be full DPs, like possessive nominals.

- (187) *Finite NC with GEN subject, accusative case and POSS enclitic*

Naraa [**minii** unta-j bai-san gedg]-iig(=**min'**) med-ne.
 N. 1SG.GEN sleep-PROG COP-PST C.NML -ACC=1SG.POSS know-NPST
 'Naraa knows that I was sleeping.'

- (188) *Nonfinite NC with GEN subject, accusative case and POSS enclitic*

Naraa [**minii** unta-j bai-sn]-iig(=**min'**) med-ne.
 N. 1SG.GEN sleep-PROG COP-PFV -ACC=1SG.POSS know-NPST
 'Naraa knows that I was sleeping.'

Strikingly, converbial adjuncts exhibit only a subset of these properties. First, while converbial adjuncts may host possessive enclitics, they may not contain genitive subjects. This is seen in (189).

- (189) *Converbial adjunct may host POSS enclitic but cannot contain GEN subject*

Chi [namaig/***minii** hicheel-Ø=ee hii-j bai-had](=**min'**)
 2SG 1SG.ACC/*1SG.GEN lesson-ACC=RP do-PROG COP-TEMP.CVB =1SG.POSS
 unt-san.
 sleep-PST
 'When I was studying (lit. doing my lesson), you fell asleep.'

Additionally, converbial adjuncts cannot be marked with any case marking, such as accusative case with a matrix transitive verb in (190), or dative case with a matrix psych verb in (191).¹⁶

(190) *Converbial adjunct cannot be marked with accusative case*

- a. *Bi [Naraa-g hicheel-Ø=ee hii-j bai-had]-iig(=n')
 1SG N.-ACC lesson-ACC=RP do-PROG COP-TEMP.CVB -ACC=3.POSS
 med-ne
 know-NPST
 Int. 'I know when Naraa was studying.'
- b. Bi [Naraa-g huushuur ide-j bai-had](*-iig)(=n') nom
 1SG N.-ACC khuushuur eat-PROG COP-TEMP.CVB (*-ACC)=3.POSS book
 unshi-j bai-san.
 read-PROG COP-PST
 'I was reading a book while Naraa was eating khuushuur.'

(191) *Converbial adjunct cannot be marked with dative case*

- a. *Bi [Naraa-g unta-j bai-had]-ad(=n') uurla-san.
 1SG N.-ACC sleep-PROG COP-TEMP.CVB -DAT=3.POSS be.angry-NPST
 Int. 'I was angry at (the time when) Naraa (was) sleeping.'
- b. Bi [Naraa-g unta-j bai-had](*-ad)(=n')
 1SG N.-ACC sleep-PROG COP-TEMP.CVB (*-DAT)=3.POSS
 Dorj-id uurla-san.
 D.-DAT be.angry-NPST
 'I was angry at Dorj when Naraa was sleeping.'

¹⁶One might argue that converbial adjuncts are not overtly case-marked because they are inherently case-marked. Umetani (2019) proposes a decompositional analysis for certain converbs, particularly those with transparent morphology. For instance, he analyses the temporal converb *-hAd* as consisting of the infinitive/futuritive aspect marker *-h* and the dative case suffix *-Ad*. If this analysis were correct, one might predict that dative-marked adjuncts could function as complements of psych predicates like *uurla-* 'be angry,' which take dative-marked arguments.

However, the example in (i), though grammatically acceptable, fails to convey the intended reading where the embedded subject *Naraa's* sleeping directly causes the first-person subject's anger. Instead, consultants report that the source of anger in (i) is unclear or attributed to a *pro*-dropped argument. This suggests that the converb does not contain an active case marker. While I leave open the possibility that some converbs may include fossilised case markers due to their etymology, I do not assume that converbial adjuncts are inherently case-marked.

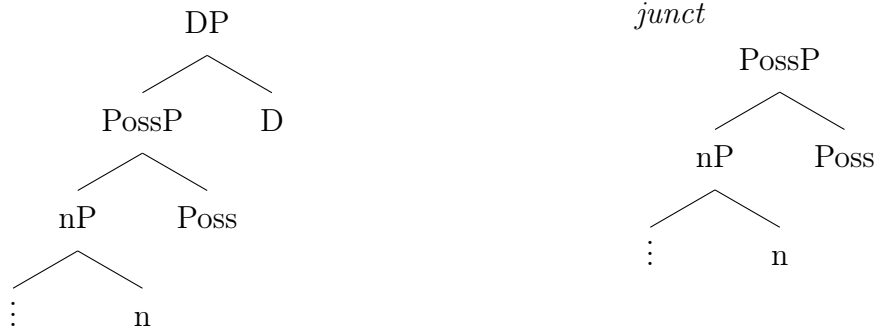
- (i) Bi [Naraa-g unta-j bai-had](=n') uurla-san.
 1SG N.-ACC sleep-PROG COP-TEMP.CVB =3.POSS be.angry-NPST
 Means: 'I was angry (at something) when Naraa was sleeping.'
 Does not mean: 'I was angry at Naraa sleeping.'

The ability of converbial adjuncts to host possessive suffixes indicates the presence of some nominal structure. However, they lack the additional DP layer found in finite and nonfinite complement NCs that enables genitive case assignment and allows the clause itself to be case-marked. This implies that converbial adjuncts are structurally smaller in the nominal domain compared to finite and nonfinite complement NCs.

3.4.4 Two types of nominal structure in Khalkha NCs

To summarise, finite and nonfinite NCs exhibit the full range of nominal properties found in possessive nominals, while converbial adjuncts exhibit only a subset – namely, the ability to host possessive enclitics. I propose that this variation arises because finite and nonfinite NCs are full DPs that contain PossP and nP (182), whereas converbial adjuncts are reduced PossPs that contain nP but lack DP (183) (structures repeated below). Note that I assume n^0 to be phonologically null in Khalkha NCs, possessive enclitics (e.g. third possessive *=n'*) to occupy Poss^0 as a result of Agree with the embedded subject, and case suffixes (e.g. accusative *-iig*) to occupy D^0 . Additionally, as discussed in Chapter 2, I assume that a postsyntactic reordering operation also applies to NCs to derive the POSS-CASE order, although this reordering is not depicted in (182) and (183). Under this analysis, I propose to call converbial adjuncts small nominalised clauses (SNCs), under similarity with small nominals (Pereltsvaig 2006, et seq.).

- (182) *Nominal structure in complement NC* (183) *Nominal structure in converbial adjunct*



3.4.5 On genitive case assignment

Before concluding this section, I briefly highlight how the analysis of nominal structure impacts our understanding of genitive case assignment in Khalkha. Recall that the subject of a finite or nonfinite argument NC may receive genitive case, while the subject of a converbial adjunct NC may not. I have argued that argument NCs contain a full DP layer, whereas converbial adjunct NCs are reduced PossPs lacking a DP layer. This suggests that the DP layer is crucial for genitive case assignment in the nominal domain. This observation aligns with an Agree-based, functional-head theory of genitive case where it is assigned via Agree by D^0 (e.g., Asarina 2011; Baker and Vinokurova 2010; Bošković 2008; Hale 2002; Kornfilt 2003; Laszakovits 2018b; Ótrott-Kovács 2020, a.m.o.).

While D^0 appears to be important in assigning genitive case in Khalkha, the data in this section show that the Agree operation responsible for genitive case is distinct from the one that yields possessor agreement morphology. Crucially, argument NCs and converbial adjuncts in Khalkha exhibit a double dissociation between genitive case assignment and possessor agreement (realised as possessive enclitics). First, non-genitive subjects, such as accusative subjects, are compatible with possessive enclitics in converbial adjuncts and argument NCs, suggesting that the appearance of genitive case does not always co-occur with possessor agreement. Second, genitive case cannot be assigned to subjects of converbial adjuncts despite the presence of possessive enclitics on these clause types. In Chapter 5, I provide an analysis of genitive case assignment that involves an Agree relation with D^0 but does not yield overt possessor agreement morphology.

3.4.6 Summary

This section has demonstrated that while finite and nonfinite complement NCs exhibit the full range of nominal properties shared with possessive nominals, converbial adjuncts exhibit only a subset of these properties. Following the view that nominal properties in nominalisations correspond to the presence of nominal structure (Kornfilt and Whitman

2011), I have proposed that complement NCs are full DPs containing PossP, whereas converbial adjuncts are reduced PossPs lacking a DP layer. Table 3.5 summarises these findings:

Table 3.5: Nominal properties of Khalkha embedded clauses

Morphosyntactic properties	Clause type			
	Finite bare CP	Finite NC	Nonfinite NC	Converbial adjunct
<u>Correlates of nominal structure</u>				
Able to host possessive enclitics	✗	✓	✓	✓
Able to contain genitive subject	✗	✓	✓	✗
Able to be assigned case	✗	✓	✓	✗

Interestingly, the DP/PossP distinction aligns with the argument/adjunct distinction in Khalkha. While Khalkha arguments can be full DPs or pseudo-incorporated nouns (Guntsetseg 2016; Driemel 2020, 2023; Barrie and Kang 2022; Kang 2023; Barrie 2024), it remains unclear whether standalone small nominals can serve as arguments in Khalkha. As Pereltsvaig and Lyutikova (2014) argue for Tatar, both DPs and PossPs may serve as arguments in the language, see, e.g., (179). In §3.6 and §3.7, I offer some predictions as to where one may find small nominals or SNCs in a given language. Before doing so, in the next section, I provide sketches of the structures of the four embedded clause types under investigation.

3.5 Putting it together: The structures of Khalkha embedded clauses

In §3.3 and §3.4, I have shown how four types of embedded clauses in Khalkha vary in their morphosyntactic properties correlated with (non)finiteness and the amount of nominal structure present in them. The findings are summarised in Table 3.6, and an example of each embedded clause type (122)-(125) is repeated below.

Table 3.6: Morphosyntactic properties of Khalkha embedded clauses

Morphosyntactic properties	Clause type			
	Finite bare CP	Finite NC	Nonfinite NC	Converbial adjunct
<u>Correlates of finiteness</u>				
Contains overt C <i>gej/ged(e)g</i>	✓	✓	✗	✗
Able to contain overt tense suffixes	✓	✓	✗	✗
Able to contain WHQ <i>be/ve</i>	✓	✓	✗	-
Able to contain thematic topics	✓	-	✗	-
<u>Correlates of nominal structure</u>				
Able to host possessive suffixes	✗	✓	✓	✓
Able to contain genitive subject	✗	✓	✓	✗
Able to be assigned case	✗	✓	✓	✗

(122) *Finite bare CP*

Bi [Naraa margaash ir-ne gej] hel-sen.
 1SG N. tomorrow come-NPST C say-PST
 ‘I said that Naraa will come tomorrow.’

(123) *Finite complement NC*

Bi [Naraa-giin margaash ir-ne gedg]-iig(=n’) mart-san.
 1SG N.-GEN tomorrow come-NPST C.NML ACC=3.POSS forget-PST
 ‘I forgot that Naraa will come tomorrow.’

(124) *Nonfinite complement NC*

Bi [Naraa-giin margaash ire-h]-iig(=n’) mart-san.
 1SG N.-GEN tomorrow come-INF -ACC=3.POSS forget-PST
 ‘I forgot that Naraa is coming tomorrow.’

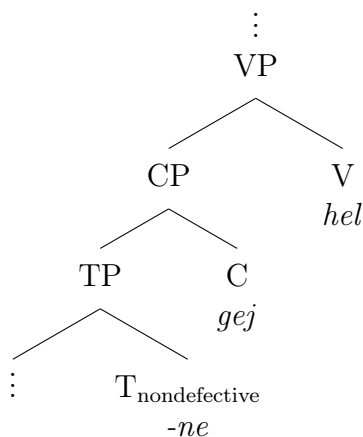
(125) *Converbial adjunct*

Bi [ter hicheel-Ø=ee hii-j bai-had](=n’) unt-san.
 1SG 3SG lesson-ACC=RP do-PROG COP-TEMP.CVB =3.POSS sleep-PST
 ‘When she/he was studying, I fell asleep.’

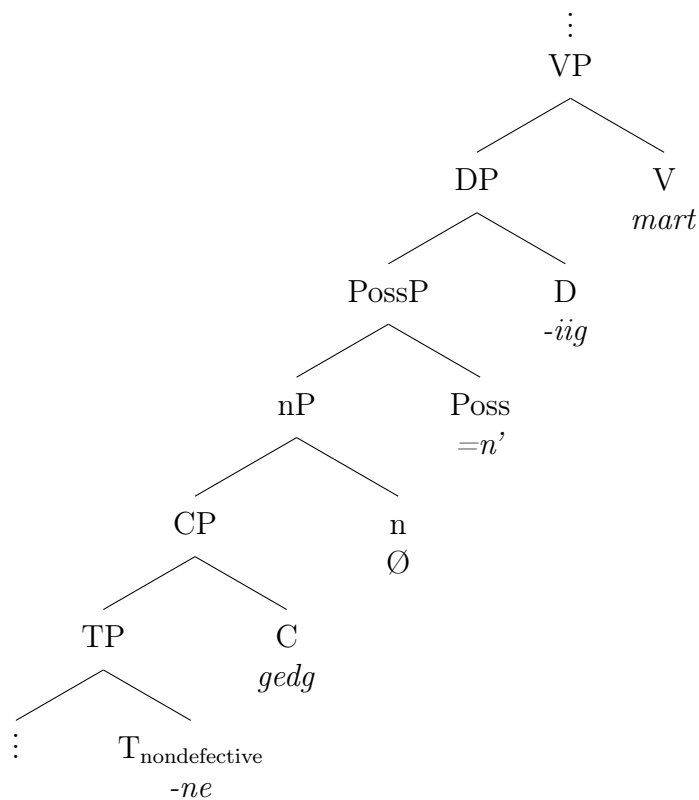
In (192)-(195), I provide partial structures for each embedded clause type, illustrating only the TP, CP and nominal layers within them. For argument clauses, I include the matrix predicate which selects for them, and for converbial adjuncts, I include the matrix vP projection to which they are adjoined. A finite bare CP contains a full CP with nondefective

T^0 , but lacks any nominal structure at all, as seen in (192). A finite NC contains a full CP with nondefective T^0 , and is nominalised by a full DP structure (193). A nonfinite NC contains no CP structure and defective T^0 , but is nominalised by a full DP (194). Lastly, a converbial adjunct, which adjoins to vP, contains no CP structure, contains an overt defective T^0 (i.e. a converb), and is nominalised by a reduced PossP structure (195). In other words, converbial adjuncts are SNCs.

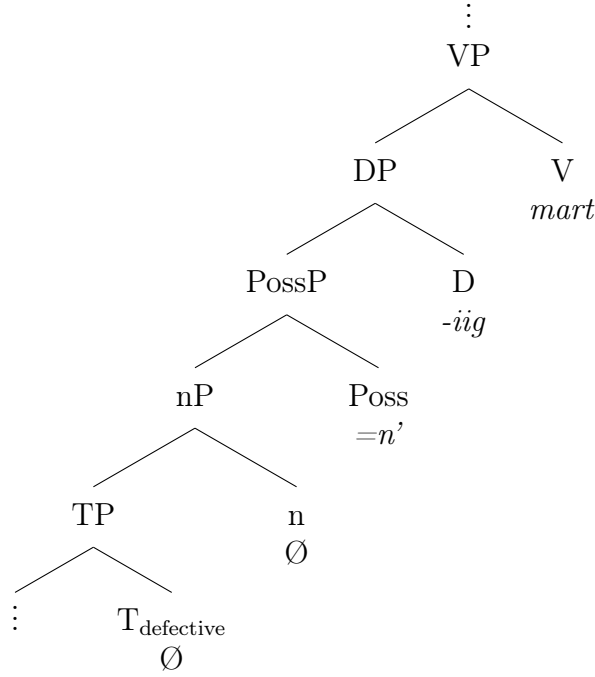
(192) *Finite bare CP* (122)



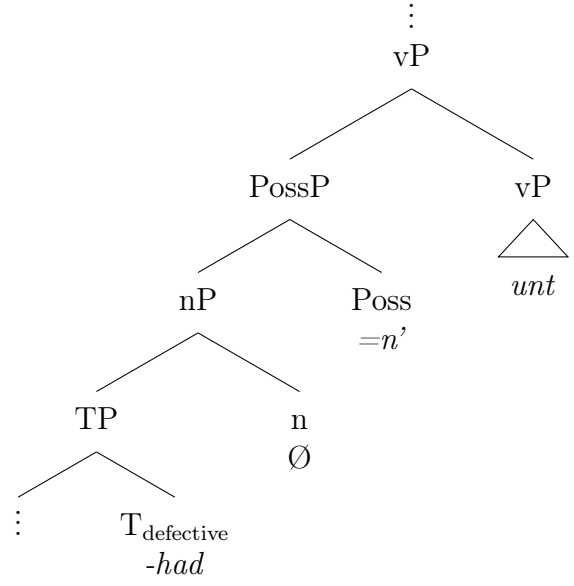
(193) *Finite NC* (123)



(194) *Nonfinite NC* (124)



(195) *Converbial adjunct* (125)



3.6 A gap in the typology of nominalised clauses

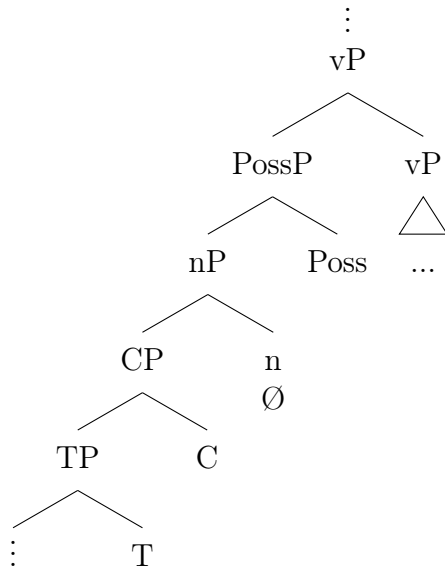
Strikingly, if one considers all logically possible combinations of clausal and nominal structure, a gap in the data emerges, represented by the bottom left cell of Table 3.1 (repeated below). To the best of my knowledge, there is no evidence in Khalkha of a structure where a full CP is nominalised by a reduced PossP structure *lacking a DP structure over it*.

Table 3.1: Types of NCs in Khalkha

	PossP	DP
TP	Converbial adjunct	Nonfinite argument
CP	(Unattested)	Finite argument

The unattested structure from Table 3.1 is represented in (196) as a type of adjunct clause adjoined to vP, similar to converbial adjuncts in Khalkha.

(196) *Unattested mixed extended projection:* (no DP)>PossP>nP>CP



Under the present account which takes there to be intermediate nominal projections between CP and DP, the missing structure (196) cannot be understood as a selectional restriction between two adjacent heads, e.g. that C^0 must be selected by D^0 . In this section, I argue that the gap is not accidental, and follows from more general principles of structure building.

3.6.1 Hierarchies of Domains

In fact, the gap resembles a prediction by Svenonius (2021), who proposes that mixed extended projections are constrained by what he calls Hierarchies of Domains (HoD):

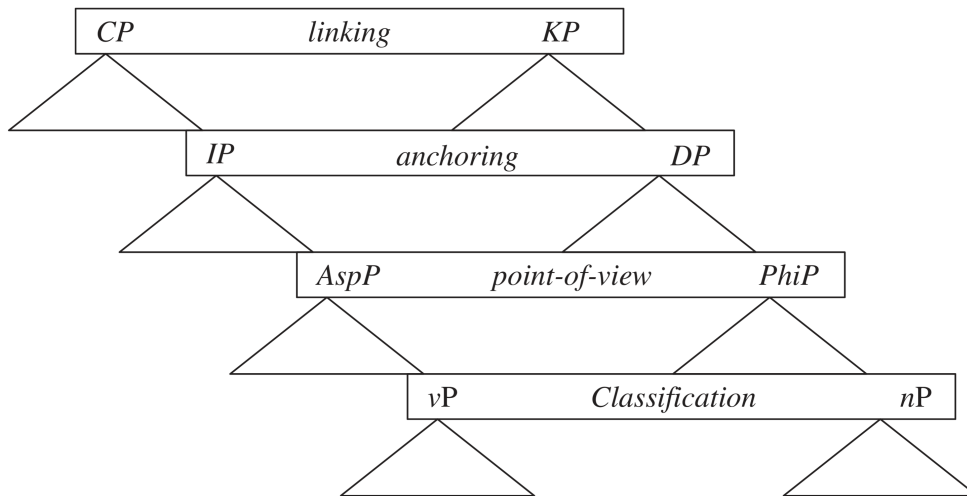
HoD allows high nominal over high verbal (Spanish), high nominal over intermediate verbal (English gerund), or low nominal over low verbal (nominalizations like *destruction*), **but not low nominal over high verbal structure** [...] in which relatively high verbal morphology like tense or the perfect could be dominated by relatively low nominal characteristics...

(Svenonius 2021, p.29; emphasis mine)

Svenonius' notion of HoD combines two ideas. The first idea is the common assumption that there exists a hierarchy of projections within the verbal and nominal extended

projections (e.g. CP>TP>AspP>vP) and is familiar from works such as Cinque 1999; Abney 1987; Rizzi 1997; Szabolcsi 1994. The second idea comes from Wiltschko’s (2014) Universal Spine Hypothesis, which posits that each verbal/nominal projection may be associated with a category-neutral domain along a more abstract functional spine, illustrated in (197). For instance, Wiltschko argues that the ‘anchoring’ function is expressed by person-number marking in Blackfoot, while the same function is expressed by locative auxiliaries in Halkomelem (see also Ritter and Wiltschko 2014).

(197) *The parallelism between the nominal and verbal spine (Wiltschko 2014, p.78)*



The Universal Spine Hypothesis offers a framework to conceptualise parallels between the CP and DP domains, an idea also explored by Abney 1987; Grimshaw 1991; Giusti 2002, 2006; Hiraiwa 2005, among many others. Svenonius argues that it further constrains how domains combine: a verbal functional projection within a given domain cannot be dominated by a verbal or nominal functional projection from a lower domain. This restriction applies to extended and mixed extended projections but excludes, for example, complements or adjuncts. I restate Svenonius’ constraint as in (198).

(198) *HoD constraint on (mixed) extended projections*

A verbal functional projection in domain α may not be dominated by a verbal/nominal functional projection in domain β , if $\alpha > \beta$.

(198) implies that a TP-sized clause belonging to the ‘anchoring’ domain cannot be nominalised by a functional projection belonging to the lower domains of ‘point-of-view’ and ‘classification’; it can only be nominalised by a nominal projection belonging to the ‘anchoring’ (e.g. DP) or ‘linking’ (e.g. KP – following Wiltschko 2014, or PP – following Svenonius 2021) domain.^{17,18}

While Svenonius’s (2021) analysis is similar in spirit to my proposal outlined below, it does not appear to make correct predictions for the Khalkha data shown above, as it predicts that structures where nP or PossP is projected above CP – regardless of whether nP or PossP is dominated by DP or not – should be ill-formed. This is because n^0 and $Poss^0$, by virtue of them being lower than D^0 , would belong to lower functional domains than C^0 . However, in Khalkha finite NCs, nP and PossP may be projected above CP *if* nP and PossP are dominated by DP. The unattested structure involves nP and PossP projected above CP *without* DP projected above those nominal layers. More generally, Svenonius’s analysis is incompatible with my analysis of mixed extended projections where lower nominal projections must be projected *in addition to* DP, in order to account for the variation in nominal properties that we find across different NC types in Khalkha. The HoD constraint in (198) as it stands thus appears to be too restrictive.

¹⁷In a similar vein, Alexiadou 2020 and Iordăchioaia 2020 argue that clausal structure containing at least TP must be directly nominalised by D^0 . For example, Iordăchioaia (2020) argues that, assuming DP is the counterpart of CP, we would not expect functional projections below DP(=CP) to be projected above TP. Further, given that AspP is “in between” (p.17) vP and TP, it may be directly nominalised by DP (e.g., the Romanian supine; see (174a) and surrounding discussion) or involve nominalising nP.

¹⁸Svenonius (2021, p.13, fn.13) acknowledges that some flexibility must be allowed with regards to which projection maps onto which domain across languages. For example, as some languages contain CPs directly nominalised by D^0 – e.g., in Ndebele (Pietraszko 2019) – this suggests that D^0 and C^0 belong to the same domain in these languages, instead of K^0 and C^0 as seen in (197).

3.6.2 Improper mixed extended projections

Building on Svenonius 2021’s core insight, I propose instead that the relevant constraint is a constraint on the *maximal* projection of the nominal and clausal functional layers within a mixed extended projection, given in (199).

- (199) *HoD constraint on mixed extended projections (revised)*
Given a HoD where $\alpha > \beta$, a verbal/clausal functional projection whose *maximal* projection belongs to $v_\alpha P$ cannot be dominated by a nominal functional projection whose *maximal* projection belongs to $n_\beta P$.

I further argue that (199) can be understood in terms of the relative timings of clausal and nominal embedding, which indirectly restrict the relative size of the clausal and nominal extended projections in a mixed extended projection. My proposal builds on similar types of asymmetries that have been subsumed under the general heading of Williams Cycle phenomena, named after an original account of movement asymmetries proposed by Williams (2003). The basic intuition behind Williams’ account is that a syntactic operation cannot take place between two sites, X and Y, if Y is higher than X in an assumed functional sequence (see, e.g., Starke 2001). Accordingly, such an operation is described as *improper*. While traditionally proposed to account for cases of improper movement (e.g. Williams 2003; Müller 2014), it has also been extended to account for parallel properties of agreement and case assignment, respectively called improper agreement (Keine 2020) and improper case (Poole 2023). In this vein, I shall refer to the illicit structure in (196) as an *improper mixed extended projection*.

Before discussing the constraint on mixed extended projections, it is useful to consider how Williams (2003) accounts for the impossibility of improper movement under a timing-based account of clausal embedding. Williams proposes the Level Embedding Conjecture (LEC), a constraint on the timing of clausal embedding, which indirectly relates to the size of the embedded clause. A version of the LEC as phrased in Poole 2023 is given in (200).

- (200) *Level Embedding Conjecture (LEC)*
An XP can only be embedded in a structure that is built up to an XP.

The basic idea is that a clause of size XP can only be embedded within a clause of size XP or larger according to an assumed clausal functional sequence (see, e.g., Starke 2001). Under the LEC, both the matrix and embedded clauses are simultaneously built up in separate workspaces during the course of the derivation, which can be thought of as comprising multiple stages along what Williams calls a “F(unctional)-clock”. For concreteness, I will follow Poole 2023 in adopting the use of a placeholder (PH) device and a substitution operation as a means to formalise how embedding works under the Williams Cycle. Additionally, the placeholder contains a “pointer” in the sense of Williams 2011 that specifies the construction that will be substituted for the placeholder; I will represent the pointer as a subscript to PH, e.g. PH_{TP1} . Consider first the derivation of a licit structure $[_{\text{CP}} \textit{Alex believes } [_{\text{TP}} \textit{Sam to be sick}]]$ under Level Embedding (vertical dashed lines are used to separate workspaces):

(201) Ruled in by the LEC: $[_{\text{CP}} \dots [_{\text{TP}} \dots]]$

	F-clock	Matrix root clause	Embedded clause 1
a.	F_{VP}	V_{M}	V_1
b.	$\text{F}_{\text{VP}(\text{insert PH})}$	$[\text{V}_{\text{M}} \text{PH}_{\text{TP1}}]$	$[\text{V}_1]$
c.	F_{TP}	$[\text{T}_{\text{M}} [\text{V}_{\text{M}} \text{PH}_{\text{TP1}}]]$	$[\text{T}_1 [\text{V}_1]]$
d.	$\text{F}_{\text{TP}(\text{substitution})}$	$[\text{T}_{\text{M}} [\text{V}_{\text{M}} [\text{T}_1 [\text{V}_1]]]]$	
e.	F_{CP}	$[\text{C}_{\text{M}} [\text{T}_{\text{M}} [\text{V}_{\text{M}} [\text{T}_1 [\text{V}_1]]]]]$	

In (201), the matrix clause and the embedded clause are simultaneously built up in separate workspaces, starting from the VP level (201a). At F_{VP} (201b), the matrix verb is merged with a placeholder PH_{TP1} that specifies TP1 as the construction it will be substituted for at the TP stage. At F_{TP} , both clauses are built up to the TP level (201c), following which the embedded TP1 is embedded into the matrix clause (201d). At F_{CP} (201e), the matrix clause, now containing the embedded TP1, is built up to the CP level.

Next, consider an illicit structure where a CP is embedded within a matrix root TP, e.g. $*[_{\text{TP}} \textit{Alex to believe } [_{\text{CP}} \textit{that Sam is sick}]]$. This is ruled out by the LEC as there

is no grammatical derivation in which the embedded clause but not the matrix root clause has been built up to the CP level. However, such a structure is possible if the matrix TP is contained within a root CP, as in $[_{CP} \text{Taylor wanted } [_{TP} \text{Alex to believe } [_{CP} \text{that Sam is sick }]]]$. Consider (202):

(202) Ruled in by the LEC: $[_{CP} \dots [_{TP} \dots [_{CP} \dots]]]$

	F-clock	Matrix root clause	Emb. clause 1	Emb. clause 2
a.	F_{VP}	V_M	V_1	V_2
b.	$F_{VP(INS. PH)}$	$[V_M PH_{TP1}]$	$[V_1 PH_{CP2}]$	$[V_2]$
c.	F_{TP}	$[T_M [V_M PH_{TP1}]]$	$[T_1 [V_1 PH_{CP2}]]$	$[T_2 [V_2]]$
d.	$F_{TP(sub.)}$	$[T_M [V_M [T_1 [V_1 PH_{CP2}]]]]$		$[T_2 [V_2]]$
e.	F_{CP}	$[C_M [T_M [V_M [T_1 [V_1 PH_{CP2}]]]]]$		$[C_2 [T_2 [V_2]]]$
f.	$F_{CP(sub.)}$	$[C_M [T_M [V_M [T_1 [V_1 [C_2 [T_2 [V_2]]]]]]]]]$		

In (202), the middle TP contains a placeholder that will be substituted with the most embedded CP. At F_{TP} (202d), the middle TP is embedded into the root clause. The most embedded clause continues to be built in parallel with the root clause up to CP at F_{CP} (202e). Finally, the most embedded clause is embedded into the matrix root clause, as shown in (202f). Essentially, a CP can be embedded into a TP if, at the time of embedding, the TP is already in the matrix root clause. Thus, the LEC does not block any specific type of embedding, e.g. $TP > CP$, but dictates the *timing* of the embedding (which is indirectly related to the size of the embedding).

Going back to NCs, I assume that some form of parallelism between the CP and DP domains obtains at a more general or abstract level, e.g. the categorial parallelism posited by Hiraiwa (2005) or the functional parallelism under Wiltschko's (2014) Universal Spine Hypothesis. I adopt Svenonius's (2021) idea that the parallelism between the CP and DP domains can be mapped onto a HoD. For concreteness, I propose that Khalkha possesses the HoD in (203), where d stands for an abstract domain to which a particular clausal and

nominal projection is mapped. Note that some clausal projections (e.g. VoiceP, AspP) are not represented in (203) but may be mapped to other nominal projections (e.g. NumP) under a more comprehensive conception of the HoD in Khalkha.

(203) HoD in Khalkha

	d_1	d_2	d_3	d_4
Clausal	V	v	T	C
Nominal	N	n	Poss	D

Once we accept the idea of a HoD like (203), we predict that not only is nominal structure building subject to the LEC, but also that any interaction between clausal and nominal layers will be subject to the LEC. For instance, we would predict that a clausal layer may only be embedded under a nominal layer belonging to a parallel or higher domain, but not under a nominal layer belonging to a lower domain.

Williams (2011, 2013) also further differentiates nominal embedding from clausal embedding. While clausal embedding occurs through Level Embedding, nominal embedding operates via Co-generation, where a verbal category is merged with a placeholder to be replaced by a nominal construction. Notably, clausal embedding takes place once the embedded clause reaches the size required by the embedding predicate, whereas nominal embedding occurs *only at the end of the derivation*, e.g., at F_{CP} . Following Williams, I assume that once a structure is embedded (i.e. substituted for its placeholder), it can no longer participate in structure-building operations, as only top-level elements in a single workspace (i.e. the outermost nominal or clausal layers) are eligible for further derivation (Williams 2011, p.164). Additionally, at F_{VP} , I assume that NP is not constructed within the nominal layer of Khalkha NCs (as Khalkha NCs are not N-CP structures); instead, a placeholder is inserted with a pointer indicating the size of the clausal structure to be embedded under the nominal layer.

With this background in mind, let's consider a licit derivation of an embedded nominalised clause, a finite NC in Khalkha with the structure $[_{CP} \dots [_{DP} \dots [_{CP} \dots]]]$:

(204) Ruled in by Co-generation and LEC: $[_{CP} \dots [_{DP} \dots [_{CP} \dots]]]$

	F-clock	Matrix root clause	Embedded NC	
			Nominal layer 1	Clause 2
a.	F_{VP}	V_M		V_2
b.	$F_{VP(INS. PH)}$	$[V_M PH_{DP1}]$	PH_{CP2}	V_2
c.	F_{vP}	$[v_M [V_M PH_{DP1}]]$	$[n_1 PH_{CP2}]$	$[v_2 V_2]$
d.	F_{TP}	$[T_M [v_M [V_M PH_{DP1}]]]$	$[POSS_1 [n_1 PH_{CP2}]]$	$[T_2 [v_2 V_2]]$
e.	F_{CP}	$[C_M [T_M [v_M [V_M PH_{DP1}]]]]$	$[D_1 [POSS_1 [n_1 PH_{CP2}]]]$	$[C_2 [T_2 [v_2 V_2]]]$
f.	$F_{CP(sub.)}$	$[C_M [T_M [v_M [V_M PH_{DP1}]]]]$	$[D_1 [POSS_1 [n_1 [C_2 [T_2 [v_2 V_2]]]]]]$	
g.	$F_{CP(sub.)}$	$[C_M [T_M [v_M [V_M [D_1 [POSS_1 [n_1 [C_2 [T_2 [v_2 V_2]]]]]]]]]$		

At F_{VP} (204a), both the matrix root clause and the most embedded clause (Clause 2) are built up to VP; NP is not built within the nominal layer. Next, as shown in (204b), a placeholder PH_{DP1} is inserted into the matrix root VP, while another placeholder PH_{CP2} is inserted into the nominal layer of the embedded NC, specifying that Clause 2 is to be embedded into the nominal layer when it has been built to CP. All three layers continue to be built up to vP/nP, TP/PossP and CP/DP at F_{vP} (204c), F_{TP} (204d), and F_{CP} (204e). After all three structures have been built up to CP or DP, two embedding operations occur. First, as shown in (204f) Clause 2 is embedded into the middle nominal layer via Level Embedding, as it has reached the size required by the embedding predicate. Second, in the final step of the derivational cycle (204g), the entire nominal layer containing Clause 2 is embedded into the matrix root clause via Co-generation.

For a nonfinite NC with the structure $[_{CP} \dots [_{DP} \dots [_{TP} \dots]]]$, the middle nominal layer contains PH_{TP2} instead of PH_{CP2} . At F_{TP} , the most embedded clause, built up to TP_2 , embeds into the middle nominal layer. At F_{CP} , the matrix root clause and the middle

nominal layer are built up to CP and DP respectively. In the final substitution step, the nominal layer, which already contains TP₂, is embedded into the matrix root clause.

Essentially, once we assume some form of parallelism between the clausal and nominal domains, the logics of clausal and nominal embedding under the Williams Cycle derive the revised HoD constraint (199), repeated below:

- (199) *HoD constraint on mixed extended projections (revised)*
 Given a HoD where $\alpha > \beta$, a verbal/clausal functional projection whose *maximal* projection belongs to $v_\alpha P$ cannot be dominated by a nominal functional projection whose *maximal* projection belongs to $n_\beta P$.

I illustrate how (199) can be derived via the derivation of a SNC under the Williams Cycle. (205) shows a converbial adjunct embedded in a matrix root clause, which has the structure [CP ... [PossP ... [TP ...]]]. Since converbial adjuncts adjoin to vP, I propose that the placeholder for converbial adjuncts – PH_{PossP} – is inserted at F_{vP}.

- (205) Ruled in by Co-generation and LEC: [CP ... [PossP ... [TP ...]]]
 Ruled out by LEC: [CP ... [PossP ... [CP ...]]]

F-clock	Matrix root clause	Embedded NC	
		Nominal layer 1	Clause 2
a. F _{VP}	V _M		V ₂
b. F _{VP(ins. PH)}	V _M	PH _{TP2}	V ₂
c. F _{vP}	[V _M V _M]	[n ₁ PH _{TP2}]	[v ₂ V ₂]
d. F _{vP(ins. PH)}	[PH _{PossP1} [V _M V _M]]	[n ₁ PH _{TP2}]	[v ₂ V ₂]
e. F _{TP}	[T _M [PH _{PossP1} [V _M V _M]]]	[Poss ₁ [n ₁ PH _{TP2}]]	[T ₂ [v ₂ V ₂]]
f. F _{TP(sub.)}	[T _M [PH _{PossP1} [V _M V _M]]]	[Poss ₁ [n ₁ [T ₂ [v ₂ V ₂]]]]	
g. F _{CP}	[C _M [T _M [PH _{PossP1} [V _M V _M]]]]	[Poss ₁ [n ₁ [T ₂ [v ₂ V ₂]]]]	
		↑ <i>CP cannot be built</i>	
h. F _{CP(sub.)}	[C _M [T _M [Poss ₁ [n ₁ [T ₂ [v ₂ V ₂]]]] [V _M V _M]]]		

At F_{VP} (205a), VP is built in both the matrix root clause and Clause 2. Then, PH_{TP2} is inserted into the middle nominal layer (205b), indicating that Clause 2 will be embedded under the nominal layer once it reaches the size of TP.

At F_{vP} (205c), vP is built in the matrix root clause and Clause 2, and nP is built in the middle nominal layer. Next, PH_{PossP1} is inserted into the matrix root clause (205d), specifying that $PossP_1$ will embed into the matrix root clause at the end of the derivation.

At F_{TP} (205e), TP is constructed in both the matrix root clause and Clause 2, while $PossP$ is built in the middle nominal layer. At this stage, as Clause 2 has reached the required size, it substitutes for PH_{TP2} in the middle nominal layer via Level Embedding. This is shown in (205f).

At F_{CP} (205g), only the CP of the matrix root clause is constructed. Finally, at (205h), $PossP_1$ is embedded into the matrix root clause.

Consider what it would take to generate an improper mixed extended projection where the middle nominal layer – $PossP$ – contains a CP-sized clause. One possibility is that CP may be built above TP after the most embedded clause – Clause 2 – is embedded into the nominal layer. However, as shown in (205g), this is ruled out as only top-level structures within a single workspace can undergo further derivational steps.

Another possibility is to delay the substitution of the clause into the nominal layer until after CP has been built at F_{CP} . However, this would run afoul of the LEC, as shown in (206g). Essentially, at the point where the most embedded clause CP_2 is to be embedded into the middle nominal layer (206g), the LEC dictates that it must be of an equal size to CP (i.e. DP). If the nominal layer were already embedded into the matrix root clause, which has been built up to CP, this would satisfy the LEC. However, this is ruled out by Co-generation: because nominals are only embedded at the end of the derivation, the size of the matrix root clause is not relevant at (206g). Since the nominal layer has only been built up to $PossP$ at (206g), embedding CP_2 into $PossP$ violates LEC and is ruled out.

(206) Ruled out by LEC : $[_{CP} \dots [_{PossP} \dots [_{CP} \dots]]]$

F-clock	Matrix root clause	Embedded NC	
		Nominal layer 1	Clause 2
a. F_{VP}	V_M		V_2
b. $F_{VP(Ins. PH)}$	V_M	PH_{CP2}	V_2
c. F_{vP}	$[v_M V_M]$	$[n_1 PH_{CP2}]$	$[v_2 V_2]$
d. $F_{vP(Ins. PH)}$	$[PH_{PossP1} [v_M V_M]]$	$[n_1 PH_{CP2}]$	$[v_2 V_2]$
e. F_{TP}	$[T_M [PH_{PossP1} [v_M V_M]]]$	$[Poss_1 [n_1 PH_{CP2}]]$	$[T_2 [v_2 V_2]]$
f. F_{CP}	$[C_M [T_M [PH_{PossP1} [v_M V_M]]]]$	$[Poss_1 [n_1 PH_{CP2}]]$	$[C_2 [T_2 [v_2 V_2]]]$
g. $F_{CP(sub.)}$	$[C_M [T_M [PH_{PossP1} [v_M V_M]]]]$	$[Poss_1 [n_1 PH_{CP2}]]$	$[C_2 [T_2 [v_2 V_2]]]$

$\uparrow$ X $\downarrow$
LEC violation

In addition to improper mixed extended projections of the type illustrated above, embedding under the Williams Cycle also rules out *any* mixed extended projection where the clausal structure's maximal projection belongs to a higher abstract domain than the nominal structure's maximal projection, predicted by (199). For example, in a language with a HoD like Khalkha, a CP can be dominated by a nominal superstructure only if the superstructure contains DP, a TP can be dominated by a nominal superstructure only if the superstructure contains PossP, and so forth. The resulting typology of possible and impossible nominalised structures is illustrated in Table 3.7.

Table 3.7: (Im)possible nominalised structures

Nominal structure	Clausal structure			
	CP	TP	vP	VP
nP	✗	✗	✓	✓
PossP > nP	✗	✓	✓	✓
DP > PossP > nP	✓	✓	✓	✓

3.6.3 Summary

To summarise, in this section, I have discussed a gap in the empirical landscape of Khalkha NCs: the absence of a structure in which a full CP is nominalised by reduced PossP. I have shown that this unattested structure, which I term an improper mixed extended projection, basically falls out under Williams’ (2003; 2011; 2013) ideas about clausal and nominal embedding once we assume some form of parallelism between the clausal and nominal domain (e.g. Wiltschko 2014). The present account also derives a revised version of Svenonius’s (2021) HoD constraint on mixed extended projections. Finally, the present account predicts that there should be no nominalised structures in which the maximal projection within the nominal exterior belongs to a lower level in a given HoD than the maximal projection of the clausal structure being nominalised.

However, recall from §3.4.1 that Nasu et al. (2024) argue that Japanese contains NCs in which a reduced nP lacking a DP layer dominates a full CP. This is shown in (176), repeated below.

(176) Japanese

- a. John-wa [(***ano**) nani-o Mary-ga katta ka]-o {tazuneta/touta}.
 J.-TOP (*that) what-ACC M. bought Q -ACC {asked/questioned}
 ‘John {asked/questioned} what Mary had bought.’
- b. Kyoo-no kaigi-de [**ano** dooyatte kakaku-o osaeru ka]-ga
 today-GEN meeting-IN that how price-ACC control Q -NOM
 hanasiaw-are-ta.
 discuss-PASS-PST
 ‘That how we control the price was discussed in today’s meeting’
 (Nasu et al. 2024, p.13, exx.28-29)

Both embedded clauses (176a) and (176b) contain the question particle *ka*, suggesting that they contain a full CP structure. Both embedded clauses are also marked with case, suggesting that they are nominalised. Nasu et al. (2024) argue that the anaphoric demonstrative *ano* ‘that’ occupies Spec,DP in Japanese. Since (176a) may not contain *ano* but (176b) may do so, the authors argue that the NC in (176a) involves a full CP dominated by a nP exte-

rior without a DP layer, while the NC in (176b) involves a full CP dominated by a nominal exterior containing nP and DP.

However, the data presented by Nasu et al. (2024) also exhibit some unexplained properties that warrant further investigation. First, the presence of case marking on the embedded clause in (176a) suggests that there may be additional nominal functional structure above nP within the nominal superstructure above the clause. Second, for clauses that may contain the demonstrative *ano*, the authors only show subject clauses marked with nominative case *-ga*, e.g. (176b); for clauses that may not contain the demonstrative *ano*, the authors only show complement clauses marked with accusative case *-o*, e.g. (176a). Given that subject clauses are commonly noted to contain more nominal structure than complement clauses cross-linguistically (see, e.g., Davies and Dubinsky 1998, 2001, 2009; Hartman 2012), it may not be surprising that subject clauses in Japanese may contain an overt demonstrative while complement clauses may not. For example, Pietraszko (2019) notes that subject clauses in Polish obligatorily contain a demonstrative, while complement clauses may not contain one (although see (171)). Given that there may be other ways to think about the Japanese data in (176b), I will assume that the Japanese data does not constitute a counterexample to the central claim advanced in this section: that a smaller nominal layer may not embed a larger clausal layer.

3.7 Conclusion

In this chapter, I have shown that Khalkha NCs may vary in two key aspects. In terms of clausal structure, they may contain a full CP, which correlates with the presence of an overt complementiser and finite tense morphology, or they may lack CP structure, which correlates with the lack of an overt complementiser and a lack of finite tense morphology. In terms of nominal structure, they may be full DPs, paralleling the morphosyntactic properties of possessive nominals, or they may be reduced PossPs, thus sharing only a subset of nominal

properties with possessive nominals. In showing that converbial adjuncts contain a reduced PossP nominal structure, I have shown how the full DP-SN distinction can also be observed in the clausal domain, even though this distinction is not observed in the nominal domain of Khalkha. This chapter thus offers a case study on how to look for SNs in a language which may not straightforwardly exhibit reduced nominal structures in the nominal domain.

The question remains as to why there is an asymmetry in Khalkha as to where small nominals can and cannot be found. As seen in Russian and Tatar, small nominals can serve as arguments of verbs. As Khalkha does not seem to contain argument small nominals that are not (pseudo-)incorporated, one would have to look at non-arguments, e.g. adjuncts, which cannot be simplex nominals in the language and only appear as clause-sized constituents. This is why the full DP-SN distinction maps onto the argument-adjunct clause distinction in Khalkha. A prediction here is that a language that allows argument small nominals may also allow argument SNCs, if nominalized clauses are possible in that language to begin with. If this turns out to be correct, then it suggests that the argument-adjunct split between full DPs and SNCs in Khalkha is not due to a universal constraint pertaining to the argument vs adjunct distinction, but a mere coincidence owing to the lack of non-(pseudo-)incorporated SN arguments. It also suggests that adjunct small nominals may be found in a language where simplex nominals could function as adverbials.

Finally, I have proposed a typology of possible and impossible nominalised structures based on Williams' (2003; 2011; 2013) theory of clausal and nominal embedding, which derives a revised version of Svenonius's (2021) HoD constraint on building mixed extended projections. This proposed typology further yields two interesting predictions. First, assuming that languages that do not contain DPs in the purely nominal domain also do not contain DP-sized nominalisations, the proposed typology predicts that such languages will not be able to nominalise CPs. This may be true of so-called NP-languages, which have been argued to lack DP-sized nominals (Bošković 2008, et seq.). This requires the further

assumption that CPs in NP-languages completely lack a parallel in the nominal domain.¹⁹ Second, the proposed typology predicts that if a language contains only CP-sized embedded clauses, wholly lacking reduced embedded clauses, in such a language we will not find SNCs.

Building on the findings in this chapter, in the next chapter, I show how they can be useful for understanding the locality constraints on two cross-clausal operations in Khalkha: subject extraction and binding.

¹⁹Alternatively, it may be the case that the highest non-DP nominal projection serves as the parallel for CP in a NP-language. I leave this issue for future research.

Chapter 4

The position of embedded subjects

4.1 Introduction

The previous chapter has established that four embedded clause types in Khalkha – finite bare CPs, finite NCs, nonfinite NCs, and converbial adjuncts – can each be uniquely understood in terms of finiteness (i.e. the presence of CP and the type of T^0), the presence of nominalisation, and the size of the nominal superstructure above the clause (i.e. whether a full DP or a reduced PossP lacking DP is projected). Additionally, it was shown that subjects of the four embedded clause types may be differentially case-marked: while accusative and unmarked subjects can be found in all clause types, only finite and nonfinite NCs may contain genitive subjects. These findings are recapitulated in Table 4.1.

Table 4.1: Structure of four embedded clause types in Khalkha

Clause type	Clausal structure		Nominal structure		Embedded subject		
	Type of T^0	CP	PossP	DP	ACC	GEN	UNM
Finite bare CP	Nondefective	✓	✗	✗	✓	✗	✓
Finite NC	Nondefective	✓	✓	✓	✓	✓	✓
Nonfinite NC	Defective	✗	✓	✓	✓	✓	✓
Converbial adjunct	Defective	✗	✓	✗	✓	✗	✓

This chapter takes a closer look at the subjects of the four embedded clause types and establishes a link between the cases they bear (or lack) and their syntactic positions. Building on previous work by Fong 2019; Aravind 2021; Gong 2022; Peters 2024, I propose that accusative subjects occupy the highest phase edge of the embedded clause – Spec,CP in finite bare CPs, Spec,DP in finite and nonfinite NCs, and Spec,nP in converbial adjuncts; genitive subjects occupy Spec,nP in finite and nonfinite NCs; and unmarked (i.e. “nominative”) subjects occupy Spec,TP. My argument rests on a constellation of syntactic diagnostics: i) the distribution of subject case across the four clause types, ii) the extractability of embedded subjects, and iii) the visibility of embedded subjects for binding by the matrix subject. As I show by the end of this chapter, the structure of each clause type and the positions of subjects within them are schematised in (207): accusative subjects are located in the highest phase edge of the embedded (nominalised) clause, genitive subjects are located in Spec,nP within a DP-sized superstructure found in finite and nonfinite NCs, and case-unmarked subjects are located in Spec,TP.¹

(207) *Positions of subjects in four embedded clause types in Khalkha*

- a. *Finite bare CP*
 $[_{CP} \text{Subj}(\overline{\text{ACC}}) \text{ } [_{TP} \text{Subj}(\overline{\text{UNM}}) \dots]]$
- b. *Finite NC*
 $[_{DP} \text{Subj}(\overline{\text{ACC}}) \text{ } [_{POSSP} [_{nP} \text{Subj}(\overline{\text{GEN}}) \text{ } [_{CP} [_{TP} \text{Subj}(\overline{\text{UNM}}) \dots]]]]]]$
- c. *Nonfinite NC*
 $[_{DP} \text{Subj}(\overline{\text{ACC}}) \text{ } [_{POSSP} [_{nP} \text{Subj}(\overline{\text{GEN}}) \text{ } [_{TP} \text{Subj}(\overline{\text{UNM}}) \dots]]]]$
- d. *Converbial adjunct*
 (no DP) $[_{POSSP} [_{nP} \text{Subj}(\overline{\text{ACC}}) \text{ } [_{TP} \text{Subj}(\overline{\text{UNM}}) \dots]]]]$

¹In principle, accusative subjects in converbial adjuncts may also be located in Spec,PossP; genitive subjects in finite and nonfinite NCs may also be located in Spec,PossP; and case-unmarked subjects in finite NCs may also be located in Spec,CP. To my knowledge, there is no clear diagnostic that may be used to distinguish between the abovementioned possibilities and the positions I have proposed in (207). I leave this to future work. In this dissertation, I shall abstract away from the abovementioned alternative subject positions and focus on the positions identified in (207).

Additionally, this chapter demonstrates that long-distance extraction of the embedded subject and binding of the embedded subject by the matrix subject are subject to different locality domains.² Specifically, I show that i) extraction into the matrix clause may only target accusative subjects, and ii) binding by the matrix subject may target both accusative and genitive subjects but not case-unmarked subjects. The contrast suggests that in Khalkha long-distance extraction is subject to phase-locality, while nominal domains are transparent to binding irrespective of the presence of nominal phases. More broadly, the contrast implies that within a language, the locality domains of different syntactic operations such as extraction and binding may not be straightforwardly reduced to a single domain such as a standard phasal spellout domain.

This chapter is organised as follows. In §4.2, I discuss some evidence that supports the idea that (i) CP, vP, DP, and nP constitute phases in Khalkha, and that (ii) accusative and genitive subjects are not proleptic. I also review the distribution of subject case across the four embedded clause types, which will provide some preliminary ideas about the position of embedded subjects in each clause. In §4.3, I investigate the extractability of subjects in each clause type, showing that accusative subjects, but not genitive and case-unmarked subjects, are eligible for extraction into the matrix clause. In §4.4, I turn to binding of embedded subjects across the four clause types, demonstrating that accusative and genitive, but not case-unmarked, subjects are accessible to the matrix subject for binding. In §4.5, I discuss a staggered spellout approach to modelling the non-isomorphism of extraction and binding domains in Khalkha. §4.6 concludes.

²An interesting parallel exists in the literature on islands, in which it has long been noted that binding dependencies are typically not sensitive to island effects (see, e.g., Sprouse and Villata 2021).

4.2 Theoretical assumptions

In this section, I outline my theoretical assumptions about phases and the positions of embedded subjects. Specifically, I argue that CP, vP, DP, and nP constitute phases in Khalkha.³ I also argue that accusative subjects and genitive subjects are not proleptic (i.e. they are not merged as a direct object in the matrix clause).

4.2.1 On phases in Khalkha

In current Minimalist syntax (Chomsky 2000 et seq.), it is commonly argued that syntactic structure is built in chunks called *phases*. Once a phase is complete, only a portion of the phase remains accessible to further syntactic operations, while the rest is spelled-out to the phonological (PF) and semantic (LF) interfaces and is no longer accessible to the narrow syntax. This is known as the *Phase Impenetrability Condition* (PIC) (Chomsky 2000, 2001). I adopt the following definition of the PIC in (208) (where H refers to a phase head):⁴

- (208) *Phase Impenetrability Condition* (Chomsky 2001, p.13)
The domain of H is not accessible to operations outside HP; only H and its edge are accessible to such operations.

(208) states that once the domain of a phase head H^0 is complete, its complement is spelled-out (i.e. sent to the PF and LF interfaces), whereas H^0 and its specifier, Spec,HP, remain accessible to further syntactic operations. Thus, the only way for a constituent in HP to be accessible to operations outside HP is to move to Spec,HP. This also implies that movement to Spec,HP precedes completion of the HP phase.

³Given my proposal in Chapter 2 that the extended verbal domain in Khalkha contains distinct projections VoiceP and vP, in this section I am specifically arguing that vP but not VoiceP constitutes a phase (see, e.g., van Urk 2015; Hedding and Yuan 2024, 2025) in Khalkha.

⁴(208) is also known as the Strong PIC, in contrast to the Weak PIC, also proposed by Chomsky (2001), in which the domain of H except H and its specifier is not accessible to operations at ZP, where Z is the next higher phase head. Since Chomsky 2000, 2001, other variants of the PIC have been proposed, including the idea that every phrase is a phase (e.g. Müller 2004) or that the phasehood of a head depends on its syntactic context (e.g. Bošković 2014).

Since Chomsky 2000, 2001, it is commonly assumed that CP and vP are phases. Additionally, many works have argued that DP is a phase as well; see, e.g., Svenonius 2004; Matushansky 2005; Bošković 2005, 2008, 2012; Aravind 2021. Finally, since Marantz 2001, it is also commonly assumed that categorising heads such as n^0 are phasal. In this section, I discuss movement-related evidence which support CP, vP and DP being phases in Khalkha.⁵ I will also assume that nP is a phase in Khalkha, although key evidence for its phasehood will only be discussed in Chapter 5.

Starting with the clausal domain in Khalkha, Fong (2019) shows that only accusative subjects may be displaced from the embedded clause to a position within the matrix clause (209). This observation is also independently reported by Guntsetseg (2016), as shown in (210).⁶

- (209) ACC *but not* UNM *subject may be extracted from finite bare CP*
 Bat **nohoi-g**/***nohoi**₁ changaar [₁ gaihal-tai gej] hel-sen.
 B. dog-ACC/*dog loudly wonder-COM C say-PST
 ‘Bat said loudly that dogs are wonderful.’ (Fong 2019, p.3, ex.5)

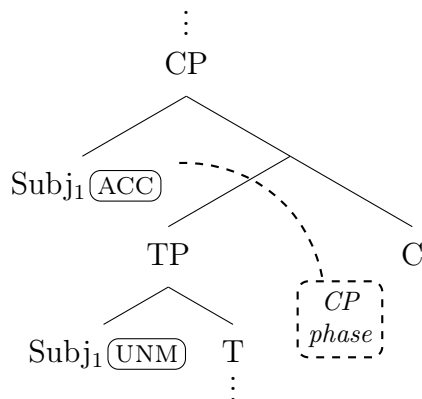
- (210) ACC *but not* UNM *subject may be extracted from finite bare CP*
 a. **Tuyaa-g**₁ Dorj [₁ ene nom-iig unsh-san gej] hel-sen.
 T.-ACC D. this book-ACC read-PST C say-PST
 ‘Dorj said that Tuyaa read this book.’
 b. **Tuyaa** Dorj ene nom-iig unsh-san gej hel-sen.
 T. D. this book-ACC read-PST C say-PST
 Means: ‘Tuyaa said that Dorj read this book.’
 Does not mean: ‘Dorj said that Tuyaa read this book.’
 (Guntsetseg 2016, p.33, ex.72)

⁵While previous authors have also used binding to diagnose the phasehood of CP, vP, DP, I show later in this chapter that binding does not actually diagnose phase boundaries.

⁶To clarify, Guntsetseg (2016) is agnostic as to the nature of subject extraction, while Fong (2019) analyses data such as (209) as hyperraising into the matrix clause. I shall abstract away from any differences across these constructions, including *bol*-topicalisation (Aravind 2021) which I discuss below, since they all show movement of the embedded subject across a clause boundary.

The contrast between (209) and (210) can be accounted for under the view that CP is a phase,⁷ and the embedded subject is assigned accusative case in Spec,CP. (209) and (210) show that the embedded subject cannot be extracted into the matrix clause without first having moved to embedded Spec,CP, where it is assigned accusative case. This also implies that the apparent long-distance movement of the embedded subject proceeds successive-cyclically through embedded Spec,CP (i.e. via a series of local movement steps). Thus, building on Fong 2019, I propose that accusative and case-unmarked subjects in a finite bare CP are separated by a CP phase boundary, as schematised in (211).

(211) *CP phase and positions of ACC and UNM subjects in finite bare CP*



A similar contrast can be found between accusative-marked and case-unmarked direct objects in matrix root clauses. If a direct object precedes the subject, it must bear accusative case, as shown in (212).

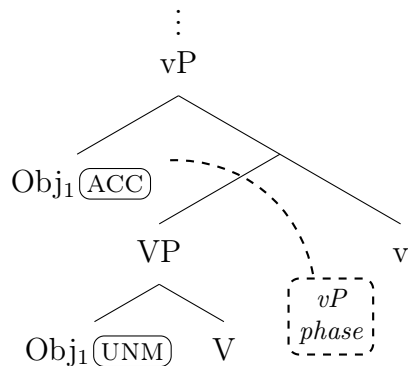
(212) *Object must be ACC if it precedes subject*

- a. **Muur-iig₁** bi ₁ har-san.
 Cat-ACC 1SG see-PST
 'I saw a cat.'
- b. *Muur bi har-san.
 Cat 1SG see-PST
 Int. 'I saw a cat.'

⁷Fong (2019) assumes the Weak PIC, wherein the domain of a phase head is not spelled-out until the next higher phase head is merged. Presumably, this is because Fong treats v^0 as a phase head, but requires the embedded subject in Spec,CP to remain visible to the matrix subject in Spec,TP for dependent accusative case assignment. Under the Strong PIC, this issue may be circumvented if we assume that embedded clauses obligatorily shift to Spec,vP, as I propose in Chapter 5 (see also Gong 2022, p.143).

This can be accounted for under the assumption that a phasal boundary exists in Khalkha's extended verbal domain. I propose that the relevant phase is vP (instead of VoiceP), following the view that categorising heads are phasal (Marantz 2001). (212) shows that a direct object must first shift to Spec,vP, where it is assigned accusative case, before it can move to a position above the matrix subject in Spec,TP. The position of a direct object in a matrix root clause is shown in (213).

(213) *vP phase and positions of ACC and UNM objects in root clause*



Turning to the nominal domain, Aravind (2021) also argues that DP constitutes a phasal domain in Khalkha. She demonstrates that if an embedded subject is displaced from a nonfinite complement NC into the matrix clause, it obligatorily bears accusative case. This is shown in (214) with *bol*-topicalisation, which Aravind shows displays properties of A'-movement.

(214) *ACC but not GEN/UNM subject may undergo bol-topicalisation from nonfinite NC*
Bat-iig/*Bat-iin/*Bat₁ bol bi [₁ ene nom-iig unsh-sn]-iig ol-j
 B.-ACC/*B.-GEN/*B. TOP 1SG this book-ACC read-PFV -ACC find-CVB
 med-sen.
 know-PST
 'As for Bat, I found out that he read this book.' (Aravind 2021, p.386, ex.20)

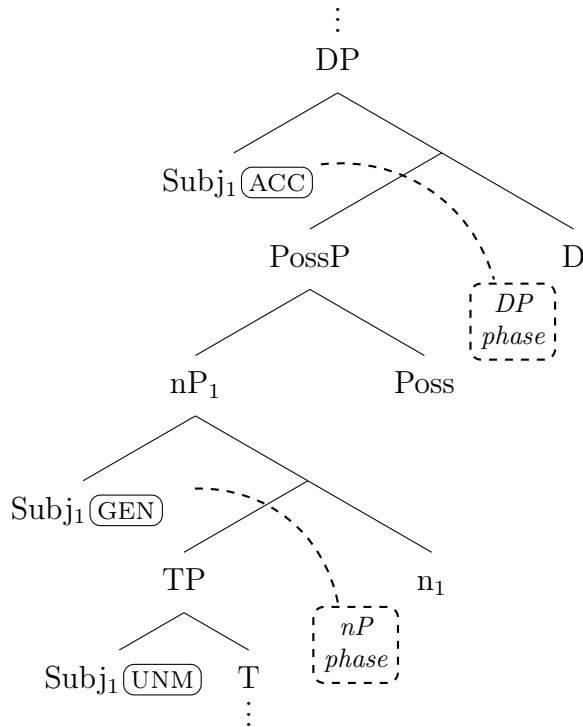
(214) can be accounted for under the assumption that DP constitutes a phasal domain, and that the embedded subject in nonfinite NCs is assigned accusative case in Spec,DP. (214) shows that the embedded subject must also move to Spec,DP, where it is assigned accusative case, before it may move to a position in the matrix clause.

In Chapters 2 and 3, I showed that genitive subjects are found in finite and nonfinite NCs but not finite bare CPs and converbial adjuncts. I linked the availability of genitive subjects to the presence of a DP layer in the embedded clause, suggesting that D^0 is responsible for genitive case assignment – a point I return to in Chapter 5. Under the broader view that categorising heads are phasal (Marantz 2001), I propose that nP also constitutes a phase within the extended nominal domain of Khalkha. Under this view, genitive subjects cannot be located in Spec,TP or Spec,CP. If they were, in a finite or nonfinite NC, they would be spelled out within the nP phase and rendered inaccessible to D^0 , which merges above nP only after the nP phase is complete. This predicts that subjects located below nP, e.g. in Spec,TP, would not be marked with any case. This also implies that genitive subjects must occupy a structurally higher position than unmarked ones, which I show in §4.4 is correct based on binding asymmetries.

Moreover, the fact that genitive subjects cannot be extracted into the matrix clause, as demonstrated by Aravind (2021) in (214), suggests that they are not located at the edge of the DP phase. Instead, their inability to undergo extraction supports the view that genitive subjects are spelled out within the DP phase, rendering them inaccessible to matrix syntactic operations such as extraction. Building on Aravind 2021, I propose the following structural distribution of embedded subjects in nonfinite NCs: accusative subjects occupy Spec,DP, genitive subjects occupy Spec,nP, and unmarked subjects occupy Spec,TP. These positions are crucially separated by phase boundaries, as schematised in (215).⁸

⁸(215) is identical to the structure proposed by Aravind (2021) for nonfinite NCs, with the exception of an additional PossP layer in my structure. Aravind (2021) also does not assume that nP constitutes a phase.

(215) *DP phase, nP phase, and positions of ACC, GEN, and UNM subjects in nonfinite NC*



To recap, this section has shown that only accusative subjects in finite bare CPs and nonfinite NCs may be extracted into the matrix clause. I have also shown that within the same clause, a direct object may only precede the subject if it is marked with accusative case. Based on these pieces of evidence, I have argued that CP, vP, and DP constitute phases in Khalkha. Further, I have argued that accusative subjects occupy the highest phase edge of the embedded clause – Spec,CP in finite bare CPs and Spec,DP in nonfinite NCs, genitive subjects are located in Spec,nP in nonfinite NCs, and unmarked subjects remain in Spec,TP. I have also proposed that nP is a phase, under the broader view that categorising heads are phasal (Marantz 2001); I provide further arguments for the phasehood of nP in Chapter 5.

4.2.2 Against a proleptic analysis of accusative and genitive subjects

Next, I argue that accusative subjects are not proleptic objects, that is, they are not object arguments base-generated in the matrix clause and coreferential with a null subject in the embedded clause; see Salzmann 2017 for an overview of theoretical treatments of prolepsis. Under such a view, an accusative subject does not originate from a lower position within the embedded clause (e.g. Spec,VoiceP for agentive subjects), from which it moves to a higher position. Instead, the accusative-marked argument is base-generated as an object in the matrix clause or at the edge of the embedded clause, where it binds a null *pro* within the embedded clause (Lohninger et al. 2022; see also Fong 2019, p.5). This is schematised in (216).

- (216) *Proleptic analysis of accusative subjects*
 [CP₂ ... [TP₂ Matrix.Subj ... Proleptic.Obj;ACC] ... [CP₁/DP ... *pro*_j ...]]

Fong (2019) extensively argues against a proleptic analysis of accusative subjects in finite bare CPs (see also Gong 2023a). In this subsection, I extend two of her main arguments to include accusative subjects in finite NCs, nonfinite NCs, and converbial adjuncts. I show that i) accusative subjects preserve their idiomatic meanings in subject-containing idioms, and ii) accusative negative polarity item (NPI) subjects can be licensed by embedded negation. Crucially, these two tests show that accusative subjects behave like unmarked ones, which are unambiguously in the embedded clause.

First, under the assumption that idiomatic meanings require constituency in syntax, Fong notes that the availability of the idiomatic reading with an accusative subject in (217) provides evidence that such subjects are not proleptic.

- (217) Dorj changaar **Bat-iin nüd-iig** oroi deer=ee gar-san ge*j* hel-sen.
 D. loudly B.-GEN eye-ACC top on=RP climb-PST C say-PST
 ‘Dorj said loudly that Bat was very surprised.’
 Lit. ‘Dorj said loudly that Bat’s eyes climbed on top of themselves.’
 (Fong 2019, p.5, ex.11)

The second argument comes from NPI licensing. In Khalkha, one way of forming NPIs involves using the focus particle *ch*, e.g., *hen ch* ‘anyone/no one’ and *neg ch hün* ‘not one person/no one’. In matrix root clauses, these NPI arguments must be licensed by negation, as shown in (218).

- (218) a. Dulmaa **hen-iig ch** har-aa-**güi**.
 D. who-ACC FOC see-IPFV-NEG
 ‘Dulmaa did not see anyone.’
 b. *Dulmaa **hen-iig ch** har-san.
 D. who-ACC FOC see-PST
 Int. ‘Dulmaa did not see anyone.’
 c. **Hen ch** Dulmaa-g har-aa-**güi**.
 Who FOC D.-ACC see-IPFV-NEG
 ‘Nobody saw Dulmaa.’
 d. ***Hen ch** Dulmaa-g har-san.
 Who FOC D.-ACC see-PST
 Int. ‘Nobody saw Dulmaa.’ (Fong 2019, p.7, exx.21-22)

In finite bare CPs, Fong shows that NPI objects must be licensed by embedded negation (219a) and not matrix negation (219b).

- (219) a. Naraa [Dulmaa-g/Dulmaa **hen-iig ch** har-aa-**güi** gej] hel-sen.
 N. D.-ACC/D. who-ACC FOC see-IPFV-NEG C say-PST
 ‘Naraa said that Dulmaa did not see anyone.’
 b. *Naraa [Dulmaa-g/Dulmaa **hen-iig ch** har-san gej] hel-ee-**güi**.
 N. D.-ACC/D. who-ACC FOC see-PST C say-IPFV-NEG
 Int. ‘Naraa said that Dulmaa did not see anyone.’ (Fong 2019, p.8, ex.23)

Fong further shows that NPI embedded subjects bearing accusative case or no case marking must also be licensed by embedded negation, as shown in (220). Thus, she argues that NPI arguments must be licensed by clausemate negation. She explains that if accusative subjects were proleptic arguments that originate in the matrix clause, this would incorrectly predict (220b) should be grammatical.

- (220) a. Naraa [**hen-iig/hen ch** ir-ee-**güi** gej] hel-sen.
 N. who-ACC/who FOC come-IPFV-NEG C say-PST
 ‘Naraa said that nobody came.’

- b. *Naraa [**hen-iig/hen ch** ir-sen gej] hel-ee-**güi**.
 N. who-ACC/who FOC come-PST C say-IPFV-NEG
 Int. ‘Naraa said that nobody came.’ (Fong 2019, p.8, ex.24)

Below, I demonstrate that Fong’s 2019 findings are consistent with my three consultants’ judgements regarding accusative subjects in finite bare CPs. I further show that accusative subjects in finite NCs, nonfinite NCs, and converbial adjuncts behave like unmarked subjects with respect to the two tests – idiom preservation and NPI licensing – which provides evidence against a proleptic analysis of accusative subjects.

For the idiomatic test, I will use a different subject-containing idiom, as shown in (221).

- (221) **Tüünii süns** zail-san.
 3SG.GEN soul run.away-PST
 ‘She/he was very scared. (Lit. Her/his soul ran away.)’

The examples below show that the idiomatic interpretation is preserved when the embedded subject bears accusative case or no case in a finite bare CP (222), finite NC (223), nonfinite NC (224), and converbial adjunct (225).⁹

- (222) *Idiomatic reading preserved with ACC/UNM subject in finite bare CP*
 Naraa [**Bat-iin süns-iig/süns** zail-san gej] hel-sen.
 N. B.-GEN soul-ACC/soul run.away-PST C say-PST
 ‘Naraa said that Bat was very scared.’

- (223) *Idiomatic reading preserved with ACC/UNM subject in finite NC*
 Naraa [**Bat-iin süns-iig/süns** zail-san gedg]-iig sons-son.
 N. B.-GEN soul-ACC/soul run.away-PST C.NML -ACC hear-PST
 ‘Naraa heard that Bat was very scared.’

- (224) *Idiomatic reading preserved with ACC/UNM subject in nonfinite NC*
 Naraa [**Bat-iin süns-iig/süns** zail-sn]-iig med-ne.
 N. B.-GEN soul-ACC/soul run.away-PFV -ACC know-NPST
 ‘Naraa knows that Bat was very scared.’

⁹My consultants note, however, that the versions of (222)-(225) containing an accusative subject, while acceptable, sound more marked than the versions containing a case-unmarked subject.

- (225) *Idiomatic reading preserved with ACC/UNM subject in converbial adjunct*
 Naraa [**Bat-iin süns-iig/süns** zail-had]
 N. B.-GEN soul-ACC/soul run.away-TEMP.CVB
 tüüniig taivsh-r-uul-san.
 3SG.ACC be.calm-CAUS-PST
 ‘When Bat was very scared, Naraa calmed him.’

To test the behaviour of embedded NPI subjects with respect to embedded vs. matrix negation, I use the NPI *neg ch hün* ‘not one person; nobody’. In matrix root clauses, *neg ch hün* must be licensed by negation, as shown in (226).

- (226) a. **Neg ch hün** Misheel-iig har-aa-güi.
 one FOC person M.-ACC see-IPFV-NEG
 ‘Not one person saw Misheel.’
 b. ***Neg ch hün** Misheel-iig har-san.
 one FOC person M.-ACC see-PST
 Int. ‘Not one person saw Misheel.’

The next four examples show that an accusative or a case-unmarked NPI subject can be licensed by embedded negation in a finite bare CP (227a), finite NC (228a), nonfinite NC (229a), and converbial adjunct (230a). The same sentences are unacceptable if there is no negation in the embedded clause or the matrix clause, as shown in the (b) examples below.¹⁰

- (227) *ACC/UNM NPI subject may be licensed by embedded negation in finite bare CP*
 a. Naraa [**neg ch hün-iig** / **neg ch hün** ene nom-iig unsh-aa-güi
 N. one FOC person-ACC / one FOC person this book-ACC read-IPFV-NEG
 gej] hel-sen.
 C say-PST
 ‘Naraa said that not one person read this book.’
 b. *Naraa [**neg ch hün-iig** / **neg ch hün** ene nom-iig unsh-san gej
 N. one FOC person-ACC / one FOC person this book-ACC read-PST C
] hel-sen.
 say-PST
 Int. ‘Naraa said that not one person read this book.’

¹⁰The previous literature contains conflicting judgements as to whether embedded NPI subjects may bear accusative case. Using the NPI *hen ch* ‘nobody’, Fong (2019, pp.7-8) finds accusative and case-unmarked NPI subjects to be possible in finite bare CPs containing clausemate negation, as shown in (220). Using the NPI *neg ch hün* ‘not one person; nobody’, Aravind (2021, p.381) finds accusative and genitive NPI subjects to be ungrammatical in nonfinite NCs containing clausemate negation. The data presented here confirms that accusative NPI subjects are possible in all embedded clause types containing clausemate negation, but that genitive NPI subjects are degraded, as I discuss below.

- (228) ACC/UNM *NPI subject may be licensed by embedded negation in finite NC*
- a. Bilgüün [**neg ch hün-iig** / **neg ch hün** ene nom-iig unsh-aa-güi
 B. one FOC person-ACC / one FOC person this book-ACC read-IPFV-NEG
 gedg]-iig sons-son.
 C.NML -ACC hear-PST
 ‘Bilguun heard that not one person read this book.’
- b. *Bilgüün [**neg ch hün-iig** / **neg ch hün** ene nom-iig unsh-san
 B. one FOC person-ACC / one FOC person this book-ACC read-PST
 gedg]-iig sons-son.
 C.NML -ACC hear-PST
 Int. ‘Bilguun heard that not one person read this book.’
- (229) ACC/UNM *NPI subject may be licensed by embedded negation in nonfinite NC*
- a. Naraa [**neg ch hün-iig** / **neg ch hün** ene nom-iig unsh-aa-güi
 N. one FOC person-ACC / one FOC person this book-ACC read-IPFV-NEG
]-g med-ne.
 -ACC know-NPST
 ‘Naraa knows not one person read this book.’
- b. *Naraa [**neg ch hün-iig** / **neg ch hün** ene nom-iig unsh-sn
 N. one FOC person-ACC / one FOC person this book-ACC read-PFV
]-iig med-ne.
 -ACC know-NPST
 Int. ‘Naraa knows not one person read this book.’
- (230) ACC/UNM *NPI subject may be licensed by embedded negation in converbial adjunct*
- a. Gerelee [**neg ch hün-iig** / **neg ch hün** har-aa-güi bai-had
 G. one FOC person-ACC / one FOC person see-IPFV-NEG COP-TEMP.CVB
] jignemeg id-chih-sen.
 cookie eat-COMPL-PST
 ‘Gerelee ate cookies when no one was watching.’
- b. *Gerelee [**neg ch hün-iig** / **neg ch hün** hara-had] jignemeg
 G. one FOC person-ACC / one FOC person see-TEMP.CVB cookie
 id-chih-sen.
 eat-COMPL-PST
 Int. ‘Gerelee ate cookies when no one was watching.’

However, in contrast to the reported judgements in Fong 2019 as seen in (220b), for my consultants, embedded NPI subjects in complement clauses may be licensed by matrix negation, as shown in (231) for a finite bare CP:

- (231) ACC/UNM NPI subject may be licensed by matrix negation in finite bare CP
 Naraa [**neg ch hün-iig** / **neg ch hün** ene nom-iig unsh-san gej]
 N. one FOC person-ACC / one FOC person this book-ACC read-PST C
 hel-ee-güi.
 say-IPFV-NEG
 ‘Naraa didn’t say that anyone read this book.’

Overall, not only do my consultants allow embedded NPI subjects to be licensed by embedded negation, which c-commands the NPI subject in its base position (Spec,VoiceP), they also allow NPI subjects to be licensed by matrix negation, which c-commands the accusative NPI subject at the edge of the embedded clause. A more accurate generalisation for my consultants appears to be that NPI arguments must be licensed by c-commanding negation, and that negation may originate from within the embedded clause or the matrix clause (cf. Fong 2019). Crucially, the fact that negation within the embedded clause can license both accusative and case-unmarked subjects provides strong evidence against a proleptic analysis, and instead supports the view that accusative subjects in Khalkha originate from a position within the embedded clause.¹¹

In sum, I have shown that accusative subjects pattern with case-unmarked ones, which are unambiguously located in the embedded clause, with respect to both idiom interpretation and NPI licensing. I conclude that a proleptic analysis of accusative subjects cannot be maintained. Instead, I will continue to analyse accusative subjects as originating from a thematic position inside the embedded clause before moving to the highest phase edge of the embedded clause where they are assigned accusative case.

While I have demonstrated above that idiom preservation and NPI-licensing provide evidence against a proleptic analysis of accusative subjects, the two tests are inconclusive

¹¹If NPI subjects can be licensed by c-commanding negation, we would expect that NPI subjects in converbial adjuncts should be able to be licensed by matrix negation. However, the examples that I have tested are judged ungrammatical, as shown in (i):

- (i) *Gerelee [**neg ch hün-iig** / **neg ch hün** hara-had] jignemeg id-ee-güi.
 G. one FOC person-ACC / one FOC person see-TEMP.CVB cookie eat-IPFV-NEG
 Int. ‘Gerelee didn’t eat cookies when no one was watching.’

I tentatively conclude that NPI subjects may be licensed by c-commanding negation only if the subjects are found in complement clauses but not in adjunct clauses. I leave this for future investigation.

with respect to genitive subjects. When a genitive subject is construed as a part of a subject-containing idiom (232)-(233) or as an NPI subject (234)-(235), my consultants report the constructions to be ungrammatical.

(232) GEN *subject ungrammatical in subject-containing idiom in finite NC*

* Naraa [**Bat-iin süns-nii** zail-san gedg]-iig sons-son.
 N. B.-GEN soul-GEN run.away-PST C.NML -ACC hear-PST
 Int. ‘Naraa heard that Bat was very scared.’

(233) GEN *subject ungrammatical in subject-containing idiom in nonfinite NC*

* Naraa [**Bat-iin süns-nii** zail-sn]-iig med-ne.
 N. B.-GEN soul-GEN run.away-PFV -ACC know-NPST
 Int. ‘Naraa knows that Bat was very scared.’

(234) GEN *NPI subject ungrammatical in finite NC with clausemate negation*

* Bilgüün [**neg ch hün-ii** ene nom-iig unsh-aa-güi gedg]-iig
 B. one FOC person-GEN this book-ACC read-IPFV-NEG C.NML -ACC
 sons-son.
 hear-PST
 Int. ‘Bilguun heard that not one person read this book.’

(235) GEN *NPI subject ungrammatical in nonfinite NC with clausemate negation*

* Naraa [**neg ch hün-ii** ene nom-iig unsh-aa-güi]-g med-ne.
 N. one FOC person-GEN this book-ACC read-IPFV-NEG -ACC know-NPST
 Int. ‘Naraa knows not one person read this book.’

I do not have a clear explanation for why genitive subjects are degraded in (232)-(235). One possibility is that genitive subjects with a genitive possessor are generally disallowed in Khalkha, which may explain the idiom interpretation facts in (232)-(233); see also the discussion surrounding (271)-(272) in §4.4.2.2. For my present purposes, I will assume that the degradedness of (232)-(235) is orthogonal to the structural position of genitive subjects. Nevertheless, it is implausible that genitive subjects are proleptic objects in the matrix clause, as object arguments are never marked with genitive case in Khalkha, as shown in (236).

- (236) Bi muur-iig/***muur-iin** har-san.
 1SG cat-ACC/*cat-GEN see-PST
 ‘I saw a cat.’

Thus, I conclude that genitive subjects also originate from the embedded clause, although this cannot be directly tested with idiom preservation and NPI-licensing.

4.2.3 Summary

In this section, I have outlined my theoretical assumptions about phases and the position of embedded subjects in Khalkha. Specifically, I have argued that CP and DP constitute phases in Khalkha, and that vP and nP may be treated as phases too under the broader view that categorising heads are phasal (Marantz 2001). I have also argued that accusative and genitive subjects originate from within the embedded clause. Discussing evidence from previous works (e.g. Guntsetseg 2016; Fong 2019; Aravind 2021; Peters 2024), I have proposed that in finite bare CPs accusative subjects occupy Spec,CP and case-unmarked subjects occupy Spec,TP, whereas in nonfinite NCs accusative subjects occupy Spec,DP, genitive subjects occupy Spec,nP, and case-unmarked subjects occupy Spec,TP. I turn next to an investigation of the extractability of subjects across the four embedded clause types under consideration in this dissertation.

4.3 Subject extraction and phase-based locality in Khalkha

This section examines the extractability of embedded subjects across the four embedded clause types under investigation in Khalkha. As discussed in the previous section, while previous authors have investigated subject extraction out of embedded finite bare CPs (Fong 2019) and nonfinite NCs (Aravind 2021), it has not, to my knowledge, been demonstrated for finite NCs and converbial adjuncts.

Anticipating the key findings of this section, I show that, except in converbial adjuncts, only accusative-marked subjects can be extracted to a position in the matrix clause, while non-accusative subjects cannot undergo such movement. I argue that subject extraction from converbial adjuncts, which results in ungrammaticality regardless of the case of the subject, is due to the fact that adjuncts are islands for movement. Overall, I argue that the extraction facts support the idea that CP is the highest phase boundary in finite bare CPs, while DP is the highest phase boundary in finite and nonfinite NCs.

4.3.1 Subject extraction from embedded clauses

Focusing on finite bare CPs, finite NCs, and nonfinite NCs, I demonstrate that accusative subjects can be displaced from within the embedded clause to matrix clause-initial position, while non-accusative subjects may not be. This is shown in (237)-(239).

(237) *ACC but not UNM subject may be extracted from finite bare CP*

- a. **Dorj-iig₁** öchigdör Naraa [margaash ₁ ir-ne gej] hel-sen.
D.-ACC yesterday N. tomorrow come-NPST C say-PST
'Yesterday Naraa said that Dorj will come tomorrow.'
- b. **Dorj** öchigdör Naraa margaash ir-ne gej hel-sen.
D. yesterday N. tomorrow come-NPST C say-PST
Means: 'Yesterday Dorj said that Naraa will come tomorrow.'
Does not mean: 'Yesterday Naraa said that Dorj will come tomorrow.'

(238) *ACC but not GEN/UNM subject may be extracted from finite NC*

- a. **Dorj-iig/*Dorj-iin₁** öchigdör Naraa [margaash ₁ ir-ne gedg]-iig
D.-ACC/*D.-GEN yesterday N. tomorrow come-NPST C.NML -ACC
sons-son.
hear-PST
'Yesterday Naraa heard that Dorj will come tomorrow.'
- b. **Dorj** öchigdör Naraa margaash ir-ne gedg-iig sons-son.
D. yesterday N. tomorrow come-NPST C.NML-ACC hear-PST
Means: 'Yesterday Dorj heard that Naraa will come tomorrow.'
Does not mean: 'Yesterday Naraa heard that Dorj will come tomorrow.'

- (239) *ACC but not GEN/UNM subject may be extracted from nonfinite NC*
- a. **Dorj-iig**/***Dorj-iin**₁ öchigdör Naraa [margaash ₁ ir-eh]-iig
D.-ACC/*D.-GEN yesterday N. tomorrow come-INF -ACC
mart-chih-san.
forget-COMPL-PST
‘Yesterday Naraa forgot that Dorj will come tomorrow.’
- b. **Dorj** öchigdör Naraa margaash ir-eh-iig mart-chih-san.
D. yesterday N. tomorrow come-INF-ACC forget-COMPL-PST
Means: ‘Yesterday Dorj forgot that Naraa will come tomorrow.’
Does not mean: ‘Yesterday Naraa forgot that Dorj will come tomorrow.’

(237) is consistent with an analysis where the embedded subject is assigned accusative case in Spec,CP, while it is assigned no case at all if it remains in Spec,TP (within the CP phase). (238) and (239) are consistent with an analysis where the embedded subject is assigned accusative case in Spec,DP, while it is assigned genitive case in Spec,nP (within the DP phase) and no case at all if it remains in Spec,TP (within the nP and/or CP phase). Under both analyses, for embedded subjects to be extracted into the main clause, they must first move to the highest phase edge of the embedded clause – Spec,CP in finite bare CPs and Spec,DP in finite and nonfinite NCs – where they are assigned accusative case. The extraction facts in (237)-(239) thus obey a standard conception of phase-locality.

Subject extraction from converbial adjuncts is impossible regardless of whether the subject is accusative or case-unmarked (240).¹² I propose that this is because adjuncts are islands for movement in Khalkha; for further discussion on the Adjunct Island Condition, see, e.g., Huang 1982; Stepanov 2007; Truswell 2007; Ernst 2022; Privoznov 2022; a.m.o.

- (240) *Converbial adjuncts are islands for movement*
- a. ***Dorj-iig**₁ Naraa [₁ nom unshi-j bai-had] unt-san.
D.-ACC N. book read-PROG COP-TEMP.CVB sleep-PST
Int. ‘Naraa slept while Dorj was reading a book.’
- b. **Dorj** Naraa nom unshi-j bai-had unt-san.
D. N. book read-PROG COP-TEMP.CVB sleep-PST
Means: ‘Dorj slept while Naraa was reading a book.’
Does not mean: ‘Naraa slept while Dorj was reading a book.’

¹²Note that the ungrammaticality of (240a) provides additional evidence against a proleptic object analysis of accusative subjects in converbial adjuncts, under which we would predict that accusative subjects in converbial adjuncts may move to the matrix clause edge since they are base-generated in the matrix clause.

4.3.2 Summary

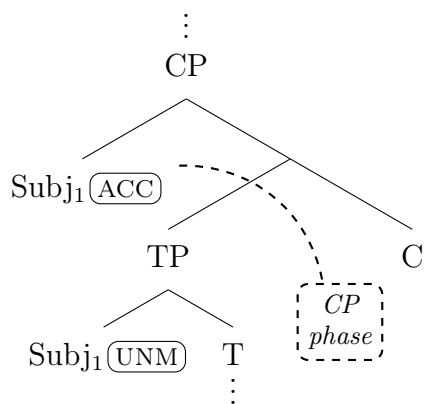
Summarising the subject extraction facts in this section, Table 4.2 shows that non-accusative (i.e. genitive and case-unmarked) subjects cannot be displaced from a position within the embedded clause to a position in the matrix clause, while accusative subjects may do so. In other words, long-distance subject extraction in Khalkha displays an ACC vs. GEN/UNM contrast. The last row in Table 4.2 – representing converbial adjuncts – has been greyed out due to the fact that subject extraction cannot occur because of an independent reason – the Adjunct Island Condition.

Table 4.2: Subject extraction across four embedded clause types in Khalkha

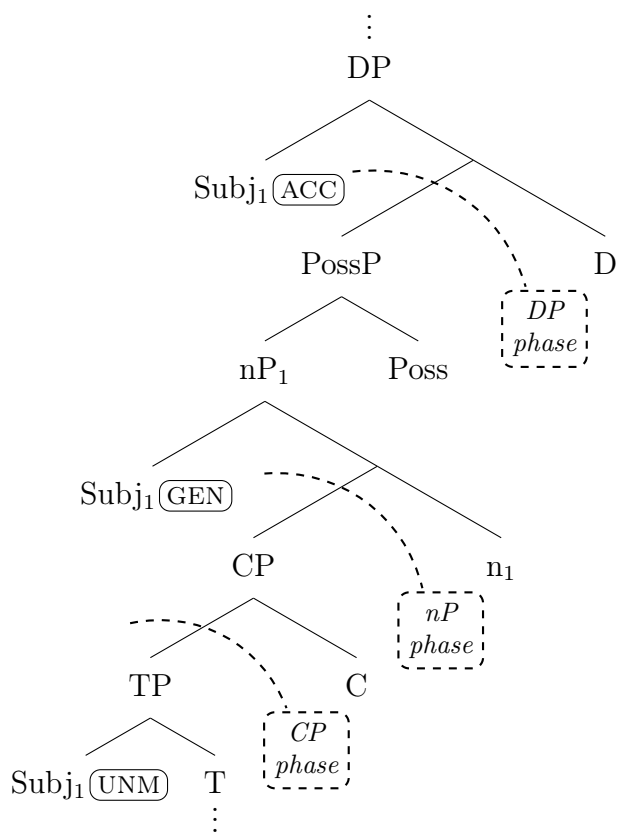
	ACC	GEN	UNM
Finite bare CP	✓	-	✗
Finite NC	✓	✗	✗
Nonfinite NC	✓	✗	✗
Converbial adjunct	✗	-	✗

These facts provide support for accusative subjects being located in the highest phase edge of the embedded clause – Spec,CP in finite bare CPs and Spec,DP in finite and nonfinite NCs – and for genitive and case-unmarked subjects being located in a lower position in the embedded clause. At the same time, the facts discussed in this section support the idea that CP and DP constitute phases in Khalkha, such that a subject that extracts into the matrix clause must move to Spec,CP or Spec,DP where it is assigned accusative case. In Spec,CP or Spec,DP, the embedded subject escapes spell-out within the embedded clause, and remains accessible to further syntactic operations, such as extraction, in the matrix clause. Before turning to an investigation of cross-clausal binding in Khalkha in the next section, I provide trees below that show the positions of accusative subjects vs. genitive and case-unmarked subjects in finite bare CPs, finite NCs, and nonfinite NCs.

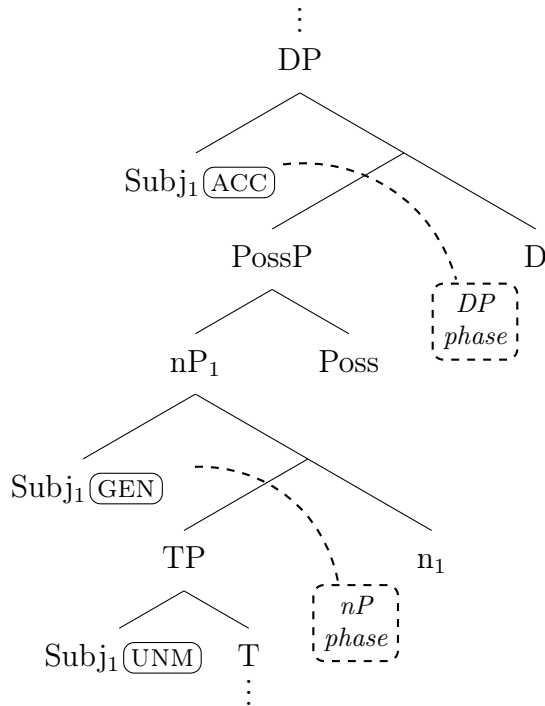
(241) *Positions of ACC and UNM subjects in finite bare CP*



(242) *Positions of ACC, GEN, and UNM subjects in finite NC*



(243) *Positions of ACC, GEN, and UNM subjects in nonfinite NC*



4.4 Cross-clausal binding in Khalkha

This section investigates the behaviour of Binding Conditions A and B across the four embedded clause types under investigation in Khalkha. While previous authors have investigated cross-clausal binding in finite bare CPs (Fong 2019) and nonfinite NCs (Aravind 2021), it has not, to my knowledge, been demonstrated for finite NCs and converbial adjuncts. The goal of this section is to demonstrate the behaviour of cross-clausal binding across the four embedded clause types using three types of subject-oriented anaphors introduced in Chapter 2: reflexives, reciprocals, and reflexive possessives.

Anticipating the key findings of this section, I show that, strikingly, accusative- and genitive-marked subjects are accessible to the matrix subject for binding, while case-unmarked subjects are not. This ACC/GEN vs. UNM contrast that I show for cross-clausal binding into the embedded subject differs from the ACC vs. GEN/UNM contrast demonstrated previously in §4.3 for long-distance embedded subject extraction. I argue that this

provides evidence that accusative and genitive subjects are located in a higher position in the embedded clause than case-unmarked subjects.

This section is organised as follows. In §4.4.1, I briefly recap the behaviour of Binding Conditions A and B in Khalkha matrix root clauses, and discuss the morphological properties of certain reciprocals and reflexive possessives that make them suitable for testing cross-clausal binding. §4.4.2 extends the three types of anaphor binding to subjects of the four types of embedded clauses examined in this dissertation. I demonstrate that reflexive and reciprocal subject anaphors, as well as reflexive possessors within subjects, can be bound by the matrix subject without violating Binding Condition A only when they are accusative or genitive, but not when they are case-unmarked. Additionally, I show that a pronominal possessor of an embedded subject triggers a Condition B violation when the subject is accusative or genitive, but not when it is case-unmarked. §4.4.3 summarises the findings in this section and discusses their implications for our understanding of binding domains.

4.4.1 Binding Conditions A and B in matrix root clauses

As discussed in Chapter 2, reflexive pronouns and referential pronouns are found in complementary distribution in Khalkha. An object pronominal cannot be bound by a clause-mate subject as it violates Binding Condition B, as shown in (244a) and (245a). Reflexive pronouns must be bound by a local subject, as shown in (244b) and (245b). Further, reflexive objects must be marked with accusative case; without case marking, they are interpreted as emphatic reflexives (246), which I analyse as modifiers that are adjoined to the subject DP.

- (244) a. *Bi **namaig** har-san.
 1SG 1SG.ACC see-PST
 Int. ‘I saw me.’
 b. Bi **öör-iig=öö** har-san.
 1SG self-ACC=RP see-PST
 ‘I saw myself.’

- c. Bi **tüüniig** har-san.
 1SG 3SG.ACC see-PST
 ‘I saw her/him/it.’
- (245) a. Naraa_j **tüüniig**_{k/*j} har-san.
 N. 3SG.ACC see-PST
 ‘Naraa_j saw her_{k/*j}/him/it.’
 b. Naraa **öör-iig=öö** har-san.
 N. self-ACC=RP see-PST
 ‘Naraa saw herself.’
 c. Naraa **namaig** har-san.
 N. 1SG.ACC see-PST
 ‘Naraa saw me.’
- (246) Bi/Naraa_j **öör=öö** *pro*_{k/*j} har-san.
 1SG/N. self=RP *pro* see-PST
 Means: ‘I/Naraa_j saw her_{k/*j}/him/it myself/herself_{j/*k}’
 Does not mean: ‘I_j/Naraa_k saw myself_j/herself_k.’

In addition to reflexive pronouns, Khalkha also has reciprocal pronouns, as shown in (247)-(248). The reciprocal *biye.biye* ‘each other’ in object position may surface with hidden-*n*, accusative case, and the RP-enclitic, as shown in (247a). The accusative case suffix may also surface as zero in the context of the RP enclitic, giving rise to the form found in (247b). Finally, hidden-*n* is also optional on *biye.biye*, resulting in the form found in (247c).

- (247) ‘We saw each other.’
 a. Bid **biye.biye-n-iig=ee** har-san.
 1PL each.other-N-ACC=RP see-PST
 b. Bid **biye.biye-n-Ø=ee** har-san.
 1PL each.other-N-ACC=RP see-PST
 c. Bid **biye.biy-Ø=ee** har-san.
 1PL each.other-ACC=RP see-PST

The appearance of hidden-*n* on *biye.biye* ‘each other’ in the context of accusative may be related to the fact that it is a pronominal; hidden-*n* is also found on certain personal pronoun stems in the context of accusative, e.g. *bid-n-iig* 1PL-N-ACC and *tüü-n-iig* 3SG-N-ACC.¹³

¹³The fact that hidden-*n* is found on pronominal stems in the context of accusative is unusual as on common noun stems hidden-*n* is only found in the context of genitive, dative, and ablative in Khalkha. For further discussion, see, e.g., Thompson 2008, p.40, 44 and Tserenpil and Kullmann 2008, p.78.

Additionally, accusative may be zero on non-kinship terms in the context of the RP-enclitic; see discussion surrounding (250)-(251) below. Thus, zero expression of the accusative morpheme is morphological and not syntactic, i.e. it is not due to the reciprocal object being located within the vP phase. Together, these three factors account for the form in (247b) in which hidden-*n* is overt but accusative is not, and the form in (247c) in which both hidden-*n* and accusative are not overt.

The reciprocal *neg.neg* ‘each other/one another’ in object position may also surface with hidden-*n* in the context of accusative case (248a); it thus patterns like other pronominals in Khalkha. The accusative case suffix may surface as zero in the context of the RP enclitic, giving rise to the form found in (248b). However, unlike the optionality of hidden-*n* on *biye.biye* (247c), hidden-*n* is obligatory on *neg.neg* in the context of the RP enclitic, even when accusative is zero (248b); the form in (248c) without hidden-*n* is ungrammatical.

(248) ‘We saw each other.’

- a. Bid **neg.neg-n-iig=ee** har-san.
1PL each.other-N-ACC=RP see-PST
- b. Bid **neg.neg-n-Ø=ee** har-san.
1PL each.other-N-ACC=RP see-PST
- c. *Bid **neg.neg-Ø=ee** har-san.
1PL each.other-ACC=RP see-PST

The different behaviours of hidden-*n* between *biye.biye* and *neg.neg* may not come as a surprise, as Thompson (2008) finds that hidden-*n* nouns in Khalkha may be classified into two groups: stems that obligatorily take hidden-*n* and stems that optionally take hidden-*n* in certain case contexts. Thompson (2008, p.147) also notes that hidden-*n* declension is optional on *biye* ‘body’, which is the root of *biye.biye*. Under this view, I treat *biye.biye* as an optional hidden-*n* stem and *neg.neg* as an obligatory hidden-*n* stem in the context of accusative. Thus, I take the presence of hidden-*n* on *neg.neg* to indicate the presence of accusative case underlyingly, regardless of whether it is overt (248a) or null (248b).

Reciprocal subjects are not possible in matrix root clauses (249).

- (249) * **Biye.biy=ee/Neg.neg=ee** bidniig har-san.
 Each.other=RP/Each.other=RP 1PL.ACC see-PST
 Int. ‘Each other saw us.’

Finally, in Khalkha, referential possessive pronouns are also found in complementary distribution with the use of the RP enclitic (250c) or a genitive-marked reflexive possessive pronoun (250d).

- (250) a. Naraa **minii** nom-iig unsh-san
 N. 1SG.GEN book-ACC read-PST
 ‘Naraa read my book.’
 b. Naraa_j **tüünii_{k/*j}** nom-iig unsh-san.
 N. 3SG.GEN book-ACC read-PST
 ‘Naraa_j read her_{k/*j} book.’
 c. Naraa **nom-Ø=oo** unsh-san.
 N. book-ACC=RP read-PST
 ‘Naraa_j read her_{j/*k} book.’
 d. Naraa **öör-iin-h=öö** nom-iig unsh-san.
 N. self-GEN-H=RP book-ACC read-PST
 ‘Naraa_j read her_{j/*k} (own) book.’

On non-kinship terms, the accusative case suffix may surface as zero in the context of the RP-enclitic, as seen in (250c). However, the accusative case suffix is obligatorily overt on kinship terms, as shown in (251).

- (251) Bi **egch*(-iig)=ee** har-san.
 1SG older.sister*(-ACC)=RP see-PST
 ‘I saw my older sister.’

RP-marked subjects (252a) and subjects containing a genitive-marked reflexive possessive pronoun (252b) are also not allowed in matrix root clauses:

- (252) a. * **Egch=ee** namaig har-san.
 Older.sister=RP 1SG.ACC see-PST
 Int. ‘My older sister saw me.’
 b. * **Öör-iin-h=öö** egch namaig har-san.
 Self-GEN-H=RP older.sister 1SG.ACC see-PST
 Int. ‘My older sister saw me.’

In the next subsection, I turn to the behaviour of subject anaphors and subjects containing reflexive possessors across the four embedded clause types under investigation. Due to the morphological properties of *biye.biye* ‘each other’ and RP-marked non-kinship terms, it is not possible to tease apart the accusative vs. case-unmarked forms of these anaphors if accusative case is zero and hidden-*n* is not triggered. Instead, I will focus on the reciprocal *neg.neg* ‘each other/one another’, where hidden-*n* indicates the presence of underlying accusative case, and RP-marked kinship terms, where accusative case must be overt.

4.4.2 Binding Conditions A and B in embedded clauses

This subsection focuses on the behaviours of Binding Conditions A and B across four types of embedded clauses – finite bare CPs, finite NCs, nonfinite NCs, and converbial adjuncts. I first investigate the behaviours of reflexive and reciprocal embedded subjects and reflexive possessors in embedded subjects in the four embedded clause types. Next, I turn to an investigation of the behaviour of pronominal possessors of subjects across the four embedded clause types.

4.4.2.1 Binding Condition A in embedded clauses

I show that reflexive and reciprocal subjects, and subjects containing reflexive possessors, can appear in finite bare CPs, finite NCs, nonfinite NCs, and converbial adjuncts if and only if they bear accusative or genitive case (when available). I present the relevant data for each embedded clause type in turn.

Finite bare CPs allow ACC or case-unmarked subjects, but not GEN subjects. The examples below illustrate that in finite bare CPs, a reflexive subject (253a), a reciprocal subject (254a), or a reflexive possessor within a subject (255a), can be bound by the matrix subject if and only if it bears accusative case. When the embedded subject is case-unmarked, the result is ungrammatical on the intended reading, as shown in the (b) examples. The

data here corroborates the data in Fong 2019, pp.11-13, who reaches the same conclusion based on the behaviour of reflexive subjects and reflexive possessors inside subjects in finite bare CPs.

(253) *Reflexive subject may be ACC but not UNM in finite bare CP*

- a. Dorj [margaash **öör-iig=öö** ir-ne gej] hel-sen.
D. tomorrow self-ACC=RP come-NPST C say-PST
‘Dorj_j said that he_j is coming tomorrow.’
- b. *Dorj [margaash **öör=öö** ir-ne gej] hel-sen.
D. tomorrow self=RP come-NPST C say-PST
Int. ‘Dorj_j said that he_j is coming tomorrow.’¹⁴

(254) *Reciprocal subject may be ACC but not UNM in finite bare CP*

- a. Misheel Naraa hoyor [**neg.neg-n-iig=ee/neg.neg-n-Ø=ee** shine daashinz
M. N. two each.other-N-ACC=RP/e.o.-N-ACC=RP new dress
av-san gej] hel-sen.
buy-PST C say-PST
‘Misheel_j and Naraa_k said that each other_{j+k} bought a new dress.’
- b. *Misheel Naraa hoyor [**neg.neg=ee** shine daashinz av-san gej] hel-sen.
M. N. two each.other=RP new dress buy-PST C say-PST
Int. ‘Misheel_j and Naraa_k said that each other_{j+k} bought a new dress.’

(255) *Subject containing reflexive possessive may be ACC but not UNM in finite bare CP*

- a. Bat [margaash **egch-iig=ee** ir-ne gej] hel-sen.
B. tomorrow older.sister-ACC=RP come-NPST C say-PST
‘Bat_j said that his_j older sister is coming tomorrow.’
- b. *Bat [margaash **egch=ee** ir-ne gej] hel-sen.
B. tomorrow older.sister=RP come-NPST C say-PST
Int. ‘Bat_j said that his_j older sister is coming tomorrow.’

Next, finite NCs allow ACC, GEN, or case-unmarked subjects. The examples below illustrate that a reflexive subject (256a), a reciprocal subject (257a), or a reflexive possessor inside a subject (258a), can be bound by the matrix subject only if it is marked with

¹⁴(253b) is grammatical when it the embedded clause is understood to contain a *pro* subject, and *ööröö* is interpreted as an emphatic reflexive modifying *pro*:

- (i) Dorj [margaash *pro* **öör=öö** ir-ne gej] hel-sen.
D. tomorrow *pro* self=RP come-NPST C say-PST
‘Dorj_j said that he_j is coming himself_j tomorrow.’

accusative or genitive case. When the embedded subject is case-unmarked, the result is ungrammatical on the intended reading, as shown in the (b) examples.

(256) *Reflexive subject may be ACC/GEN but not UNM in finite NC*

- a. Dorj [margaash **öör-iig=öö/öö-iin-h=öö** yav-na gedg]-iig sons-son.
 D. tomorrow self-ACC=RP/self-GEN-H=RP go-NPST C.NML -ACC hear-PST
 ‘Dorj_j heard that he_j will go tomorrow.’
- b. *Dorj [margaash **öör=öö** yav-na gedg]-iig sons-son.
 D. tomorrow self=RP go-NPST C.NML -ACC hear-PST
 Int. ‘Dorj_j heard that he_j will go tomorrow.’¹⁵

(257) *Reciprocal subject may be ACC/GEN but not UNM in nonfinite NC*

- a. Misheel Naraa hoyor [**neg.neg-n-iig=ee/neg.neg-n-Ø=ee/**
 M. N. two each.other-N-ACC=RP/e.o.-N-ACC=RP/
neg.neg-nii-h=ee shine daashinz av-san gedg]-iig sons-son.
 e.o.-GEN-H=RP new dress buy-PST C.NML -ACC hear-PST
 ‘Misheel_j and Naraa_k heard that each other_{j+k} bought a new dress.’
- b. *Misheel Naraa hoyor [**neg.neg=ee** shine daashinz av-san gedg]-iig
 M. N. two each.other=RP new dress buy-PST C.NML -ACC
 sons-son.
 hear-PST
 Int. ‘Misheel_j and Naraa_k heard that each other_{j+k} bought a new dress.’

(258) *Subject containing reflexive possessive may be ACC/GEN but not UNM in finite NC*

- a. Misheel [**egch-iig=ee/egch-iin-h=ee** shine daashinz av-san
 M. older.sister-ACC=RP/older.sister-GEN-H=RP new dress buy-PST
 gedg]-iig sons-son.
 C.NML -ACC hear-PST
 ‘Misheel_j heard that her_j sister bought a new dress.’
- b. *Misheel [**egch=ee** shine daashinz av-san gedg]-iig sons-son.
 M. older.sister=RP new dress buy-PST C.NML -ACC hear-PST
 Int. ‘Misheel_j heard that her_j older sister bought a new dress.’

Nonfinite NCs also allow ACC, GEN, or case-unmarked subjects. The examples below illustrate that a reflexive subject (259a), a reciprocal subject (260a), or a reflexive possessor

¹⁵(256b) is grammatical under the reading where the embedded clause contains a *pro* subject, and *ööröö* is interpreted as an emphatic reflexive modifying *pro*:

- (i) Dorj [margaash *pro* **öör=öö** yav-na gedg]-iig sons-son.
 D. tomorrow *pro* self=RP go-NPST C.NML -ACC hear-PST
 ‘Dorj_j heard that he_j himself_j will go tomorrow.’

inside a subject (261a), can be bound by the matrix subject only when it is marked with accusative or genitive case. When the embedded subject is case-unmarked, the result is ungrammatical on the intended reading, as shown in the (b) examples.¹⁶

(259) *Reflexive subject may be ACC/GEN but not UNM in nonfinite NC*

- a. Dorj [**öör-iig=öö/öör-iin-h=öö** nom unsh-ih]-iig har-san.
D. self-ACC=RP/self-GEN-H=RP book read-INF -ACC see-PST
‘Dorj_j saw himself_j reading a book.’
- b. *Dorj [**öör=öö** nom unsh-ih]-iig har-san.
D. self=RP book read-INF -ACC see-PST
Int. ‘Dorj_j saw himself_j reading a book.’

(260) *Reciprocal subject may be ACC/GEN but not UNM in nonfinite NC*

- a. Misheel Naraa hoyor [**neg.neg-n-iig=ee/neg.neg-n-Ø=ee/**
M. N. two each.other-N-ACC=RP/e.o.-N-ACC=RP/
neg.neg-nii-h=ee shine daashinz av-ah]-iig har-san.
e.o.-GEN-H=RP new dress buy-INF -ACC see-PST
‘Misheel_j and Naraa_k saw each other_{j+k} buying a new dress.’
- b. *Misheel Naraa hoyor [**neg.neg=ee** shine daashinz av-ah]-iig har-san.
M. N. two each.other=RP new dress buy-INF -ACC see-PST
Int. ‘Misheel_j and Naraa_k saw each other_{j+k} buying a new dress.’

(261) *Subject containing reflexive possessive may be ACC/GEN but not UNM in nonfinite NC*

- a. Misheel [**egch-iig=ee/egch-iin-h=ee** shine daashinz av-ah
M. older.sister-ACC=RP/older.sister-GEN-H=RP new dress buy-INF
]-iig har-san.
-ACC see-PST
‘Misheel_j saw her_j sister is buying a new dress.’
- b. *Misheel [**egch=ee** shine daashinz av-ah]-iig har-san.
M. older.sister=RP new dress buy-INF -ACC see-PST
Int. ‘Misheel_j saw her_j sister is buying a new dress.’

Finally, converbial adjuncts may contain ACC or case-unmarked subjects but not GEN subjects. The examples below show that a reflexive subject (262a), a reciprocal subject (263a), or a reflexive possessor inside a subject (264a), can be bound by the matrix subject

¹⁶Aravind (2021, p.388, fn.12) writes that, for her consultants, “GEN-marked subjects seem to be marginally acceptable in cross-phasal binding configurations”. However, they are fully acceptable for my consultants.

only when it is marked with accusative case. When the embedded subject is case-unmarked, the result is ungrammatical on the intended reading, as shown by the (b) examples.

(262) *Reflexive subject may be ACC but not UNM in converbial adjunct*

- a. Dorj [**öör-iig=öö** unt-tal] nom unsh-san.
 D. self-ACC=RP sleep-TERM.CVB book read-PST
 ‘Dorj_j read a book until he_j fell asleep.’
- b. *Dorj [**öör=öö** unt-tal] nom unsh-san.
 D. self=RP sleep-TERM.CVB book read-PST
 Int. ‘Dorj_j read a book until he_j fell asleep.’

(263) *Reciprocal subject may be ACC but not UNM in converbial adjunct*

- a. Misheel Naraa hoyor [**neg.neg-n-iig=ee/neg.neg-n-Ø=ee**
 M. N. two each.other-N-ACC=RP/e.o.-N-ACC=RP
 inee-tel] onigoo yar'-san.
 laugh-TERM.CVB joke tell-PST
 ‘Misheel_j and Naraa_k told jokes until each other_{j+k} laughed.’
- b. *Misheel Naraa hoyor [**neg.neg=ee** inee-tel]
 M. N. two each.other=RP laugh-TERM.CVB
 onigoo yar'-san.
 joke tell-PST
 Int. ‘Misheel_j and Naraa_k told jokes until each other_{j+k} laughed.’

(264) *Subject containing reflexive possessive may be ACC but not UNM in converbial adjunct*

- a. Odgerel [**egch-iig=ee** ene nom-iig unshi-j bai-had]
 O. older.sister-ACC=RP this book-ACC read-PROG COP-TEMP.CVB
 unt-san.
 sleep-PST
 ‘When her_j older sister was reading this book, Odgerel_j fell asleep.’
- b. *Odgerel [**egch=ee** ene nom-iig unshi-j bai-had]
 O. older.sister=RP this book-ACC read-PROG COP-TEMP.CVB
 unt-san.
 sleep-PST
 Int. ‘When her_j older sister was reading this book, Odgerel_j fell asleep.’

Summarising the data above, Table 4.3 shows that while accusative and genitive subjects across the four embedded clause types under investigation are visible to the matrix clause for Binding Condition A, case-unmarked subjects are not.

Table 4.3: Anaphoric binding by matrix subject across four embedded clause types in Khalkha

	ACC	GEN	UNM
Finite bare CP	✓	-	✗
Finite NC	✓	✓	✗
Nonfinite NC	✓	✓	✗
Converbial adjunct	✓	-	✗

The facts in Table 4.3 suggest that accusative and genitive subjects and possessors inside such subjects belong to the same binding domain as the matrix subject while case-unmarked subjects and possessors inside such subjects are evaluated in a separate binding domain from the matrix subject. Thus, cross-clausal binding displays an ACC/GEN vs. UNM contrast, in contrast to a ACC vs. GEN/UNM contrast found with respect to long-distance subject extraction. As such, accusative and genitive subject anaphors, as well as reflexive possessors within these subjects, are able to be bound by the matrix subject, thereby satisfying Binding Condition A. Conversely, the facts suggest that no sufficiently local binder exists for case-unmarked subject anaphors or reflexive possessors within case-unmarked subjects, which leads to a Binding Condition A violation. Next, I discuss additional evidence from Binding Condition B that supports the abovementioned picture of cross-clausal binding locality in Khalkha.

4.4.2.2 Pronominal possessors of embedded clause subjects

If accusative and genitive subjects belong to the same binding domain as the matrix subject, we would expect accusative and genitive pronominal subjects as well as pronominal possessors within these subjects to violate Binding Condition B when they are coreferential with the matrix subject. Conversely, if case-unmarked subjects belong to a separate binding

domain from the matrix subject, we would also expect case-unmarked pronominal subjects as well as pronominal possessors of case-unmarked subjects to not violate Binding Condition B when they are coreferential with the matrix subject. In other words, we predict that Binding Condition B effects between the matrix subject and an embedded subject or its possessor should look the reverse of Table 4.3.

I shall first demonstrate that the use of pronominal embedded subjects to test for Binding Condition B should be ruled out for independent reasons. As discussed in §4.2.1, Fong (2019) shows that a third person pronominal subject can be coreferential with the matrix subject only when it surfaces without any case marking (265a) but not when it bears accusative case (265b):

- (265) *Judgements reported by Fong's (2019) consultants*
- a. Odgerel_j [margaash **ter**_{j/k} ir-ne gej] hel-sen.
 O. tomorrow 3SG come-NPST C say-PST
 ‘Odgerel_j said that she_{j/k}/he_k is coming tomorrow.’
 - b. Odgerel_j [margaash **tüüniig**_{k/*j} ir-ne gej] hel-sen.
 O. tomorrow 3SG.ACC come-NPST C say-PST
 ‘Odgerel_j said that she_{k/*j}/he_k is coming tomorrow.’ (Fong 2019, p.13, ex.40)

Fong (2019) takes the contrast between (265a) and (265b) to provide evidence that accusative subjects occupy a higher position than case-unmarked subjects, such that accusative subjects are evaluated in the same binding domain as the matrix subject while case-unmarked subjects are evaluated in a different binding domain than the matrix subject. However, I am unable to replicate the judgements in (265a) with my three consultants, for whom the coreferential reading is unavailable even when the embedded subject is case-unmarked (266a). For my consultants, the coreferential reading can be achieved with the use of an accusative reflexive subject (266b) or a null subject (266c).¹⁷

¹⁷Fong's consultants also accept the use of an accusative-marked reflexive subject (i) or a dropped subject (ii) to express coreference.

- (i) Tuyaa [margaash **öör-iig=öö** Mongol ruu yav-na gej] hel-sen.
 T. tomorrow self-ACC=RP Mongolia DIR go-NPST C say-PST
 ‘Tuyaa_j said that she_j is going to Mongolia tomorrow.’ (Fong 2019, p.13, ex.39b)

(266) *Judgements reported by my consultants*

- a. Odgerel_j [margaash **ter**_{k/*j} ir-ne gej] hel-sen.
 O. tomorrow 3SG come-NPST C say-PST
 ‘Odgerel_j said that she_{k/*j}/he_k is coming tomorrow.’
- b. Odgerel_j [margaash **öör-iig=öö**_j ir-ne gej] hel-sen.
 O. tomorrow self-ACC=RP come-NPST C say-PST
 ‘Odgerel_j said that she_j is coming tomorrow.’
- c. Odgerel_j [margaash **pro**_j ir-ne gej] hel-sen.
 O. tomorrow *pro* come-NPST C say-PST
 ‘Odgerel_j said that she_{j/k}/he_k is coming tomorrow.’

In finite NCs (267), nonfinite NCs (268), and converbial adjuncts (269), my consultants also reject the use of an overt pronominal subject to express coreference with the matrix subject, as shown in the (a) examples below. To express coreference of the embedded subject with the matrix subject, they may use (i) an accusative-marked reflexive subject, as I have shown in §4.4.2.1, (ii) a dropped subject, as shown in the (b) examples, or (iii) a dropped subject with RP-marking on the embedded clause to disambiguate in favour of a coreferential reading, as shown in the (c) examples. For further discussion of RP-marking on embedded nominalised clauses, see Gong 2023b; Gong and Despić 2024.

(267) *Cross-clausal subject coreference in finite NCs*

- a. Naraa_j [margaash **ter**_{k/*j} ir-ne gedg]-iig mart-san.
 N. tomorrow 3SG come-NPST C.NML -ACC forget-PST
 ‘Naraa_j forgot that she_{k/*j}/he_k will come tomorrow.’
- b. Naraa_j [margaash **pro**_{j/k} ir-ne gedg]-iig mart-san.
 N. tomorrow *pro* come-NPST C.NML -ACC forget-PST
 ‘Naraa_j forgot that she_{j/k}/he_k will come tomorrow.’
- c. Naraa_j [margaash **pro**_j ir-ne gedg]-iig=**ee** mart-san.
 N. tomorrow *pro* come-NPST C.NML -ACC=RP forget-PST
 ‘Naraa_j forgot that she_j will come tomorrow.’

(ii) Bat_j [**pro**_{j/k} muur-and durtai gej] hel-sen.
 B. *pro* cat-DAT like C say-PST
 ‘Bat_j said that he_{j/k}/she/they like(s) cats.’

(Fong 2019, p.5, ex.9d)

(268) *Cross-clausal subject coreference in nonfinite NCs*

- a. Naraa_j [margaash **ter**_{k/*j} ir-eh]-iig mart-san.
 N. tomorrow 3SG come-INF -ACC forget-PST
 ‘Naraa_j forgot that she_{k/*j}/he_k will come tomorrow.’
- b. Naraa_j [margaash **pro**_{j/k} ir-eh]-iig mart-san.
 N. tomorrow *pro* come-INF -ACC forget-PST
 ‘Naraa_j forgot that she_{j/k}/he_k will come tomorrow.’
- c. Naraa_j [margaash **pro**_j ir-eh]-iig=**ee** mart-san.
 N. tomorrow *pro* come-INF -ACC=RP forget-PST
 ‘Naraa_j forgot that she_j will come tomorrow.’

(269) *Cross-clausal subject coreference in converbial adjuncts*

- a. Bilgüün_j [**ter**_{k/*j} duul-j bai-had] büjigle-j bai-na.
 B. 3SG sing-PROG COP-TEMP.CVB dance-PROG COP-NPST
 ‘Bilguun_j is dancing while he_{k/*j}/she_k is singing.’
- b. Bilgüün_j [**pro**_{j/k} duul-j bai-had] büjigle-j bai-na.
 B. *pro* sing-PROG COP-TEMP.CVB dance-PROG COP-NPST
 ‘Bilguun_j is dancing while he_{j/k}/she_k is singing.’
- c. Bilgüün_j [**pro**_j duul-j bai-hd]=**aa** büjigle-j bai-na.
 B. *pro* sing-PROG COP-TEMP.CVB =RP dance-PROG COP-NPST
 ‘Bilguun_j is dancing while he_j is singing.’

I suggest that for my consultants, the unacceptability of using an overt pronominal embedded subject when it is coreferential with the matrix subject is due to independent reasons such as the Avoid Pronoun Principle (Chomsky 1981), which states that in a context where both an overt pronoun or a non-overt pronoun may be used, the non-overt pronoun will be chosen over the overt pronoun. Exploring this idea would take us too far afield, and I leave this to future research.

To recap, I predicted, following the findings of §4.4.2.1, that Binding Condition B will be violated if i) an embedded pronominal subject or a pronominal possessor within an embedded subject is coreferential with the matrix subject, and ii) the embedded subject bears accusative or genitive case. While it is not possible to confirm these predictions for pronominal subjects, as I have explained above, I will now show that they are borne out in the case of pronominal possessors within embedded subjects.

Finite bare CPs may contain ACC and case-unmarked subjects, but not GEN ones. The pair of examples below show that a pronominal possessor of an embedded subject bearing accusative case (270a) cannot be coreferential with the matrix subject, while a pronominal possessor of a case-unmarked embedded subject may be coreferential with the matrix subject (270b).¹⁸

(270) *Pronominal possessor of UNM but not ACC subject can be coreferential with matrix subject in finite bare CP*

- a. Tuyaa_j [öchigdör **tüünii**_{k/*j} **naiz-iig** nomin san-d och-son gej] hel-sen.
T. yesterday 3SG.GEN friend-ACC library-DAT go-PST C say-PST
‘Tuyaa_j said that her_{k/*j} friend went to the library yesterday.’
- b. Tuyaa_j [öchigdör **tüünii**_{j/k} **naiz** nomin san-d och-son gej] hel-sen.
T. yesterday 3SG.GEN friend library-DAT go-PST C say-PST
‘Tuyaa_j said that her_{j/k} friend went to the library yesterday.’

Finite NCs may contain ACC, GEN and case-unmarked subjects. The examples below show that a pronominal possessor of a subject bearing accusative (271a) or genitive (271b) case cannot be coreferential with the matrix subject, while a pronominal possessor of a case-unmarked subject may be coreferential with the matrix subject (271c).¹⁹

(271) *Pronominal possessor of UNM but not ACC/GEN subject can be coreferential with matrix subject in finite NC*

- a. Dorj_j [**tüünii**_{k/*j} **muur-iig** öchigdör har nohoi maa_j-san gedg]-iig
D. 3SG.GEN cat-ACC yesterday black dog scratch-PST C.NML -ACC
sons-son.
hear-PST
‘Dorj_j heard that yesterday his_{k/*j}/her_k cat scratched a black dog.’

¹⁸In (270) and the following examples, note once again that even though the embedded subject – regardless of its case – surfaces to the right of a temporal adverb in the embedded clause, I have argued in Chapter 2 that the ordering of temporal adverbs relative to subjects does not accurately diagnose the position of the subject, possibly due to the fact that such adverbs may scramble to the edge of the embedded clausal left periphery.

¹⁹For reasons that are unclear to me, pronominal possessors of GEN subjects, e.g. (271b) and (272b), are judged as quite degraded for all three of my consultants regardless of their binding possibilities, which is why I have marked them with ?/?*. Speaker G (marked *) rejects these examples, while Speakers B and M (marked ??) find them marginally acceptable. To the extent that they are acceptable for two consultants (Speakers B and M), the pronominal possessor cannot be coreferential with the matrix subject. This is not strictly due to the presence of multiple GEN-marked arguments, but may be due to the presence of multiple GEN-marked arguments associated with different theta roles, e.g. a possessor and an agent theta role. For instance, it is possible to have multiple GEN-marked arguments indicating multiple possessors, e.g. *minii ah-iin nohoi* 1SG.GEN older.brother-GEN dog ‘my older brother’s dog’. See also (232)–(235).

- b. ??/* Dorj_j [**tüünii**_{k/*j} **muur-iin** öchigdör har nohoi maa_j-san gedg]-iig
 D. 3SG.GEN cat-GEN yesterday black dog scratch-PST C.NML -ACC
 sons-son.
 hear-PST
 ‘Dorj_j heard that yesterday his_{k/*j}/her_k cat scratched a black dog.’
 Comment: Sounds like ‘Dorj heard that yesterday his cat’s black dog scratched something.’
- c. Dorj_j [**tüünii**_{j/k} **muur** öchigdör har nohoi maa_j-san gedg]-iig sons-son.
 D. 3SG.GEN cat yesterday black dog scratch-PST C.NML -ACC hear-PST
 Int. ‘Dorj_j heard that yesterday his_{j/k}/her_k cat scratched a black dog.’

Nonfinite NCs may also contain ACC, GEN and case-unmarked subjects. The examples below show that a pronominal possessor of a subject bearing accusative (272a) or genitive (272b) case cannot be coreferential with the matrix subject, while a pronominal possessor of a case-unmarked subject may be coreferential with the matrix subject (272c).

(272) *Pronominal possessor of UNM but not ACC/GEN subject can be coreferential with matrix subject in nonfinite NC*

- a. Misheel_j [**tüünii**_{k/*j} **egch-iig** shine daashinz av-ah]-iig har-san.
 M. 3SG.GEN older.sister-ACC new dress buy-INF -ACC see-PST
 ‘Misheel_j saw her_{k/*j}/his_k older sister buying a new dress.’
- b. ??/* Misheel_j [**tüünii**_{k/*j} **egch-iin** shine daashinz av-ah]-iig har-san.
 M. 3SG.GEN older.sister-GEN new dress buy-INF -ACC see-PST
 ‘Misheel_j saw her_{k/*j}/his_k older sister buying a new dress.’
- c. Misheel_j [**tüünii**_{j/k} **egch** shine daashinz av-ah]-iig har-san.
 M. 3SG.GEN older.sister new dress buy-INF -ACC see-PST
 ‘Misheel_j saw her_{j/k}/his_k older sister buying a new dress.’

Finally, converbial adjuncts may contain ACC and case-unmarked but not GEN subjects. The pair of examples below show that a pronominal possessor of an accusative subject cannot be coreferential with the matrix subject (273a), while a pronominal possessor of a case-unmarked subject may be coreferential with the matrix subject (273b).

(273) *Pronominal possessor of UNM but not ACC subject can be coreferential with matrix subject in converbial adjunct*

- a. Odgerel_j [**tüünii**_{k/*j} **naiz-iig** hicheel-Ø=ee hii-j bai-had]
 O. 3SG.GEN friend-ACC lesson-ACC=RP do-PROG COP-TEMP.CVB
 unt-san.
 sleep-PST
 ‘When her_{k/*j}/his_k friend was studying, Odgerel_j fell asleep.’
- b. Odgerel_j [**tüünii**_{j/k} **naiz** hicheel-Ø=ee hii-j bai-had] unt-san.
 O. 3SG.GEN friend lesson-ACC=RP do-PROG COP-TEMP.CVB sleep-PST
 ‘When her_{j/k}/his_k friend was studying, Odgerel_j fell asleep.’

Summarising the data above, Table 4.4 shows that pronominal possessors of accusative and genitive subjects cannot corefer with the matrix subject across the four embedded clause types under investigation, while pronominal possessors of case-unmarked subjects may do so. For genitive subjects, the ungrammatical coreferential readings are represented as (**X**) to indicate that the constructions themselves were judged by Speakers B and M to be marginally acceptable under the noncoreferential reading.

Table 4.4: Coreference between matrix subject and pronominal possessor of embedded subject across four embedded clause types in Khalkha

	ACC	GEN	UNM
Finite bare CP	X	-	✓
Finite NC	X	(X)	✓
Nonfinite NC	X	(X)	✓
Converbial adjunct	X	-	✓

As predicted, Table 4.3 is the mirror image of Table 4.4, which illustrates that anaphors and pronouns are also found in complementary distribution across the four types of embedded clauses in Khalkha. These findings provide support for the idea that accusative and genitive subjects, as well as possessors inside such subjects, belong to the same binding domain as the matrix subject. As such, pronominal possessors of accusative and genitive subjects violate Binding Condition B when they are coreferential with the matrix subject. On the other hand, the findings also provide support for the idea that case-unmarked subjects, as well as possessors inside such subjects, belong to a separate binding domain from

the matrix subject. As such, pronominal possessors of case-unmarked subjects do not violate Binding Condition B when interpreted as coreferential with the matrix subject.

4.4.3 Summary and implications

Summarising the findings in §4.4.1 and §4.4.2, Table 4.5 shows that across the four types of embedded clauses examined herein, accusative and genitive subjects are accessible to the matrix subject for binding, while case-unmarked subjects are inaccessible to the matrix subject for binding. Recall that finite bare CPs lack any nominal superstructure above the clause, finite and nonfinite NCs contain DP>PossP>nP above the clause, and converbial adjuncts contain PossP>nP above the clause. The following generalisation emerges: except when Spec,CP is the highest phase edge of the embedded clause (i.e. in finite bare CPs) and the embedded subject is located in Spec,CP, an embedded subject is inaccessible for binding by the matrix subject if it is located in a separate *clausal* domain than the matrix subject. *Nominal* domains that are part of the superstructure of an embedded clause do not appear to count as a separate binding domain than the matrix clause.

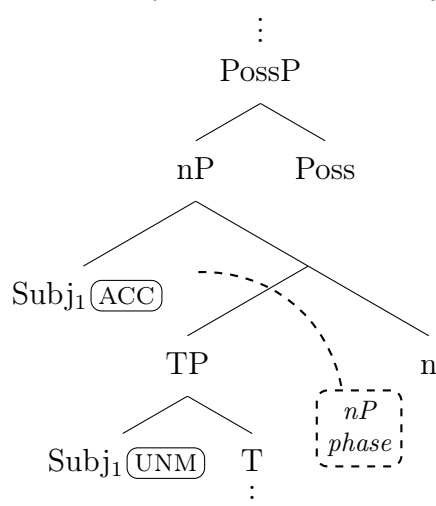
Table 4.5: Accessibility of embedded subject to matrix subject for binding across four embedded clause types in Khalkha

	ACC	GEN	UNM
Finite bare CP	✓	-	✗
Finite NC	✓	✓	✗
Nonfinite NC	✓	✓	✗
Converbial adjunct	✓	-	✗

Two notable conclusions emerge from the findings in this section. First, recall from the preceding section that it was not possible to deduce the position of accusative subjects in converbial adjuncts from long-distance subject extraction, since extraction out of adjuncts is not possible in Khalkha. The binding asymmetry with respect to accusative and case-unmarked subjects in converbial adjuncts nonetheless informs us that accusative subjects are located in a higher position than case-unmarked subjects. Specifically, accusative sub-

jects must be located within the nominal domain of the converbial adjunct, since they are accessible for binding in the matrix clause, while case-unmarked subjects must be located in Spec,TP of the converbial adjunct, since they are not visible for binding in the matrix clause. Given that converbial adjuncts lack DP and CP, I propose that accusative subjects in them occupy at least Spec,nP – the highest phase edge of the adjunct clause – while case-unmarked subjects occupy Spec,TP, where they are spelled out within the nP phase. This is schematised in (274).

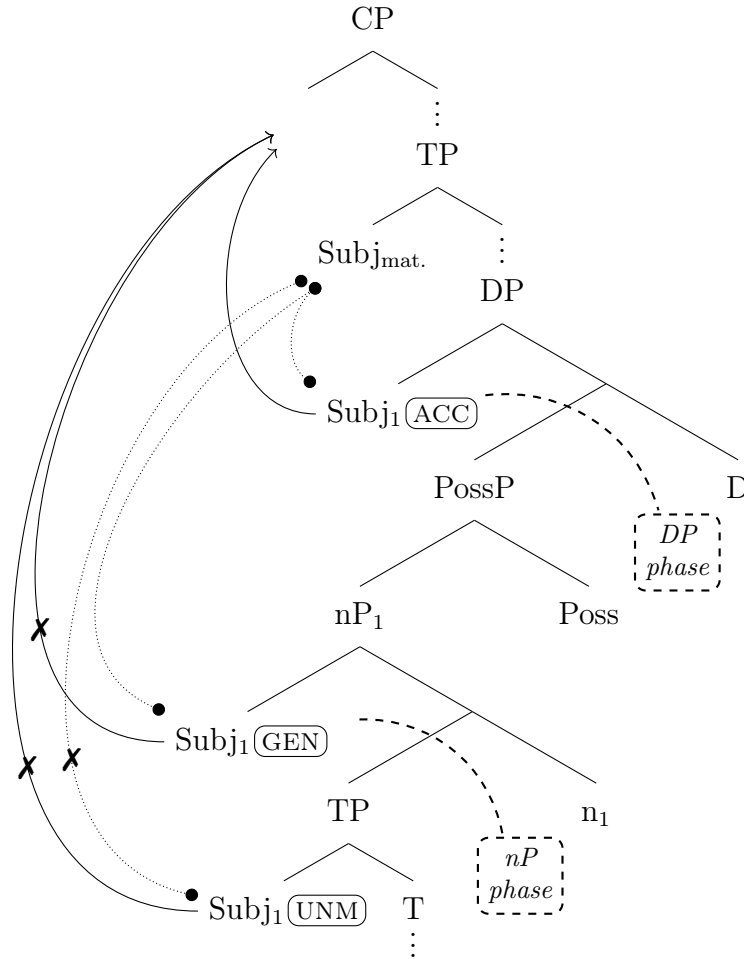
(274) *Positions of ACC and UNM subjects in converbial adjunct*



Second, my proposal that genitive subjects occupy Spec,nP in finite and nonfinite NCs, together with the finding in this section that genitive subjects remain accessible for binding by matrix subjects, might initially suggest that DP does not constitute a phase in Khalkha. However, evidence from §4.3 indicates that DP acts as a phasal boundary for long-distance extraction. In other words, while genitive subjects located in Spec,nP within a DP phase cannot be extracted, they remain accessible for cross-clausal binding. This asymmetry is illustrated in (275) for a nonfinite NC. Assuming that the extracted embedded subject lands in a position in the matrix CP, the arrows indicate that long-distance extraction may only take place from Spec,DP – the highest phase edge of the nonfinite NC. The dotted lines show that binding by the matrix subject may target the embedded subject in Spec,DP or

Spec,nP – both of which are positions within the nominal superstructure above the clause – but not Spec,TP.

(275) *Extraction vs. binding asymmetry in nonfinite NC*



The most plausible conclusion, therefore, is that the locality domains relevant for extraction and for binding are not necessarily identical and do not uniformly align with locality domains imposed by a standard conception of phases. The comparison of long-distance extraction and binding across the four embedded clause types in Khalkha thus reveals a non-isomorphism in their locality domains – a finding that crucially emerged only via an investigation of complement NCs.

4.5 A staggered spellout approach to extraction and binding domains in Khalkha

Before concluding this chapter, I sketch a possible approach to formally model the mismatch between binding and extraction domains in Khalkha without abandoning the notion of phases. Hicks (2009, pp.51-52), building on Marušič 2005 and Matushansky 2005, proposes that when a phase is complete, it may trigger spellout of its domain to PF but not LF, to LF but not PF, or to both PF and LF.²⁰ In this sense, a phasal spellout domain may be a PF-phase, an LF-phase, or both, and spell out to PF or LF may be “staggered”. For example, Hicks (2009, p.51) quotes the following from Marušič 2005, p.9:

If for example a certain head is an LF phase head but not a PF phase head, let’s call it an **LF-only phase**, its completion would freeze/ship all the features that must end up at LF, but not those that are relevant for PF. Then, at the next (full) phase, when the derivation reaches e.g. *vP*, the structure ready to be shipped to PF would be twice the size of the structure ready to get shipped to LF, since part of the structure has been already shipped to LF at an earlier point of LF-only Spell-Out. (Marušič 2005, p.9)

Hicks (2009) proposes that i) C^0 , v^0 , D^0 , and n^0 are phase heads, although they differ in triggering spellout to PF only, LF only, or both, and ii) an anaphor must be bound within its minimal LF-phase while a pronominal must be free within its minimal LF-phase. Let’s consider how such a view may be applied to understand the asymmetry between extraction and binding in Khalkha. As I discuss later in Chapter 5, I take complement NCs to shift out of the *vP* phase, given that they must bear accusative case – assigned in Spec,*vP* under my analysis – if they are not already assigned a lexical case (e.g. dative). Additionally, I have

²⁰This is similar to the theory of modular PIC proposed by D’Alessandro and Scheer (2015), in which it is proposed that a phase head may trigger a syntactic PIC effect, a phonological PIC effect, both, or neither. However, D’Alessandro and Scheer (2015) propose that any functional head may be a phase head, and that this choice is language-specific. In this sense, Hicks’ (2009) staggered spellout approach is different in adopting a more conservative set of phase heads (i.e. C^0 , v^0 , D^0 , and n^0).

argued in Chapter 2 that converbial adjuncts adjoin to vP. In short, both complement NCs and converbial adjuncts are not spelled out within the vP phase. Lastly, in this chapter, I have argued that case-unmarked subjects in embedded clauses are located in Spec,TP.

The fact that the nominal superstructure above the embedded clause is accessible for binding by the matrix subject suggests that nP constitutes an LF-phase but DP does not constitute an LF-phase in Khalkha.²¹ This accounts for why embedded subjects and possessors within embedded subjects located below nP are inaccessible for binding by the matrix subject, whereas such elements located within the nominal superstructure above the clause remain accessible. Specifically, upon completion of the nP phase – which I propose is an LF-phase – embedded subjects and possessors within embedded subjects must be bound (if they are anaphors) or free (if they are pronominals) within that phase; failure to do so results in ungrammaticality. In contrast, upon completion of the DP phase, these elements are not yet spelled out within an LF-phase and therefore remain accessible for binding within the next higher LF-phase, namely, the matrix CP phase. Since complement NCs and converbial adjuncts are not spelled out within the matrix vP phase, embedded subjects and possessors within embedded subjects located in Spec,nP or Spec,DP of embedded NCs need to be bound or free only upon completion of the matrix CP phase. Given that the matrix subject (located in Spec,VoiceP at some point in the derivation; see §2.1.1.2) is contained within this phase, it is able to serve as a binder for such embedded elements.

4.6 Conclusion

Drawing on a constellation of syntactic evidence, this chapter has argued for the positions of embedded subjects across four types of embedded clauses in Khalkha: finite bare CPs, finite NCs, nonfinite NCs, and converbial adjuncts. I first considered the distribution

²¹This view is slightly different than Hicks' (2009) account. He proposes that CPs and transitive vPs both constitute PF- and LF-phases (pp.124-126), while DPs and “subjectless” nPs (e.g. nPs that do not introduce a possessor argument, on par with transitive vPs) are PF- but not LF-phases (pp.145-157). For my sketch proposal, I have to stipulate that nPs are always LF-phases in Khalkha.

of subject case across clause types, noting that genitive subjects are restricted to NCs that contain a full DP layer – specifically, finite and nonfinite NCs. This suggests that genitive subjects are licensed within the DP-sized nominal exterior of NCs. I proposed that subjects of finite and nonfinite NCs are assigned genitive case if they move to and remain in Spec,nP.

Second, I investigated long distance extraction of subjects across the four embedded clause types. I found that in finite bare CPs, finite NCs, and nonfinite NCs, only accusative subjects may be displaced from within the embedded clause to a position in the matrix clause. This suggests that accusative subjects occupy the highest phase edge of these embedded clause types – Spec,CP in finite bare CPs and Spec,DP in finite and nonfinite NCs. Subject extraction from converbial adjuncts is independently banned in Khalkha, which may be accounted for under the view that adjuncts are islands for movement.

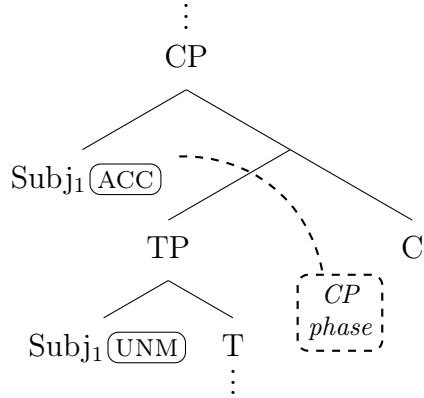
Third, I investigated binding patterns across the four embedded clause types, focusing on i) the satisfaction of Binding Condition A with embedded subject anaphors and embedded subjects containing reflexive possessors; and ii) violations of Binding Condition B with pronominal possessors of embedded subjects. I found that accusative and genitive subjects and possessors within such subjects are accessible to the matrix subject for binding purposes, whereas case-unmarked subjects and possessors within such subjects are not. This suggests that accusative and genitive subjects occupy higher structural positions than case-unmarked subjects. These findings are consistent with the proposal that accusative subjects occupy the highest phase edge within the embedded clause, while genitive subjects occupy Spec,nP within the DP-sized nominal superstructure in finite and nonfinite NCs. The subject extraction and binding facts are summarised in Table 4.6.

Table 4.6: Extraction and binding across four embedded clause types in Khalkha

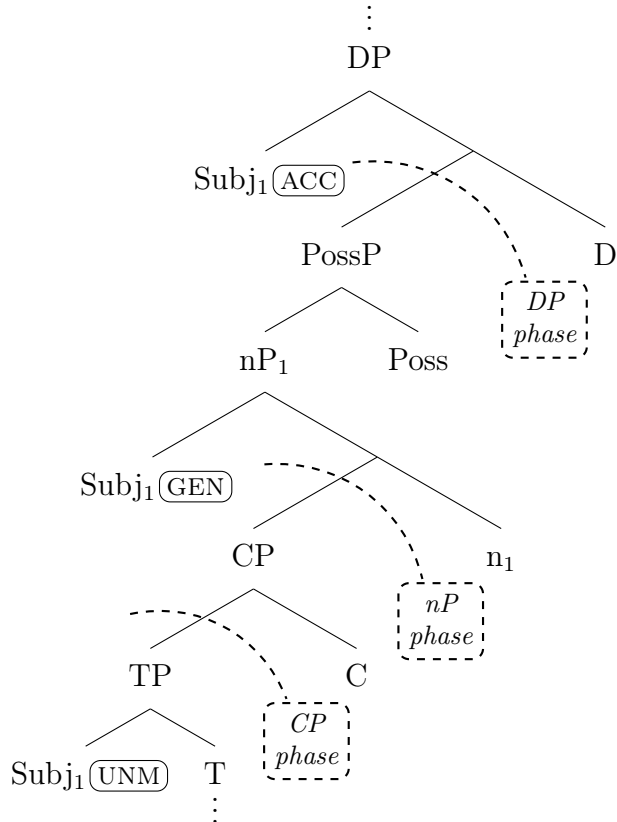
	Embedded subject		
	ACC	GEN	UNM
	(in finite & nonfinite NCs)		
May be extracted into matrix clause?	✓	✗	✗
Visible to matrix subject for binding?	✓	✓	✗

The proposed positions of embedded subjects across the four clause types are shown once again in (276)–(279) below.

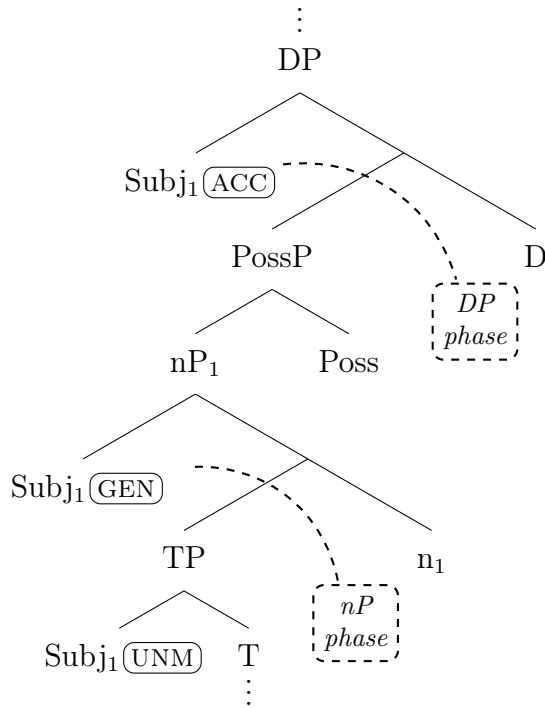
(276) *Positions of ACC and UNM subjects in finite bare CP*



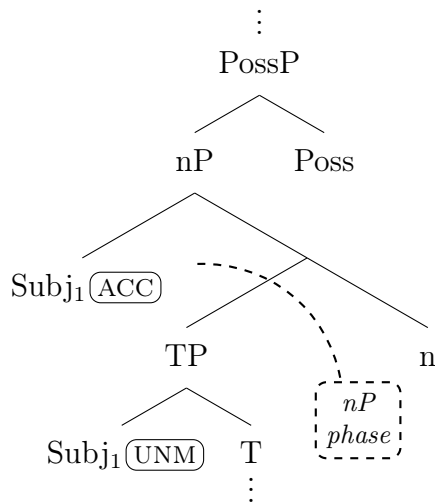
(277) *Positions of ACC, GEN, and UNM subjects in finite NC*



(278) *Positions of ACC, GEN, and UNM subjects in nonfinite NC*



(279) *Positions of ACC and UNM subjects in converbial adjunct*



Beyond establishing the positions of embedded subjects, this chapter has also demonstrated that extraction domains and binding domains are not identical in Khalkha. Crucially, this finding would not have been possible without examining the binding properties of genitive subjects in finite and nonfinite NCs. Had the investigation focused solely on the extraction and binding behaviours of accusative subjects, it would have appeared that binding

obeys the same locality constraints as extraction – constraints that could straightforwardly be modelled under a standard phase-based approach to syntactic locality. However, the behaviour of genitive subjects reveals a different picture: while the entire nominal domain (i.e. DP>PossP>nP) is transparent to binding, only the highest phase edge (e.g. Spec,DP) is accessible for syntactic movement. This suggests that although a standard conception of phase locality may be applied to understand extraction domains, it cannot be straightforwardly applied to binding domains, at least not without modifications. I have discussed one such modification under the staggered spell-out approach proposed by Hicks (2009) (building on Marušič 2005; Matushansky 2005), where phase heads may trigger spellout to PF, LF, or both, and where DP is not an LF-phase. Given that binding dependencies are resolved only within a minimal LF-phase, the completion of the DP phase (a PF- but not LF-phase) is predicted to have no effect on anaphors and pronominals located within that DP, provided they have not already been spelled out within an earlier LF-phase.

The next chapter discusses differential argument marking across the four embedded clause types. Building on those findings and the findings in this chapter, I also propose a dependent account of case assignment that can capture the distribution of accusative and genitive case, as well as of case-unmarked arguments, in Khalkha.

Chapter 5

Differential argument marking and case assignment in Khalkha

This chapter begins with a detailed investigation of accusative-based differential argument marking (DAM) in matrix root clauses, finite bare CPs, finite NCs, nonfinite NCs, and converbial adjuncts. I show that differential object marking (DOM) and differential subject marking (DSM) in Khalkha are regulated by similar referentiality scales that differ only in what types of arguments are obligatorily accusative-marked. These findings contribute to a broader understanding of how case interacts with DAM in Khalkha, as well as what DSM looks like in accusative languages.

Building on the findings from DAM and from Chapters 3 and 4, this chapter then provides an account of accusative and genitive case assignment, as well as of the treatment of case-unmarked nominals, in Khalkha. Adopting a movement-based approach to DAM, I show that accusative is best analysed as a dependent case in Khalkha. I then introduce data on subjects of subject clauses, which provides novel support for an Agree-based approach (Clem and Deal 2024; Poole 2024) over a configurational approach (e.g. Yip et al. 1987; Marantz 1991; Baker and Vinokurova 2010; Baker 2015; Levin and Preminger 2015) to dependent case assignment. Building on the Agree-based approach of Clem and Deal 2024, I argue that

genitive may also be understood as a case assigned by D^0 under Agree. Finally, I propose a unified explanation for the absence of case on objects spelled out within the vP phase, matrix subjects, and embedded subjects that remain in Spec,TP.

This chapter is organised as follows. §5.1 recaps the core case-marking patterns in Khalkha, focusing on subject case in the four types of embedded clauses under consideration: finite bare CPs, finite NCs, nonfinite NCs, and converbial adjuncts. In the same section, I also introduce the idea that differential argument marking may be modelled along a hierarchy known as a referentiality scale. §5.2 provides some background on the notions of definiteness and indefiniteness. §5.3 discusses DAM patterns in Khalkha and shows that the patterns can be fruitfully modelled along a referentiality scale. A significant finding is that the referentiality scales that regulate DOM and DSM in Khalkha turn out to be similar, in contrast to characterisations of DSM in ergative languages, which show the inverse of DOM. §5.4 argues that accusative should be treated as a dependent case in Khalkha, and shows how the positions of accusative subjects can be understood as the interplay between a movement-based approach to DAM and dependent case assignment. Novel data from subjects of subject clauses in Khalkha is introduced in §5.5, which provide some preliminary support for an Agree-based approach to dependent case assignment over a configurational case approach. I show how both accusative subjects in subject clauses – which are challenging for a configurational case approach – and complement clauses can be accounted for under a dependent case via Agree approach. In §5.6, I argue that genitive case may also be modelled as a dependency between a nominal and a distinct D^0 under the Agree-based approach to dependent case. I also propose that case-unmarked nominals in Khalkha – which includes “nominative” subjects – are due to a lack of a specific vocabulary item for case that targets the features present on objects within the vP phase, matrix subjects, and embedded subjects in Spec,TP. §5.7 considers alternative models of case assignment that may be applied to Khalkha. §5.8 concludes.

5.1 Introduction

The preceding chapters have established three interrelated findings. First, as I argued in Chapter 3, the four types of embedded clauses that I investigate in Khalkha may each be uniquely defined in terms of their structure: finite bare CPs contain a full CP layer but lack any nominal structure (280a); finite NCs contain a full CP and DP layer (280b); nonfinite NCs lack a CP layer but contain a full DP layer (280c); and converbial adjuncts lack a CP layer and contain a reduced nominal layer that lacks DP (280d).

(280) *Structure of four embedded clause types in Khalkha*

- a. *Finite bare CP*
[CP [TP ...]]
- b. *Finite NC*
[DP [PossP [nP [CP [TP ...]]]]]
- c. *Nonfinite NC*
[DP [PossP [nP [TP ...]]]]
- d. *Converbial adjunct*
(no DP) [PossP [nP [TP ...]]]

In Chapter 4, by systematically comparing the extraction and binding possibilities for accusative, genitive and case-unmarked subjects in finite bare CPs, finite NCs, nonfinite NCs, and converbial adjuncts, I have argued these subjects are located in different positions within each clause type. These findings are summarised in (281).

(281) *Subjects in four types of embedded clauses in Khalkha*

- a. *Finite bare CP*
[CP Subj^{ACC} [TP Subj^{UNM} ...]]
- b. *Finite NC*
[DP Subj^{ACC} [PossP [nP Subj^{GEN} [CP [TP Subj^{UNM} ...]]]]]
- c. *Nonfinite NC*
[DP Subj^{ACC} [PossP [nP Subj^{GEN} [TP Subj^{UNM} ...]]]]
- d. *Converbial adjunct*
(no DP) [PossP [nP Subj^{ACC} [TP Subj^{UNM} ...]]]

(281) shows that while there is no one unique structural position associated exclusively with accusative subjects, I have proposed that they are located in the specifier of the highest

phase head of the (nominalised) clause – Spec,CP of finite bare CPs, Spec,DP of finite and nonfinite NCs, and Spec,nP of converbial adjuncts. Genitive subjects are located in Spec,nP within a DP superstructure, which is found only in finite and nonfinite NCs. Case-unmarked subjects are located in Spec,TP of all four embedded clause types.

An example of each clause type containing the different subject case possibilities is shown again in (282).¹ These facts have also been noted in previous literature, e.g. Guntsetseg 2010, 2016; Klein et al. 2012; Fong 2019; Aravind 2021; Gong 2022, 2023a, 2025; Peters 2020, 2024.

(282) *Case alternation in four types of embedded clauses in Khalkha*

a. *Finite bare CP*

Naraa Dorj-id [margaash **Tuyaa-g/Tuyaa** ir-ne gej] hel-sen.
 N. D.-DAT tomorrow T.-ACC/T. come-NPST C say-PST
 ‘Naraa told Dorj that tomorrow Tuyaa will come.’

b. *Finite NC*

Naraa [margaash **Tuyaa-g/Tuyaa-giin/Tuyaa** ir-ne gedg]-iig
 N. tomorrow T.-ACC/T.-GEN/T. come-NPST C.NML -ACC
 sons-son.
 hear-PST
 ‘Naraa heard that Tuyaa will come tomorrow.’

c. *Nonfinite NC*

Naraa [margaash **Tuyaa-g/Tuyaa-giin/Tuyaa** ir-eh]-iig med-ne.
 N. tomorrow T.-ACC/T.-GEN/T. come-INF -ACC know-NPST
 ‘Naraa knows that Tuyaa will come tomorrow.’

d. *Converbial adjunct*

Naraa [**Tuyaa-g/Tuyaa** hool hii-hed] video togloom toglo-son.
 N. T.-ACC/T. food do-TEMP.CVB video game play-PST
 ‘Naraa played video games while Tuyaa cooked.’

As noted by Guntsetseg 2008, 2010, 2016; von Heusinger et al. 2011; Klein et al. 2012, Khalkha is a language that displays *differential subject marking* (DSM) and *differential object marking* (DOM). DSM and DOM fall under the umbrella of *differential argument marking* (DAM), which refers to the fact that some languages may mark the same nominal expression

¹As discussed in Chapter 2, the position of temporal adverbs does not accurately diagnose the position of subjects, as these adverbs can freely precede or follow subjects, unlike aspectual and manner adverbs (cf. Gong 2023a). For example, in (282a), I will simply assume that temporal adverbs may scramble to an outer Spec,CP above the accusative-marked subject.

differently (e.g. via case marking) based on various factors, including but not limited to definiteness, specificity, and animacy. For typological overviews and analyses of DAM, see, e.g. Aissen 1999, 2003; Sinnemäki 2014; Witzlack-Makarevich and Seržant 2018; de Hoop and de Swart 2009; Bárány and Kalin 2020a. In this chapter, I will show that Khalkha DAM can be fruitfully understood as a process that is sensitive to types of nominal and subtypes of definiteness and specificity. Before delving into DAM in Khalkha, I discuss how DAM may be conceptualised along a referentiality scale.

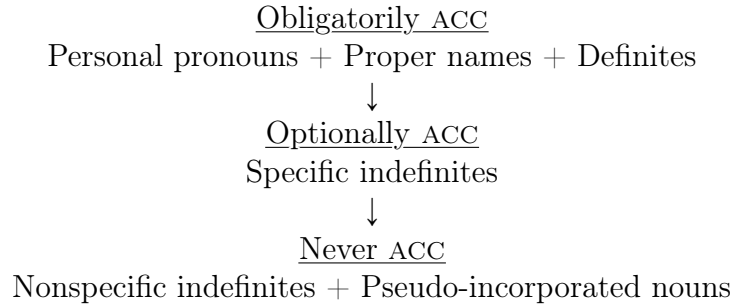
5.1.1 DAM and referentiality scales

The idea of a referentiality scale originated in works such as Silverstein 1976; Croft 1988; Comrie 1989, and has been adopted and developed by many, e.g., Aissen 2003; Von Heusinger 2008; Dryer 2014; Ahn 2017; Becker 2021. A referentiality scale is conceptualised as a hierarchy in which more referential expressions outrank less referential expressions. Such hierarchies are often used to explain DAM patterns. For example, in Turkish, definite and specific objects occupy the top of the hierarchy and are marked with accusative case while nonspecific indefinite objects may not be marked accusative (Enç 1991; Von Heusinger and Kornfilt 2005).

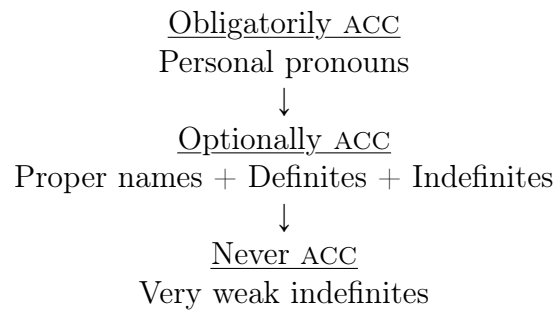
Guntsetseg (2016) has applied the idea of a referentiality scale to explain DAM in Khalkha. Adopting an expanded scale which includes personal pronouns and proper names (Von Heusinger 2008), Guntsetseg proposes the following scales for Khalkha DOM and DSM, shown in (283) and (284).²

²Guntsetseg (2016) also considers a separate scale for DSM in Khalkha that considers adjacency between the matrix subject and the embedded subject. For example, the embedded subject may be non-adjacent to the matrix subject if there is an intervening adverb between them, or if the entire complement clause is scrambled in front of the matrix subject. Based on an experimental study, she finds that participants prefer accusative case over no case marking on embedded subjects if they are immediately adjacent to the matrix subject, while they judge accusative or no case marking on embedded subjects as equally acceptable when they are non-adjacent to the matrix subject. I set aside adjacency as a factor for DAM in Khalkha in this chapter.

(283) *Guntsetseg's (2016, p.112) referentiality scale for Khalkha DOM*



(284) *Guntsetseg's (2016, p.150) referentiality scale for Khalkha DSM*



By ‘pseudo-incorporated noun’, Guntsetseg refers to the direct object in examples like (285), which may appear as a bare noun (285a) or a modified noun (285b). Guntsetseg argues that these nouns cannot bear accusative case marking.

- (285) a. Tuyaa bainga **nom(*-iig)** unsh-dag.
 T. always book(*-ACC) read-HAB
 ‘Tuyaa always reads books (i.e. does book-reading).’

(Guntsetseg 2016, p.84, ex.230)

- b. Tuyaa bainga **zuzaan nom(*-iig)** unsh-dag.
 T. always thick book(*-ACC) read-HAB
 ‘Tuyaa always reads thick books (i.e. does thick-book-reading).’

(Guntsetseg 2016, p.85, ex.232)

Barrie and Kang (2022) show that a bare (i.e. lacking plural or case marking) object noun is compatible with a pseudo-incorporated reading (286a) and a non-pseudo-incorporated reading (286b), and that the two are additionally distinguishable based on their prosodic profiles. Barrie and Kang (2022) propose that all bare object nouns that have

a narrow scope reading are pseudo-incorporated (under the view that they are reduced nPs), while bare object nouns that have a wide scope reading are not pseudo-incorporated.

- (286) a. Bi **güü** saa-maar bai-na ... al' ch güü hamaagüi.
 1SG mare milk-WANT COP-NPST which FOC mare irrelevant
 'I want to milk a mare... any mare will do.' WANT > $\exists$
- b. Bi **güü** saa-maar bai-na ... ter tsagaan güü.
 1SG mare milk-WANT COP-NPST that white mare
 'I want to milk a mare... that white mare.' $\exists$ > WANT

(Barrie and Kang 2022, p.415, ex.15)

For the purposes of this dissertation, I assume that not all case-unmarked object nouns with narrow scope readings are necessarily pseudo-incorporated. Further investigation into the prosody of such nouns may clarify whether true non-pseudo-incorporated examples exist. See Guntsetseg (2016), Driemel (2020, 2023), Barrie and Kang (2022); Kang (2023), and Barrie (2024) for further discussion of pseudo-noun incorporation in Khalkha.

By 'very weak indefinite NPs', Guntsetseg refers to indefinite expressions formed by combining a *wh*-item, the numeral *neg* 'one', and an NP, as seen in (287).

- (287) Tuyaa [**hen neg hün(*-iig)** ger-t=n' hulgai hii-sn]-iig med-sen.
 T. who one person(*-ACC) house-DAT=3.POSS theft do-PFV -ACC know-PST
 'Tuyaa knew that someone committed a theft at her home.'

(Guntsetseg 2016, p.150, ex.345)

Other possible candidates for 'very weak indefinite NPs' are NPIs; Aravind (2021) reports that NPIs such as the one in (288), formed by the numeral *neg* 'one', the focus marker *ch*, and an NP, cannot be accusative-marked:

- (288) Bi [**neg ch hün(*-iig)** Naraa-d tustl-aa-güi]-g chamd hel-sen.
 1SG one FOC person(*-ACC) N.-DAT help-IPFV-NEG -ACC 2SG.DAT say-PST
 'I told you that no one helped Naraa.' (Aravind 2021, p.381, ex.11)

However, I have not been able to replicate Guntsetseg's and Aravind's consultants' judgements with my three consultants. As I have showed in §4.2.2, embedded NPI subjects may

be accusative-marked for my consultants. Further, an embedded *wh*-item, which is regarded as nonspecific indefinite, may be an accusative-marked subject:

(289) *Embedded wh-subjects may be accusative-marked*

Context: Bilguun and his friends are having a meal at a restaurant. At the end of the meal, Bilguun leaves the table for awhile. When he returns, someone has already paid the bill.

Bilgüün [**hen-iig/hen** tootsoo(-g) hii-sn]-iig asuu-san.

B. who-ACC/who bill(-ACC) do-PFV -ACC ask-PST

‘Bilguun asked who paid the bill.’

Guntsetseg (2016) investigates Khalkha DAM using an experimental approach, considering factors such as the referentiality and animacy of the direct object, linear adjacency between the matrix and embedded subjects, and embedded clause type. However, for embedded clause types, it appears that Guntsetseg distinguishes only between nonfinite NCs and postpositional adjunct clauses, based on the example sentences she provides for her test conditions (p.183). Thus, one of the goals of this chapter is to offer a finer-grained picture of DOM and DSM in Khalkha by examining the correlations between case marking and different subtypes of definiteness and specificity across four types of embedded clause: finite bare CPs, finite NCs, nonfinite NCs, and converbial adjuncts.

5.2 Definiteness and indefiniteness

Before I discuss definiteness and specificity in Khalkha, I provide some background on these concepts, specifically in relation to the referential functions of nominal expressions. The notion of definiteness can be further divided into unique definite and familiar definite uses. The notion of indefiniteness can be further divided into specific vs. nonspecific and wide scope vs. narrow scope uses. I discuss each of these subtypes in turn.

5.2.1 Definiteness

There are two prominent approaches to understanding definite descriptions in natural language. One approach argues that definite descriptions should be understood as imposing a uniqueness requirement on the entity that is being referred to (Frege 1892/1997, Russell 1905; Strawson 1950; Abbott 1999; Roberts 2003, a.m.o.). Another approach argues that the relevant requirement should be understood in terms of familiarity or anaphoricity (Christophersen 1939; Karttunen 1968; Heim 1982, 1983, a.m.o.). There are also approaches that argue that neither approach is correct (e.g. Birner and Ward 1994; Lyons 1999), or that both are needed (e.g. Schwarz 2009).

While it is a matter of continued debate whether either approach can be reduced to the other, I will follow Schwarz 2009 in adopting the view that both perspectives are logically independent and compatible with each other. Below, I discuss several subtypes of definite descriptions that have been identified, as shown in Table 5.1 (see also Hawkins 1978; Schwarz 2009; Becker 2021; Dawson and Jenks 2023):

Table 5.1: Types of definite descriptions

Type	Subtype	Example
Familiarity/ Anaphoricity	Anaphoric	<i>I adopted a cat and a dog. The cat ran away.</i>
	Producer-product bridging	<i>I bought a painting. The painter is from Mongolia.</i>
	Donkey anaphora	<i>Every farmer who owns a donkey beats the donkey.</i>
Uniqueness	Immediate situation	<i>The desk is dirty.</i> (uttered in a room with only one desk)
	Larger situation	<i>The king is old.</i> (uttered in the UK)
	Part-whole bridging	<i>I went to a wedding. The bride wore a suit.</i>

Under familiarity/anaphoricity accounts of definiteness, an anaphoric referent is understood as definite as it is identical to a previously mentioned referent, and thus unambiguously identifiable by both the speaker and hearer. Under an account that treats definiteness as encoding uniqueness, a referent can be unique in an immediate situational context (e.g.

in a room where it is the only referent of its kind) or a larger situational context (e.g. in a country where it is the only referent of its kind). Some researchers (e.g. Becker 2021) define a further category of absolutely unique referents, which are understood as the only unique referent of its kind (in our physical world), such as *the sun* or *the moon*. Following Hawkins 1978 and Schwarz 2009, I take these to fall under larger situation unique definites.

The inclusion of producer-product bridging definites as a subtype of familiar/anaphoric definiteness and part-whole bridging definites as a subtype of unique definiteness deserves further comment. Schwarz (2009) shows that in German and Fering (North Frisian), producer-product bridging referents take the so-called strong article – the same form used to mark anaphoric definites – while part-whole bridging referents take the so-called weak article – the same form used to mark unique definites. The shared morphosyntactic patternings suggest that anaphoric definites and producer-product bridging definites form a natural subclass, whereas unique definites and part-whole bridging definites form another natural subclass, as shown in Table 5.1. Schwarz (2009) explains this distinction by way of analogy with a situational containment relationship. For part-whole bridging definites, a previously mentioned entity (the “whole”) establishes a context in which it is uniquely identifiable, and reference to a “part” of that entity – where it is the only one of its kind in relation to the whole – entails its uniqueness as well. Put differently, the “part” is licensed by a containment relationship shared with the previously-mentioned “whole”. However, for producer-product bridging definites, prior mention of a “product” does not necessarily evoke a context in which the “producer” is uniquely identifiable. For instance, the speaker of the utterance containing the producer-product bridging referent in Table 5.1 could have bought the painting in a store where no painter is present. Thus, Schwarz argues that producer-product bridging definites are licensed not by a containment relation that entails uniqueness, but by an anaphoric dependency.

As an aside, donkey anaphora refer to constructions in which a definite expression covaries with a preceding indefinite expression. In many languages such as English and

Mandarin (Jenks 2018), donkey anaphora pattern morphosyntactically with anaphoric definites, suggesting that these expressions may form a natural subclass. Since Khalkha uses a reflexive-possessive strategy for expressing donkey anaphora, I will leave them out of the following discussion.³

5.2.2 Indefiniteness

Indefinite descriptions may be described in terms of their specificity or scopal properties, as shown in Table 5.2 (English examples from Becker 2021):

Table 5.2: Types of indefinite descriptions

Type	Subtype	Example
Specificity	Specific	<i>I saw a new neighbour move in yesterday.</i>
	Nonspecific	<i>Lea wants to buy a bike, but she doesn't know if she will find one.</i>
Scope	Wide scope	See (290a) below.
	Narrow scope	See (290a) and (290b) below.

³In Khalkha, only (iv) can have a covarying, donkey anaphoric reading:

- (i) Mor' bai-dag hün bür **mor'** una-dag.
Horse be-HAB person every horse ride-HAB
'Every person who has a horse rides a horse (in general). (non-covarying reading)'
- (ii) Mor' bai-dag hün bür **mor-iig** una-dag.
Horse be-HAB person every horse-ACC ride-HAB
'Every person who has a horse rides a (certain) horse. (non-covarying reading)'
- (iii) Mor' bai-dag hün bür **mor-iig=n'** una-dag.
Horse be-HAB person every horse-ACC=3.POSS ride-HAB
'Every person_j who has a horse rides his/her_k horse. (non-covarying reading)'
- (iv) Mor' bai-dag hün bür **mor-Ø=io** una-dag.
Horse be-HAB person every horse-ACC=RP ride-HAB
'Every person_j who has a horse rides his/her_j horse. (covarying reading)'

As I show in §5.3, accusative-marked objects are compatible with anaphoric readings in Khalkha. This may seem at odds with (ii) and (iii), where an accusative-marked *mor-iig* 'horse' does not license a donkey anaphoric reading. However, donkey anaphora differ from classical anaphora (e.g. pronominals) in that their covarying readings typically involve a binding dependency between the antecedent and the anaphor. While donkey anaphora may pattern morphosyntactically with anaphoric definites in languages such as English and Mandarin (Jenks 2018), this parallel does not entail that they should be treated like classical anaphors.

In the specific indefinite example above, both the speaker and hearer understand “a new neighbour” to refer to a particular referent, even though the speaker but not the hearer is able to unambiguously identify the referent. Becker (2021) also discusses specific indefinites where both the speaker and hearer are not able to unambiguously identify a particular referent, e.g. *Sarah wanted to talk to a colleague of mine, but I don’t know which one.* (example from Ionin 2006, p.182). While a particular referent is evoked by “a colleague of mine”, the identity of the colleague is unknown by the speaker and hearer.

Nonspecific indefinites differ from specific indefinites in that no particular referent is evoked in the speaker and hearer’s minds by the expression, such as “a bike” in the example in Table 5.2.

Indefinites are also often characterised in relation to scope. While specific indefinites typically correlate with a wide scope reading and nonspecific indefinites with a narrow scope reading, these are distinct concepts which I keep separate. As an illustration of scopal indefinites, in Spanish, an object marked with *a*⁴ is ambiguous between a wide scope and narrow scope reading (290a), while an unmarked object can only be interpreted with narrow scope (290b).

(290) Spanish

- a. Todo hombre amó **a** una mujer.
 every man loved ACC a woman
 ‘For every man, there was a woman that he loved.’ $\forall > \exists$
 ‘There was a woman that every man loved’ $\exists > \forall$
- b. Todo hombre amó una mujer
 every man loved a woman
 ‘For every man, there was a woman that he loved.’ $\forall > \exists$
 ‘There was a woman that every man loved’ $\exists > \forall$

(López 2012, pp.13-14)

In (290a), the *a*-marked *una mujer* ‘a woman’ may take wide scope or narrow scope with respect to *todo hombre* ‘every man’. In (290b), *una mujer* lacks *a*-marking, and can only

⁴There are various treatments of Spanish *a*, e.g. as a preposition, an accusative case marker, or a DOM marker. For instance, Von Heusinger (2008, p.2) writes that “[i]n European Spanish, human direct objects can be *a*-marked if they have certain referential properties, while inanimate direct objects are generally unmarked.”

be interpreted with narrow scope with respect to *todo hombre*. López argues that *a*-marked nominals may undergo short scrambling to a VP-external position but nominals that lack *a* may not. In this VP-external position, *a*-marked nominals can be targeted by a choice function which allows them to be interpreted with wide scope over other scope-taking operators (e.g. *todo hombre* ‘every man’) that they do not c-command.

In the following discussion on indefinites in Khalkha, I will focus on wide scope vs. narrow scope indefinites, leaving an investigation of specificity to future work.

5.3 Differential argument marking in Khalkha

In this section, I investigate DAM in Khalkha, i.e. how accusative case marking on direct objects and embedded subjects correlates with different types of definite and indefinite interpretations. Previous research on Khalkha has discussed the interaction of DAM with factors such as nominal type, definiteness, indefiniteness, animacy, the relative definiteness and animacy of matrix subject and embedded subject, and the adjacency of the matrix subject and embedded subject (e.g. Guntsetseg 2008, 2010, 2016; von Heusinger et al. 2011; Klein et al. 2012). However, this section shows that paying attention to both nominal types and the different definiteness and indefiniteness distinctions discussed in §5.2 allows for a finer-grained view of DAM in Khalkha.

Before delving into the patterns, I briefly discuss another type of nominal marking which I take to be orthogonal to the case marking patterns that are the present focus of this chapter. As noted in previous literature (e.g. Hammar 1983; Guntsetseg 2016; Gong 2022), the 3.POSS marker *=n'* appears to have an additional non-possessive, partitive-like function in Khalkha, as shown in (291).

(291) Context: There are some fruits on the table.

Hojoṛ alim=**n'** amttai bai-san.

Two apple=3.POSS tasty be-PST

‘Two apples (of those fruits) were tasty.’

(Guntsetseg 2016, p.53, ex.139; translation modified)

I will henceforth refer to such uses of n' as non-possessive $=n'$ or simply n' -marking. As I demonstrate below, three types of definite descriptions – part-whole bridging definites, producer-product bridging definites, and anaphoric definites referring to a member of a previously-mentioned non-singleton set – allow the use of the non-possessive $=n'$ seen in (291). While a full account of non-possessive n' -marking is beyond the scope of this dissertation, I offer two key observations regarding its felicitous use. First, non-possessive $=n'$ requires plurality in the antecedent, either in the form of multiple entities or a single entity that can be conceptualised as comprising multiple subparts. Second, its use depends on the n' -marked nominal sharing a contextually salient relation with a plural antecedent, either through co-occurrence in discourse or via world knowledge. In this way, non-possessive $=n'$ serves to highlight the two-place relation linking the n' -marked nominal to its plural antecedent.

In the following few subsections, I will discuss the interaction of DAM with different types of definiteness and indefiniteness distinctions in the following order: Pronouns > Proper names > Anaphoric definites > Bridging definites > Unique definites > Indefinites. I begin by discussing DOM in matrix root clauses before discussing DSM across four embedded clause types: finite bare CPs, finite NCs, nonfinite NCs, and converbial adjuncts. I present the judgements of my three main consultants – speakers B, G, and M – on the use of accusative-marked arguments and arguments without case marking with various types of definite and indefinite descriptions. Where interspeaker variation occurs, I indicate the judgements of individual consultants; otherwise, the examples reflect converging judgements across all three consultants.

5.3.1 Differential object marking

In direct object position, pronouns and proper names are obligatorily accusative-marked (see also Guntsetseg 2016, pp.81-82 and Aravind 2021), as shown in (292). The case-unmarked form of pronouns and a case-unmarked proper name cannot appear in direct object position, as shown in (292b).

- (292) *Case marking of nominal types in matrix root clauses*
- a. Naraa **namaig/chamaig/tüüniig/Bat-iig** har-san.
 N. 1SG.ACC/2SG.ACC/3SG.ACC/B.-ACC see-PST
 ‘Naraa saw me/you/her/him/Bat.’
 - b. *Naraa bi/chi/ter/Bat har-san.
 N. 1SG/2SG/3SG/B. see-PST
 Int. ‘Naraa saw me/you/her/him/Bat.’

When we consider anaphoric definites, Khalkha makes a distinction between whether an anaphoric referent is identical to a previously mentioned entity (293), or whether the anaphoric referent is identical to a member of a previously-mentioned, non-singleton set of entities (294). In the former, which I refer to as *anaphoric* (=1) contexts, accusative case marking is obligatory for all three speakers, as shown in (293b). In the latter, which I refer to as *anaphoric* (>1) contexts, accusative and non-possessive *n'* are obligatory for all three speakers, as shown in (294b).

- (293) *Anaphoric (=1) object in matrix root clause*
- a. Bi öchigdör ger-t=ee hulgana har-san.
 1SG yesterday house-DAT=RP mouse see-PST
 ‘I saw a mouse in my house yesterday.’
 - b. # Önöödör manai ah **hulgana** bar'-san.
 Today 1PL.GEN older.brother mouse catch-PST
 ‘Today my older brother caught a mouse.’
 - c. Önöödör manai ah **hulgan-iig** bar'-san.
 Today 1PL.GEN older.brother mouse-ACC catch-PST
 ‘Today my older brother caught the mouse.’

(294) *Anaphoric (>1) object in matrix root clause*

- a. Bi tüürüü doloo honog-t neg muur, neg nohoi, neg tuulai av-san.
1SG past seven day-DAT one cat, one dog, one rabbit buy-PST
'Last week I bought a cat, a dog, and a rabbit.'
- b. # Gehdee Naraa **tuulai(=n')** al-chih-san.
But N. rabbit(=3.POSS) kill-COMPL-PST
'But Naraa has killed the rabbit.'
- c. # Gehdee Naraa **tuulai-g** al-chih-san.
But N. rabbit-ACC kill-COMPL-PST
'But Naraa has killed the rabbit.'
- d. Gehdee Naraa **tuulai-g=n'** al-chih-san.
But N. rabbit-ACC=3.POSS kill-COMPL-PST
'But Naraa has killed the rabbit.'

Turning next to bridging definites, in producer-product bridging contexts, all three consultants accept *n'*-marked accusative objects (295d), while speaker G also accepts accusative objects without *n'*-marking (295c). Put differently, regardless of whether *n'*-marking is present or not, all three consultants require accusative case marking on producer-product bridging objects.

(295) *Producer-product bridging object in matrix root clause*

Context: You and your friend are looking at a painting. You say to your friend...

- a. Ene zurag üzesgelentei yum.
This painting beautiful EVID.COP
'This painting is beautiful.'
- b. # Bi **zuraach(=n')** rvestoran-d har-san.
1SG painter.(=3.POSS) restaurant-DAT see-PST
'I saw the painter at a restaurant.'
- c. Bi **zuraach-iig** rvestoran-d har-san.
1SG painter-ACC restaurant-DAT see-PST
'I saw the painter at a restaurant.'
- d. Bi **zuraach-iig=n'** rvestoran-d har-san.
1SG painter-ACC=3.POSS restaurant-DAT see-PST
'I saw the painter at a restaurant.'

B:✗, G:✓, M:✗

In part-whole bridging contexts, all consultants accept *n'*-marked accusative objects (296d), while speaker G also accepts accusative objects without *n'*-marking (296c). The same pattern found in producer-product bridging definites is observed: regardless of whether *n'*-marking is present or not, all three consultants require accusative case marking on part-whole bridging objects.

(296) *Part-whole bridging object in matrix root clause*

Context: Bat's rat was ripped to pieces by a cat. Tuyaa tells you...

- a. Bat-iin hulgana muur-and tasd-uul-san.

B.-GEN mouse cat-DAT tear-PASS-PST

'Bat's mouse was torn up by a cat.'

- b. # Bat **süül(=n')** hai-j bai-na.

B. tail(=3.POSS) look.for-PROG COP-NPST

'Bat is looking for the tail.'

- c. Bat **süül-iig** hai-j bai-na.

B. tail-ACC look.for-PROG COP-NPST

'Bat is looking for the tail.'

B:✗, G:✓, M:✗

- d. Bat **süül-iig=n'** hai-j bai-na.

B. tail-ACC=3.POSS look.for-PROG COP-NPST

'Bat is looking for the tail.'

Turning next to unique definites, (297) shows that accusative case marking is compatible with but optional for a larger situation unique referent such as *the moon*. This is unlike the previous examples where the object is obligatorily marked accusative.

(297) *Larger situation unique object in matrix root clause*

Context: There was a lunar eclipse last night. The next day, you ask your friend...

Chi öchigdör **sar(-iig)** har-san uu?

2SG yesterday moon(-ACC) see-PST Q

'Did you see the moon yesterday?'

Accusative case marking is also compatible but optional with immediate situation unique referents, as shown in (298).

(298) *Immediate situation unique object in matrix root clause*

Context: You, your friend, and a dog are in the same room. You and your friend are reading a book. After one hour, you say to your friend...

Bi odoo **nohoi(-g)** hoollo-no.

1SG now dog(-ACC) feed-NPST

'I'm going to feed the dog now.'

Turning to indefinites, a case-unmarked object may interact with other scope-taking expressions in a root clause, e.g. *ödör бүр* ‘every day’, giving rise to ambiguity as seen in (299a). On the other hand, accusative-marked objects are obligatorily interpreted with wide-scope over other scope-taking operators (299b).⁵ In short, case-unmarked objects are compatible as wide or narrow scope indefinites, while accusative-marked objects are compatible as wide but not narrow scope indefinites.

(299) *Wide vs. narrow scope indefinite object in matrix root clause*

- a. *Ödör бүр* *Naraa har muur* *hoollo-dog*.
 Day every N. black cat feed-HAB
 Reading 1: ‘Every day, Naraa feeds a (different) black cat.’ $\forall > \exists$
 Reading 2: ‘Naraa feeds a (certain) black cat every day.’ $\exists > \forall$
- b. *Ödör бүр* *Naraa har muur-iig* *hoollo-dog*.
 Day every N. black cat-ACC feed-HAB
 #Reading 1: ‘Every day, Naraa feeds a (different) black cat.’ $\# \forall > \exists$
 Reading 2: ‘Naraa feeds a (certain) black cat every day.’ $\exists > \forall$

There is interspeaker variation as to the relevant notion of specificity associated with accusative-marked objects. For (299b) to be felicitous, speakers B and M share the intuition that the black cat must be known by both the speaker and hearer, while speaker G possesses the intuition that the speaker but not necessarily the hearer knows a certain black cat. See also Driemel (2023, pp.73-74), who finds that DOM does not correlate with epistemic specificity in Khalkha. At this point, a potential alternative would be to treat the object in (299b) as a definite. However, this can be ruled out because objects modified by a strong quantifier, e.g. *bür* ‘every’, in Khalkha obligatorily bear accusative marking, as shown in (300); see also Guntsetseg 2016, pp.83-84 and Driemel 2023, p.69. This is unexpected if accusative case solely marks definites in Khalkha.

⁵Under the movement-based analysis proposed by López (2012) for Spanish, *a*-marked objects optionally scramble to a VP-external position where they can be interpreted with wide scope via a choice function, as shown in (290a). The same analysis, applied to Khalkha, would suggest that accusative-marked objects obligatorily scramble to a vP-external position where they can be interpreted with wide scope via a choice function, as shown in (299b). Thus, accusative-marked objects need not c-command an external argument subject, which is introduced in Spec,VoiceP, in order to take scope over it.

- (300) Context: There are many cats in the neighbourhood. Naraa likes black cats, so she feeds only the black cats.
 Naraa (oir haviin) **har muur бүр*(-iig)** hoollodog.
 N. (nearby) black cat every*(-ACC) feed-HAB
 ‘Naraa feeds every black cat (nearby).’

Second, with the exception of *hen* ‘who’ (301a), accusative case marking is optional on *wh*-items (301b) and negative indefinites (302) – objects that are typically considered non-specific.⁶ This is also unexpected if accusative case solely marks definites.

- (301) a. Chi **hen*(-iig)** har-san be?
 2SG who*(-ACC) see-PST WHQ
 ‘Who did you see?’
 b. Chi **yuu(-g)** har-san be?
 2SG what(-ACC) see-PST WHQ
 ‘What did you see?’
- (302) a. Dulmaa **neg ch hün(-iig)** har-aa-güi
 D. one FOC person(-ACC) see-IPFV-NEG
 ‘Dulmaa did not see anyone.’
 b. Dulmaa **yuu(-g) ch** har-aa-güi
 D. what(-ACC) FOC see-IPFV-NEG
 ‘Dulmaa did not see anything.’

In this chapter, I focus on the scopal properties of accusative case marking on indefinite nominals, excluding *wh*-items and NPIs. Thus, the generalisations I propose regarding the relationship between DAM and the scopal behaviour of indefinites apply only to non-*wh*, non-NPI indefinites. I leave for future research the question of why *wh*-items and NPIs pattern differently from regular indefinites with respect to DAM, as well as how DAM interacts with other factors such as animacy and specificity. The DOM patterns observed across all three speakers are represented in Table 5.3 below.

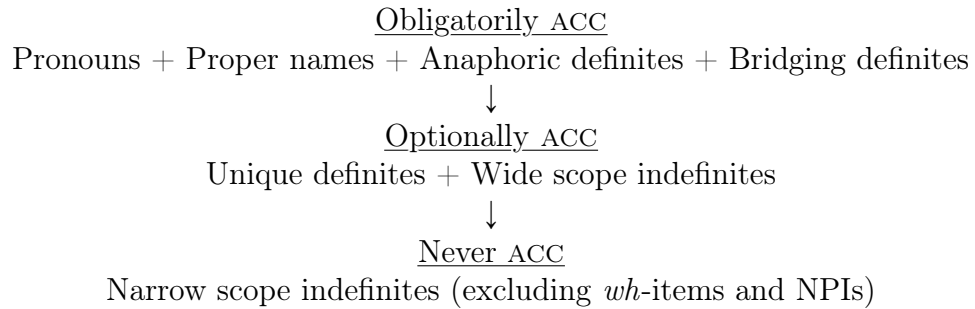
⁶While I do not have an explanation for the contrast in (301), the data suggest that animacy might be a relevant factor in deriving DOM of *wh*-items.

Table 5.3: DOM in Khalkha

	Nominal type											
	1st person			2nd person			3rd person			Proper name		
	B	G	M	B	G	M	B	G	M	B	G	M
UNM	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗
ACC	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
	Immediate anaphora						Bridging					
	=1 mentioned			>1 mentioned			Producer-product			Part-whole		
	B	G	M	B	G	M	B	G	M	B	G	M
UNM	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗
ACC	✓	✓	✓	✓ ⁺	✓ ⁺	✓ ⁺	✓ ⁺	✓ ⁽⁺⁾	✓ ⁺	✓ ⁺	✓ ⁽⁺⁾	✓ ⁺
	Unique						Indefinite					
	Larger situation			Immediate situation			Wide scope			Narrow scope		
	B	G	M	B	G	M	B	G	M	B	G	M
UNM	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
ACC	✓	✓	✓	✓	✓	✓	✓	✓	✓	✗	✗	✗
✓: Acceptable ✗: Unacceptable +: 3.POSS obligatory (+): 3.POSS optional												

The patterns above may be organised into a referentiality scale for Khalkha DOM as shown in (303). Pronouns, proper names, anaphoric definites, and bridging definites, which are obligatorily accusative-marked, are ranked higher than unique definites and wide scope indefinites, which may be accusative-marked or case-unmarked. Ranked lowest are narrow scope indefinites, which may only be case-unmarked.

(303) *Referentiality scale for Khalkha DOM*



In the following subsections, I investigate how DSM interacts with the different types of definite and indefinite distinctions across the four types of embedded clauses under consideration: finite bare CPs, finite NCs, nonfinite NCs and converbial adjuncts. As we will

see, while the DSM patterns resemble the DOM pattern discussed above, they differ in what types of arguments may be optionally accusative-marked.

5.3.2 DSM in finite bare CPs

Finite bare CPs are embedded CPs without a nominal exterior. The first difference that can be observed between DOM and DSM in finite bare CPs concerns the marking of nominal types. In finite bare CPs, 1st and 2nd person subjects are obligatorily marked with accusative case, while 3rd person pronominal and proper name subjects are optionally accusative-marked or case-unmarked. This is different than what was observed previously for DOM, where all pronouns and proper names are obligatorily accusative-marked.

(304) *Case marking of nominal types in Khalkha finite bare CPs*

- a. Naraa Dorj-id [margaash **namaig**/***bi** ir-ne gej] hel-sen.
 N. D.-DAT tomorrow 1SG.ACC/*1SG come-NPST C say-PST
 ‘Naraa said to Dorj that I will come tomorrow.’
- b. Naraa Dorj-id [margaash **chamaig**/***chi** ir-ne gej] hel-sen.
 N. D.-DAT tomorrow 2SG.ACC/*2SG come-NPST C say-PST
 ‘Naraa said to Dorj that you will come tomorrow.’
- c. Naraa_j Dorj-id_k [margaash **tüüniig**/**ter**_{l/*j/*k} ir-ne gej] hel-sen.
 N. D.-DAT tomorrow 3SG.ACC/3SG come-NPST C say-PST
 ‘Naraa_j said to Dorj_k that she_{l/*j}/he_{l/*k} will come tomorrow.’
- d. Naraa [margaash **Dorj-iig**/**Dorj** ir-ne gej] hel-sen.
 N. tomorrow D.-ACC/D. come-NPST C say-PST
 ‘Naraa said that Dorj will come tomorrow.’

Turning to anaphoric definites, in an anaphoric (=1) context (305), all three consultants accept the use of an accusative or case-unmarked subject (305b). This is different from what was observed in Khalkha DOM previously, where anaphoric (=1) objects must bear accusative case.

(305) *Anaphoric (=1) subject in finite bare CP*

- a. Margaash manai surguul'-d aldartai hiilch ir-ne.
 Tomorrow 1PL.EXCL.GEN school-DAT famous violinist come-PST
 'Tomorrow a famous violinist will come to our school.'
- b. Manai bagsh [**hiilch/hiilch-iig** tavan duu toglo-no gej]
 1PL.EXCL.GEN teacher violinist/violinist-ACC five song play-NPST C
 hel-sen.
 say-PST
 'Our teacher said that the violinist will play five songs.'

In an anaphoric (>1) context (306), all three consultants accept accusative-marked (306b) and case-unmarked (306c) subjects, with or without *n'*-marking. This is a different pattern than what was observed in Khalkha DOM, where anaphoric (>1) direct objects obligatorily bear accusative case and *n'*-marking.

(306) *Anaphoric (>1) subject in finite bare CP*

- a. Margaash manai surguul'-d aldartai hiilch tögöldör huurch hoyor
 Tomorrow 1PL.EXCL.GEN school-DAT famous violinist pianist two
 ir-ne.
 come-PST
 'Tomorrow a famous violinist and pianist will come to our school.'
- b. Manai bagsh [**hiilch(=n')** tavan duu toglo-no gej] hel-sen.
 1PL.EXCL.GEN teacher violinist(=3.POSS) five song play-NPST C say-PST
 'Our teacher said that the violinist will play five songs.'
- c. Manai bagsh [**hiilch-iig(=n')** tavan duu toglo-no gej]
 1PL.EXCL.GEN teacher violinist-ACC(=3.POSS) five song play-NPST C
 hel-sen.
 say-PST
 'Our teacher said that the violinist will play five songs.'

Turning next to bridging definites, in a producer-product bridging context (309), my consultants differ in whether *n'*-marking is required on the bridging referent. All three consultants accept *n'*-marked case-unmarked (307c) and accusative (307e) subjects. However, speaker G also accepts accusative-marked subjects without *n'*-marking (307d). Setting aside *n'*-marking, this is also a different pattern than what was observed in Khalkha DOM, where producer-product bridging direct objects must be accusative-marked.

(307) *Producer-product bridging subject in finite bare CP*

Context: You and your friend are looking at a painting in a museum. You say to your friend...

- a. Ene zurag üzesgelentei yum.
This painting beautiful EVID.COP
'This painting is beautiful.'
- b. # Bi [**zuraach** daraa 7 honog-t Mongol-d ir-ne gej]
1SG painter next 7 day-DAT Mongolia-DAT come-NPST C
bodo-j bai-na.
think-PROG COP-NPST
'I think that the painter will come to Mongolia next week.'
- c. Bi [**zuraach=n'** daraa 7 honog-t Mongol-d ir-ne gej]
1SG painter=3.POSS next 7 day-DAT Mongolia-DAT come-NPST C
bodo-j bai-na.
think-PROG COP-NPST
'I think that the painter will come to Mongolia next week.'
- d. Bi [**zuraach-iig** daraa 7 honog-t Mongol-d ir-ne gej]
1SG painter-ACC next 7 day-DAT Mongolia-DAT come-NPST C
bodo-j bai-na.
think-PROG COP-NPST
'I think that the painter will come to Mongolia next week.' B: ✗, G: ✓, M: ✗
- e. Bi [**zuraach-iig=n'** daraa 7 honog-t Mongol-d ir-ne gej]
1SG painter-ACC=3.POSS next 7 day-DAT Mongolia-DAT come-NPST C
bodo-j bai-na.
think-PROG COP-NPST
'I think that the painter will come to Mongolia next week.'

In part-whole bridging contexts, all three consultants accept accusative and case-unmarked subjects, with or without *n'*-marking, as shown in (308). Again, this is a different pattern than what was observed in Khalkha DOM, where part-whole bridging direct objects must be accusative-marked.

(308) *Part-whole bridging subject in finite bare CP*

Context: Your friend asks you what you did over the weekend. You say...

- a. Egch=maan öchigdör hurim-and yav-san.
Older.sister=1PL.POSS yesterday wedding-DAT go-PST
'My older sister went to a wedding yesterday.'
- b. Ter [**süit büsgüi(=n')** kostyum öms-sön gej] hel-sen.
3SG bride(=3.POSS) suit wear-PST C say-PST
'She said that the bride wore a suit.'
- c. Ter [**süit büsgüi-g(=n')** kostyum öms-sön gej] hel-sen.
3SG bride-ACC(=3.POSS) suit wear-PST C say-PST
'She said that the bride wore a suit.'

Case-unmarked and accusative-marked subjects in finite bare CPs are compatible with larger situation and immediate situation uniqueness contexts, as shown in (309) and (310) respectively. This is the same pattern observed for definite objects previously.

(309) *Larger situation unique subject in finite bare CP*

Context: You and your friend live in the UK. One day your friend asks you what's new. You say...

- Bi [**haan/haan-iig** emneleg-t hevt-sen gej] sons-son.
1SG king/king-ACC hospital-DAT lay-PST C hear-PST
'I heard the king is hospitalised (lit. laying in a hospital).'

(310) *Immediate situation unique subject in finite bare CP*

Context: You and your roommate own a dog. One day your roommate comes home from work. You say to him...

- Bi [öglöö **nohoi/nohoi-g** minii gutl-iig mer-chih-sen gej]
1SG morning dog/dog-ACC 1SG.GEN shoe-ACC gnaw-COMPL-PST C
bodo-j bai-na.
think-PROG COP-NPST
'I think that the dog gnawed on my shoe this morning.'

Finally, as shown in (311a), a case-unmarked subject may take wide scope or narrow scope relative to other scope-taking operators in the clause, resulting in scope ambiguity. However, as shown in (311b), an accusative-marked subject must take wide scope relative to other scope-taking operators. This is the same pattern observed in Khalkha DOM previously.

(311) *Wide vs. narrow scope indefinite subject in finite bare CP*

- a. Bilgüün [ireh doloo honog-t ödör бүр **oyutan** iltgel tavi-na gej]
 B. next seven day-DAT day every student presentation make-NPST C
 hel-sen.
 say-PST
 ‘Bilguun said that for every day next week, a (different) student will give a presentation.’ $\forall > \exists$
 ‘Bilguun said that a (certain) student will give a presentation every day next week.’ $\exists > \forall$
- b. Bilgüün [ireh doloo honog-t ödör бүр **oyutn-iig** iltgel tavi-na
 B. next seven day-DAT day every student-ACC presentation make-NPST
 gej] hel-sen.
 C say-PST
 #‘Bilguun said that for every day next week, a (different) student will give a presentation.’ $\# \forall > \exists$
 ‘Bilguun said that a (certain) student will give a presentation every day next week.’ $\exists > \forall$

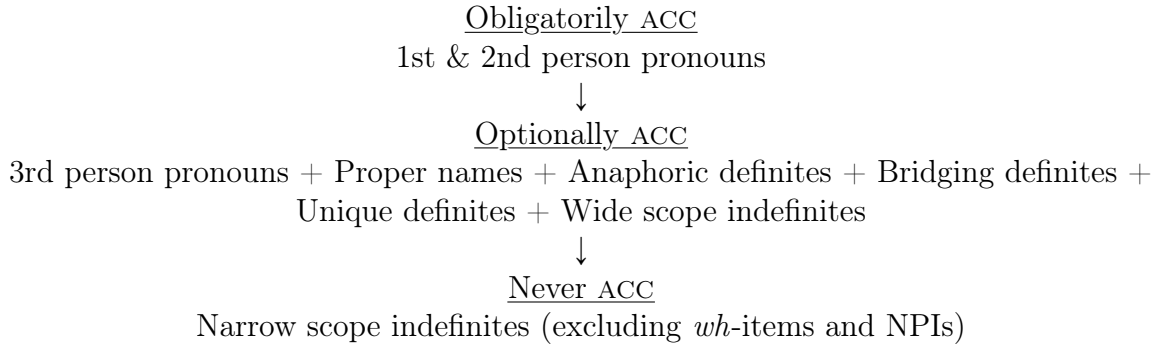
The judgements of all three consultants are represented in Table 5.4. There are two key distinctions between Table 5.3 and Table 5.4. First, considering the effect of nominal type on DAM, while 3rd person pronouns and proper names are obligatorily accusative-marked in direct object position, they can be accusative-marked or case-unmarked in embedded subject position in finite bare CPs. Second, in anaphoric and bridging contexts, all three consultants allow both case-unmarked and accusative-marked subjects in finite bare CPs, but disallow case-unmarked objects in matrix root clauses.

Table 5.4: DSM in Khalkha finite bare CPs

	Nominal type											
	1st person			2nd person			3rd person			Proper name		
	B	G	M	B	G	M	B	G	M	B	G	M
UNM	✗	✗	✗	✗	✗	✗	✓	✓	✓	✓	✓	✓
ACC	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
	Immediate anaphora						Bridging					
	=1 mentioned			>1 mentioned			Producer-product			Part-whole		
	B	G	M	B	G	M	B	G	M	B	G	M
UNM	✓	✓	✓	✓(+)	✓(+)	✓(+)	✓+	✓+	✓+	✓(+)	✓(+)	✓(+)
ACC	✓	✓	✓	✓(+)	✓(+)	✓(+)	✓+	✓(+)	✓+	✓(+)	✓(+)	✓(+)
	Unique						Indefinite					
	Larger situation			Immediate situation			Wide scope			Narrow		
	B	G	M	B	G	M	B	G	M	B	G	M
UNM	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
ACC	✓	✓	✓	✓	✓	✓	✓	✓	✓	✗	✗	✗
✓: Acceptable ✗: Unacceptable +: 3.POSS obligatory (+): 3.POSS optional												

The patterns in Table 5.4 may be organised along a referentiality scale for Khalkha DSM as shown in (312). In this scale, 1st and 2nd person pronouns, which are obligatorily accusative-marked, are ranked higher than 3rd person pronouns, proper names, anaphoric definites, bridging definites, unique definites and wide scope indefinites, which may be accusative-marked or case-unmarked. Ranked lowest are narrow scope indefinites, which may only be case-unmarked.

(312) *Referentiality scale for Khalkha DSM*



5.3.3 DSM in finite NCs

In finite NCs – NCs in which a full DP exterior embeds a CP-sized clause – 1st and 2nd person subjects are obligatorily marked with accusative case, while 3rd person and proper name subjects may be accusative-marked or case-unmarked, as shown in (313). This is the same DSM pattern observed in finite bare CPs previously.

(313) *Case marking of nominal types in finite NCs*

- a. Naraa [margaash **namaig**/***bi** ir-ne gedg]-iig sons-son.
N. tomorrow 1SG.ACC/*1SG come-NPST C.NML -ACC hear-PST
'Naraa heard that I will come tomorrow.'
- b. Naraa [margaash **chamaig**/***chi** ir-ne gedg]-iig sons-son.
N. tomorrow 2SG.ACC/*2SG come-NPST C.NML -ACC hear-PST
'Naraa heard that you will come tomorrow.'
- c. Naraa_j [margaash **tüüüig**/**ter_{k/*j}** ir-ne gedg]-iig sons-son.
N. tomorrow 3SG.ACC/3SG come-NPST C.NML -ACC hear-PST
'Naraa_j heard that she_{k/*j}/he will come tomorrow.'
- d. Naraa [margaash **Dorj-iig**/**Dorj** ir-ne gedg]-iig sons-son.
N. tomorrow D.-ACC/D. come-NPST C.NML -ACC hear-PST
'Naraa heard that Dorj will come tomorrow.'

Turning to anaphoric definites, in an anaphoric (=1) definite context, my three consultants accept an accusative-marked or case-unmarked subject (314). This is the same DSM pattern found in finite bare CPs, but a different one from DOM.

(314) *Anaphoric (=1) subject in finite NC*

- a. Margaash manai surguul'-d aldartai hiilch ir-ne.
Tomorrow 1PL.GEN school-DAT famous violinist come-NPST
'Tomorrow a famous violinist will come to our school.'
- b. Bi [**hiilch**/**hiilch-iig** tavan duu toglo-no gedg]-iig sons-son.
1SG violinist/violinist-ACC five song play-NPST C.NML -ACC hear-PST
'I heard that the violinist will play five songs.'

In an anaphoric (>1) context, all three consultants accept an accusative-marked or case-unmarked subject, with or without *n'*-marking, as shown in (315). This is the same DSM pattern observed in finite bare CPs, but a different one from DOM.

(315) *Anaphoric (>1) subject in finite NC*

- a. Margaash manai surguul'-d aldartai hiilch tögöldör huurch hoyor
 Tomorrow 1PL.GEN school-DAT famous violinist pianist two
 ir-ne.
 come-NPST
 'Tomorrow a famous violinist and pianist will come to our school.'
- b. Bi [**hiilch**(=n') tavan duu toglo-no gedg]-iig sons-son.
 1SG violinist(=3.POSS) five song play-NPST C.NML -ACC hear-PST
 'I heard that the violinist will play five songs.'
- c. Bi [**hiilch-iig**(=n') tavan duu toglo-no gedg]-iig sons-son.
 1SG violinist-ACC(=3.POSS) five song play-NPST C.NML -ACC hear-PST
 'I heard that the violinist will play five songs.'

Turning next to bridging definites, in a producer-product bridging context, all three consultants reject a case-unmarked subject without *n'*-marking (316b) but accept a case-unmarked (316c) or accusative (316e) subject with *n'*-marking. Additionally, only speaker G accepts an accusative-marked subject without *n'*-marking (316d). This is the same DSM pattern and same interspeaker variation previously observed in finite bare CPs, but a different pattern than that observed in Khalkha DOM.

(316) *Producer-product bridging definite in a finite NC*

Context: You and your friend are looking at a painting in a museum. You say to your friend...

- a. Ene zurag üzesgelentei yum.
 This painting beautiful EVID.COP
 'This painting is beautiful.'
- b. # Bi [**zuraach** daraa 7 honog-t Mongol-d ir-ne gedg]-iig
 1SG painter next 7 day-DAT Mongolia-DAT come-NPST C.NML -ACC
 sons-son.
 hear-PST
 'I heard that the painter will come to Mongolia next week.'

- c. Bi [**zuraach=n'** daraa 7 honog-t Mongol-d ir-ne gedg]-iig
 1SG painter=3.POSS next 7 day-DAT Mongolia-DAT come-NPST C.NML -ACC
 sons-son.
 hear-PST
 'I heard that the painter will come to Mongolia next week.'
- d. Bi [**zuraach-iig** daraa 7 honog-t Mongol-d ir-ne gedg]-iig
 1SG painter-ACC next 7 day-DAT Mongolia-DAT come-NPST C.NML -ACC
 sons-son.
 hear-PST
 'I heard that the painter will come to Mongolia next week.' B: ✗, G:✓, M:✗
- e. Bi [**zuraach-iig=n'** daraa 7 honog-t Mongol-d ir-ne gedg
 1SG painter-ACC=3.POSS next 7 day-DAT Mongolia-DAT come-NPST C.NML
]-iig sons-son.
 -ACC hear-PST
 'I heard that the painter will come to Mongolia next week.'

In a part-whole bridging definite context, all three consultants accept an accusative-marked or a case-unmarked subject with or without *n'*-marking in finite NCs, as shown in (317). This is the same DSM pattern observed previously in finite bare CPs, but different than that observed in Khalkha DOM.

(317) *Part-whole bridging subject in finite NC*

Context: You are telling your friend what you did yesterday.

- a. Bi öchigdör hurim-and yav-san.
 1SG yesterday wedding-DAT go-PST
 'I went to a wedding yesterday.'
- b. Bi [**süit büsgui(=n')** Singapur-t ajilla-dag gedg]-iig sons-son.
 1SG bride(=3.POSS) S.-DAT work-HAB C.NML -ACC hear-PST
 'I heard the bride works in Singapore.'
- c. Bi [**süit büsgui-g(=n')** Singapur-t ajilla-dag gedg]-iig sons-son.
 1SG bride-ACC(=3.POSS) S.-DAT work-HAB C.NML -ACC hear-PST
 'I heard the bride works in Singapore.'

Turning next to unique definites, (318) and (319) show that both accusative-marked and case-unmarked subjects in finite NCs are compatible with larger situation and immediate situation definites respectively. This is the same DSM pattern observed in finite bare CPs, and the same DOM pattern seen in matrix root clauses.

(318) *Larger situation unique subject in finite NC*

Context: You and your friend live in the UK. One morning your mother tells you that the king has been hospitalised. Later that day, you meet your friend and tell them...

Bi [**haan/haan-iig** emneleg-t hevt-sen gedg]-iig sons-son.
 1SG king/king-ACC hospital-DAT lay-PST C.NML -ACC hear-PST
 ‘I heard that the king is hospitalised (lit. laying in a hospital).’

(319) *Immediate situation unique subject in finite NC*

Context: Yesterday’s school performance started late. Today at school, you ask a teacher why. They say...

Bi [**zahiral/zahiral-iig** hotsor-son gedg]-iig sons-son.
 1SG director/director-ACC be.late-PST C.NML -ACC hear-PST
 ‘I heard that the principal was late.’

Finally, case-unmarked subjects may take wide or narrow scope relative to other scope-taking operators (320a), while accusative-marked subjects only take wide scope (320b). This is the same DSM pattern previously seen in finite bare CPs, and the same DOM pattern seen in matrix root clause.

(320) *Wide vs. narrow indefinite subject in finite NC*

- a. Dorj [ireh doloo honog-t **zaluu professor** iltgel tavi-na gedg
 D. next seven day-DAT young professor presentation make-NPST C.NML
]-iig sons-son.
 -ACC hear-PST
 ‘Dorj heard that for every day next week, a (different) young professor will give a talk.’ $\forall > \exists$
 ‘Dorj heard that a (certain) young professor will give a talk every day next week.’ $\exists > \forall$
- b. Dorj [ireh doloo honog-t **zaluu professor-iig** iltgel tavi-na gedg
 D. next seven day-DAT young professor-ACC presentation make-NPST C.NML
]-iig sons-son.
 -ACC hear-PST
 #‘Dorj heard that for every day next week, a (different) young professor will give a talk.’ $\# \forall > \exists$
 ‘Dorj heard that a (certain) young professor will give a talk every day next week.’ $\exists > \forall$

The judgements of my three consultants are represented in Table 5.5. DSM in finite NCs patterns identically to DSM in finite bare CPs and thus adheres to the referentiality scale for DSM in (312).

Table 5.5: DSM in Khalkha finite NCs

	Nominal type											
	1st person			2nd person			3rd person			Proper name		
	B	G	M	B	G	M	B	G	M	B	G	M
UNM	✗	✗	✗	✗	✗	✗	✓	✓	✓	✓	✓	✓
ACC	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
	Immediate anaphora						Bridging					
	=1 mentioned			>1 mentioned			Producer-product			Part-whole		
	B	G	M	B	G	M	B	G	M	B	G	M
UNM	✓	✓	✓	✓(+)	✓(+)	✓(+)	✓+	✓+	✓+	✓(+)	✓(+)	✓(+)
ACC	✓	✓	✓	✓(+)	✓(+)	✓(+)	✓+	✓(+)	✓+	✓(+)	✓(+)	✓(+)
	Unique						Indefinite					
	Larger situation			Immediate situation			Wide scope			Narrow scope		
	B	G	M	B	G	M	B	G	M	B	G	M
UNM	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
ACC	✓	✓	✓	✓	✓	✓	✓	✓	✓	✗	✗	✗
✓: Acceptable ✗: Unacceptable +: 3.POSS obligatory (+): 3.POSS optional												

5.3.4 DSM in nonfinite NCs

We turn our attention to nonfinite NCs – NCs in which a full DP exterior embeds a TP-sized clause. Identical to the DSM patterns seen in finite bare CPs and finite NCs, 1st and 2nd person subjects are obligatorily marked with accusative case, while 3rd person pronominal and proper name subjects may be accusative or case-unmarked. This is shown in (321).

(321) *Case marking of nominal types in nonfinite NCs*

- a. Naraa [margaash **namaig**/***bi** ir-eh]-iig mart-san.
N. tomorrow 1SG.ACC/*1SG come-INF -ACC forget-PST
'Naraa forgot that I am coming tomorrow.'
- b. Naraa [margaash **chamaig**/***chi** ir-eh]-iig mart-san.
N. tomorrow 2SG.ACC/*2SG come-INF -ACC forget-PST
'Naraa forgot that you are coming tomorrow.'
- c. Naraa [margaash **tüüniig**/**ter** ir-eh]-iig mart-san.
N. tomorrow 3SG.ACC/3SG come-INF -ACC forget-PST
'Naraa_j forgot that she_k/*_j/he is coming tomorrow.'

- d. Naraa [margaash **Dorj-iig/Dorj** ir-eh]-iig mart-san.
 N. tomorrow D.-ACC/D. come-INF -ACC forget-PST
 ‘Naraa forgot that Dorj is coming tomorrow.’

Turning to anaphoric definites, in anaphoric (=1) contexts, all three of my consultants accept accusative or case-unmarked subjects (322b). This is the same DSM pattern observed previously, but a different one from DOM.

(322) *Anaphoric (=1) subject in nonfinite NC*

- a. Öchigdör manai surguul'-d aldartai hiilch ir-sen.
 Yesterday 1PL.EXCL.GEN school-DAT famous violinist come-PST
 ‘Yesterday a famous violinist came to our school.’
- b. Bi önöö öglöö [**hiilch/hiilch-iig** nom-in san-d nom
 1SG today morning violinist/violinist-ACC book-GEN center-DAT book
 unsh-ij bai-h]-iig har-san.
 read-PROG COP-INF -ACC see-PST
 ‘This morning I saw the violinist reading a book in the library.’

In anaphoric (>1) contexts, all three of my consultants accept case-unmarked (323b) and accusative (323c) subjects, with or without *n'*-marking. This is also the same pattern observed for DSM previously, but different from DOM.

(323) *Anaphoric (>1) subject in nonfinite NC*

- a. Öchigdör manai surguul'-d aldartai hiilch tögoöldör huurch hoyor
 Yesterday 1PL.EXCL.GEN school-DAT famous violinist pianist two
 ir-sen.
 come-PST
 ‘Yesterday a famous violinist came to our school.’
- b. Bi önöö öglöö [**hiilch(=n')** nom-in san-d nom unsh-ij
 1SG today morning violinist(=3.POSS) book-GEN center-DAT book read-PROG
 bai-h]-iig har-san.
 COP-INF -ACC see-PST
 ‘This morning I saw the violinist reading a book in the library.’
- c. Bi önöö öglöö [**hiilch-iig(=n')** nom-in san-d nom
 1SG today morning violinist-ACC(=3.POSS) book-GEN center-DAT book
 unsh-ij bai-h]-iig har-san.
 read-PROG COP-INF -ACC see-PST
 ‘This morning I saw the violinist reading a book in the library.’

Turning next to bridging definites, in producer-product bridging contexts, my three consultants accept *n'*-marked case-unmarked (324c) and accusative subjects (324e). Additionally, speaker G accepts accusative-marked subjects without *n'*-marking (324d). This is the same pattern observed in finite NCs and finite bare CPs, but a different pattern than that observed for direct objects.

(324) *Producer-product bridging subject in nonfinite NC*

Context: You and a friend are in a café. A song starts to play over the speakers. You tell your friend...

- a. Ene minii durtai duul!
This 1SG.GEN favourite song
'This is my favourite song!'
- b. # Bi türüü jil [**duuchin** ene duu-g duula-h]-iig toglolt deer sons-son.
1SG last year singer this song-ACC sing-INF -ACC concert at hear-PST
'I heard the singer singing this song at a concert last year.'
- c. Bi türüü jil [**duuchin=n'** ene duu-g duula-h]-iig toglolt deer
1SG last year singer=3.POSS this song-ACC sing-INF -ACC concert at
sons-son.
hear-PST
'I heard the singer singing this song at a concert last year.'
- d. Bi türüü jil [**duuchin-iig** ene duu-g duula-h]-iig toglolt deer sons-son.
1SG last year singer-ACC this song-ACC sing-INF -ACC concert at hear-PST
'I heard the singer singing this song at a concert last year.' B: ✗, G: ✓, M: ✗
- e. Bi türüü jil [**duuchin-iig=n'** ene duu-g duula-h]-iig toglolt deer
1SG last year singer-ACC=3.POSS this song-ACC sing-INF -ACC concert at
sons-son.
hear-PST
'I heard the singer singing this song at a concert last year.'

In part-whole bridging contexts, all three consultants accept case-unmarked and accusative-marked subjects, with or without *n'*-marking, as shown in (325). This is the same DSM pattern observed for finite NCs and finite bare CPs, but a different pattern than that observed in Khalkha DOM.

(325) *Part-whole bridging subject in nonfinite NC*

Context: You are telling your friend what you did yesterday.

- a. Bi öchigdör hurim-and yav-san.
1SG yesterday wedding-DAT go-PST
'I went to a wedding yesterday.'

- b. Bi [**süit büsgui(=n')** uil-j bai-h]-iig har-san.
 1SG bride(=3.POSS) cry-PROG COP-INF -ACC see-PST
 'I saw the bride crying.'
- c. Bi [**süit büsgui-g(=n')** uil-j bai-h]-iig har-san.
 1SG bride-ACC(=3.POSS) cry-PROG COP-INF -ACC see-PST
 'I saw the bride crying.'

Both accusative-marked and case-unmarked subjects are compatible with larger situation unique definites (326) and immediate situation unique definites (327). These observations are identical to the one previously reported for finite NCs, finite bare CPs, and direct objects.

(326) *Larger situation unique subject in nonfinite NC*

Context: You went to a Naadam festival in Mongolia. When you come home, your mother asks if you saw anything interesting. You say...

- Bi [**yörönhiilögch/yörönhiilögch-iig** mor' una-h bai-h]-iig har-san.
 1SG president/president-ACC horse ride-PROG COP-INF -ACC see-PST
 'I saw the president riding a horse.'

(327) *Immediate situation unique subject in nonfinite NC*

Context: You and Naraa are in a room with a cat. Both of you are reading a book. Suddenly, Naraa says to you...

- Bi [**muur/muur-iig** buidan-g maaji-h]-iig har-san.
 1SG cat/cat-ACC sofa-ACC scratch-INF -ACC see-PST
 'I saw the cat scratching the sofa.'

Finally, case-unmarked subjects in nonfinite NCs may take wide scope or narrow scope relative other scope-taking operators, as shown in (328a), while accusative-marked ones may only take wide scope, as shown in (328b). This is the same DSM pattern found for subjects in finite NCs and finite bare CPs, and for DOM.

(328) *Wide vs. narrow scope indefinite subject in nonfinite NC*

- a. Bilgüün ödör бүр [**har muur** mod-ond avir-ch bai-h]-iig har-dag.
 B. every day black cat tree-DAT climb-PROG COP-INF -ACC see-HAB
 'Bilguun sees a (different) black cat climbing a tree every day.' $\forall > \exists$
 'Bilguun sees a (certain) black cat climbing a tree every day.' $\exists > \forall$
- b. Bilgüün ödör бүр [**har muur-iig** mod-ond avir-ch bai-h]-iig har-dag.
 B. every day black cat-ACC tree-DAT climb-PROG COP-INF -ACC see-HAB
 # 'Bilguun sees a (different) black cat climbing a tree every day.' $\# \forall > \exists$
 'Bilguun sees a (certain) black cat climbing a tree every day.' $\exists > \forall$

The DSM patterns in nonfinite NCs across all three speakers are reflected in Table 5.6. The patterns are identical to those found in finite NCs and finite bare CPs.

Table 5.6: DSM in Khalkha nonfinite NCs

	Nominal type											
	1st person			2nd person			3rd person			Proper name		
	B	G	M	B	G	M	B	G	M	B	G	M
UNM	✗	✗	✗	✗	✗	✗	✓	✓	✓	✓	✓	✓
ACC	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
	Immediate anaphora						Bridging					
	=1 mentioned			>1 mentioned			Producer-product			Part-whole		
	B	G	M	B	G	M	B	G	M	B	G	M
UNM	✓	✓	✓	✓(+)	✓(+)	✓(+)	✓+	✓+	✓+	✓(+)	✓(+)	✓(+)
ACC	✓	✓	✓	✓(+)	✓(+)	✓(+)	✓+	✓(+)	✓+	✓(+)	✓(+)	✓(+)
	Unique						Indefinite					
	Larger situation			Immediate situation			Wide scope			Narrow scope		
	B	G	M	B	G	M	B	G	M	B	G	M
UNM	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
ACC	✓	✓	✓	✓	✓	✓	✓	✓	✓	✗	✗	✗
✓: Acceptable ✗: Unacceptable +: 3.POSS obligatory (+): 3.POSS optional												

5.3.5 DSM in converbial adjuncts

Finally, we turn our attention to converbial adjuncts, which I have argued in Chapter 2 that they are adjoined to vP, and in Chapter 3 that they are NCs that lack both a CP layer and a DP layer. First, observe that 1st and 2nd person subjects are obligatorily accusative as the subjects of converbial adjuncts (329a)-(329b), while 3rd person pronominal and proper name subjects may be accusative or case-unmarked (329c)-(329d). This is the same DSM pattern observed in all previous embedded clause types, but different than that observed for direct objects.

(329) *Case marking of nominal types in converbial adjuncts*

- a. Naraa [**namaig**/***bi** nom unshi-j bai-had] unt-san.
N. 1SG.ACC/*1SG book read-PROG COP-TEMP.CVB sleep-PST
‘Naraa fell asleep while I was reading a book.’
- b. Naraa [**chamaig**/***chi** nom unshi-j bai-had] unt-san.
N. 2SG.ACC/*2SG book read-PROG COP-TEMP.CVB sleep-PST
‘Naraa fell asleep while I was reading a book.’
- c. Naraa [**tüüniig**/**ter** nom unshi-j bai-had] unt-san.
N. 3SG.ACC/3SG book read-PROG COP-TEMP.CVB sleep-PST
‘Naraa_j fell asleep while he/she_k/*_j was reading a book.’
- d. Naraa [**Dorj-iig**/**Dorj** nom unshi-j bai-had] unt-san.
N. D.-ACC/D. book read-PROG COP-TEMP.CVB sleep-PST
‘Naraa fell asleep while Dorj was reading a book.’

Turning to anaphoric definites, in anaphoric (=1) contexts, all three consultants accept case-unmarked or accusative-marked subjects, as shown in (330). This is the same DSM pattern found previously in finite bare CPs, finite NCs, and nonfinite NCs.

(330) *Anaphoric (=1) subject in converbial adjunct*

Context: You are telling a friend about your performance tomorrow...

- a. Bi margaash hiilch-tei tog_l-no.
1SG tomorrow violinist-COM play-NPST
‘I will perform with a violinist tomorrow.’
- b. Bi [**hiilch**/**hiilch-iig** tog_{lo}-j ehle-tel] duul-na.
1SG violinist/violinist-ACC play-CVB start-TERM.CVB sing-NPST
‘I will sing until the violinist starts playing.’

In anaphoric (>1) contexts, all three consultants accept case-unmarked or accusative-marked subjects regardless of the presence of *n*’-marking, as shown in (331). This is also the same DSM pattern found previously.

(331) *Anaphoric (>1) subject in converbial adjunct*

Context: You are telling a friend about your performance tomorrow...

- a. Bi margaash hiilch tögöldör huurch hoyur-toi toglo-no.
1SG tomorrow violinist pianist two-COM play-NPST
'I will perform with a violinist and a pianist tomorrow.'
- b. Bi [**hiilch(=n')** toglo-j ehle-tel] duul-na.
1SG violinist(=3.POSS) play-CVB start-TERM.CVB sing-NPST
'I will sing until the violinist starts playing.'
- c. Bi [**hiilch-iig(=n')** toglo-j ehle-tel] duul-na.
1SG violinist-ACC(=3.POSS) play-CVB start-TERM.CVB sing-NPST
'I will sing until the violinist starts playing.'

Turning next to bridging definites, in producer-product bridging contexts, all three consultants only accept *n'*-marked case-unmarked (332b) or accusative (332c) subjects. Setting aside *n'*-marking, this is the same case-marking pattern found for the three embedded clause types examined previously.

(332) *Producer-product bridging subject in converbial adjunct*

Context: You are talking to a friend...

- a. Bi sayahan shine nom unsh-san.
1SG recently new book read-PST
'I recently read a new book.'
- b. Bi [**zohiolch#(=n')** shine nom bich-tel] öör nom
1SG author#(=3.POSS) new book write-TERM.CVB other book
unsh-ih-güi.
read-INF-NEG
'I won't read another book until the author writes a new book.'
- c. Bi [**zohiolch-iig#(=n')** shine nom bich-tel] öör nom
1SG author-ACC#(=3.POSS) new book write-TERM.CVB other book
unsh-ih-güi.
read-INF-NEG
'I won't read another book until the author writes a new book.'

In part-whole bridging contexts, all three consultants accept case-unmarked and accusative-marked subjects, regardless of whether they are *n'*-marked, as shown in (333). This is the same pattern found for the three previous embedded clause types.

(333) *Part-whole bridging subject in converbial adjunct*

Context: You are telling a friend what you did yesterday.

- a. Bi öchigdör hurim-and yav-san.
1SG yesterday wedding-DAT go-PST
'I went to a wedding yesterday.'
- b. Hümüüs [**süit büsgüi(=n')** ir-tel] yörööl hel-sen.
People bride(=3.POSS) come-TERM.CVB well.wishing say-PST
'People said well-wishings until the bride arrived.'
- c. Hümüüs [**süit büsgüi-g(=n')** ir-tel] yörööl hel-sen.
People bride-ACC(=3.POSS) come-TERM.CVB well.wishing say-PST
'People said well-wishings until the bride arrived.'

Turning next to unique definites, case-unmarked and accusative-marked subjects in converbial adjuncts are compatible as larger situation (334) and immediate situation (335) unique definites. This is the same pattern as was found in all previous embedded clause types and for direct objects.

(334) *Larger situation unique subject in converbial adjunct*

Context: You are at a performance in Mongolia, where the guest-of-honour is the president. After some time, the performance hasn't started. You ask someone why and they say...

- Toglolt [**yörönhiilögch/yörönhiilögch-iig** ir-tel] ehle-h-güi.
Performance president/president-ACC come-TERM.CVB start-INF-NEG
'The performance will not start until the president arrives.'

(335) *Immediate situation unique subject in converbial adjunct*

Context: You are at a school performance. After some time, the performance hasn't started. You ask someone why and they say...

- Toglolt [**zahirai/zahirai-iig** ir-tel] ehle-h-güi.
Performance director/director-ACC come-TERM.CVB start-INF-NEG
'The performance will not start until the principal arrives.'

There is some variation as to whether case-unmarked and accusative subjects in converbial adjuncts are compatible as wide or narrow scope readings, as shown in (336). For speaker B, case-unmarked subjects are compatible with both wide and narrow scope, while

accusative-marked subjects are incompatible with narrow scope indefinites – this is the same pattern found in preceding embedded clause types. For speaker G, both case-unmarked and accusative-marked subjects are compatible with wide or narrow scope readings. Based on the current available data, speaker M does not allow converbial adjunct subjects to have narrow scope regardless of case marking. The latter two consultants’ patterns are different than what was observed in preceding embedded clause types.

(336) *Wide vs. narrow scope indefinite subject in converbial adjunct*

- a. Naraa ödoör бүр [**oyutan** tuüünii offis deer ir-tel]
 N. day every student 3SG.GEN office in come-TERM.CVB
 nom unsh-dag.
 book read-HAB
 ‘Every day Naraa_j reads a book until a student comes to her_j office.’ $\forall > \exists$
 B: ✓, G:✓, M:✗
 ‘Every day Naraa_j reads a book until a (certain) student comes to her_j office.’
 $\exists > \forall$
 B: ✓, G:✓, M:✓
- b. Naraa ödör бүр [**oyutn-iig** tuüünii offis deer ir-tel]
 N. day every student-ACC 3SG.GEN office in come-TERM.CVB
 nom unsh-dag.
 book read-HAB
 ‘Every day Naraa_j reads a book until a student comes to her_j office.’ $\forall > \exists$
 B: ✗, G:✓, M:✗
 ‘Every day Naraa_j reads a book until a (certain) student comes to her_j office.’
 $\exists > \forall$
 B: ✓, G:✓, M:✓

While the source of variation in the scopal properties of converbial adjunct subjects is unclear to me – particularly in contrast to my consultants’ judgements in the preceding three embedded clause types – I suggest that the adjunct status of converbial adjuncts may influence how their subjects interact with scope-taking operators in the matrix clause. For these reasons, diagnosing the scopal properties of converbial adjunct subjects may not be as straightforward as in the other three complement clause types.

The DSM patterns in converbial adjuncts across all three speakers are reflected in Table 5.7. With producer-product bridging definites, all three speakers require the presence

of n' -marking, whereas in the previous three embedded clause types, speaker G optionally require n' -marking on accusative subjects. For nonspecific indefinites, we observe interspeaker variation not present in the other embedded clause types.

Table 5.7: DSM in Khalkha converbial adjuncts

	Nominal type											
	1st person			2nd person			3rd person			Proper name		
	B	G	M	B	G	M	B	G	M	B	G	M
UNM	✗	✗	✗	✗	✗	✗	✓	✓	✓	✓	✓	✓
ACC	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
	Immediate anaphora						Bridging					
	=1 mentioned			>1 mentioned			Producer-product			Part-whole		
	B	G	M	B	G	M	B	G	M	B	G	M
UNM	✓	✓	✓	✓(+)	✓(+)	✓(+)	✓+	✓+	✓+	✓(+)	✓(+)	✓(+)
ACC	✓	✓	✓	✓(+)	✓(+)	✓(+)	✓+	✓+	✓+	✓(+)	✓(+)	✓(+)
	Unique						Indefinite					
	Larger situation			Immediate situation			Wide scope			Narrow scope		
	B	G	M	B	G	M	B	G	M	B	G	M
UNM	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✗
ACC	✓	✓	✓	✓	✓	✓	✓	✓	✓	✗	✓	✗
✓: Acceptable ✗: Unacceptable +: 3.POSS obligatory (+): 3.POSS optional												

5.3.6 DAM and referentiality scales in Khalkha

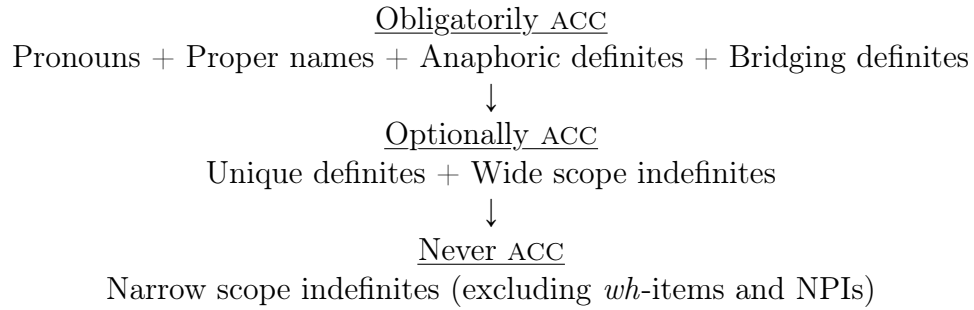
A summary of the DOM and DSM patterns discussed above is provided in Table 5.8. While similar, DOM in Khalkha differs from DSM in where the cutoff point occurs for optional accusative marking. For DOM, the cutoff point is between producer-product (PP)/part-whole (PW) bridging definites and immediate situation (IS)/larger situation (LS) unique definites. However, for DSM, the cutoff point is between local person (i.e. 1st and 2nd) person and 3rd person pronouns. Finally, the characterisations of converbial adjunct subjects as wide scope (WS) or narrow scope (NS) are greyed out, reflecting the possibility that independent syntactic properties of converbial adjuncts may render diagnoses of their subjects' scopal properties unreliable.

Table 5.8: DOM and DSM in Khalkha

	DOM	DSM			
		Finite bare CP	Finite NC	Nonfinite NC	Converbial adjunct
1/2P pronoun	ACC	ACC	ACC	ACC	ACC
3P pronoun	ACC	ACC/UNM	ACC/UNM	ACC/UNM	ACC/UNM
Proper name	ACC	ACC/UNM	ACC/UNM	ACC/UNM	ACC/UNM
Anaphoric (=1/>1)	ACC	ACC/UNM	ACC/UNM	ACC/UNM	ACC/UNM
PP/PW bridging	ACC	ACC/UNM	ACC/UNM	ACC/UNM	ACC/UNM
IS/LS unique	ACC/UNM	ACC/UNM	ACC/UNM	ACC/UNM	ACC/UNM
WS indefinite	ACC/UNM	ACC/UNM	ACC/UNM	ACC/UNM	ACC/UNM
NS indefinite	UNM	UNM	UNM	UNM	%ACC/UNM

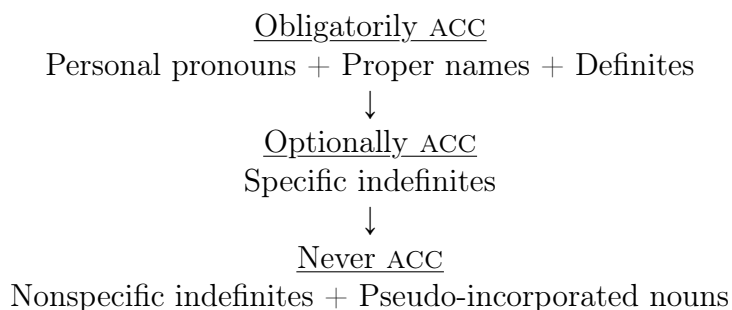
The DOM patterns suggest that in a referentiality scale for Khalkha DOM, e.g. (303) (repeated below), pronouns, proper names, anaphoric definites, and bridging definites occupy the top of the scale as they are always accusative-marked. Ranked below these types of objects are unique definites and wide scope indefinites, as they may be accusative-marked or case-unmarked. Finally, narrow scope indefinites (with the exception of *wh*-items and NPIs) occupy the bottom of the scale, as they are always case-unmarked.

(303) *Referentiality scale for Khalkha DOM*



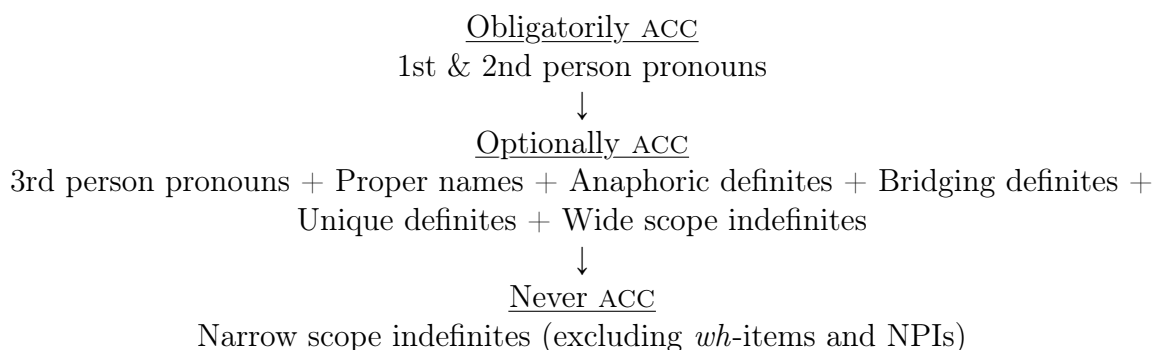
Compare (303) to the DOM referentiality scale proposed by Guntsetseg (2016), repeated below.

(283) *Guntsetseg's (2016, p.112) referentiality scale for Khalkha DOM*



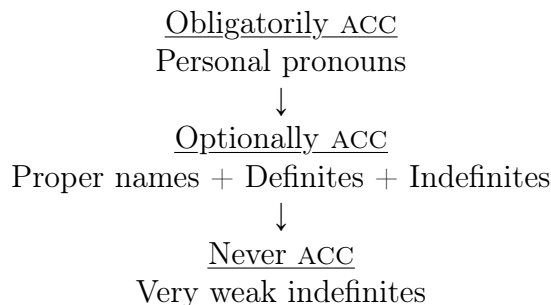
The DSM patterns suggest that in a referentiality scale for Khalkha DSM, e.g. (312) (repeated below), local (i.e. 1st and 2nd) person pronouns are at the top of the scale, given that they are always marked accusative in embedded subject position. This is followed by 3rd person pronouns, proper names, anaphoric definites, bridging definites, unique definites, and wide scope indefinites, which may be accusative-marked or case-unmarked in embedded subject position. Finally, narrow scope indefinites (except *wh*-items and NPIs) occupy the bottom of the scale, as they are always case-unmarked.

(312) *Referentiality scale for Khalkha DSM*



Compare (312) to the DSM referentiality scale proposed by Guntsetseg (2016), repeated below.

(284) *Guntsetseg's (2016, p.150) referentiality scale for Khalkha DSM*



Comparing my (303) and (312) with Guntsetseg's (2016) (283) and (284), there are some notable differences as to where the cutoff points for optional accusative marking are. For DOM, according to Guntsetseg, the cutoff point is between definites and specific indefinites,⁷ whereas I have found that the cutoff point is between bridging definites and unique definites. A possible explanation is that Guntsetseg did not test for different subtypes of definiteness. By doing so, as my investigation has shown, we observed that anaphoric and bridging definites are ranked higher than unique definites on the DOM scale. In other words, a consideration of different subtypes of definiteness has led to a finer-grained view of DOM in Khalkha.

For DSM, according to Guntsetseg, the cutoff point is between personal pronouns on one hand and proper names, definites, and indefinites on the other,⁸ whereas I have found that the cutoff point is between local person pronouns on one hand, and 3rd person pronouns, proper names, definites, and wide scope indefinites on the other. A possible explanation is that Guntsetseg (2016) did not actually test the DSM properties of 3rd person pronouns,

⁷While I have discussed indefinites in terms of their scopal properties, Guntsetseg (2016, p.94) considers scopal, epistemic, partitive, and relative specificity in classifying specific vs. nonspecific indefinites in her scale for DOM. In terms of scopal specificity, she shows that case-unmarked objects may bear wide or narrow scope, while accusative-marked objects may only bear wide scope; this is in line with my findings. I thus assume that Guntsetseg's category of 'specific indefinites' corresponds to my category of 'wide scope indefinites', and her category of 'nonspecific indefinites' corresponds to my category of 'narrow scope indefinites'.

⁸Unfortunately, a strong parallel cannot be drawn between 'wide/narrow scope indefinites' under my DSM scale (312) and the category of 'indefinites' under Guntsetseg's DSM scale (284) as she does not discuss the scopal properties of DSM.

as the test sentences she considered only contained local person pronouns.⁹ By explicitly testing for 3rd person pronominal subjects in embedded clauses, I have shown that they actually rank lower than local person subjects on Khalkha’s DSM scale.

5.3.7 Analyses of DAM

The many influential analyses of DAM can be grouped into five camps: i) Optimality-Theoretic (OT) accounts; ii) morphological accounts; iii) structural size accounts; iv) movement accounts; and v) licensing accounts. For an overview of these accounts, see Bárány and Kalin 2020b.

OT accounts of DAM can be found in works such as Aissen 1999, 2003; de Hoop and Malchukov 2007; Woolford 2008. Under OT accounts, the surface form of an argument is regulated by a set of markedness and faithfulness constraints. For example, Woolford (2008) proposes a set of contextually-restricted markedness constraints, any of which (e.g. *ERGATIVE/LOCAL) could be ranked above a faithfulness constraint, MAX(ERGATIVE), that requires faithful spell-out of the Case feature that has been assigned to a nominal in syntax. Thus, *ERGATIVE/LOCAL > MAX(ERGATIVE) predicts that there is a language in which 1st and 2nd person (i.e. local) subjects cannot bear ergative case marking whereas 3rd person subjects can. Such a language is Dyirbal (Silverstein 1976).

While OT accounts of DAM may be successful in capturing patterns in languages where DSM behaves like the mirror image of DOM, such as ergative systems (Silverstein 1976; Aissen 2003; Woolford 2008), it is less successful in accounting for patterns in languages where the same case used in DOM is also used in DSM, as we saw in the previous section for Khalkha. This is because OT accounts are developed based on the principle of harmonic alignment of grammatical functions between different hierarchies, e.g. a grammatical relations hierarchy and a definiteness hierarchy. For instance, a human object (low on

⁹Guntsetseg (2016) may not have tested for 3rd person pronouns as she does not seem to consider them to exist in Khalkha. As she writes on p.63, “[i]n Mongolian, there are personal pronouns for first and second person, but not for the third person. Instead the simple demonstrative pronoun[s] *ene* /*ter* ‘this / that’ are used” (see also p.66).

the grammatical relations hierarchy but high on the animacy hierarchy) may be considered marked in a language, and must thus be differentially case-marked. Accordingly, a subject, high on the grammatical relations hierarchy, that is also high on any other hierarchy (e.g. an animacy hierarchy) is not predicted to be case-marked. This is contrary to what we find for embedded subjects in Khalkha, where first and second person pronouns are obligatorily accusative. Perhaps a more pertinent question to ask is whether in a language where DSM involves accusative marking (i.e. the case of objects), DOM and DSM follow distinct hierarchies relevant for objects and subjects. The answer seems to be yes in the case of Khalkha: while the hierarchies for DOM and DSM are similar, they are distinct in where the cutoff is for obligatory accusative marking. Thus, I will set aside OT accounts of DAM in this chapter.

Morphological accounts of DAM take all nominal expressions to bear an abstract Case feature, but the realisation of said Case feature may be affected by morphological operations such as allomorphy (Nuger 2016, p.150) or Impoverishment (Keine and Müller 2010, 2011, 2015).¹⁰ However, if the Khalkha data discussed in the previous subsection were purely morphologically conditioned, we would expect DOM and DSM to behave uniformly across all five syntactic environments that I have investigated in the previous section. Instead, I have shown that DSM exhibits a relatively consistent pattern across all four embedded clause types under consideration, while DOM behaves slightly differently from DSM. This asymmetry suggests that syntactic factors may play a more central role in determining DAM. I therefore set aside a purely morphological analysis of Khalkha DAM.

¹⁰A plausible candidate for morphological DOM in Khalkha may be the alternation between overt accusative case and zero accusative in the context of the RP enclitic. As discussed in Chapter 2 and also by Guntsetseg 2016, pp.58-59, 82-83, in Khalkha the accusative case morpheme is obligatory on RP-marked kinship terms while it may be zero for all other RP-marked nouns:

(i) *Overt vs. zero ACC in Khalkha*

a. $eej^{*}(-iig)=ee$
 mother^{*}(-ACC)=RP
 ‘one’s mother’

b. $muur(-iig)=aa$
 cat(-ACC)=RP
 ‘one’s cat’

The next three approaches fall under syntactic analyses of DAM. First, some accounts argue for a difference in the size of nominals (e.g. DPs vs. NPs that lack a DP layer) or their featural make-up (e.g. some nominals bear certain features that render them visible to an Agree probe). This leads to some but not all nominals being viable targets for case assignment. For size-based accounts of DOM, see, e.g., Massam 2001; Danon 2006; López 2012; Lyutikova and Pereltsvaig 2015, a.m.o.

Second, movement approaches to DAM argue that certain nominals need to move out of a certain domain. For example, under the Mapping Hypothesis of Diesing 1992, definite and specific nominals need to move out of VP to escape existential closure. There is also recent work that proposes to extend the Mapping Hypothesis to CPs to account for DSM; under such accounts, it is proposed that definite and specific nominals move out of CP (see, e.g., Predolac 2017; Ótrott-Kovács 2023). Once a nominal has moved to a higher position, it may be assigned case due to locality with a higher case-assigning head (e.g. Torrego 1998; Rodríguez-Mondoñedo 2007; López 2012), or with another DP, which induces case assignment under case competition (e.g. Baker and Vinokurova 2010; Baker 2015; Levin and Preminger 2015). See also Xu 2021 for a movement-based analysis of DSM in Delaware.

Third, licensing approaches to DAM posit that certain nominals must enter into a syntactic dependency with some functional head; in other words, they need to be exceptionally licensed. For licensing-based approaches to DOM, see Kalin 2018 on Senaya and Levin 2019 on Palauan. For an analysis of DSM involving the Person Licensing Condition (PLC) (Béjar and Rezac 2003), see Chandra and Kaur 2017 on Punjabi.

Note that size-based, movement, and licensing approaches do not exclude one another. For instance, one may imagine that in a certain language, definite nominals contain more structure or a certain feature which requires exceptional licensing. Further, in order to be local to a suitable licensor, such definite nominals have to undergo movement.

In Khalkha, a movement-based approach is independently motivated. As demonstrated in Chapter 4, accusative subjects occupy the highest phase edge of the embedded

(nominalised) clause – a position structurally higher than that of case-unmarked subjects – where they are accessible for matrix clause operations such as extraction and binding. Additionally, genitive subjects occupy Spec,nP in finite and nonfinite NCs – a position structurally higher than case-unmarked subjects in those clause types – where they are accessible for binding by the matrix subject. These structural asymmetries provide strong empirical evidence that accusative and genitive subjects have undergone movement to higher positions than case-unmarked subjects.

In the next section, I propose a movement-based account of accusative-based DAM in Khalkha. On this account, certain types of nominals bear a feature – which I dub [DAM] – that correlates with the semantic properties of nominal type, definiteness, and grammatical function discussed in this section. If a nominal bears [DAM], it becomes accessible to a movement probe on a phase head (e.g. C^0 , v^0 , D^0 , or n^0), which attracts it to the specifier of that phase. It is in this higher syntactic position that nominals may be assigned accusative case, thereby deriving the effects of DAM.

It should be emphasised that the goal of positing a [DAM] feature is for it to serve as a mechanism to account for the observed height asymmetry between accusative and case-unmarked arguments, and how it interacts with case assignment. At this stage, there are two possible implementations of the optionality observed in accusative case marking. One possibility is that nominals of a certain semantic type (e.g. unique definites) optionally bear the [DAM] feature, while the relevant phase heads always bear movement-triggering probes.¹¹ An alternative is that such nominals obligatorily bear [DAM], but the movement-triggering probe on phase heads is itself optional. However, this option would run into the challenge of explaining the obligatoriness of accusative case on certain types of nominals, such as local person pronouns. As such, I will adopt the first option. Crucially, however,

¹¹However, this would predict that in an embedded clause where the subject does not bear [DAM] but the object does, the object would always move to the embedded clause edge, yielding OSV order. This prediction is not empirically supported, as SOV order is always possible in embedded clauses containing a case-unmarked subject and accusative-marked object. To circumvent this issue, I assume that case-unmarked subjects act as defective interveners for movement of [DAM]-bearing objects.

I show that the presence or absence of accusative case on a nominal reflects the fact that that nominal has moved to Spec,CP in finite bare CPs, Spec,DP in finite and nonfinite NCs, or Spec,nP in converbial adjuncts. This provides a principled explanation for the structural and interpretive differences between accusative and case-unmarked arguments.

5.3.8 Summary

In this section, I have investigated DAM in Khalkha by considering different types of nominals and subtypes of definiteness and indefiniteness. For DOM, pronouns, proper names, anaphoric definites, and bridging definites are always marked accusative, whereas unique definites and wide scope indefinites may be accusative-marked or case-unmarked, and narrow scope indefinites (except *wh*-items and NPIs) are always case unmarked. For DSM, local person pronouns are always accusative-marked, whereas 3rd person pronouns, proper names, anaphoric definites, bridging definites, and wide scope indefinites may be accusative-marked or case-unmarked, and narrow scope indefinites (except *wh*-items and NPIs) are always case-unmarked. This asymmetry suggests that DOM and DSM are sensitive to two similar but ultimately distinct referentiality scales.

In the next section, I propose an analysis of case in Khalkha that combines a movement-based approach to DAM and dependent approach to accusative case assignment.

5.4 Dependent accusative case in Khalkha: A movement-based approach

The preceding section has established that certain kinds of objects and embedded subjects may be accusative in Khalkha. In this section, I argue that accusative in Khalkha is best analysed as a dependent case – a case assigned to the lower of two DPs in a local domain – and propose a movement-based approach to DAM that accounts for the extraction and binding facts in Chapter 4.

This section is organised as follows. In §5.4.1, I provide some background on dependent case theory and how it contrasts with a more traditional view of case as a feature assigned by functional heads via Agree. I also discuss two prominent approaches to modelling dependent accusative case: a configurational approach, under which accusative is assigned via configurational case rules, and a dependent case via Agree approach, under which a nominal that is the second goal of Agree receives accusative case marking. In §5.4.2, I motivate the idea that accusative case should be treated as a dependent case in Khalkha. Remaining neutral towards a specific implementation of dependent case until §5.5, I show in §5.4.3 that the distribution of accusative objects and embedded subjects in Khalkha can be captured via a movement-based approach to DAM and a dependent case approach to accusative case assignment. §5.4.4 summarises this section.

5.4.1 Background on dependent case assignment

In this subsection, I discuss earlier work that have proposed that matrix transitive v^0 , which also introduces the external argument, assigns accusative case. Under the expanded verbal functional domain which I adopt in this dissertation, the equivalent verbal functional head that introduces the external argument is Voice⁰. As the works I discuss below refer to matrix transitive v^0 as the head that assigns accusative case, I shall follow that terminology with the understanding that the equivalent functional head under my analysis would be Voice⁰.

Traditionally, case on a nominal has been understood as a feature that is assigned by a dedicated functional head via an Agree relation (Chomsky 2000, 2001). I refer to this as a functional-head account of case assignment. Using nominative case in English to illustrate, (337) shows that there is a strong correlation between nominative case on subjects, ϕ -agreement, and finite T⁰.

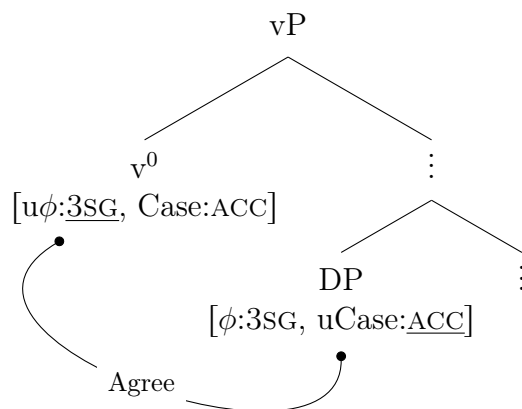
- (337) a. I believe that **he**/***him** has played the piano for two hours.
 b. I believe **him**/***he** to have played the piano for two hours.

c. **He**/***Him** was believed to have played the piano for two hours.

(337a) shows that a nominative embedded subject is possible when embedded T^0 is finite and Agrees with the subject. (337b) shows that when embedded T^0 is nonfinite and does not Agree with the embedded subject, the subject surfaces with accusative case. (337c) shows that when the embedded T^0 is nonfinite and the matrix verb is passivised, accusative is not possible on the embedded subject (as matrix passive v^0 does not assign accusative case). Instead, the embedded subject raises to matrix subject position, where it is assigned nominative case. An interpretation of these facts is to say that in (337b) and (337c), nonfinite T^0 is defective in a sense that it does not establish an Agree relation with the embedded subject. Under the view that case is assigned via an Agree relation, the embedded subject has to receive case from a different source, e.g. matrix transitive v^0 (337b) or matrix finite T^0 (337c), which assign accusative and nominative case respectively.

A functional-head account of accusative case assignment is schematised in (338) below: a functional head v^0 (Voice⁰ under my analysis) bears unvalued ϕ -features and an (abstract) ACC Case feature, while a DP in v^0 's c-command domain bears an unvalued Case feature and 3SG ϕ -features. To value its unvalued ϕ -features, v^0 Agrees with the DP and copies its ϕ -features while simultaneously valuing the DP's unvalued Case feature with ACC.

(338) *Functional-head ACC case assignment*



In contrast to a functional-head account of accusative case, dependent case theory refers to the idea that the assignment of certain cases to a nominal depends on the presence of another nominal. The traditional view of dependent case – which I shall refer to as configurational case theory – posits a DP may receive case via one of three ways, in the following order (Yip et al. 1987; Marantz 1991; McFadden 2004; Baker and Vinokurova 2010; Baker 2015; Levin and Preminger 2015; a.m.o.):

1. *Lexical/Inherent case assignment*: A DP may be assigned lexical or inherent case by a head it is selected by.
2. *Dependent case assignment*: If lexical or inherent case has not been assigned, a DP may be assigned dependent case via configurational rules.
3. *Unmarked case assignment*: If dependent case has not been assigned, a DP may be assigned unmarked case.

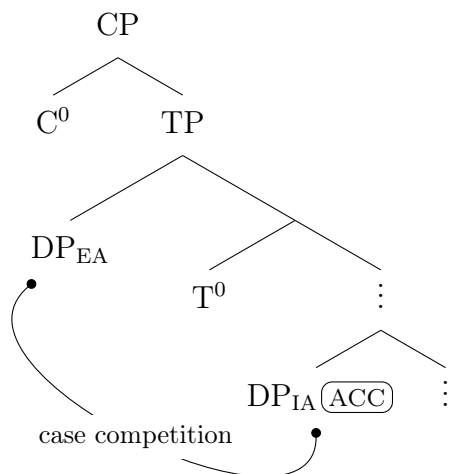
Lexical case refers to case assigned idiosyncratically by specific lexical items, while inherent case refers to case assigned by a functional head in relation to the DP’s theta role. Dependent case refers to case assigned to a DP under case competition with another DP, where two DPs are said to be in case competition if they stand in an asymmetric c-command relation in some local domain. Dependent ergative and accusative case are assigned via configurational rules, such as (339):

- (339) *Configurational case rules* (Baker 2015, p.80)
- a. If XP c-commands ZP in the same TP, then assign ergative case to XP.
 - b. If XP is c-commanded by ZP in the same TP, then assign accusative case to XP.

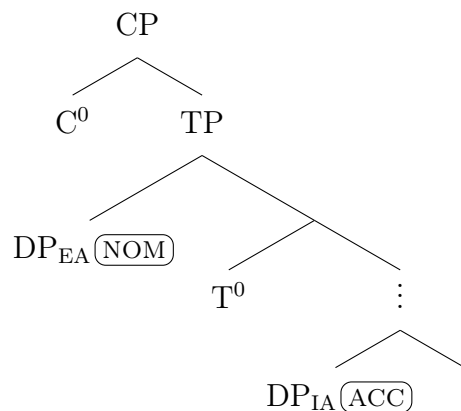
It is proposed that the directionality of case assignment is parameterised across languages: a language may either assign ergative case to the higher of two DPs (339a) or accusative case to the lower of two DPs (339b) (see also (340)). Finally, any DP that has not been assigned any case by the end of the derivation will received an unmarked (not to be confused with morphologically unmarked) case depending on the domain in which it is

located. Baker (2015) proposes that the unmarked case within TP is nominative case (see (341)), while the unmarked case within DP is genitive case.

(340) *Dependent ACC assignment*



(341) *Unmarked NOM assignment*



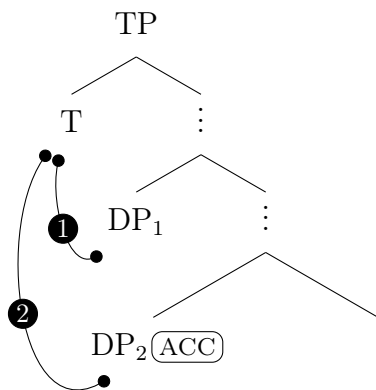
It is often assumed that case competition and unmarked case assignment happen upon phasal spellout (see, e.g., McFadden 2004; Baker 2015; Preminger 2014; Levin and Preminger 2015). Assuming C^0 is a phase head that spells out its complement upon merging and that there is no other phasal boundary between DP_{EA} and DP_{IA} , let's consider how (340) and (341) proceed derivationally. First, at the point where TP has been built, DP_1 and DP_2 have not been assigned lexical or inherent case. When the phase head C^0 is merged, this triggers spellout of its TP complement, at which point dependent case is calculated. Since DP_{EA} asymmetrically c-commands DP_{IA} and both are in the same spell-out domain, DP_{IA} is assigned accusative case via case competition, as shown in (340). Next, since DP_{EA} has not been assigned any case after the application of dependent case rules, it is assigned unmarked case within TP, i.e. nominative case, as shown in (341).¹²

¹²It is a matter of continued debate whether a language can contain two modalities of case assignment, i.e. both dependent case assignment and functional-head case assignment. For example, while Baker and Vinokurova (2010) argue that Sakha (Turkic) possesses a hybrid case assignment system where accusative is assigned via configurational case rules and nominative is assigned by T^0 , Levin and Preminger (2015) argue that the facts can also be accounted for under a fully configurational treatment.

Configurational approaches to dependent case rely on configurational case rules for case assignment, e.g. (339). One downside of this is that it requires an entirely separate operation not seen elsewhere in syntax; see, e.g., chapter 7 of Baker 2015 and Bárány and Sheehan 2024 for some discussion. In recent work, however, Clem and Deal (2024) and Poole (2024) argue that Agree – an operation that is independently needed in syntax – can be leveraged to account for patterns of case dependency without recourse to configurational rules.

Under Clem and Deal’s (2024) account, an Agreeing head may Agree with multiple goals, and certain cases like accusative and ergative are the spell out of the features of a previous goal of Agree on another DP. In contrast, Poole (2024) proposes that an Agreeing head may contain multiple ordered probes; when the first probe successfully Agrees with a goal, it activates the second probe and it is the second probe that assigns accusative or ergative case. Crucially, both works converge on the idea that in accusative systems, the Agreeing head – T^0 – locally c-commands and Agrees with two goal DPs, and the second goal of Agree surfaces with accusative case. The two approaches are schematised in (342).

(342) *Dependent ACC case via Agree (Clem and Deal 2024; Poole 2024)*



Since accusative case on DP_2 depends on T^0 having first Agreed with DP_1 , Agree-based approaches to dependent case derive the core intuition that accusative case on a nominal is dependent on the presence of another nominal. I discuss Clem and Deal’s (2024) and Poole’s (2024) approaches in more detail in §5.5.

To summarise, the main difference between a functional-head approach vs. a dependent case approach to accusative case assignment resides in the source of accusative case. Under a functional-head approach, accusative case is assigned by transitive v^0 under Agree. Under a dependent case approach, I have introduced two implementations: a configurational approach (e.g. Yip et al. 1987; Marantz 1991; McFadden 2004; Baker and Vinokurova 2010; Baker 2015; Levin and Preminger 2015) or a dependent case via Agree approach (Clem and Deal 2024; Poole 2024). A configurational approach assigns accusative case to the lower of two nominals in a local c-command relationship via configurational case rules. An Agree-based approach requires a distinct head to Agree with two nominals in its c-command domain, with the second goal of Agree surfacing with accusative case. Despite their differences, dependent case approaches claim that accusative case depends on the presence of another higher nominal regardless of the presence of transitive v^0 . In the next subsection, I motivate the idea that accusative case in Khalkha should be treated as a dependent case and not a case assigned by transitive v^0 .

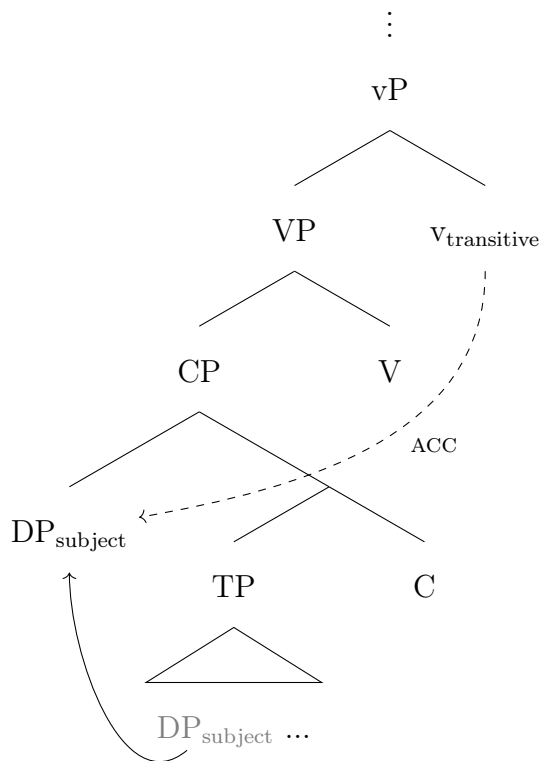
5.4.2 Accusative as a dependent case in Khalkha

As shown in this dissertation thus far, Khalkha displays an exceptionally wide distribution of accusative case – accusative appears not only on direct objects but also on subjects of the four embedded clause types under consideration. Following many previous works which have argued that accusative case behaves like a dependent case in Khalkha (Aravind 2021; Peters 2020, 2024; Gong 2022, 2025; cf. Fong 2019), the goal of this subsection is to establish that a functional-head approach like in (338) cannot successfully capture the distribution of accusative case in Khalkha.

Adopting a functional-head approach, Fong (2019) proposes that accusative case on subjects of finite bare CPs is assigned by matrix transitive v^0 (Voice⁰ under my account). She proposes that C^0 may bear a ϕ -probe that attracts the embedded subject to Spec,CP.

The subject in Spec,CP escapes spellout within the CP phase and remains visible to matrix transitive v^0 for accusative case assignment. This is schematised in (343).

(343) ACC assigned by matrix transitive v^0 (Fong 2019)



The main argument against such a functional-head approach relies on showing that accusative subjects are acceptable in contexts where one would not expect them to be. I discuss two such environments: i) the availability of accusative when the matrix verb assigns a non-accusative case, e.g. psych predicates; and ii) the availability of accusative when the matrix verb is unaccusative or passivised.

First, as noted by Aravind (2021); Gong (2022, 2025); Peters (2024), a psych predicate such as *uurla-h* ‘to be angry’ assigns dative but not accusative case to its internal argument (344a). When *uurla-h* selects for a nonfinite complement NC, the clause is marked with dative case as expected, but the embedded subject of the clause may surface with accusative but not dative case (344b). Given that the subject of a psych predicate receives an experiencer theta role, I assume that psych predicates are unaccusative predicates, i.e. they do not

contain transitive Voice⁰ that assigns an agent theta role. This suggests that the source of accusative on the embedded subject in (344b) cannot be a matrix transitive Voice⁰ within the psych predicate *uurla-h*, but the presence of the matrix subject *Naraa*.

- (344) *ACC subject possible in nonfinite complement NC selected by a psych predicate*
- a. Naraa **hüühed-ed**/*-iig uurla-san.
 N. child-DAT/*-ACC be.angry.PST
 ‘Naraa got angry at the child.’
- b. Naraa [**hüühed-iig**/*-ed bagsh-iig=aa haz-san]-d uurla-san.
 N. child-ACC/*-DAT teacher-ACC=RP bite-PST -DAT be.angry-PST
 ‘Naraa got angry because the child bit his teacher.’ (Aravind 2021, p.383)

Similarly, Gong (2022) shows that the embedded subject of a finite complement NC may surface with accusative case, even though the matrix predicate *itgeltei bai-h* ‘to believe (lit. to be with belief)’ assigns dative case to the clause (345).

- (345) *ACC subject possible in finite complement NC selected by non-ACC-assigning predicate*
- Bi [**Chang-iig** amid bai-gaa gedeg]-t itgel-tei bai-na.
 1SG C.-ACC alive COP-IPFV C.NML -DAT belief-COM COP-NPST
 ‘I believe that Chang is alive.’ (Gong 2022, p.141, ex.212)

(346) shows that accusative may be assigned to the embedded subject within a converbial adjunct. This is in spite of the fact that the matrix predicate *uulz-ah* ‘to meet’ selects for a comitative but not an accusative internal argument (347), which suggests that *uulz-ah* does not contain a transitive Voice⁰ which assigns accusative case.

- (346) *ACC subject possible in converbial adjunct*
- [**Tüüniig** Mongol-d ir-hleer]₁ bi ₁ tüüntei uulz-na
 3SG.ACC M.-DAT come-SUCC.CVB 1SG 3SG.COM meet-NPST
 ‘When he comes to Mongolia, I’ll meet him.’ (Tserenpil and Kullmann 2008, p.165)
- (347) *Uulz-ah ‘to meet’ does not assign ACC*
- Bi **tüüntei**/*tüüniig uulz-na.
 1SG 3SG.COM/3SG.ACC meet-NPST
 ‘I will meet her/him.’

Second, Guntsetseg (2016, p.158) and Peters (2024) note that in converbial adjuncts, the embedded subject can surface with accusative case even when the matrix predicate is

intransitive (348)-(349). Note that the converbial adjunct in (349) has also been displaced to clause-initial position.

- (348) *ACC subject possible with matrix unergative verb*
 Tuyaa [**Dorj-iig** ir-megts] yav-san.
 T. D.-ACC come-IMM.CVB go-PST
 ‘Tuyaa went as soon as Dorj came.’ (Guntsetseg 2016, p.140, ex.318b)

- (349) *ACC subject possible with matrix unaccusative verb*
 [**Dorj-iig** ir-megts]₁ Tuyaa ₁ övd-sön.
 D.-ACC come-IMM.CVB T. become.sick-PST
 ‘As soon as Dorj came, Tuyaa fell ill.’ (Peters 2024, p.5, ex.7)

Accusative subjects are also possible in converbial adjuncts when the matrix verb is passivised (350).

- (350) *ACC subject possible with matrix passivised verb*
 Minii eej [**namaig** nohoi ger-t=ee avchra-nguut]
 1SG.GEN mother 1SG.ACC dog home-DAT=RP bring-CCMT.CVB
 tüünd haz-agd-san.
 3SG.DAT bite-PASS-PST
 ‘Immediately after I brought a dog home, my mother was bitten by it.’

Under the view that Voice⁰ of unaccusative intransitive and passivised verbs does not assign accusative case, the data in (348)-(350) indicate that the source of accusative cannot be from matrix Voice⁰, but rather the presence of a higher nominal – the matrix subject.

In sum, this subsection has shown that accusative case may be assigned to embedded subjects in environments in which we expect accusative case assignment to be impossible. This compels us to treat accusative as a dependent case in Khalkha – a case that is assigned to the lower of two (or more) nominals in a local domain. In the next subsection, I propose a dependent analysis of accusative case in Khalkha that incorporates a movement-based account of DAM to capture the highest-phase-edge positions of accusative arguments, as I have argued for in Chapter 4.

5.4.3 Deriving dependent accusative case in Khalkha

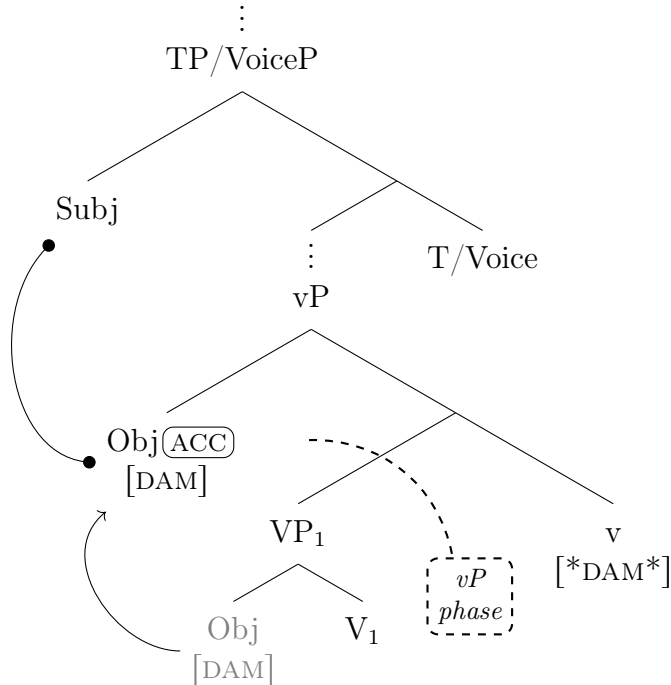
Building on the findings in §5.3, I propose that in Khalkha, certain nominals bear a feature [DAM] correlated with the properties of DOM and DSM. Additionally, building on the findings in Chapter 4, I propose that the phase heads v^0 , C^0 , n^0 , and D^0 (see, e.g., Chomsky 2000, 2001, 2008; Marantz 2001; Svenonius 2004; Bošković 2005) in Khalkha contain a movement probe that targets a [DAM]-bearing nominal to the specifier of the respective phase head. I will represent this probe as [$*_{DAM}$]. As discussed in §5.3.7, I will adopt the view that [DAM] may be optional on certain types of nominals, and that the relevant phase heads always bear the probe [$*_{DAM}$].

In what follows, I will demonstrate how accusative can be assigned dependently to a [DAM]-bearing nominal in five contexts – matrix root clauses, finite bare CPs, finite NCs, nonfinite NCs, and converbial adjuncts – while remaining neutral to the exact implementation of dependent case. I also abstract away from the exact position of the matrix subject in my trees below – it may be located in Spec,VoiceP or have moved to Spec,TP; in either position, the matrix subject locally c-commands a matrix object located in Spec,vP and a subject located in the highest phase edge of the embedded (nominalised) clause.

To begin with, consider how accusative case is assigned to [DAM]-bearing objects in matrix root clauses. As shown in §5.3.1, since pronoun, proper name, anaphoric definite, and bridging definite objects are obligatorily accusative, I propose that these objects always bear [DAM]; since unique definite and wide scope indefinite objects are optionally accusative, I propose that they optionally bear [DAM]; and since narrow scope indefinite objects (with the exception of *wh*-items and NPIs) may not be accusative, I propose that they never bear [DAM]. If an object bears [DAM], it will be targeted by the movement probe on v^0 , which moves it to its specifier before spelling out its VP complement. Subsequently, the agentive subject is merged in Spec,VoiceP and moves to Spec,TP. The [DAM]-bearing object is assigned dependent accusative case due to its proximity to the agentive subject. This is

illustrated in (351), where $\bullet \longrightarrow \bullet$ represents the dependency between two nominals under which accusative case is assigned.

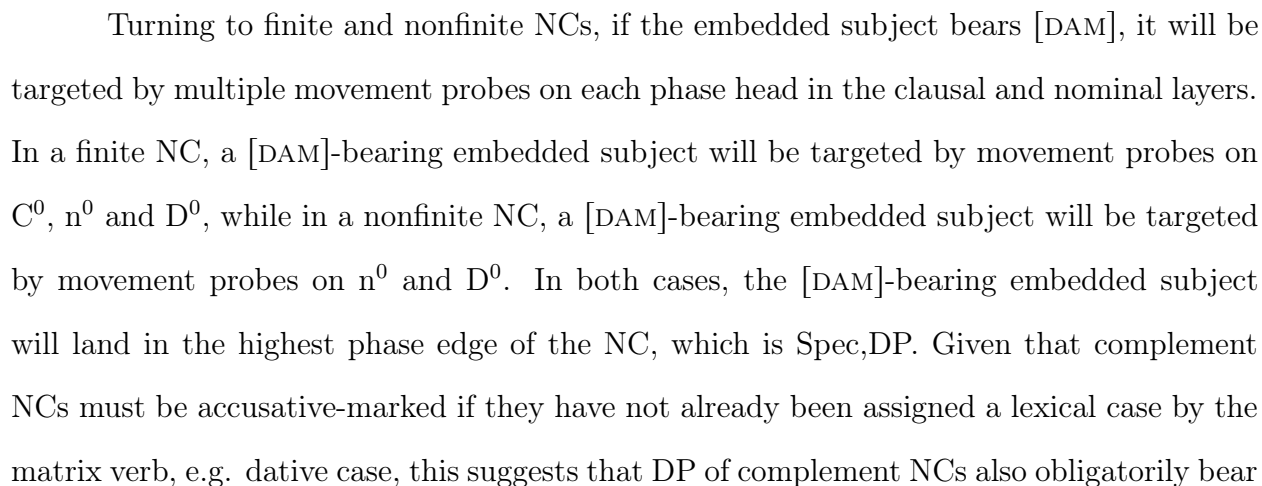
(351) *Dependent ACC assignment to object in matrix root clause*



Next, consider how accusative case is assigned to [DAM]-bearing subjects in each of the four embedded clause types under consideration. As shown in §5.3.2-§5.3.5, local person embedded subjects are obligatorily accusative and I propose that they always bear [DAM]; 3rd person, proper name, anaphoric, bridging, unique definite and wide scope indefinite embedded subjects are optionally accusative, thus I propose that they optionally bear [DAM]; finally, narrow scope indefinite embedded subjects (except for *wh*-items and NPIs) may not be accusative, and I propose that they do not bear [DAM].

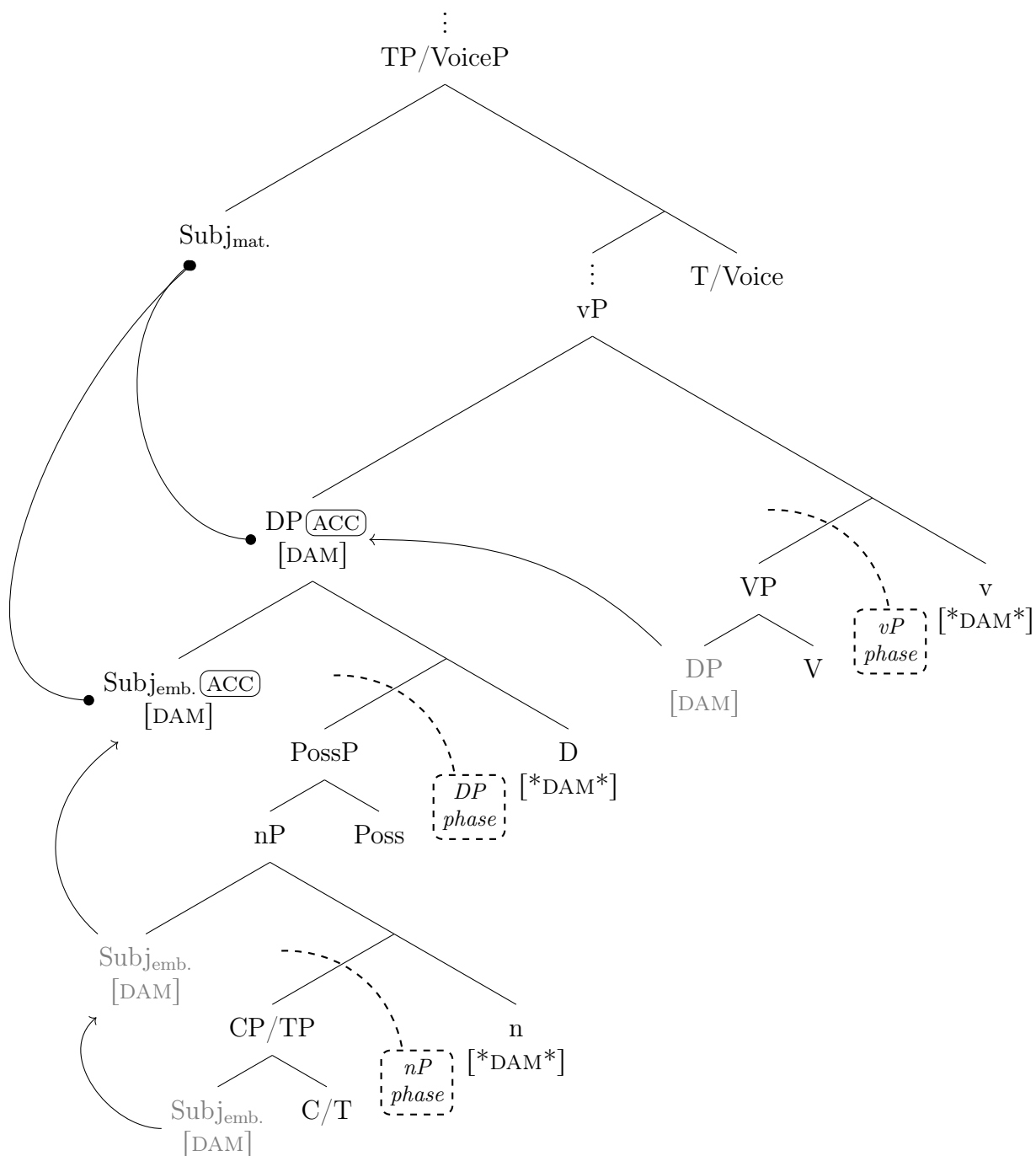
In a finite bare CP, if the embedded subject bears [DAM], it will be targeted by the movement probe on C^0 , which moves it to Spec,CP before the CP phase is spelled out. Following Gong 2022, p.143, fn.10 and Baker and Vinokurova 2010, p.617, fn.19, I assume that finite bare CPs obligatorily shift out of the matrix VP to Spec,vP, thereby escaping spellout within the matrix vP phase, which implies that CPs also obligatorily bear [DAM]. Next, the matrix subject is merged in Spec,VoiceP and subsequently moves to Spec,TP. The

(352) *Dependent ACC assignment to subject in finite bare CP*



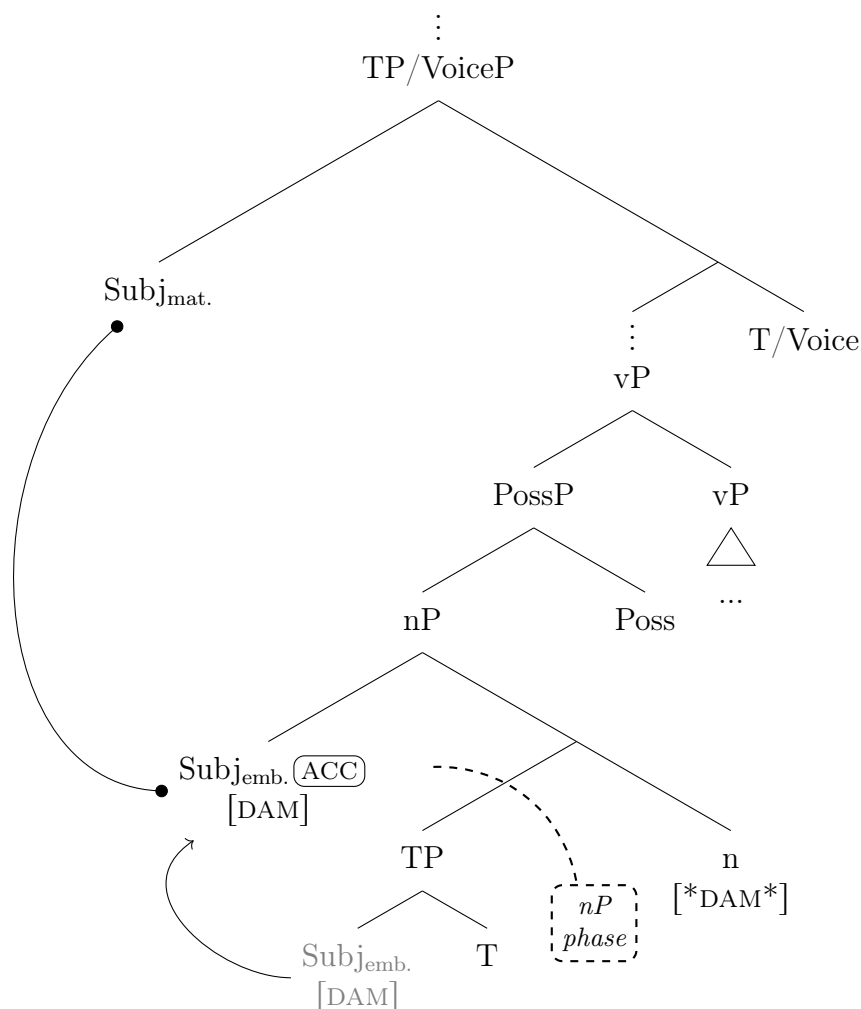
[DAM]. Thus, complement NCs also shift out of the matrix VP to Spec,vP, thereby escaping spellout within the matrix vP phase. Next, the matrix subject is merged in Spec,VoiceP and subsequently moves to Spec,TP. The [DAM]-bearing embedded subject is assigned dependent accusative case due to its proximity with the matrix subject. This is illustrated in (353).

(353) *Dependent ACC assignment to finite/nonfinite NC and its subject*



Finally, in converbial adjuncts, if the embedded subject bears [DAM], it will be targeted by a movement probe on n^0 , which moves the subject to Spec,nP before spellout of the nP phase. The converbial adjunct is adjoined to matrix vP, as I argued in Chapter 2. Next, the matrix subject is merged in Spec,VoiceP and subsequently moves to Spec,TP. The [DAM]-bearing embedded subject is assigned dependent accusative case due to its proximity with the matrix subject. This is illustrated in (354).

(354) *Dependent ACC assignment to subject in converbial adjunct*



In sum, this subsection has shown how a movement-based approach to DAM, combined with dependent case assignment, is able to derive the wide distribution of accusative case in Khalkha. Crucially, the present account derives the highest-phase-edge positions of accusative subjects, as argued for in Chapter 4, while demonstrating that accusative case

assignment depends on the presence of another higher argument and is independent of the properties of Voice⁰.

5.4.4 Summary

This section began by contrasting two approaches to accusative case assignment: a functional-head approach, in which accusative is assigned by transitive Voice⁰ via Agree, and a dependent case approach, in which accusative arises in the presence of a higher c-commanding nominal. I have argued that accusative in Khalkha behaves like a dependent case (Aravind 2021; Peters 2020, 2024; Gong 2022, 2025), contra analyses that treat it as assigned by a functional head (Fong 2019). This conclusion is supported by evidence showing that accusative subjects are attested in environments where (i) the matrix predicate assigns a non-accusative case (e.g. psych predicates), and (ii) matrix Voice⁰ is unaccusative or passive.

I have shown that the broad distribution of accusative case in Khalkha – on objects and on embedded subjects of the four embedded clause types under consideration – can be captured by combining a dependent approach to accusative case assignment and a movement-based account of DAM. I have argued that certain nominals bear a feature [DAM], which is correlated with the interpretive properties of DAM. These [DAM]-bearing nominals are targeted by movement probes on the phase heads v^0 , C^0 , n^0 , and D^0 . [DAM]-bearing objects thus move to Spec,vP, while [DAM]-bearing embedded subjects move to the highest phase edge of the embedded (nominalised) clause, which may be Spec,CP for finite bare CPs, Spec,DP for finite and nonfinite NCs, or Spec,nP for converbial adjuncts. In these positions, the object or embedded subject is sufficiently local to the matrix subject to receive accusative case under a dependent case approach.

While the evidence provided in this section clearly favours a dependent analysis of accusative case in Khalkha, I have remained agnostic about its formal implementation – whether it arises via configurational case rules (e.g. Yip et al. 1987; Marantz 1991; McFadden 2004; Baker and Vinokurova 2010; Baker 2015; Levin and Preminger 2015) or via Agree (Clem

and Deal 2024; Poole 2024). A configurational approach to dependent case in Khalkha has been proposed by Peters 2020, 2024; Aravind 2021; Lim 2022a, 2024a; Gong 2022, 2025. In the next section, I turn to novel data from subjects of subject clauses in Khalkha, which provide preliminary evidence in favour of the Agree-based approach.

5.5 Subject clauses in Khalkha and dependent case via Agree

The previous section has argued for a dependent treatment of accusative case in Khalkha combined with a movement-based account of DAM. In this section, I introduce novel data from subjects of subject clauses in Khalkha, demonstrating that they provide some preliminary evidence that favours an Agree-based approach (Clem and Deal 2024; Poole 2024) to dependent case assignment over one that relies on configurational case rules (e.g. Yip et al. 1987; Marantz 1991; McFadden 2004; Baker and Vinokurova 2010; Baker 2015; Levin and Preminger 2015).

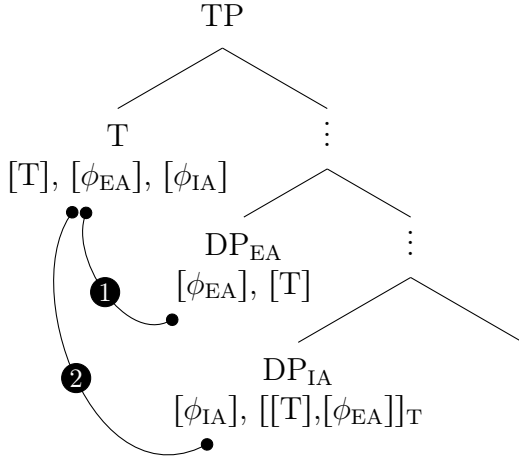
This section is organised as follows. In §5.5.1, I compare and contrast two Agree-based approaches to dependent case proposed by Clem and Deal (2024) and Poole (2024). In §5.5.2, I identify a novel empirical domain which has the potential to adjudicate between a configurational vs. an Agree-based account of dependent case assignment. In §5.5.3, I discuss data from subject clauses in Khalkha, in which one of my consultants consistently accepts accusative subjects, and the other two accept them to varying degrees. While the availability of accusative subjects in subject clauses is problematic for dependent case theories that rely on configurational rules, I show in §5.5.4 how they can be accounted for under a dependent case via Agree approach, following Clem and Deal 2024. §5.5.5 summarises this section.

5.5.1 Two Agree-based approaches to dependent case

As I have discussed in §5.4.1, Clem and Deal (2024) and Poole (2024) propose that dependent accusative case can be modelled using the operation Agree. Under either of their approaches, they derive the core intuition that dependent case on a nominal depends on it being the second goal of Agree by the same functional head. Further, both approaches converge on the idea that in accusative languages, the relevant functional head that c-commands the two nominals is T^0 , and that the lower of the two nominals is marked accusative. In this subsection, I discuss each approach in more detail.

Clem and Deal (2024) propose that certain cases such as accusative and ergative are the spell out of features of a prior goal of Agree on a subsequent goal of Agree by the same functional head. The authors adopt an Interaction/Satisfaction model of Agree, under which an Agree probe is specified with an Interaction and a Satisfaction Condition (Deal 2015, 2024). A probe’s Interaction condition specifies what features are copied back onto the probe, while its Satisfaction condition specifies what features will cause the probe to stop searching. A probe may lack a Satisfaction condition altogether, which entails that it will Agree with every goal within its search domain, i.e. it is insatiable. Clem and Deal also propose that Agree involves a bidirectional transfer of features between a probe and a goal. Thus, at each cycle of Agree, if the probe copies features of the goal back onto the probe (determined by the probe’s Interaction condition), the probe will also transfer its features to the goal. This latter process is dubbed “goal flagging” by Deal (to appear). Assuming that T^0 Interacts with ϕ and is insatiable (i.e. it lacks a Satisfaction condition); this implies that T^0 will Agree with all ϕ -bearing goals it locally c-commands. This is illustrated in (355).

(355) *Dependent ACC case via Agree, following Clem and Deal 2024*



The derivation in (355) proceeds as follows:

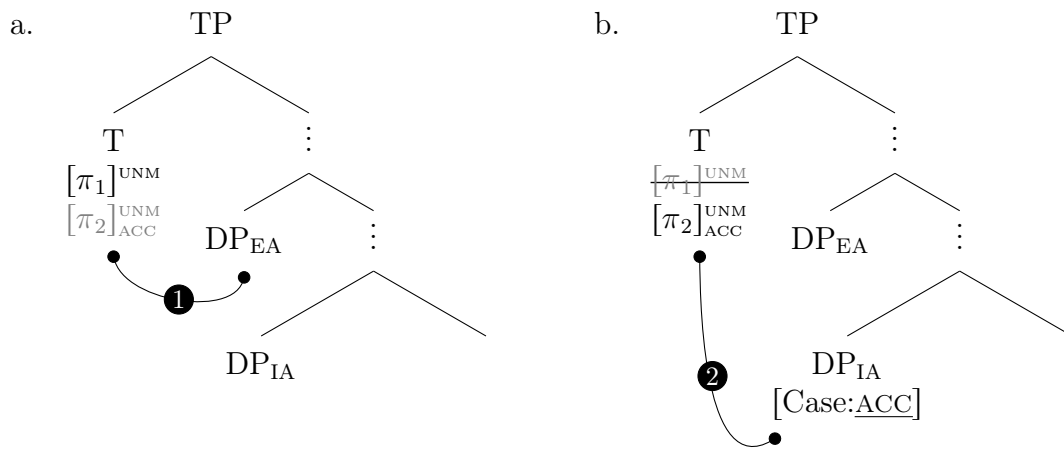
- Step ❶: T^0 Agrees with DP_{EA} and copies its features $[\phi_{EA}]$ back onto T^0 (Interaction). The features of T^0 , e.g. $[T]$ are also transferred onto DP_{EA} (goal flagging).
- Step ❷: As the probe on T^0 is insatiable, T^0 continues to probe its c-command domain and Agrees with the next active goal, DP_{IA} , copying $[\phi_{IA}]$ back onto itself (Interaction). Concurrently, T^0 transfers its features – $[T]$ and $[\phi_{EA}]$, bundled into $[[T], [\phi_{EA}]]_T$ – onto DP_{IA} (goal flagging).

At spellout, the ϕ -features of DP_{EA} that are transferred to DP_{IA} via goal-flagging will be realised as the accusative case morpheme via the Vocabulary Insertion (VI) rule in (356). Essentially, (356) states that the accusative case morpheme is the realisation of another DP's ϕ -features in the context of a DP's own ϕ -features. This derives the core intuition that accusative case on a nominal is dependent on the presence of another nominal.

(356) *ACC vocabulary item*
 $ACC \leftrightarrow [\phi_2] / [D, \phi_1] _$

Poole (2024) proposes an alternative account in which an Agreeing head contains an ordered “probe stack” (i.e. an ordered set of probes). Given a probe stack $\pi_1 > \pi_2 > \dots > \pi_n$, a probe π_m only becomes active when probe π_{m-1} is valued. The dependent nature of accusative case can thus be derived if i) the first probe π_{m-1} does not assign case, and ii) the second probe assigns case but its ability to Agree depends on whether the first probe has been valued. Consider (357):

(357) *Dependent accusative case via Agree, following Poole 2024*



Following Poole, let’s assume that probes are case discriminating (represented as a superscript on the probe) and that they may assign case (represented as a subscript on the probe). For example, a probe $[\pi_2]^{\text{UNM}_{\text{ACC}}}$ will target case-unmarked (UNM) DPs for Agree and assign them accusative (ACC) case. In (357a), $[\pi_1]^{\text{UNM}}$ Agrees with DP_{EA} , which is case-unmarked, but does not assign it any case. This first Agree step activates the second probe in T^0 ’s probe stack, $[\pi_2]^{\text{UNM}_{\text{ACC}}}$. Since DP_{EA} has already established an Agree relation with T^0 on the first cycle of Agree, it will no longer be able to participate in subsequent cycles of Agree. Thus, on the second cycle of Agree, $[\pi_2]^{\text{UNM}_{\text{ACC}}}$ Agrees with DP_{IA} , which is case-unmarked, and values it accusative.

In Khalkha, it is possible for multiple arguments to be accusative. In (358) for example, both the embedded subject of the converbial adjunct and the matrix object are assigned accusative case. Note that the converbial adjunct has been displaced to clause-initial position.

- (358) [**Namaig** hicheel-Ø=ee duus-tal]₁ ter __₁ minii huu-**g** har-dag.
 1SG.ACC lesson-ACC=RP finish-TERM.CVB 3SG 1SG.GEN son-ACC watch-HAB
 ‘She/he looks after my son until I have finished my lessons.’
 (Tserenpil and Kullmann 2015, p.167, cited in Gong 2022, p.142, ex. 213)

The appearance of multiple accusative cases is predicted under Clem and Deal’s approach (as well as under a standard configurational approach): an insatiable probe may initiate multiple cycles of Agree, resulting in multiple goals bearing the ϕ -features of a previously-Agreed-with goal, which are later realised as the accusative case morpheme via VI rules. However, under Poole’s (2024) implementation (although he does not explicitly address this issue), this would imply the presence of as many accusative-assigning probes as there are accusative-marked nominals. This raises the question as to whether there is any principled limit as to how many probes a probe stack may contain, and how one can ensure that are always enough probes for all the goals. Thus, the appearance of multiple accusative cases may provide some support for adopting Clem and Deal’s (2024) implementation of dependent case via Agree over Poole’s (2024).¹³

In the next subsection, I discuss a novel empirical domain – previously unnoticed by Clem and Deal (2024) and Poole (2024) – that has the potential to adjudicate between adopting a configurational vs. Agree-based approach to dependent case assignment.

¹³As an aside, in ergative case systems, Clem and Deal (2024) and Poole (2024) propose that a probe on v^0 undergoes cyclic expansion (Béjar and Rezac 2009; Clem 2023) after Agreeing with the internal argument. Subsequently, the intermediate projection of the probe may enter a second cycle of Agree with the external argument, thus deriving the dependent nature of ergative case marking via Agree.

5.5.2 Where dependent case theories come apart

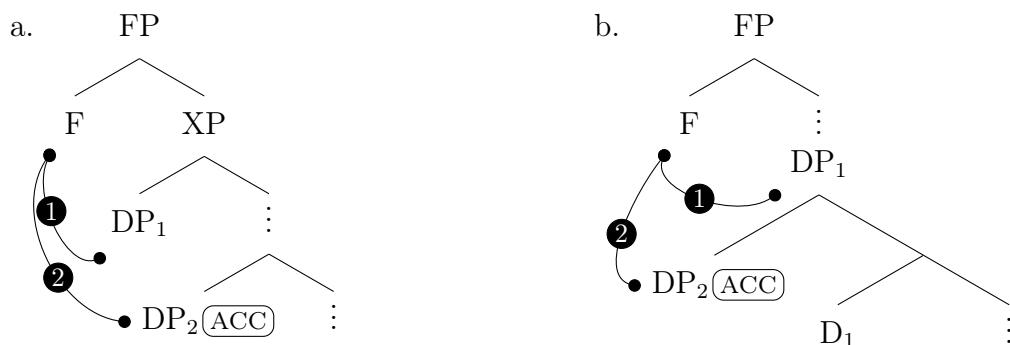
In §5.4, I have shown that accusative case assignment in Khalkha arises from a local c-command relationship between the matrix subject and either a direct object or an embedded subject. This dependency is straightforwardly encoded in configurational case rules, as shown in (339), repeated below.

- (339) *Configurational case rules* (Baker 2015, p.80)
- a. If XP c-commands ZP in the same TP, then assign ergative case to XP.
 - b. If XP is c-commanded by ZP in the same TP, then assign accusative case to XP.

Under a configurational approach, the assignment of dependent case strictly requires that one nominal locally c-command another, as explicitly dictated by rules such as (339).

In contrast, under a dependent case via Agree approach, the relationship between two nominals is mediated by a functional head that locally c-commands both. While the contexts examined in §5.4 involved two nominals in a local c-command relationship and are compatible with the Agree-based approach, the Agree-based approach also allows for the theoretical possibility that the two nominals are *not* in a local c-command relation with each other, as long as both fall within the local domain of the same functional head F^0 . This distinction is illustrated in (359).

- (359) *Two types of dependency between nominals under dependent case via Agree*



In (359a), both DP₁ and DP₂ are in a local c-command relationship and are c-commanded by F^0 – this structure exemplifies all the cases examined in §5.4.3. In contrast, in (359b),

DP2 is structurally contained within DP1 and is not c-commanded by it. Nevertheless, both DPs are still c-commanded by F^0 , suggesting that under a dependent case via Agree approach, DP2 should remain a viable target for dependent accusative case assignment via a second cycle of Agree.

The structure in (359b) constitutes a novel empirical domain, previously unnoticed in Clem and Deal 2024 or Poole 2024, which may serve to adjudicate between configurational and Agree-based approaches to dependent case. A configurational account would predict that accusative case should not be licensed in (359b), while an Agree-based account does predict its availability. In the following subsection, I present novel data from subject clauses in Khalkha, demonstrating that for at least one of my consultants, accusative case may be consistently assigned in precisely the type of structure found in (359b).

5.5.3 Subject clauses in Khalkha

Previous literature has noted that Khalkha subject clauses can be non-nominalised (i.e. finite bare) CPs or NCs (Tserenpil and Kullmann 2008, p.395; Guntsetseg 2016; Aravind 2021; Gong 2022, pp.22-23; Peters 2024).¹⁴ (360) shows a finite bare subject CP, (361) a finite NC subject, and (362) a nonfinite NC subject. These subject clauses originate as

¹⁴Aravind (2021) shows that finite bare CPs cannot be subjects, e.g. (i), while Fong (2019) and Peters (2024) show that they can be subjects, e.g. (ii). Note that both examples involve the use of a passivised matrix verb. I suggest that the discrepancy may be due to the fact that passive verbs and subject clauses are relatively uncommon in Khalkha (see Tserenpil and Kullmann 2008, p.124, 395 for discussion). My consultants indeed find both (i) and (ii) strange; they remark that *hel-egd-eh* ‘to be said’ is usually used to express universal truths. For one consultant, G, he uses *med-egd-eh* ‘(lit.) to be known’ interchangeably with its causative counterpart *med-üül-eh* ‘to notify (lit. cause to know)’, suggesting that the passive meaning has been completely lost in his grammar. In my examples of subject clauses, I have settled on a set of morphologically passive verbs that are accepted and used by all three of my consultants in passive constructions.

- (i) * [Bat höl-Ø=öö hölgäl-san gej](=n') med-egd-sen.
 B. leg-ACC=RP break-PST C (=3.POSS) know-PASS-PST
 Int. ‘That Bat broke his leg is known.’ (judgement reported in Aravind 2021, p.379, ex.5)
- (ii) [Dulmaa/*Dulmaa-g sain seheeten gej] hel-egd-sen.
 D./*D.-ACC good noble C say-PASS-PST
 ‘It was said that Dulmaa is good and noble.’ (judgement reported in Fong 2019, p.9, ex.28)

internal arguments of the matrix predicate, e.g. a passivised verb *sonsogd-oh* ‘to be heard’ in (360) or a passivised predicate *chuhai sanagdah* ‘to feel/seem important (lit. to be thought important)’ in (361)-(362).

(360) *Finite bare subject CP*

[**Bilgüün** hural-d yav-san gej] sons-ogd-son.
 B. meeting-DAT go-PST C hear-PASS-PST
 ‘It was heard that Bilguun went to a meeting/conference.’

(361) *Finite NC subject*

[**Naraa-giin/Naraa** hural-d yav-san gedeg]*(=n’)
 N.-GEN/N. meeting-DAT go-PST C.NML *(=3.POSS)
 Bilgüün-d chuhai sana-gd-san.
 B.-DAT important think-PASS-PST
 ‘It felt/seemed important to Bilguun that Naraa went to a meeting/conference.’

(362) *Nonfinite NC subject*

[**Naraa-giin/Naraa** hural-d yav-san]*(=n’)
 N.-GEN/N. meeting-DAT go-PST *(=3.POSS)
 Bilgüün-d chuhai sana-gd-san.
 B.-DAT important think-PASS-PST
 ‘It felt/seemed important to Bilguun that Naraa went to a meeting/conference.’

Tserenpil and Kullmann (2008, p.395) note that the subjects of (nominalised) subject clauses are *usually* realised in the genitive case; since finite bare CPs cannot contain genitive subjects, the subject is case-unmarked (see also Fong 2019). My consultants consistently accept case-unmarked subjects in subject NCs, as shown in (361)-(362) (see Aravind 2021; Gong 2022, p.150, fn.14; and Peters 2024 for similar observations).

Subject clauses may also originate as an external argument of the matrix predicate, e.g. with the causative verb *gaishruul-ah* ‘to surprise (tr.)’ shown in (363) or a transitive verb *notl-oh* ‘to prove’ shown in (364) below. In (363), the subject NC triggers accusative case on *Dorj*, while in (364), the subject NC triggers accusative case on the complement NC.

- (363) [Minii sugalaa-nd hoj-son]*(=**n'**/**%**=**min'**) Dorj-iig gaihsh-ruul-san.
 1SG.GEN lottery win-PFV *(=3.POSS/**%**1SG.POSS) D.-ACC surprise-CAUS-PST
 'That I won the lottery surprised Dorj.'
- (364) [Misheel-iin/Misheel üdeshleg-t yav-san]*(=**n'**)
 M.-GEN/M. party-DAT go-PFV *(=3.POSS)
 [tüüniig ger-t=ee bai-gaa-güi]-g notol-j bai-na.
 3SG.ACC home-DAT=RP be-IPFV-NEG -ACC prove-PROG COP-NPST
 'That Misheel went to a party proves that he wasn't at home.'

Subject clauses such as (363)-(364) are particularly significant as they do not raise from the complement position of the matrix verb. As such, these clauses are not c-commanded by a higher nominal – a point to which I return shortly.

An interesting property of all subject NCs is the obligatoriness of ϕ -morphology on the clause, whereas in complement clauses such morphology is optional. Gong (2022, p.150) suggests that 3.POSS morphology “is used to nominalize clauses of various size[s], with no possessive interpretation”. However, it is not just 3.POSS morphology, but ϕ -morphology more generally that is obligatory on subject clauses. For example, (363) shows that the ϕ -morphology can optionally covary (for some speakers) with the ϕ -features of the embedded subject. I take the obligatoriness of ϕ -morphology on all subject NCs to be a diagnostic property of their subjecthood, regardless of whether they originate as an internal or external argument.¹⁵ For consistency, I will represent all subject clauses below with obligatory 3.POSS morphology, which appears to be the default option across all my consultants.

¹⁵I do not currently have a deeper explanation regarding the obligatoriness vs. optionality of ϕ -morphology on subject NCs vs. complement NCs.

While the previous literature has reported that subject clauses may not contain accusative subjects (Fong 2019; Aravind 2021; Gong 2022, 2025; Peters 2024),¹⁶ data collected across my three consultants suggests that there is some variation as to whether accusative subjects are permissible in subject clauses.

I begin with a discussion of constructed examples that I tested with my consultants. While I only show nonfinite subject NCs below, the same patterns I report also hold for finite subject NCs. Focusing on speakers B and M first, they mostly disallow accusative subjects in all types of subject clauses: a nonfinite NC subject with a complement nominal (365a);

¹⁶Peters (2024, p.10) presents some interesting data where some speakers of Southern Mongolian also allow subjects of subject clauses to surface with accusative case, as seen in (i). Even more puzzling is that such speakers also require the purported subject clause to surface with accusative case, as seen in (ii).

- (i) *čagdaa-d* [**hulgaič-ig** *yüm* *hulgail-san* *gež*] *mede-gd-sen*
 police-DAT thief-ACC thing steal-VRN.PST C know-PASS-VRN.PST
 ‘By the police, it was discovered that the thief stole something.’ (Peters 2024, p.10, ex.17a)
- (ii) *surguul-iin* *jaxirul-d* [*Tuyaa* *ir-sen*] ***(-iig)** *mede-gd-sen*.
 school-GEN director-DAT Tuyaa come-VRN.PST *(-ACC) know-PASS-VRN.PST
 ‘That Tuyaa had come was discovered by the school director.’ (Peters 2024, p.10, ex.17b)

Peters (2024) suggests that the presence of another argument – the dative marked *čagdaa* ‘police’ (i) and *surguulin jaxirul* ‘school director’ in (ii) – triggers dependent accusative case assignment on the embedded subject in (i) and on the subject clause in (ii).

Note, however, that some morphologically passive verbs, such as *med-egd-eh* ‘to be known’, have grammaticalised into transitive verbs in Khalkha. For example, the most salient meaning of *med-egd-eh* is ‘to inform/notify’ for all three of my consultants. As evidence, *med-egd-eh* is able to take a case-unmarked matrix subject and an accusative-marked NC as shown in (iii). Additionally, for speaker G, the verb seems to have completely lost its passive meaning and can only be understood as transitive, while for speakers B and M, the verb can be optionally understood as passive or transitive.

- (iii) **Bi** [*Bilgüün-iig* *hural-d* *yav-sn*] **-iig** *naiz-d=aa* *med-egd-sen*.
 1SG B.-ACC meeting-DAT go-PFV -ACC friend-DAT=RP know-PASS-PST
 ‘I notified my friend that Bilguun went to a meeting.’

Not all morphologically passive verbs have undergone this change. For instance, none of my consultants accept a case-unmarked matrix subject for predicates such as *sons-ogd-oh* ‘to be heard’, as shown in (iv).

- (iv) *Nadad/*Bi* [*Bilgüün* *hural-d* *yav-san* *gej*] *sons-ogd-sen*.
 1SG.DAT/*1SG B. meeting-DAT go-PST C hear-PASS-PST
 ‘That Bilguun went to a meeting was heard by me.’

Given the unexpected case marking patterns in (i)-(ii), I suspect that the same grammaticalisation process has taken place in Southern Mongolian. (i) may be translated as ‘Someone notified/informed the police that the thief stole something.’, and (ii) as ‘Someone notified/informed the school director that Tuyaa came.’ If that is the case, (i) and (ii) only actually provide evidence for accusative subjects in complement clauses in Southern Mongolian.

a nonfinite NC subject with a raising predicate (365b); and a nonfinite NC subject with a complement NC (365c). The only variation I have found, in the examples I have elicited, is that speaker M sometimes accepts accusative case on proper name subjects such as *Naraa* in (365b), while speaker B consistently rejects such sentences.¹⁷

(365) *Speakers B & M: No accusative subjects in subject NCs*

- a. [Minii/Bi/***Namaig** sugalaa-nd hoj-son]=n'
 1SG.GEN/1SG/*1SG.ACC lottery win-PFV =3.POSS
 Dorj-iig gaihsh-ruul-san.
 D.-ACC surprise-CAUS-PST
 'That I won the lottery surprised Dorj.'
- b. [Naraa-giin/Naraa/%**Naraa-g** hural-d yav-san]=n'
 N.-GEN/N./%N.-ACC meeting-DAT go-PFV =3.POSS
 Bilgüün-d chuhal sana-gd-san.
 B.-DAT important think-PASS-PST
 'It felt/seemed important to Bilguun that Naraa went to a meeting/conference.'
- c. [Bilgüün-ii/Bilgüün/***Bilgüün-iig** üdeshleg-t yav-san]=n'
 B.-GEN/B./*B.-ACC party-DAT go-PFV =3.POSS
 [tüüniig hicheel-Ø=ee hii-gee-gui]-g ilt-ge-ne.
 3SG.ACC less-ACC=RP do-IPFV-NEG -ACC appear-CAUS-PST
 'That Bilguun went to a party indicates that he did not study.'

¹⁷In general, in subject NCs, speaker M prefers local person subjects to be genitive, while 3rd person subjects may be genitive or case-unmarked. In the examples that I have elicited, M also sometimes accepts accusative case on a proper name subject in a subject clause, as shown in (365b). However, when the embedded subject is 1st person in the same example, it may not be accusative:

- (i) [Minii/***Namaig** hural-d yav-san]=n' Bilgüün-d chuhal sana-gd-san.
 1SG.GEN/*1SG.ACC meeting-DAT go-PFV =3.POSS B.-DAT important think-PASS-PST
 'It felt/seemed important to Bilguun that I went to a meeting/conference.'

Another example of a subject clause containing an accusative proper name subject, accepted by speaker M, is shown in (ii). This example, in which the subject NC is merged as an external argument and triggers accusative on *Naraa*, contrasts with the structurally-identical (365a), in which an accusative 1st person embedded subject is ungrammatical for speaker M.

- (ii) [**Bat-iig** sugalaa-nd hoj-son]=n' Naraa-g gaish-ruul-san.
 B.-ACC lottery-DAT win-PST =3.POSS N.-ACC surprise-CAUS-PST
 'That Bat won the lottery surprised Naraa.'

Additionally, as I show in (367) and (368), speaker M finds a 1PL and 3SG accusative subject acceptable in two naturally-occurring examples of subject NCs. Besides these observations, I have not found any deeper pattern or principle that explains her pattern of judgements.

However, speaker G allows accusative subjects in all types of subject NCs that I have constructed, as shown in (366).

(366) *Speaker G: Accusative subjects always available in subject NCs*

- a. [**Namaig**/Minii/Bi sugalaa-nd hoj-son]=n'
 1SG.ACC/1SG.GEN/1SG lottery win-PFV =3.POSS
 Dorj-iig gaihsh-ruul-san.
 D.-ACC surprise-CAUS-PST
 'That I won the lottery surprised Dorj.'
- b. [**Naraa-g**/Naraa-giin/Naraa hural-d yav-san]=n'
 N.-ACC/N.-GEN/N. meeting-DAT go-PST =3.POSS
 Bilgüün-d chuhal sana-gd-san.
 B.-DAT important think-PASS-PST
 'It felt/seemed important to Bilguun that Naraa went to a meeting/conference.'
- c. [**Bilgüün-iig**/Bilgüün-ii/Bilgüün üdeshleg-t yav-san]=n'
 B.-ACC/B.-GEN/B. party-DAT go-PFV =3.POSS
 [tüüniig hicheel-Ø=ee hii-gee-gui]-g ilt-ge-ne.
 3SG.ACC less-ACC=RP do-IPFV-NEG -ACC appear-CAUS-PST
 'That Bilguun went to a party indicates that he did not study.'

Additionally, while uncommon, I have also found some naturally-occurring examples of accusative subjects in subject clauses on the internet. Two such examples are shown in (367)-(368). While speaker G fully accepts (367), speakers B and M find it slightly degraded and prefer *bidnii* 1SG.GEN over *bidniig* 1SG.ACC in the sentence. Speakers G and M fully accept (368) while speaker B rejects it. Note that these subject NCs bear a 3.POSS enclitic and are introduced as the external argument of the clause they are embedded in. Further, while the subject NC in (368) is embedded within a larger NC, the fact that it does not bear accusative case suggests that it has not moved to Spec,DP of the larger NC, i.e. the subject of the subject NC *tüüniig* 'her' is not locally c-commanded by the matrix subject *bid* 'we'.

(367) *Finite subject NC*

[**Bidniig** hadaas üildverle-j chad-dag-güi gedeg]=n' tüüh bolon
1PL.ACC nail produce-CVB can-HAB-NEG C.NML =3.POSS history become

üild-sen.

remain-PST

‘That we cannot produce nails became history.’

(retrieved from <https://www.mass.mn/n/47740>)

B: ?, G: ✓, M: ?

(368) *Nonfinite subject NC*

... || **tüüniig** iim üildel hii-sen]=n' harin ch zöv züil bol-son
... 3SG.ACC such action do-PFV =3.POSS actually correct thing become-PST

gedeg]-t bid itgel-tei bai-na.

C.NML -DAT 1PL belief-COM COP-NPST

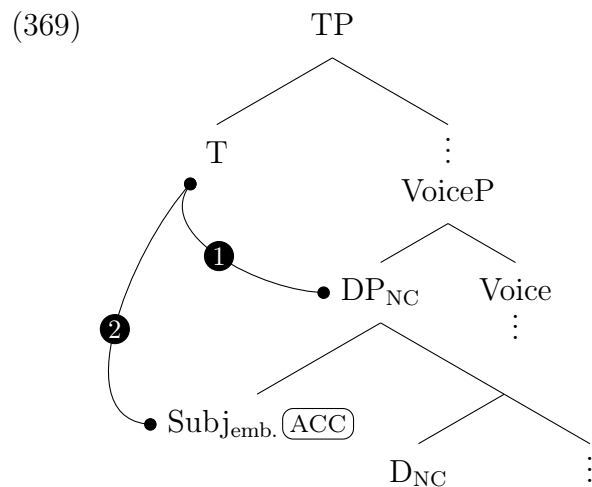
‘(The queen doesn’t speak about it, but) we believe that her taking such action was actually right.’

(retrieved from <https://www.caak.mn/post/view/251512>)

B: ✗, G: ✓, M: ✓

In sum, while speaker G fully accepts accusative subjects in subject NCs, speaker M sometimes accept them when they are proper names, although in the naturally-occurring examples, she marginally accepts a 1PL accusative subject in (367) and accepts a 3SG accusative subject in (368). Speaker B does not accept accusative subjects in most subject NC examples I have tested, except in the naturally-occurring (367), which she still finds marginal. Importantly, at least two of my speakers – G and M – accept accusative subjects in subject clauses that are introduced as external arguments of the matrix verb, e.g., (366a), (366c), (ii) in footnote 18, and (367)-(368). In these structures, there is no c-commanding nominal (e.g. a matrix subject DP) that can be responsible for accusative case on the embedded subject of the subject NC. However, these structures exemplify the containment structure (359b) in which I predicted accusative case may be assigned under an Agree-based approach following Clem and Deal 2024. This is illustrated in (369), assuming that the c-commanding probe that Agrees with both the NC and the embedded subject is located in T⁰. Crucially, under such an Agree-based approach, the “case competitor” for accusative case assignment to the embedded subject has to be the DP superstructure within the subject NC containing the

embedded subject. This is in contrast to a configurational approach, which would require a distinct DP to locally c-command the embedded subject for accusative case to be assigned.



The subject case patterns in subject NCs across my three consultants are summarised in Table 5.9. While speaker B consistently rejects accusative subjects in subject NCs (except for (367)), speaker M sometimes accepts them and speaker G consistently accepts them.

Table 5.9: Subject case in Khalkha subject NCs

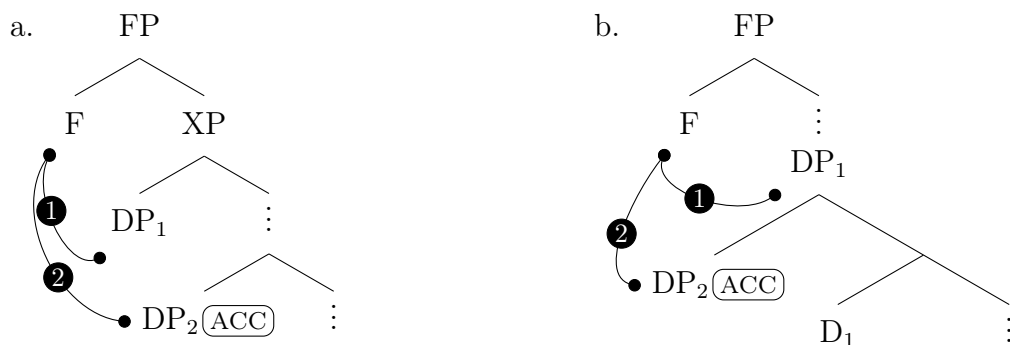
Speaker	Embedded subject		
	ACC	GEN	UNM
B	✗/(✓)	✓	✓
G	✓	✓	✓
M	✗/(✓)	✓	✓

In the next subsection, I show how accusative subjects in subject clauses in Khalkha – which are accepted to varying degrees across my consultants – provide novel support for an Agree-based analysis of dependent case in Khalkha following Clem and Deal’s (2024) implementation.

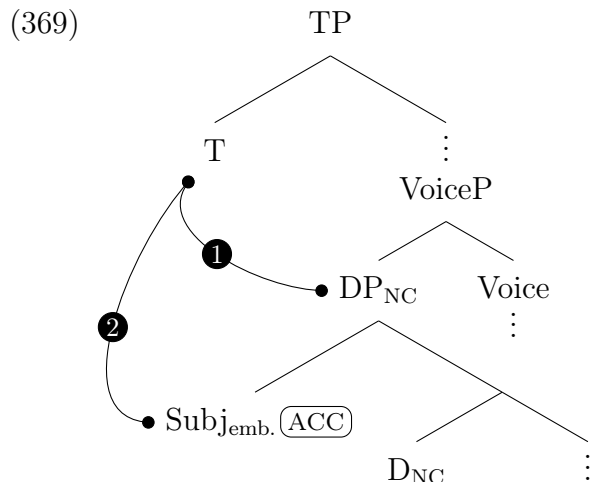
5.5.4 Deriving dependent accusative case via Agree in Khalkha

In this subsection, I show how accusative in Khalkha can be assigned in the two syntactic environments illustrated in (359), repeated below:

(359) *Two types of dependency between nominals under dependent case via Agree*



Under an Agree-based approach to dependent case, I predicted that the structure in (359b) – one in which a nominal (DP₂) is in a containment relationship with another nominal (DP₁) and both are c-commanded by an Agreeing functional head – can license accusative case. As I have shown in the previous subsection, this prediction is borne out by data from subject NCs in Khalkha. For one of my consultants – speaker G – accusative subjects are consistently available in subject NCs, even in NCs that are base-merged as external arguments in the matrix clause. With speaker M, she accepts proper name accusative subjects in subject NCs, and in the two naturally-occurring examples I have checked, she finds (367) marginal and (368) acceptable. Speaker B rejected all examples of subject NCs containing accusative subjects, only marginally accepting (367). Nevertheless, I propose that the accepted subject NC examples that contain an accusative subject and which are base-merged as an external argument exemplify the containment structure (369), repeated below.



Following Clem and Deal 2024, I propose that Khalkha T^0 – both finite and nonfinite – bears a probe with the Interaction/Satisfaction conditions in (370):¹⁸

- (370) *Probe conditions on T^0 in Khalkha*
 [INT: ϕ ,SAT:-]

(370) yields an insatiable probe on T^0 that will Agree with all ϕ -bearing goals that T^0 locally c-commands. At each step of Agree, T^0 copies the features of the ϕ -bearing goal onto itself (Interaction), and transfers its features onto the goal (goal flagging). I further propose that goal flagging involves two crucial steps:

- (371) *Goal flagging*
- a. *Bundling*
 The features on a probe P , $[F_1]$, ..., $[F_n]$, are bundled into a complex feature bundle, yielding $[[F_1], \dots, [F_n]]$.
 - b. *Transfer*
 The complex feature bundle, $[[F_1], \dots, [F_n]]$, is transferred onto the goal that P has Agreed with.

Thus, by goal flagging, any nonfirst goal will receive a complex feature bundle containing: i) the inherent features of the probe, and ii) the features of (a) previously-Agreed-with goal(s).

¹⁸If we were to extend this analysis to Kalmyk, which displays overt ϕ -agreement as shown in Chapter 2, it would imply that both finite and nonfinite T^0 do Agree with subjects but only finite T^0 but not nonfinite T^0 displays morphological agreement.

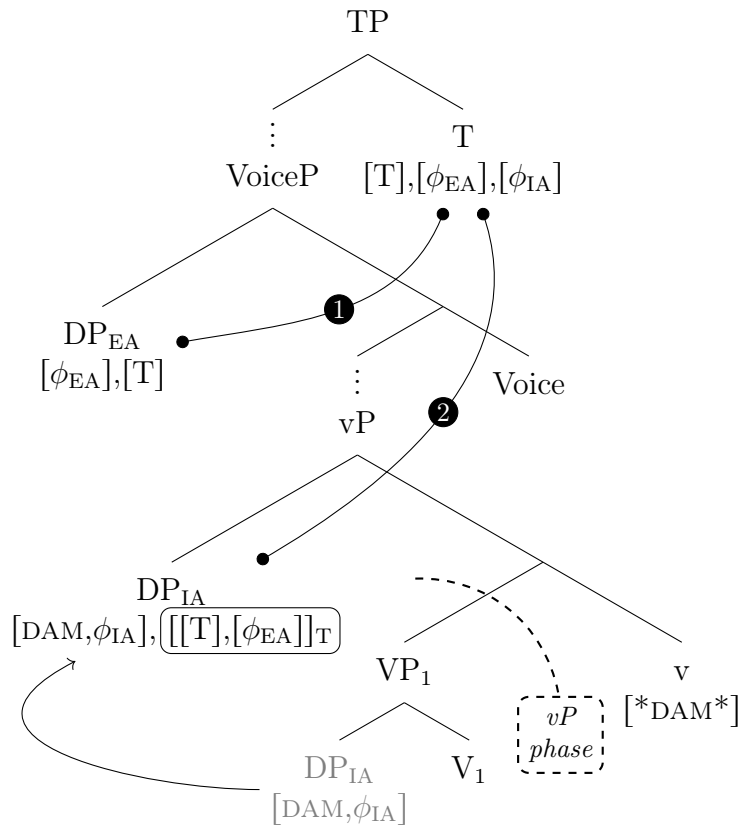
I propose the vocabulary item (372) for the accusative case morpheme in Khalkha, which states that a complex feature bundle containing $[T]$ and the features of a distinct nominal, $[\phi_2]$, will be realised as accusative in the context of a nominal's own features.

(372) *Vocabulary item for ACC in Khalkha*
 $-(ii)g \leftrightarrow [[T], [\phi_2]] / [D, \phi_1] _$

With these details in place, I first illustrate how accusative case may be assigned in structures that correspond to (359a), where two nominals are in a local c-command relationship. I provide two sketches below: dependent accusative on objects in matrix root clauses, and dependent accusative on subjects of finite/nonfinite complement NCs.

Consider how accusative case is assigned to $[DAM]$ -bearing objects in matrix root clauses. The object, DP_{IA} , bears $[DAM]$ and is targeted by $[*DAM^*]$ on v^0 , which moves it to Spec,vP before the vP phase is spelled out. Subsequently, the agentive subject, DP_{EA} , is merged in Spec,VoiceP, where it locally c-commands the $[DAM]$ -bearing object in Spec,vP. Next, T^0 is merged and Agrees with DP_{EA} first and DP_{IA} second. This is illustrated in (373).

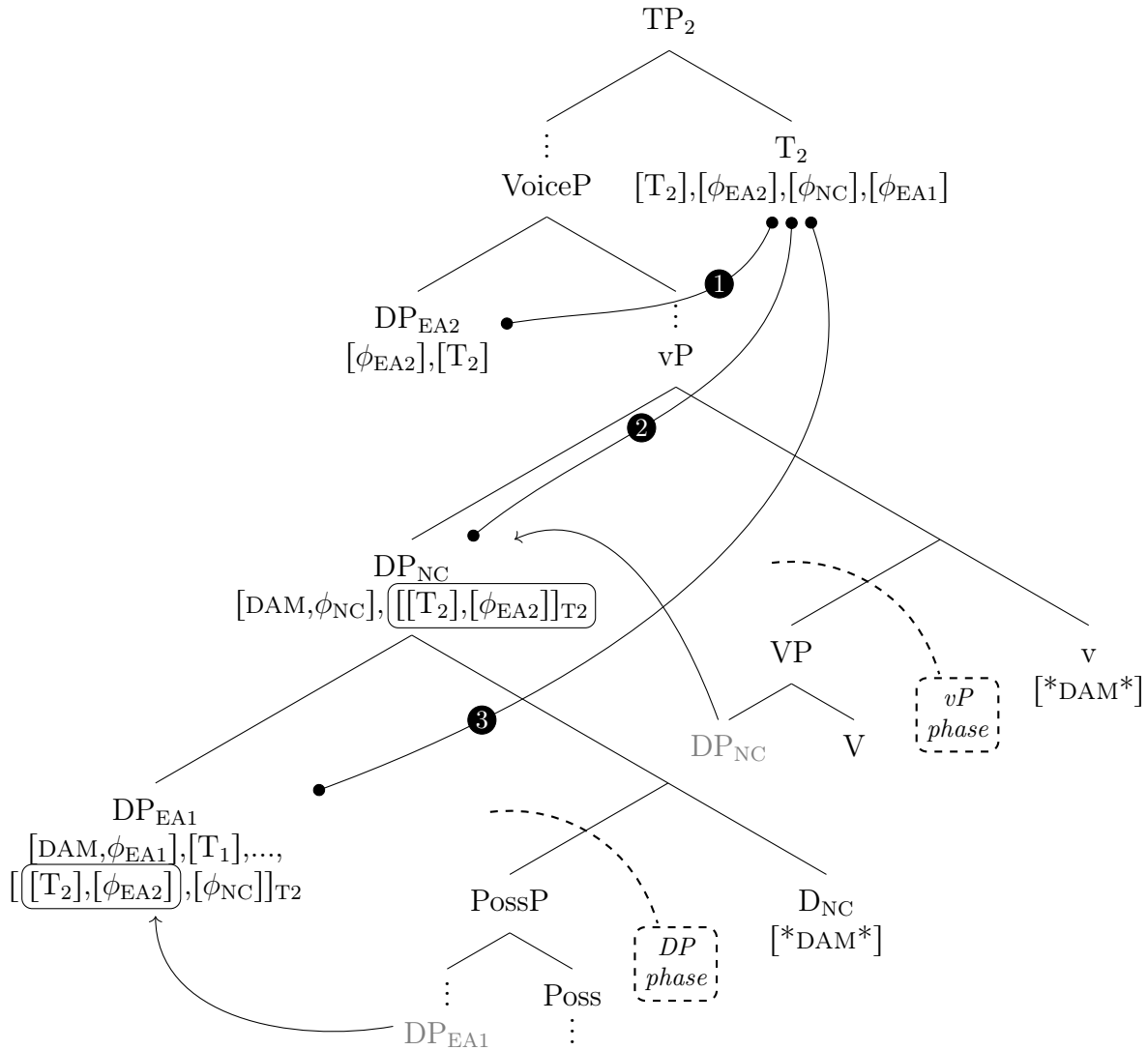
(373) *Dependent ACC via Agree on object in matrix root clause*



On the first cycle of Agree, T^0 copies the ϕ -features of DP_{EA} while simultaneously transferring its $[T]$ feature to DP_{EA} via goal flagging. At this point, T^0 bears its inherent $[T]$ feature as well as $[\phi_{EA}]$ from DP_{EA} . On the second cycle of Agree, T^0 copies the ϕ -features of DP_{IA} . At the same time, T^0 bundles its features into a complex feature bundle – $[[T], [\phi_{EA}]]$ – and transfers it to DP_{IA} via goal flagging. When the next higher phase head is merged (e.g. C^0) and TP is sent to spellout, the complex feature bundle $[[T], [\phi_{EA}]]$ will be realised as the accusative case morpheme on DP_{IA} , according to VI rule (372).

Next, consider how accusative case is assigned to $[DAM]$ -bearing subjects in finite/non-finite complement NCs. Since the embedded subject, DP_{EA1} , bears $[DAM]$, it has raised to Spec,DP of the complement NC. Additionally, the NC also bears $[DAM]$ and moves to matrix Spec,vP. Subsequently, the agentive subject, DP_{EA2} , is merged in Spec,VoiceP, where it locally c-commands both the NC and the embedded subject in Spec,DP. Next, T^0 is merged and Agrees with DP_{EA2} , DP_{NC} second, and DP_{EA1} third, as illustrated in (374).

(374) *Dependent ACC via Agree on subject of finite/nonfinite complement NC*



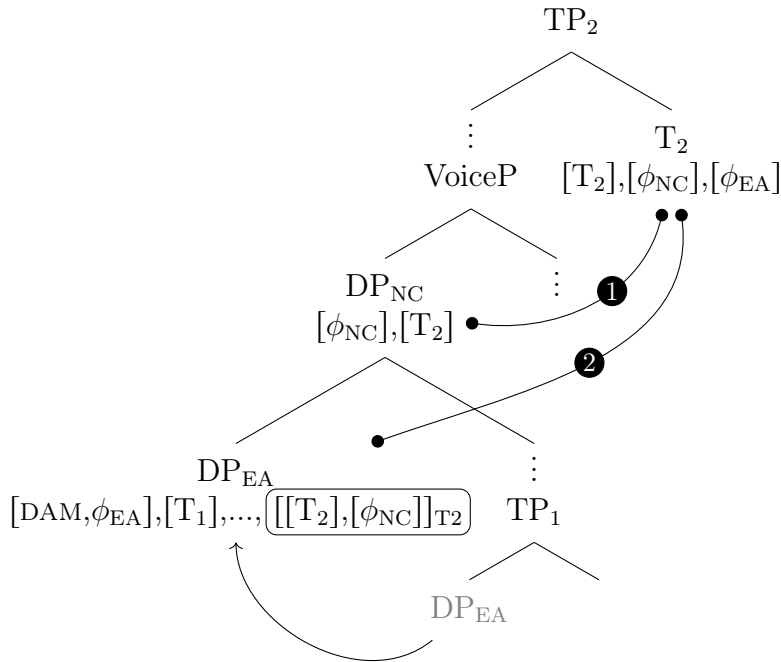
On the first cycle of Agree, T^0 copies the ϕ -features of DP_{EA2} while simultaneously transferring its $[T]$ feature to DP_{EA2} via goal flagging. At this point, T^0 bears its inherent $[T]$ feature as well as $[\phi_{EA}]$ from DP_{EA2} . On the second cycle of Agree, T^0 copies the ϕ -features of DP_{NC} while simultaneously bundling its features into a complex feature bundle – $[[T], [\phi_{EA2}]]$ – and transferring it to DP_{NC} via goal flagging. On the third cycle of Agree, T^0 copies the ϕ -features of DP_{EA1} while simultaneously bundling its features into a complex feature bundle – $[[T], [\phi_{EA2}], [\phi_{NC}]]$ – and transferring it to DP_{EA1} via goal flagging. When the next higher phase head is merged (e.g. C^0) and TP is sent to spellout, the complex

feature bundle $\boxed{[[T],[\phi_{EA}]]}$ will be realised as the accusative case morpheme on DP_{NC} and DP_{EA1} , according to VI rule (372).

Next, I illustrate how accusative case may be assigned in structures that correspond to (359b), where two nominals are in a containment relationship and both nominals are c-commanded by an Agreeing functional head. As I have shown in the preceding subsection, subjects in subject NCs exemplify such structures. For reasons that are unclear to me, I hypothesise that speaker G's grammar always permits the relevant phase heads in subject clauses to bear $[*DAM^*]$, while speaker B and M's grammars typically do not permit said phase heads to bear $[*DAM^*]$. Therefore, speaker G consistently allows the subjects of subject clauses to raise to the highest phase edge of the clause where they are assigned accusative, while speakers B and M often do not allow such movement to happen, which prevents subjects of subject clauses from being assigned accusative.

The derivation of a nonfinite subject NC – merged as an external argument and containing an accusative subject – in speaker G's grammar is sketched in (375). The embedded subject DP_{EA} bears $[DAM]$ and has raised to Spec,DP of the subject clause, DP_{NC} . DP_{NC} is merged in Spec,VoiceP of the matrix clause. When matrix T^0_2 is merged, it Agrees with DP_{NC} first and DP_{EA} second.

(375) *Dependent ACC via Agree on subject of nonfinite subject NC (Speaker G)*



On the first cycle of Agree, T^0_2 copies the ϕ -features of DP_{NC} while simultaneously transferring its $[T_2]$ feature to DP_{NC} via goal flagging. At this point, T^0_2 bears its inherent $[T_2]$ feature as well as $[\phi_{NC}]$ from DP_{NC} . On the second cycle of Agree, T^0_2 copies the ϕ -features of DP_{EA} . At the same time, T^0_2 bundles its features into a complex feature bundle – $[[T_2], [\phi_{NC}]]$ – and transfers it to DP_{EA} via goal flagging. When the next higher phase head is merged (e.g. matrix C^0) and TP is sent to spellout, the complex feature bundle $[[T_2], [\phi_{NC}]]$ will be realised as the accusative case morpheme on DP_{EA} , according to VI rule (372).

Finally, speaker G also allows a subject of a finite bare subject CP to be marked accusative while speakers B and M do not. All three speakers allow case-unmarked subjects in finite bare CPs. This is shown in (376). Note that while speaker M allows accusative proper name subjects in subject NCs, she does not accept them in finite bare subject CPs.

(376) *Finite bare subject CP*

a. *Speaker G*

[**Bilgüün-iig**/Bilgüün hural-d yav-san gej] sons-ogd-son.
 B.-ACC/B. meeting-DAT go-PST C hear-PASS-PST
 ‘It was heard that Bilguun went to a meeting/conference.’

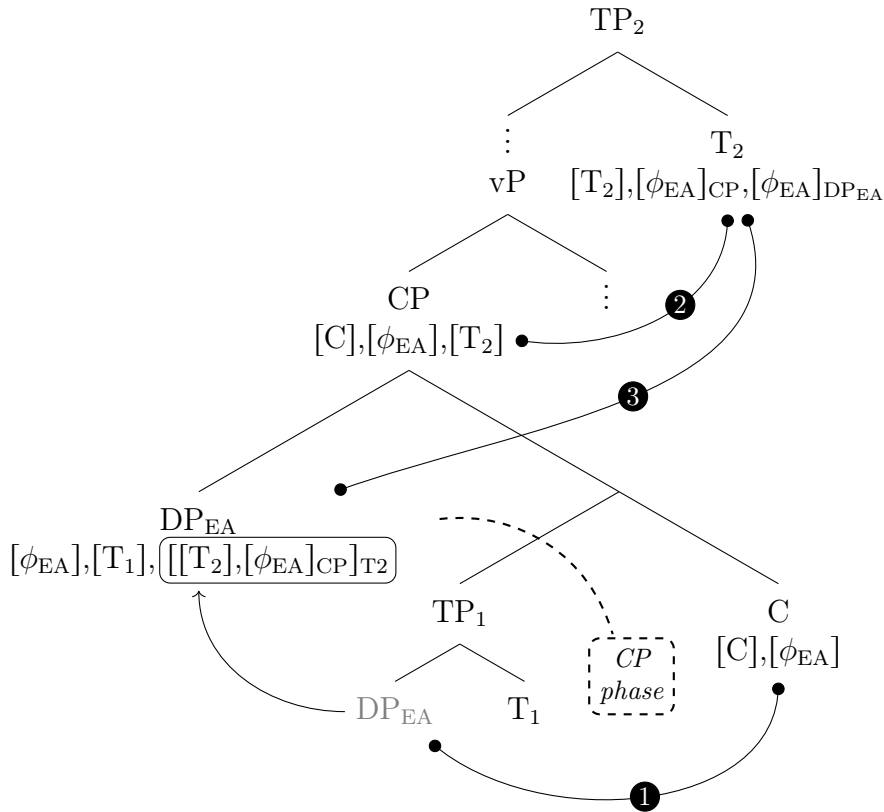
b. *Speakers B & M*

[Bilgüün/***Bilgüün-iig** hural-d yav-san gej] sons-ogd-son.
 B./*B.-ACC meeting-DAT go-PST C hear-PASS-PST
 ‘It was heard that Bilguun went to a meeting/conference.’

Following Fong (2019) and Gong (2022, 2023a), I adopt the idea that embedded C^0 also bears a ϕ -probe. Since embedded C^0 Agrees with the embedded subject, it copies the subject’s ϕ -features onto itself while simultaneously transferring its [C] feature onto the embedded subject. This implies that embedded C^0 bears ϕ -features inherited from the embedded subject. I suggest that the presence of ϕ -features on C^0 renders it a potential goal for the probe on matrix T^0 , since T^0 ’s Interaction condition is simply ϕ . When matrix T^0 probes its c-command domain and encounters embedded CP, T^0 will copy the ϕ -features of C^0 back onto itself, while simultaneously transferring its [T] feature onto CP due to goal flagging. This predicts that at a subsequent cycle of Agree, a complex feature bundle containing [T] and ϕ -features copied from a goal CP could be transferred to a goal DP, resulting in the realisation of accusative case. I propose that this is exactly what happens in (376a) in speaker G’s grammar, illustrated in (377) (note that the finite bare CP has raised from the complement position of V^0 to Spec,vP).¹⁹

¹⁹One prediction under the hypothesis that CPs may bear ϕ -features is that they should be able to trigger dependent accusative case on matrix objects. However, I have not found any cases where a finite bare CP is able to serve as the subject of a transitive clause.

(377) *Dependent ACC via Agree on subject of finite bare subject CP (Speaker G)*



5.5.5 Summary

At the start of this section, I predicted that under an Agree-based approach to dependent case, a configuration in which one nominal is structurally contained within another nominal and both are c-commanded by an Agreeing functional head should permit the assignment of accusative case. In §5.5.3, I presented novel data from subjects of subject clauses in Khalkha that provide some empirical support for my prediction. For one of my consultants, speaker G, accusative subjects are consistently permitted in subject clauses, including cases where the NC is introduced as an external argument in the matrix clause. For speaker M, proper name accusative subjects are sometimes possible in external argument subject NCs, and in two naturally-occurring examples of such subject NCs, she accepts a 1PL accusative subject (marginally) and a 3SG accusative subject. Speaker B, however, rejected most of the subject NC examples I have tested, except for one naturally-occurring example

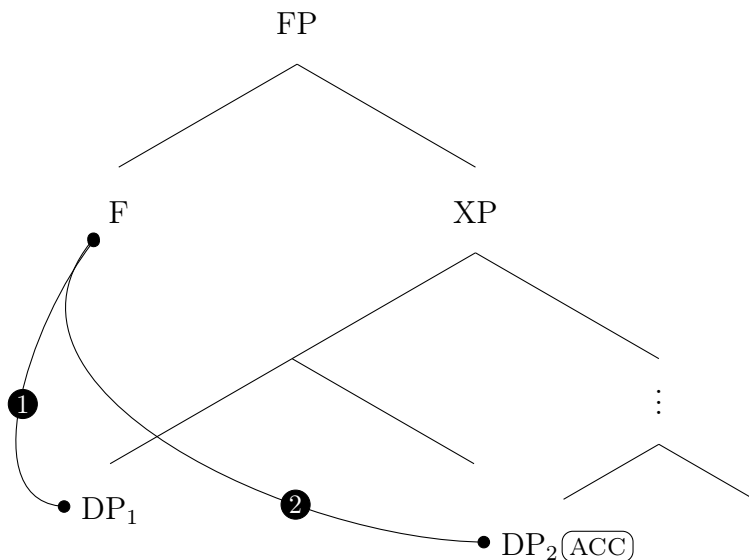
(367), which she still finds marginal. Although a large amount of variation is found across my consultants, these patterns suggest that accusative case assignment under a containment relation is possible in Khalkha.

Both configurational and Agree-based approaches to dependent case can account for the subject clause examples containing case-unmarked subjects. Under a configurational approach, this is because there is no higher nominal that c-commands the embedded subject, while under an Agree-based approach, this must be stipulated as the lack of movement to the highest phase edge of the embedded (nominalised) clause, which prevents matrix T^0 from Agreeing with the embedded subject. However, only an Agree-based approach may account for the subject clause examples in which my consultants judge accusative subjects to be grammatical. This pattern is unexpected under a configurational approach, where configurational rules dictate that accusative case is only assigned under a local c-command relationship between two nominals. The fact that accusative case may be assigned in a containment configuration in Khalkha strongly favours the Agree-based account, which also allows for accusative case to be assigned in local c-command configurations. As such, the data from subject clauses in Khalkha provide new evidence in favour of the Agree-based approach to dependent case.

More broadly, this section has shown that a dependent case via Agree framework offers additional empirical contexts for testing different types of dependencies between nominals that may underlie dependent case assignment. The Khalkha data presented here provides preliminary evidence that at least one such dependency – containment – is relevant for dependent case assignment. While this is predicted under a dependent case via Agree approach, it cannot be straightforwardly captured under a configurational approach to dependent case. This suggests that there may be empirical considerations that favour an Agree-based analysis over a configurational one to dependent case, and that the choice between these approaches may not reduce solely to considerations of “theoretical parsimony”, as suggested by Poole (2024, p.16).

Furthermore, given that what matters for an Agree-based approach to dependent case is that an Agreeing functional head c-command two nominals, we may predict the existence of additional structural configurations beyond c-command and containment that are relevant for dependent case assignment. For example, one such configuration, sketched in (378), involves two nominals that neither c-command nor contain one another but are both c-commanded by the same functional head. Further, DP_1 is dominated by a projection that asymmetrically c-commands another projection which dominates DP_2 ; thus DP_1 is higher than DP_2 . Under an Agree-based system, dependent accusative case assignment on DP_2 is predicted to be possible.

(378) *A third type of dependency predicted under dependent case via Agree*



Finally, this section has demonstrated that dependent case assignment via Agree in Khalkha obeys standard phase-locality: the embedded subject must be located in the highest phase edge of the embedded (nominalised) clause in order for it to be the second goal of Agree by matrix T^0 , which results in accusative case on the embedded subject. The highest phase edge positions of accusative subjects is supported by the long-distance extraction facts discussed in Chapter 4. If the embedded subject is located in any position below the highest phase edge, the embedded subject will be spelled out in a lower phase domain, and will not be accessible as the second goal of Agree by matrix T^0 ; therefore, these

subjects do not bear accusative case. As shown in Chapter 4, non-accusative subjects may not undergo extraction into the matrix clause, which supports the idea that they are not located in the highest phase edge of the embedded (nominalised) clause.

5.6 On the treatment of genitive and case-unmarked nominals

The previous section has argued for an Agree-based approach to dependent accusative case in Khalkha. In this section, I discuss how genitive case may be modelled as a case due to a dependency between D^0 and a goal DP, as well as the treatment of case-unmarked nominals in Khalkha.

5.6.1 On genitive case assignment in Khalkha

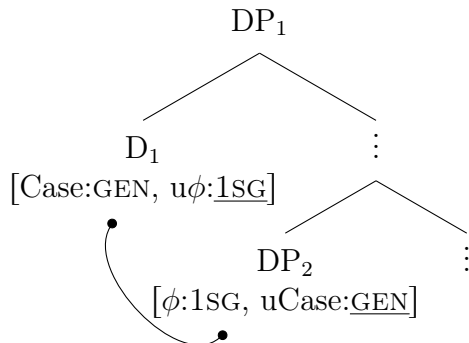
In Chapter 3, I showed that genitive subjects are only found in finite and nonfinite NCs, which contain a full DP layer on top of the clause. This is shown in (379).

- (379) a. GEN *subject in finite NC*
 Naraa [margaash **Tuyaa-giin** ir-ne gedg]-iig sons-son.
 N. tomorrow T.-GEN come-NPST C.NML -ACC hear-PST
 ‘Naraa heard that Tuyaa will come tomorrow.’
- b. GEN *subject in nonfinite NC*
 Naraa [margaash **Tuyaa-giin** ir-eh]-iig med-ne.
 N. tomorrow T.-GEN come-INF -ACC know-NPST
 ‘Naraa knows that Tuyaa will come tomorrow.’

This suggests that the availability of a genitive subject crucially depends on the presence of D^0 . Under a typical view of genitive case assignment by D^0 schematised in (380), D^0_1 , which bears a genitive Case feature and unvalued ϕ -features, locally c-commands DP_2 , which bears ϕ -features and an unvalued Case feature. D^0_1 thus Agrees with DP_2 to value its unvalued ϕ -features, while simultaneously valuing DP_2 ’s unvalued Case feature. Such a

view can be found in works such as Hale 2002; Miyagawa 2008, 2011; Bošković 2008; Baker and Vinokurova 2010; Asarina and Hartman 2011; Laszakovits 2018a; Ótrott-Kovács 2020.

(380) Genitive case assignment by D^0



Additionally, it has been observed that in Turkish, subjects obligatorily surface with genitive case in NCs that exhibit possessor agreement (Kornfilt 2020), shown in (381), while in Kyrgyz (Turkic), subjects cannot surface with genitive case without possessor agreement on NCs (Laszakovits 2018a), shown in (382). This has led to the common view that overt possessor agreement and genitive case assignment are due to the same Agree operation triggered by D^0 .

(381) *Turkish*

Herkes [sen*(-**in**) ödül-ü kazan-dığ]-**in**-ı duy-du.
 Everyone 2SG*(-GEN) prize-ACC win-FNOM -2SG.POSS-ACC hear-PST
 ‘Everyone heard that you won the award.’

(Kornfilt 2020, p.149, ex.29, glosses modified)

(382) *Kyrgyz (Turkic)*

Bakit [**Aytmatıf-tın** kitep-ti caz-gan]-*(**ı**)-n ayt-tı
 Bakit.NOM Aitmatov-GEN book-ACC write-PTCP -(3SG.POSS)-ACC say-PST
 ‘Bakit said that Aitmatov wrote the book.’

(Laszakovits 2018b, p.1, ex.3, glosses modified)

In previous work (Lim 2025), however, I have argued for a dissociation of the operation that yields overt ϕ -agreement and the operation that assigns genitive case. As I discussed in Chapter 3, Khalkha shows a double dissociation of the assumed tight link between overt ϕ -agreement and genitive case assignment. For example, in a nonfinite NC (383),

accusative subjects are always compatible with (optional) possessor agreement morphology on the clause. In converbial adjuncts such as (384), genitive subjects are impossible even with possessor agreement morphology on the clause.

- (383) *Accusative subject possible with ϕ -agreement on nonfinite NC*
 Chi [**namaig/minii** unta-j bai-sn]-iig(=**min**) med-en.
 2SG 1SG.ACC/1SG.GEN sleep-PROG COP-PFV -ACC=1SG.POSS know-NPST
 ‘You know that I was sleeping.’
- (384) *Genitive subject impossible in converbial adjunct NC with ϕ -agreement*
 Chi [**namaig/*minii** hicheel-Ø=ee hii-j bai-had](=**min**)
 2SG 1SG.ACC/*1SG.GEN lesson-ACC=RP do-PROG COP-TEMP.CVB =1SG.POSS
 unt-san.
 sleep-PST
 ‘When/While I was studying, you fell asleep.’

Instead, I have argued that overt possessor agreement is controlled by a dedicated functional head Poss^0 merged above nP but below DP.²⁰ Argument NCs contain a full DP structure above the clause, but converbial adjuncts contain a reduced PossP structure that lacks DP above the clause. This accounts for the double dissociation of genitive case and possessor agreement: i) both clause types exhibit possessor agreement due to the presence of Poss^0 , but only finite and nonfinite NCs may contain genitive subjects due to the presence of D^0 ; and ii) since overt possessor agreement does not signal that genitive case has been assigned, it is not unexpected to find it with non-genitive subjects, such as accusative subjects.

Given that the preceding section has shown how dependencies between two nominals may be modelled using the dependent case via Agree approach of Clem and Deal 2024, we may ask whether genitive case in Khalkha can be modelled in a similar fashion, e.g. as the spell out of a *goal flag* due to Agree between D^0 and a nominal. This is in contrast to a standard Chomskyan approach to genitive case assignment, in which genitive case is assigned as a *syntactic primitive* due to Agree with D^0 . In this section, I propose that genitive case may be analysed as a goal flag from D^0 .

²⁰This implies that embedded subjects that have Agreed with Poss^0 also carry features from Poss^0 . However, since these features are not associated with any vocabulary item for case, they do not yield case morphology on the subject.

In Chapter 4, I argued that genitive subjects are located in Spec,nP of finite and nonfinite NCs, while case-unmarked subjects in these clause types are located in Spec,TP. This suggests that in finite and nonfinite NCs, D^0 may target the embedded subject that has raised to Spec,nP, but not the embedded subject that is located in Spec,TP. This provides some support for the view that nP – which is projected directly above CP in finite NCs and above TP in nonfinite NCs – constitutes a phase, such that D^0 may not Agree with a DP located within the complement of n^0 .

I propose that, similar to T^0 , the D^0 found in finite and nonfinite NCs – D^0_{NC} – contains an Agree probe, with the Interaction/Satisfaction conditions specified in (385).²¹

- (385) *Probe conditions on Khalkha D^0_{NC}*
 [INT: ϕ ,SAT: ϕ]

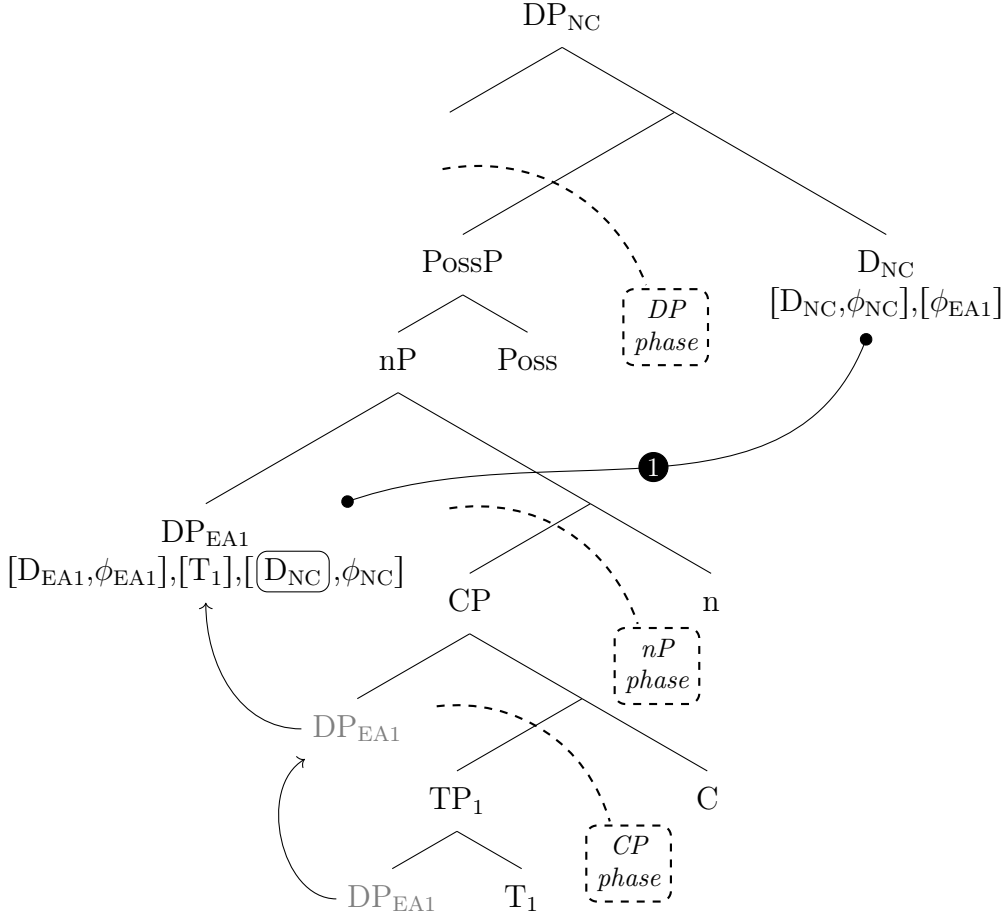
I also propose the vocabulary item (386) for the genitive case morpheme in Khalkha. (386) states that genitive case is the realisation of another nominal's [D] feature in the context of a goal DP's inherent features. This derives the core observation that genitive subjects depend on the subject DP being in the presence of another D^0 .

- (386) *Vocabulary item for GEN in Khalkha*
 -(g)iin / -(n)ii $\leftrightarrow$ [D₂] / [D₁, ϕ_1] —

²¹For concreteness, I assume that D^0_{NC} contains a Satisfaction condition, ϕ . To my knowledge, it is not possible to tell whether the probe on D^0_{NC} is insatiable, i.e. whether it lacks a Satisfaction condition, due to the presence of lower phases such as nP and CP (in the case of finite NCs) almost immediately below DP_{NC} , which significantly limits the search space of D^0_{NC} . Nothing crucial hinges on either option.

Consider the derivation of a finite NC, e.g. (379a), sketched in (387).

(387) GEN case via Agree on subject of finite complement NC



In (387), the embedded subject DP_{EA1} has agreed with embedded T_1 and raised to Spec, TP_1 . It then undergoes further movement to embedded Spec, CP and lands finally in Spec, nP.²² Next, D^0_{NC} is merged. D^0_{NC} Agrees with the closest goal, DP_{EA1} , interacting with DP_{EA1} 's $[\phi]$ -features and simultaneously transferring its $[D_{NC}]$ feature onto DP_{EA1} . At the spellout of the DP phase, $[D_{NC}]$ will be realised as the genitive case morpheme on DP_{EA1} , according to (386).

Note that the VI rule for genitive case in (386) states that genitive case is the realisation of another DP's $[D]$ feature in the context of a goal DP's inherent features. Since D_{NC}

²²A full investigation of genitive-based DSM in Khalkha is left for future work. Similar to the movement-based account of accusative-based DAM I proposed in §5.4, I will assume that genitive subjects bear some feature, and that a probe on C^0 and n^0 but not D^0 targets nominals bearing this feature for movement to their specifiers. Thus, genitive subjects do not undergo further movement to Spec, DP.

only interacts with and copies $[\phi]$ from its goal, this correctly predicts that genitive case is not realised on the NC itself. Second, even though DP_{EA1} contains $[T_1]$ as well $[\phi_{NC}]$, these features are not part of a complex feature bundle. Thus, they will not be targeted by the VI rule (372) for the realisation of accusative case.

The analysis above also implies that an accusative embedded subject in a complement NC would move successive-cyclically through all the phase edges before landing in Spec,DP. Specifically, an accusative subject in a nonfinite NC will move from Spec,TP to Spec,nP – where it would be targeted by D_{NC}^0 for Agree – and finally to Spec,DP_{NC} – where it would be targeted by matrix T^0 for second cycle Agree. In a finite NC, an accusative subject will have moved from Spec,TP to Spec,CP, next to Spec,nP, before finally landing in Spec,DP. (388) shows the features of the embedded subject as it undergoes this movement in a finite NC.

(388) *Features on embedded DP_{EA1} as it raises to Spec,DP of a finite NC*

- a. After Agree with embedded T^0_1 :
 $[D, \phi_{EA1}], [T_1]$
- b. After Agree with embedded C^0_1 :
 $[D, \phi_{EA1}], [T_1], [C_1]$
- c. After Agree with n^0 :
 $[D, \phi_{EA1}], [T_1], [C_1], [n]$
- d. After Agree with D^0_{NC} :
 $[D, \phi_{EA1}], [T_1], [C_1], [n], [D_{NC}, \phi_{NC}]$
- e. After second cycle Agree with matrix T^0_2 :
 $[D, \phi_{EA1}], [T_1], [C_1], [n], [D_{NC}, \phi_{NC}], [[T_2], [\phi_{EA2}]]$

Given (388e), a question arises as to which features should be targeted for case realisation on the embedded subject in Spec,DP of a complement NC, as both the VI rule for accusative (372) and for genitive (386) may apply. I propose that VI follows the Subset Principle (Halle and Marantz 1993, 1994), which states that only the most specific vocabulary item whose features are a subset of a morpheme will be inserted. Given that the vocabulary item for accusative case targets a more specific subset of the embedded subject's features –

$[[T_2], [\phi_{EA2}]]$ than the vocabulary item for genitive case – $[D_{NC}]$, accusative case rather than genitive case will be realised on the embedded subject in Spec,DP.

Note that according to (387), a genitive subject *must* be located in Spec,nP; it cannot be in Spec,DP or within the clausal domain, such as Spec,CP in finite NCs or Spec,TP in nonfinite NCs. If the embedded subject were to raise to Spec,DP, it would be targeted by matrix T^0 in a second cycle of Agree. Matrix T^0 would then transfer a complex feature bundle containing the features of the matrix subject onto the embedded subject, resulting in the realisation of accusative case. If the embedded subject remains in Spec,CP or Spec,TP, it will be spelled out within the nP phase, where it will be inaccessible to D^0_{NC} due to the PIC. Since the realisation of genitive case on the embedded subject depends on D^0_{NC} transferring its features via Agree, we predict that subjects spelled out within the nP phase will not bear genitive case. Instead, they will be case-unmarked, as I discuss in the next subsection. The present account of genitive case assignment is in line with the mixed behaviour of genitive subjects with respect to extraction and binding shown in Chapter 4, which provides further support for their intermediate position between accusative and case-unmarked subjects, and for the phasehood of nP. Therefore, like Agree by matrix T^0 discussed in the previous section, Agree by D^0 that results in genitive case on subjects in Khalkha can also be understood as obeying standard phase-locality.

5.6.2 On the treatment of case-unmarked arguments

Finally, I propose that case-unmarked arguments in Khalkha – which include case-unmarked objects and subjects that are traditionally called “nominative” – are due to a lack of a specific vocabulary item that corresponds to the features of an object that remains within vP or of a subject that remains within TP. In other words, I treat the surface form of these arguments as a morphological default in Khalkha (cf. Legate 2008’s analysis of absolutive case as a morphological default for nominals that are assigned either abstract nominative or accusative case).

Consider an object DP that does not move to Spec,vP, where it is accessible to T^0 for a second cycle of Agree. Since the object is spelled out within the vP phase, the only features it contains are its inherent features, i.e. $[D,\phi]$. Given the VI rules for accusative (372) and genitive (386) case in Khalkha, repeated below, no case will be realised on the object spelled out within the vP phase since none of its features correspond to those vocabulary items.

(372) *Vocabulary item for ACC in Khalkha*
 $-(ii)g \leftrightarrow [[T],[\phi_2]] / [D,\phi_1] \text{ —}$

(386) *Vocabulary item for GEN in Khalkha*
 $-(g)iin / -(n)ii \leftrightarrow [D_2] / [D_1,\phi_1] \text{ —}$

Next, consider a matrix subject or an embedded subject that remains in Spec,TP. If it is a matrix subject, it may be spelled out within the matrix CP phase or it may further raise to matrix Spec,CP via Agree (e.g. Fong 2019; Gong 2022, 2023a). If it is an embedded subject, it will be spelled out within the embedded CP phase (e.g. in finite bare CPs or finite NCs) or the nP phase (e.g. in nonfinite NCs and converbial adjuncts). Regardless of the clause type, the subject in Spec,TP will only contain its inherent features and the features of T^0 (defective or nondefective) that it has Agreed with, i.e. $[D,\phi],[T]$. A matrix subject that further raises to Spec,CP will additionally contain the features of C^0 , i.e. $[D,\phi],[T],[C]$. Given the VI rules for accusative (372) and genitive (386) case in Khalkha, no case will be realised on these subjects since none of their features correspond to those vocabulary items.

Under the present account where case features are not syntactic primitives but goal flags, the absence of case morphology on objects that are spelled out within the vP phase, matrix subjects, and embedded subjects that remain in Spec,TP, receives a unified explanation. Since their features do not correspond to specific vocabulary items for case realisation found in Khalkha, e.g. (372) and (386), they surface as a morphological default.

5.7 Alternative accounts of case assignment in Khalkha

In this penultimate section of this chapter, I briefly consider additional alternative models of accusative and genitive case assignment that have been proposed for Khalkha and other Mongolian varieties.

5.7.1 Accusative is assigned by embedded C^0

Bao et al. (2015) argue on the basis of examples like (389a) and (389b) from Khorchin Mongolian – a southern variety spoken in Inner Mongolia – that accusative case on subjects may be assigned by an embedded C^0 .

(389) *Khorchin Mongolian*

- a. ACC subject possible in ‘*gejü*’ complement clause

Bayatur [nidunun **Ulayan-i**/Ulayan nom-Ø biči-gsen gejü] öčügedür
 B. last.year U.-ACC/U. book-ACC write-PST.ADN C yesterday
 keJe-jei.
 say-PST.CON

‘Yesterday Bagatur said that Ulagan wrote a book last year.’

(Bao et al. 2015, p.150, ex.21)

- b. ACC subject possible in ‘*megče*’ adjunct clause

[**Bayatur-i**/Bayatur ende ire-megče], Ulayan surulča-ju ekile-jei.
 B.-ACC/B. here come-IMM.CVB U. study-CVB begin-PST.CON
 ‘As soon as Bagatur came, Ulagan began to study.’

(Bao et al. 2015, p.148, ex.16)

The complement clause in (389a) is headed by the complementiser *gejü* (cognate with the Khalkha complementiser *gej*), while the adjunct clause in (389b) is superficially headed by the converb *-megče* (cognate with the immediative converb *-megts* in Khalkha). Bao et al. propose that these embedded clauses contain a C^0 which assigns accusative case to the embedded subject.²³

²³While Bao et al. (2015) are not explicit on this point, under their analysis, I assume that they take converbs like *-megče* in (389b) to be C^0 , or converbial adjuncts to contain a null C^0 . Unfortunately, the authors also do not discuss if nominalised CPs (i.e. finite NCs) are possible in Khorchin Mongolian.

However, an analysis that takes accusative case to be assigned by C^0 is not tenable for Khalkha, as I have shown in Chapter 3 that nonfinite NCs and converbial adjuncts do not contain a CP layer, yet they may contain accusative subjects:

- (390) a. *ACC subject possible in nonfinite NC*
 Naraa [margaash **namaig** ir-eh]-iig mart-san.
 N. tomorrow 1SG.ACC come-INF -ACC forget-PST
 ‘Naraa forgot that I will come tomorrow.’
- b. *ACC subject possible in converbial adjunct*
 Naraa [**namaig** nom unshi-j bai-had] unt-san.
 N. 1SG.ACC book read-PROG COP-TEMP.CVB sleep-PST
 ‘Naraa slept while I was reading a book.’

Further, as noted by Fong (2019), Bao et al. (2015)’s (2015) analysis predicts that accusative case assignment to embedded subjects should *always* be possible in the context of C^0 , regardless of the properties of the matrix clause. As discussed previously, for speakers B and M (as well as Fong’s consultants), an accusative subject is not possible in a finite bare subject CP as shown in (391); however, it is possible for speaker G.

- (391) *ACC subject not possible for Khalkha speakers in finite bare subject CP*
 [Bilgüün/%**Bilgüün-iig** hural-d yav-san gej] nadad sons-ogd-son.
 B./%B.-ACC meeting-DAT go-PST C 1SG.DAT hear-PASS-PST
 ‘That Bilguun went to the meeting was heard by me.’

While Bao et al.’s analysis may account for why accusative subjects are possible in a finite bare subject CP in speaker G’s grammar, it does not account for why some Khalkha speakers do not accept accusative subjects in such clauses. I thus conclude that accusative case is not assigned by embedded C^0 in Khalkha.

5.7.2 Genitive case is licensed by a nominal and adnominal form

Maki et al. (2015) argue that in Khorchin Mongolian, embedded subjects are assigned genitive case under two conditions: i) they must be c-commanded by a nominal element, *and* ii) they must be in a local relation to an adnominal form of the predicate. An “adnominal form”, sometimes also called a “verbal noun form”, refers to what I have called aspect markers found in Mongolic languages. Their proposal does not extend straightforwardly in Khalkha as genitive subjects are available in finite NCs that demonstrably do not contain an aspect marker, e.g. as shown in (379a), repeated below:

- (379a) GEN *subject in finite NC*
Naraa [margaash **Tuyaa-giin** ir-ne gedg]-iig sons-son.
N. tomorrow T.-GEN come-NPST C.NML -ACC hear-PST
‘Naraa heard that Tuyaa will come tomorrow.’

5.8 Conclusion

This chapter has demonstrated how the positions of accusative objects and embedded accusative subjects, and their interpretive properties, can be accounted for under a movement-based approach to DAM and its interaction with dependent case assignment. I began by examining DOM and DSM patterns in Khalkha. In §5.3, I showed that DOM and DSM in Khalkha are governed by similar but distinct referentiality scales, with accusative case on direct objects and embedded subjects conditioned by factors such as nominal type, definiteness, and scopal specificity.

Building on these findings as well as the findings of Chapter 4, in §5.4, I proposed a movement-based account of DAM, in which [DAM]-bearing objects move to Spec,vP, and [DAM]-bearing subjects move to Spec,CP in finite bare CPs, Spec,DP in finite and nonfinite NCs, and Spec,nP in converbial adjuncts. Further demonstrating that accusative case in Khalkha is best analysed as a dependent case, I showed how a dependent approach to

accusative case assignment interacts with the movement-based account of DAM to yield accusative case on [DAM]-bearing arguments.

Next, in §5.5, I introduced novel data from subjects of subject clauses in Khalkha, which offer a previously-unnoticed empirical domain where the predictions of a configurational approach to dependent case and an Agree-based approach to dependent case come apart. In this domain, a nominal is contained within another nominal, and both are c-commanded by a higher Agreeing functional head. Under a dependent case via Agree approach, accusative is assigned to the lower of two nominals under multiple Agree by a c-commanding functional head; accusative case assignment is thus predicted to be possible regardless of whether the two nominals share a c-command or containment relation. Under a configurational approach, however, accusative case assignment is predicted to *not* be possible on the lower of two nominals in a containment relation, as configurational rules dictate that dependent accusative case may only be assigned to the lower of two nominals in a local c-command relation.

I demonstrated that the availability of accusative subjects – consistently by Speaker G and sometimes by Speakers B and M – represents precisely the containment relation under which accusative case assignment is predicted to be possible under the dependent case via Agree approach but not the configurational case approach. This provides some preliminary evidence that supports an Agree-based approach over a configurational approach to dependent case in Khalkha. Further, this suggests that dependent case assignment may be modelled using Agree – an independently-needed, core syntactic operation, and that configurational rules that have been proposed specifically to account for dependent case assignment may not necessarily be part of the grammar.

In §5.6, I further showed that genitive case can be modelled as being assigned under a dependency between a nominal and a distinct D^0 , which captures the intuition the genitive subjects only appear in finite and nonfinite NCs – which contain a DP layer – in Khalkha. Additionally, I proposed that case-unmarked subjects and case-unmarked objects may have

a common source in Khalkha: the lack of a specific vocabulary item for case that corresponds to the features found on these nominals.

More broadly, this chapter has provided new insights into the nature of the dependency between nominals in dependent accusative case assignment. Under traditional configurational approaches, two nominals must stand in a c-command relation within the same phasal spell-out domain for dependent case to be assigned. In contrast, under the Agree-based approach proposed here, the two nominals need only be within the same phasal spell-out domain and be goals of Agree by the same functional head; a c-command relation between them is not necessary. This opens promising avenues for future research into the types of dependencies between nominals that may be relevant for dependent case assignment.

Chapter 6

Conclusion

This dissertation began with the observation that Khalkha contains four distinct embedded clause types – finite bare CPs, finite nominalised clauses (NCs), nonfinite NCs, and converbial adjuncts – across which embedded subjects display differential case marking. Against this backdrop, I examined how the structural and interpretive properties of these embedded clauses and their subjects can shed light on key issues in syntactic theory: the architecture of clausal nominalisation and the constraints underlying it, the nature of locality across different syntactic operations, the interaction between differential argument marking (DAM) and case, and the nature of dependencies involved in dependent case assignment. In this concluding chapter, I summarise the main findings of the preceding chapters and reflect on their broader theoretical implications.

6.1 Towards a typology of nominalised clauses

In Chapter 3, I argued that the four embedded clause types under consideration can be systematically classified according to three criteria: finiteness (i.e. whether they contain CP and a nondefective vs. defective T^0), nominalisation (i.e. presence or absence of nominal structure above clausal structure), and the size of the nominal layer (i.e. whether the nominal structure is a full DP vs. a reduced PossP that lacks a DP layer). Further, I showed that

while all four clause types may contain accusative or nominative subjects, only finite and nonfinite NCs may contain genitive subjects. These findings are summarised in Table 6.1.

Table 6.1: Structure of four embedded clause types in Khalkha

Clause type	Clausal structure		Nominal structure		Embedded subject		
	Type of T ⁰	CP	PossP	DP	ACC	GEN	UNM
Finite bare CP	Nondefective	✓	✗	✗	✓	✗	✓
Finite NC	Nondefective	✓	✓	✓	✓	✓	✓
Nonfinite NC	Defective	✗	✓	✓	✓	✓	✓
Converbial adjunct	Defective	✗	✓	✗	✓	✗	✓

The classification in Table 6.1 further captures syntactic and morphological differences among the clause types. For example, finite clauses, which contain nondefective T⁰ and a CP layer, may also contain the overt complementiser *gej/ged(e)g*, finite tense suffixes, the interrogative marker *be/ve*, and thematic topics. These properties are absent in nonfinite clauses, which contain defective T⁰ and lack a CP layer. Additionally, NCs which contain PossP may host possessive enclitics, while only NCs that contain a DP layer may also contain genitive subjects and be assigned case.

A key finding from this chapter is the unattested status of a particular clause type: one that is both CP-sized and nominalised, but lacks a DP layer. This is represented by the bottom left cell in Table 6.2.

Table 6.2: Types of NCs in Khalkha

	PossP	DP
TP	Converbial adjunct	Nonfinite argument
CP	(Unattested)	Finite argument

I proposed that this gap in the typology is not accidental, but follows from general constraints on structure building formulated under the Williams Cycle (Williams 2003, 2011, 2013). Specifically, I proposed that, assuming a nominal-clausal parallelism in which DP maps onto CP, PossP maps onto TP, and vP maps onto nP, a PossP-sized nominal structure cannot

embed a CP-sized clause. This constraint predicts a typology of NCs where smaller nominal projections may nominalise smaller clausal domains, but not larger ones. A set of such predictions is shown in Table 6.3.

Table 6.3: (Im)possible nominalised structures

Nominal structure	Clausal structure			
	CP	TP	vP	VP
nP	✗	✗	✓	✓
PossP > nP	✗	✓	✓	✓
DP > PossP > nP	✓	✓	✓	✓

In addition to the unattested structure in Khalkha represented in Table 6.2, Table 6.3 predicts that we should not find NCs in which an nP layer without PossP and DP embeds a TP- or CP-sized clause. The proposed constraint and the typology it yields thus offer a novel perspective on the parallelism between nominal and clausal domains.

6.2 On the non-isomorphism of locality domains

In Chapter 4, I discussed evidence that shows that CP and vP, as well as DP and nP, are phases in Khalkha. Next, I investigated how the position of embedded subjects (as indicated by their case morphology) correlates with their acceptability with respect to long-distance extraction and cross-clausal binding. These findings are summarised in Table 6.4.

Table 6.4: Extraction and binding across four embedded clause types in Khalkha

	Embedded subject		
	ACC	GEN	UNM
	(in finite & nonfinite NCs)		
May be extracted into matrix clause?	✓	✗	✗
Visible to matrix subject for binding?	✓	✓	✗

Based on these findings, I argued that differentially case-marked subjects in each clause type are located in distinct positions within each clause. I argued that accusative subjects are

located in the highest phase edge of the embedded (nominalised) clause: Spec,CP in finite bare CPs, Spec,DP in finite and nonfinite NCs, and Spec,nP in converbial adjuncts. I showed that in these highest phase edge positions, accusative subjects are accessible for extraction into the matrix clause, and accusative subjects and their possessors are accessible to the matrix subject for binding purposes. I further argued that genitive subjects occupy Spec,nP within the nominal superstructure of finite and nonfinite NCs. I showed that in Spec,nP, genitive subjects and their possessors are accessible to the matrix subject for binding; however, genitive subjects are may not be extracted into the matrix clause. Finally, I argued that case-unmarked subjects remain in Spec,TP, which renders them inaccessible for both long-distance extraction into the matrix clause and binding by the matrix subject.

The fact that accusative subjects are visible for binding and extraction, whereas genitive subjects are visible for binding but not extraction, motivated a key conclusion in this chapter: that the locality domains for binding and extraction are not isomorphic. I showed that extraction may not target an embedded subject within a NC unless it is at the phase edge – Spec,DP – while binding of a nominal within the nominal superstructure above a clause is permitted as long as the nominal is not located within the clause itself. Additionally, the Agree-based account of accusative and genitive case assignment argued for in Chapter 5 shows that the Agree operations themselves are subject to phase-locality: the functional heads (matrix T^0 and D^0_{NC}) may only Agree with nominals they locally c-command within the same phase. Taken together, these findings reinforce the idea that extraction and Agree are constrained by phase-based locality in Khalkha, while binding is sensitive to a different locality domain that cannot be straightforwardly mapped to a standard definition of phasal spellout domains.

I proposed one way in which the non-isomorphism between extraction and binding domains may be modelled in Khalkha: a staggered approach to spellout that distinguishes between PF- and LF-phases, as proposed by Hicks 2009, building on Marušič 2005 and Matushansky 2005. Under this approach, I proposed that in Khalkha NCs, DPs constitute

PF-phases but not LF-phases, and nPs constitute both PF- and LF-phases. Given that binding dependencies are only resolved in LF-phases, the DP-sized superstructure above the clause in Khalkha finite and nonfinite NCs would remain transparent for binding by the matrix subject even after the DP phase is completed.

6.3 On the nature of differential argument marking

Chapter 5 turned to an investigation of how accusative-based differential argument marking (DAM) takes place and how it reflects the structural positions of arguments in Khalkha. In the first half of Chapter 5, I discussed the interpretation and distribution of differential argument marking (DAM), focusing on the contrast between differential object marking (DOM) in root clauses and differential subject marking (DSM) in the four types of embedded clauses under consideration. By carefully testing for different nominal types and subtypes of definiteness and specificity, I showed that accusative case on direct objects and embedded subjects is conditioned by these factors. These findings, summarised in Table 6.5, contribute to a finer-grained picture of DOM and DSM in Khalkha than previously reported.

Table 6.5: DOM and DSM in Khalkha

	DOM	DSM			
		Finite bare CP	Finite NC	Nonfinite NC	Converbial adjunct
1/2P pronoun	ACC	ACC	ACC	ACC	ACC
3P pronoun	ACC	ACC/UNM	ACC/UNM	ACC/UNM	ACC/UNM
Proper name	ACC	ACC/UNM	ACC/UNM	ACC/UNM	ACC/UNM
Anaphoric (=1/>1)	ACC	ACC/UNM	ACC/UNM	ACC/UNM	ACC/UNM
PP/PW bridging	ACC	ACC/UNM	ACC/UNM	ACC/UNM	ACC/UNM
IS/LS unique	ACC/UNM	ACC/UNM	ACC/UNM	ACC/UNM	ACC/UNM
WS indefinite	ACC/UNM	ACC/UNM	ACC/UNM	ACC/UNM	ACC/UNM
NS indefinite	UNM	UNM	UNM	UNM	⁹ %ACC/UNM

While both DOM and DSM in Khalkha involve alternations between accusative and the absence of case, the findings suggest that the two phenomena are governed by similar referentiality scales, though with distinct cutoff points that indicate where a type of nominal

may be optionally accusative. This finding is in contrast to ergative languages, in which it is commonly found that DSM is the inverse of DOM.

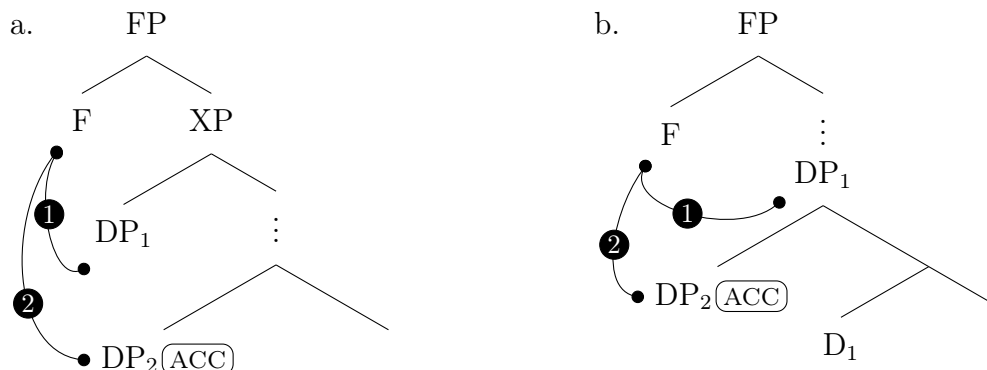
6.4 On the dependency between nominals in dependent case assignment

In the second half of Chapter 5, I proposed a movement-based account of DAM, under which nominals that bear a feature [DAM] may be targeted by probes on phase head for movement to their specifiers. Specifically, [DAM]-bearing objects move to Spec,vP, while [DAM]-bearing embedded subjects move to Spec,CP in finite bare CPs, Spec,DP in finite and nonfinite NCs, and Spec,nP in converbial adjuncts. In these positions, I proposed that accusative case is assigned to these arguments via a dependent case mechanism which crucially relies on these arguments being in a local domain with another nominal.

Next, I discussed novel data from Khalkha subject clauses, which constitute a previously unnoticed empirical domain where the predictions of two dependent case approaches – a configurational approach and an Agree-based approach – diverge. Under a configurational approach (e.g. Yip et al. 1987; Marantz 1991; McFadden 2004; Baker and Vinokurova 2010; Baker 2015; Levin and Preminger 2015; a.m.o.), dependent case is assigned via configurational rules, which dictate which of two (or more) nominals in a local c-command relation receives case marking. Under an Agree-based approach (Clem and Deal 2024; Poole 2024), dependent case on a nominal is the result of a nominal being the target of a second cycle of Agree. While both approaches predict that accusative case may be assigned to the lower of two nominals under local c-command, an Agree-based approach additionally predicts that accusative case may be assigned to a nominal *contained* within another nominal, i.e. under a containment relation. This is because the requirement of local c-command is encoded into configurational rules themselves, whereas what matters under an Agree-based approach is for two (or more) nominals to be c-commanded by an Agreeing functional head. This is a

novel prediction that has not been noticed in Clem and Deal 2024 and Poole 2024. The two dependencies are illustrated in (392).

(392) *Two types of dependency between nominals under dependent case via Agree*



Specifically, I argued that the availability of accusative subjects in subject clauses in Khalkha – which differs across my three consultants – supports the idea that containment relations (392b) are relevant for accusative case assignment. This finding provides preliminary empirical support for an Agree-based view of dependent case assignment in Khalkha. I further proposed that genitive case in Khalkha is assigned via a dependency with D⁰, which accounts for the availability of genitive subjects only in finite and nonfinite NCs – embedded clause types that contain a DP layer. I also argued that case-unmarked objects and case-unmarked subjects in Khalkha are due to the fact that these nominals do not contain features that correspond to a specific vocabulary item for case in Khalkha. As such, they surface as a morphological default.

More broadly, in arguing that a containment relation between nominals should be relevant for dependent case assignment, this chapter offered a novel perspective on the nature of the dependency between nominals in dependent case assignment. Under the Agree-based approach to dependent accusative, the only necessary requirement is that an Agreeing functional head locally c-commands and Agrees with two goal nominals. While the two nominals do still need to be located within the same spellout domain (i.e. be local to each other), the

Agree-based approach predicts that the exact nature of their relationship, e.g. whether its one of c-command or containment, is not crucial for dependent accusative assignment.

6.5 Concluding remarks

This dissertation has demonstrated the value of investigating both nominal and clausal structure in accounting for the symmetries and asymmetries observed in the morphosyntactic properties of four different embedded clause types in Khalkha and Kalmyk. Through a detailed investigation of the four embedded clause types, this dissertation has also contributed to our understanding of the types of embedded clause structures that may be found across the world's languages.

In the course of my investigation on Khalkha, the dissertation has also engaged with several issues of broader theoretical significance in syntax. I have discussed how the interactions between nominal and clausal domains may constrain the kinds of nominalised clauses we find across languages; the idea that different syntactic operations may be sensitive to different locality domains within a single language; the similarities and differences between DOM and DSM in accusative languages; how DAM interacts with dependent case assignment; and how different approaches to dependent case assignment may yield different insights and predictions regarding the nature of the dependency between nominals. Taken together, the chapters of this dissertation demonstrate how clausal nominalisation serves as a rich empirical domain for investigating varied questions in syntactic theory, and offer promising directions for further empirical and theoretical research.

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Appendix A: Mongolian Cyrillic and Romanisation

This dissertation adopts the BGN/PCGN 1964 romanisation system for Mongolian Cyrillic (source) with some modifications, as shown below.

Mongolian Cyrillic	Romanisation	Mongolian Cyrillic	Romanisation
А а	A a	Р р	R r
Б б	B b	С с	S s
В в	V v	Т т	T t
Г г	G g	У у	U u
Д д	D d	Ү ү	Ü ü
Е е	Yö yö	Ф ф	F f
Ё ё	Yo yo	Х х	H h
Ж ж	J j	Ц ц	Ts ts
З з	Z z	Ч ч	Ch ch
И и	I i	Ш ш	Sh sh
Й	i (after another vowel)	Щ щ	Shch shch
К к	K k	Ъ	”
Л л	L l	Ы	i
М м	M m	Ь	,
Н н	N n	Э э	E e
О о	O o	Ю ю	Yu yu
Ө ө	Ö ö	Я я	Ya ya
П п	P p		